

# riemann-channel: lab book

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## Contents

|           |                                                                            |           |
|-----------|----------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Frontmatter, Status Vocabulary, and Scope</b>                           | <b>6</b>  |
| 1.1       | Aim . . . . .                                                              | 6         |
| 1.2       | The organising picture . . . . .                                           | 6         |
| 1.3       | Status vocabulary . . . . .                                                | 6         |
| 1.4       | Lockstep rules . . . . .                                                   | 12        |
| <b>2</b>  | <b>Conventions and Notation</b>                                            | <b>13</b> |
| 2.1       | Standing conventions . . . . .                                             | 13        |
| 2.2       | Clashes found and resolved . . . . .                                       | 14        |
| 2.3       | Notation table . . . . .                                                   | 15        |
| <b>3</b>  | <b>Definitions</b>                                                         | <b>22</b> |
| 3.1       | The two sides . . . . .                                                    | 22        |
| 3.2       | Graphs . . . . .                                                           | 22        |
| 3.3       | Channels and matrix product states . . . . .                               | 23        |
| <b>4</b>  | <b>Definitions, Continued</b>                                              | <b>24</b> |
| 4.1       | The Riemann channel . . . . .                                              | 24        |
| 4.2       | Weil–LPS channels . . . . .                                                | 24        |
| 4.3       | Curves . . . . .                                                           | 25        |
| 4.4       | The continuous dictionary . . . . .                                        | 25        |
| <b>5</b>  | <b>Definitions, Continued: the Phantasm Campaign</b>                       | <b>27</b> |
| <b>6</b>  | <b>Definitions, Continued: Flows on Complexes and their Twists</b>         | <b>29</b> |
| <b>7</b>  | <b>Definitions, Continued: Finite BC States and Covariant Generators</b>   | <b>31</b> |
| <b>8</b>  | <b>Definitions, Continued: Boson–Fermion cMPS Rings</b>                    | <b>32</b> |
| <b>9</b>  | <b>Definitions, Continued: Graded Lattice Tensors and Lifted Frobenius</b> | <b>33</b> |
| <b>10</b> | <b>What the Riemann Hypothesis Has Become</b>                              | <b>34</b> |
| 10.1      | The one identity, and its two readings . . . . .                           | 34        |
| 10.2      | A finite permutation: the analogue is free . . . . .                       | 34        |
| 10.3      | An infinite permutation: a statement about a different matrix . . . . .    | 34        |
| 10.4      | A graph: Ramanujan . . . . .                                               | 35        |

|           |                                                                                |           |
|-----------|--------------------------------------------------------------------------------|-----------|
| 10.5      | A matrix product state: one correlation length . . . . .                       | 35        |
| 10.6      | The primes: continuous length, and the sign . . . . .                          | 36        |
| 10.7      | The Hilbert space where this is literally true . . . . .                       | 37        |
| 10.8      | Why it is still hard, in this language . . . . .                               | 37        |
| 10.9      | Erratum on words . . . . .                                                     | 37        |
| 10.10     | Summary . . . . .                                                              | 38        |
| <b>11</b> | <b>The Riemann Channel: Bost–Connes at Critical Temperature, Compressed</b>    | <b>39</b> |
| 11.1      | Setting . . . . .                                                              | 39        |
| 11.2      | The symbol is inner . . . . .                                                  | 39        |
| 11.3      | The compressed semigroup . . . . .                                             | 39        |
| 11.4      | Where the primes are . . . . .                                                 | 40        |
| 11.5      | Numerical check of the symbol . . . . .                                        | 41        |
| 11.6      | Ring norms: the trace of the semigroup is the prime measure . . . . .          | 41        |
| 11.7      | The Bost–Connes phase operators as an MPO over the prime chain . . . . .       | 42        |
| 11.8      | What is not established . . . . .                                              | 43        |
| <b>12</b> | <b>The Phantasm I: What the Prime Measure Forces</b>                           | <b>44</b> |
| 12.1      | Prior art: the grading is Deninger’s, the sign is Connes’s . . . . .           | 44        |
| 12.2      | Rigidity: the supertrace determines the graded spectrum . . . . .              | 44        |
| 12.3      | Positivity forces the parities . . . . .                                       | 45        |
| 12.4      | The no-go: numerators are odd . . . . .                                        | 46        |
| 12.5      | What exists: the compressed semigroup is the odd sector . . . . .              | 47        |
| <b>13</b> | <b>The Phantasm II: Vacuum Decay, the Prime Chain, and the Two Halves</b>      | <b>48</b> |
| 13.1      | Prior art on graded bonds . . . . .                                            | 48        |
| 13.2      | Vacuum decay: the zeros as odd coherences, the vacuum as fixed point . . . . . | 48        |
| 13.3      | The prime chain: the Gibbs fixed point, and no zeros . . . . .                 | 49        |
| 13.4      | Entropy of the bond state and its cutoffs . . . . .                            | 50        |
| 13.5      | The two halves, and the guiding observation . . . . .                          | 50        |
| <b>14</b> | <b>The Phantasm III: Stationarity, Symmetry, and Generator Freedom</b>         | <b>52</b> |
| 14.1      | The state on the finite phase algebra . . . . .                                | 52        |
| 14.2      | The complete finite feasibility problem . . . . .                              | 53        |
| 14.3      | Odd spectral freedom and coupling freedom . . . . .                            | 54        |
| 14.4      | What is still missing . . . . .                                                | 54        |
| <b>15</b> | <b>The Fermion Parity of a Boson–Fermion cMPS Ring</b>                         | <b>55</b> |
| 15.1      | Inputs . . . . .                                                               | 55        |
| 15.2      | Ordered contraction and the Fock embedding . . . . .                           | 55        |
| 15.3      | The overlap and parity theorem . . . . .                                       | 57        |
| <b>16</b> | <b>Graded cMPS Rings: Supertrace and Superdeterminant</b>                      | <b>59</b> |
| <b>17</b> | <b>Weil–LPS Channels: Exactly Ramanujan Quantum Expanders</b>                  | <b>62</b> |
| 17.1      | The object . . . . .                                                           | 62        |
| 17.2      | Exact Ramanujan behaviour . . . . .                                            | 63        |
| 17.3      | The quantum Ihara zeta of these channels . . . . .                             | 64        |
| 17.4      | What the joint spectrum looks like . . . . .                                   | 64        |

|           |                                                                                                   |           |
|-----------|---------------------------------------------------------------------------------------------------|-----------|
| 17.5      | Reading                                                                                           | 65        |
| 17.6      | Not established                                                                                   | 65        |
| <b>18</b> | <b>A Matrix-Product Formulation of the Weil Conjectures for Quadratic Artin–Schreier Curves</b>   | <b>66</b> |
| 18.1      | The shift picture                                                                                 | 66        |
| 18.2      | What was computed                                                                                 | 67        |
| 18.3      | RH is unitarity                                                                                   | 67        |
| 18.4      | Where it stops, and why that is the right place                                                   | 67        |
| 18.5      | Not established                                                                                   | 68        |
| <b>19</b> | <b>Artin–Schreier II: the Sign Law, the Super-Transfer Matrix, and the Weil Representation</b>    | <b>69</b> |
| 19.1      | The two quadratic forms and the exact sign law                                                    | 69        |
| 19.2      | The Frobenius block and the super-transfer matrix                                                 | 70        |
| 19.3      | The transfer unitary is a Weil-representation operator                                            | 70        |
| <b>20</b> | <b>Curve Counts as Twisted Ring Norms I: Inputs and the Supersingular Boundary</b>                | <b>72</b> |
| 20.1      | Inputs                                                                                            | 72        |
| 20.2      | Where the obvious Polya–Hilbert Hamiltonian lives, and why it stops                               | 73        |
| <b>21</b> | <b>Curve Counts as Twisted Ring Norms II: the Ordinary Elliptic Curve</b>                         | <b>75</b> |
| 21.1      | Tier A: the ordinary elliptic curve                                                               | 75        |
| 21.2      | Digits in base $\pi$                                                                              | 76        |
| <b>22</b> | <b>Curve Counts as Twisted Ring Norms III: the Bosonic No-Go</b>                                  | <b>77</b> |
| 22.1      | The bosonic no-go and its bookkeeping                                                             | 77        |
| <b>23</b> | <b>Curve Counts as Twisted Ring Norms IV: Genus Two</b>                                           | <b>79</b> |
| 23.1      | Inputs                                                                                            | 79        |
| 23.2      | The block equations and two impossibilities                                                       | 79        |
| 23.3      | Existence                                                                                         | 80        |
| 23.4      | Jacobian, Rosati metric, gauge                                                                    | 81        |
| 23.5      | Assessment for the Phantasm (observations, not theorems)                                          | 82        |
| <b>24</b> | <b>Curve Counts as Twisted Ring Norms V: Explicit Examples, Entanglement, Parent Hamiltonians</b> | <b>83</b> |
| 24.1      | The five states                                                                                   | 83        |
| 24.2      | Entanglement                                                                                      | 83        |
| 24.3      | Parent Hamiltonians                                                                               | 84        |
| <b>25</b> | <b>Deligne’s Proof, Told Through Regular Graphs</b>                                               | <b>85</b> |
| 25.1      | One graph, two sides                                                                              | 85        |
| 25.2      | The same statement for a curve, with one sign changed                                             | 85        |
| 25.3      | What is cheap and what is not                                                                     | 86        |
| 25.4      | The engine: power up, dominate, take roots                                                        | 86        |
| 25.5      | The pointwise theorem, in twisted-graph language                                                  | 86        |
| 25.6      | Big monodromy, and the abelian failure                                                            | 87        |
| 25.7      | The sign                                                                                          | 88        |

|           |                                                                                              |            |
|-----------|----------------------------------------------------------------------------------------------|------------|
| 25.8      | From pointwise to global: slice, bound, multiply . . . . .                                   | 88         |
| 25.9      | What the proof does not do . . . . .                                                         | 89         |
| 25.10     | What this says for the project . . . . .                                                     | 89         |
| <b>26</b> | <b>The Ihara–Bass Formula for an Arbitrary Kraus Family</b>                                  | <b>91</b>  |
| 26.1      | Setting and conventions . . . . .                                                            | 91         |
| 26.2      | Statement . . . . .                                                                          | 92         |
| 26.3      | Proof of the theorem . . . . .                                                               | 92         |
| 26.4      | Proofs of the corollaries . . . . .                                                          | 94         |
| 26.5      | Provenance and relation to the literature . . . . .                                          | 94         |
| 26.6      | Numerical checks . . . . .                                                                   | 94         |
| 26.7      | What this does and does not give . . . . .                                                   | 95         |
| <b>27</b> | <b>Weil Positivity for an Arbitrary Transfer Operator</b>                                    | <b>96</b>  |
| 27.1      | Setting . . . . .                                                                            | 96         |
| 27.2      | Positivity is the one-sided bound . . . . .                                                  | 96         |
| 27.3      | A duality makes it two-sided: the inflow identity . . . . .                                  | 96         |
| 27.4      | Kraus families: which pairing buys which half . . . . .                                      | 97         |
| <b>28</b> | <b>Weil Positivity, Continued: Continuous Time, Hilbert–Pólya, the Zeta Entry, Prior Art</b> | <b>99</b>  |
| 28.1      | Continuous time: cMPS and Lindblad generators . . . . .                                      | 99         |
| 28.2      | The Hilbert–Pólya inner product, and what positivity does not see . . . . .                  | 99         |
| 28.3      | The zeta instance and the prior art . . . . .                                                | 100        |
| 28.4      | Two observations by the orchestrator . . . . .                                               | 102        |
| <b>29</b> | <b>An Ihara Zeta for Complexes I: the Literature and the Two Flows</b>                       | <b>104</b> |
| 29.1      | The prototype: $\text{PGL}_3$ . . . . .                                                      | 104        |
| 29.2      | All ranks: an alternating product over facet dimension . . . . .                             | 104        |
| 29.3      | What happens off the buildings . . . . .                                                     | 105        |
| 29.4      | The graded reading, in the literature’s own words . . . . .                                  | 105        |
| 29.5      | Hypothesis register . . . . .                                                                | 106        |
| 29.6      | The two flows are not the same flow . . . . .                                                | 107        |
| <b>30</b> | <b>An Ihara Zeta for Complexes II: the Graded Geodesic Determinant</b>                       | <b>108</b> |
| 30.1      | Torsion, compression, and the obstruction . . . . .                                          | 108        |
| 30.2      | The graded reading and what it does not say . . . . .                                        | 108        |
| 30.3      | An incidence matrix, and why the total is a Berezinian . . . . .                             | 109        |
| 30.4      | Arbitrary weights and Kraus rings . . . . .                                                  | 109        |
| 30.5      | The three smallest exact tests . . . . .                                                     | 110        |
| <b>31</b> | <b>An Ihara Zeta for Complexes III: Twists, Quantum Expanders, Numerics</b>                  | <b>111</b> |
| 31.1      | Gauge triviality and the voltage quotient . . . . .                                          | 111        |
| 31.2      | Quantum expanders of type $\tilde{A}_{n-1}$ . . . . .                                        | 111        |
| 31.3      | Numerics on the smallest $\tilde{A}_2$ Ramanujan complex . . . . .                           | 112        |
| 31.4      | What this does and does not give . . . . .                                                   | 113        |

|                                                                                                                    |            |
|--------------------------------------------------------------------------------------------------------------------|------------|
| <b>32 Prior Art for the Quantum Ihara Zeta</b>                                                                     | <b>115</b> |
| 32.1 The object under examination . . . . .                                                                        | 115        |
| 32.2 Verdict . . . . .                                                                                             | 115        |
| 32.3 Matsuura and Ohta . . . . .                                                                                   | 116        |
| 32.4 Watanabe and Fukumizu . . . . .                                                                               | 116        |
| 32.5 Bordenave and Collins . . . . .                                                                               | 117        |
| 32.6 Hastings and Harrow . . . . .                                                                                 | 117        |
| 32.7 Iyer, Jain, Jordan and Somma . . . . .                                                                        | 117        |
| 32.8 Quantum-walk zetas and the secondary sources . . . . .                                                        | 118        |
| 32.9 What was not found, and the limits of the search . . . . .                                                    | 118        |
| 32.10 Corrections . . . . .                                                                                        | 119        |
| <b>33 The Continuous Dictionary</b>                                                                                | <b>120</b> |
| 33.1 Standard inputs . . . . .                                                                                     | 120        |
| 33.2 Cited facts . . . . .                                                                                         | 120        |
| 33.3 The circle comb and why it is blind to zeros . . . . .                                                        | 121        |
| 33.4 The Lindbladian of the vector fields is the Casimir, and the Laplacian on the $K$ -invariant sector . . . . . | 122        |
| <b>34 The Continuous Dictionary II: Selberg Tower and the Cusp Comb</b>                                            | <b>123</b> |
| 34.1 The geodesic flow: continuous Ihara–Bass . . . . .                                                            | 123        |
| 34.2 The cusp of the modular surface is the prime comb . . . . .                                                   | 123        |
| 34.3 The dictionary . . . . .                                                                                      | 124        |
| 34.4 Not established . . . . .                                                                                     | 125        |
| <b>35 Open Problems and Dead Routes</b>                                                                            | <b>126</b> |
| 35.1 Open problems, in order . . . . .                                                                             | 126        |
| 35.2 Dead routes . . . . .                                                                                         | 127        |
| <b>36 Reproducibility Map</b>                                                                                      | <b>128</b> |
| <b>37 Lab Log</b>                                                                                                  | <b>131</b> |

# 1 Frontmatter, Status Vocabulary, and Scope

## 1.1 Aim

This lab book records an attempt to formulate the Riemann hypothesis in the language of tensor networks and operational quantum mechanics: channels, Kraus operators, transfer matrices, matrix-product states. The aim is a formulation that captures the mechanisms of the proofs that exist (Weil for curves, Deligne for varieties over finite fields, Ihara–Bass and the Ramanujan property for regular graphs) and connects more directly to quantum mechanics than the Hilbert–Pólya programme does. Nothing here proves anything about the Riemann hypothesis. The Markdown notes under `notes/` are the exploratory record; this book is their distillation, and every statement here is tagged.

## 1.2 The organising picture

Every zeta function in sight has two sides. Side A is a nonnegative count of closed orbits (prime cycles of a graph, points of a curve, prime powers with weight  $\log p$ ) and its Euler product; it is a product state, the primes do not talk to each other. Side B is a small non-Hermitian operator whose power sums reproduce the counts; its eigenvalues are the zeros. The Riemann hypothesis, in every instance, is the statement that all nontrivial eigenvalues of side B share one modulus. Section 2 fixes the notation and Section 3 the definitions used for both sides throughout.

## 1.3 Status vocabulary

Every theorem-like environment in this book carries exactly one status tag, printed at its right margin, equal to the status derived by `scripts/labbook_check.py` from `db/claims.tsv`:

- **proved**: a written proof exists under `notes/` and a reviewer other than the author has returned `VERDICT <id>: VALID` in a file under `notes/reviews/`. Both author and reviewers to date are Claude models (single family); no second model family has been used.
- **sketched**: written by the author, not yet attacked.
- **numerical**: established by a script whose output is captured under `outputs/`; never more than numerical evidence.
- **cited**: a statement of the literature, backed by a quote in `db/provenance.tsv` that the gate byte-checks against the arXiv TeX source under `refs/src/`.
- **conjectured**, **open**, **refuted**, **assumed**: as the words say; a claim that depends on any of the first three prints **unsupported**, and one that depends on an assumption prints with the suffix **-conditional**.

Definitions carry no tag; they are **stipulated** (ours) or **quoted** (a source’s, with provenance) in `db/definitions.tsv`. The derived-status table generated from the claims database is reproduced here.

| id                                                  | kind        | derived status  | assumptions |
|-----------------------------------------------------|-------------|-----------------|-------------|
| <code>prop:bc-residue-state</code>                  | proposition | <b>sketched</b> | -           |
| <code>prop:bc-no-finite-phase-representation</code> | proposition | <b>sketched</b> | -           |
| <code>prop:bc-symmetry-state-extension</code>       | proposition | <b>sketched</b> | -           |
| <code>thm:bc-covariant-gks-cone</code>              | theorem     | <b>sketched</b> | -           |

| id                                    | kind        | derived status     | assumptions                |
|---------------------------------------|-------------|--------------------|----------------------------|
| thm:bc-critical-cone-dimensions       | theorem     | sketched           | -                          |
| prop:bc-weyl-covariance-collapse      | proposition | sketched           | -                          |
| prop:bc-symplectic-odd-counterexample | proposition | sketched           | -                          |
| prop:bc-compatible-prime-coupling     | proposition | sketched           | -                          |
| num:bc-symmetry-generators            | numerical   | numerical          | -                          |
| obs:one-identity                      | observation | sketched           | -                          |
| obs:rh-finite-permutation             | observation | sketched           | -                          |
| obs:rh-small-matrix                   | observation | sketched           | -                          |
| obs:rh-graph-ramanujan                | observation | sketched           | -                          |
| obs:rh-mps-correlation-length         | observation | sketched           | -                          |
| obs:rh-flow-pairing                   | observation | sketched           | -                          |
| obs:hilbert-polya-hermitian           | observation | sketched           | -                          |
| prop:scattering-inner                 | proposition | sketched           | -                          |
| num:scattering-symbol                 | numerical   | numerical          | -                          |
| prop:functional-model-modes           | proposition | sketched           | -                          |
| prop:phase-identity                   | proposition | sketched           | -                          |
| prop:ringnorm-trace                   | proposition | proved-conditional | asm:explicit-formula-trace |
| num:ringnorm-3000                     | numerical   | numerical          | -                          |
| prop:bc-mpo                           | proposition | sketched           | -                          |
| num:bc-mpo                            | numerical   | numerical          | -                          |
| conj:stinespring-jump-form            | conjecture  | open               | -                          |
| num:weil-lps-ramanujan                | numerical   | numerical          | -                          |
| num:weil-lps-rh                       | numerical   | numerical          | -                          |
| prop:weil-lps-exactly-ramanujan       | proposition | sketched           | -                          |
| conj:weil-lps-hecke                   | conjecture  | open               | -                          |
| prop:as-transfer-matrix               | proposition | proved-conditional | asm:normal-basis           |
| prop:as-unitarity                     | proposition | proved             | -                          |
| num:artin-schreier-mps                | numerical   | numerical          | -                          |
| obs:as-nonquadratic                   | observation | sketched           | -                          |
| obs:deligne-ingredients               | observation | sketched           | -                          |
| obs:abelian-obstruction               | observation | sketched           | -                          |
| thm:qihara-general                    | theorem     | proved             | -                          |
| cor:qihara-kraus                      | corollary   | proved             | -                          |
| cor:qihara-unitary                    | corollary   | proved             | -                          |
| cor:qihara-poles                      | corollary   | proved             | -                          |
| obs:rh-iff-ramanujan-channel          | observation | sketched           | -                          |
| num:qihara-unitary                    | numerical   | numerical          | -                          |
| num:qihara-general                    | numerical   | numerical          | -                          |
| cit:mo-extended-ihara                 | citedfact   | cited              | -                          |
| cit:mo-adjoint-weights                | citedfact   | cited              | -                          |
| cit:wf-arbitrary-weights              | citedfact   | cited              | -                          |
| cit:bc-nonbacktracking                | citedfact   | cited              | -                          |
| cit:harrow-transfer                   | citedfact   | cited              | -                          |
| cit:iyer-exact-ramanujan              | citedfact   | cited              | -                          |
| cit:sunada-attribution                | citedfact   | cited              | -                          |
| obs:prior-art-verdict                 | observation | sketched           | -                          |
| conj:kraus-ramanujan                  | conjecture  | open               | -                          |
| conj:assemble-over-p                  | conjecture  | open               | -                          |
| obs:prime-by-prime-blind              | observation | sketched           | -                          |
| asm:fuchsian-geometry                 | assumption  | assumed            | -                          |
| asm:laplace-spectrum-compact          | assumption  | assumed            | -                          |
| asm:selberg-zeta-entire               | assumption  | assumed            | -                          |

| id                                | kind        | derived status       | assumptions                |
|-----------------------------------|-------------|----------------------|----------------------------|
| asm:heat-semigroup                | assumption  | assumed              | -                          |
| asm:zeta-standard                 | assumption  | assumed              | -                          |
| asm:gamma-standard                | assumption  | assumed              | -                          |
| cit:dz-guillemin                  | citedfact   | cited                | -                          |
| cit:dfg-band                      | citedfact   | cited                | -                          |
| cit:dz-ruelle-selberg             | citedfact   | cited                | -                          |
| cit:marklof-trace                 | citedfact   | cited                | -                          |
| cit:fgkp-casimir-laplacian        | citedfact   | cited                | -                          |
| cit:fjs-scattering-det            | citedfact   | cited                | -                          |
| cit:mv-continuous-term            | citedfact   | cited                | -                          |
| prop:circle-comb                  | proposition | proved               | -                          |
| prop:prime-circles-orbital        | proposition | proved               | -                          |
| prop:finite-euler-spectrum        | proposition | proved-conditional   | asm:gamma-standard, asm:ze |
| prop:lindblad-sum-of-squares      | proposition | proved               | -                          |
| prop:casimir-laplacian            | proposition | proved               | -                          |
| prop:quantum-lindblad-tensor      | proposition | proved               | -                          |
| prop:poincare-jacobian            | proposition | proved               | -                          |
| prop:flat-tower                   | proposition | proved-conditional   | asm:fuchsian-geometry      |
| prop:tower-divisor-band           | proposition | proved-conditional   | asm:fuchsian-geometry, asm |
| prop:graph-tower-ruelle           | proposition | proved-conditional   | asm:fuchsian-geometry, asm |
| prop:cuspidal-prime-comb          | proposition | proved-conditional   | asm:gamma-standard, asm:ze |
| prop:damping-identity             | proposition | proved-conditional   | asm:gamma-standard, asm:ze |
| num:selberg-lindblad              | numerical   | numerical            | -                          |
| obs:continuous-dictionary         | observation | sketched-conditional | asm:fuchsian-geometry, asm |
| conj:quantum-lindblad-gap         | conjecture  | open                 | -                          |
| obs:spt-protection-is-sign-data   | observation | sketched             | -                          |
| thm:weil-positivity-finite        | theorem     | proved               | -                          |
| thm:weil-duality-pairing          | theorem     | proved               | -                          |
| prop:weil-orbit-negative          | proposition | proved               | -                          |
| prop:weil-reciprocal-only         | proposition | proved               | -                          |
| prop:kraus-conjugation-symmetry   | proposition | proved               | -                          |
| thm:kraus-inverse-pairing-duality | theorem     | proved               | -                          |
| prop:kraus-both-pairings-unitary  | proposition | proved               | -                          |
| thm:kraus-weil-criterion          | theorem     | proved               | -                          |
| prop:kraus-no-duality-example     | proposition | proved               | -                          |
| thm:weil-positivity-continuous    | theorem     | proved               | -                          |
| thm:weil-line-duality             | theorem     | proved               | -                          |
| thm:cmps-ring-norms               | theorem     | proved               | -                          |
| prop:hp-inner-product-discrete    | proposition | proved               | -                          |
| prop:hp-inner-product-continuous  | proposition | proved               | -                          |
| prop:weil-blind-jordan            | proposition | proved               | -                          |
| obs:weil-zeta-dictionary          | observation | sketched             | -                          |
| obs:huang-boundedness             | observation | proved               | -                          |
| obs:kraus-dichotomy               | observation | proved               | -                          |
| num:weil-positivity               | numerical   | numerical            | -                          |
| cit:weil-criterion                | citedfact   | cited                | -                          |
| cit:li-criterion                  | citedfact   | cited                | -                          |
| cit:huang-graph-criterion         | citedfact   | cited                | -                          |
| cit:hastings-bound                | citedfact   | cited                | -                          |
| cit:herglotz-bochner              | citedfact   | cited                | -                          |
| cit:suzuki-kernel                 | citedfact   | cited                | -                          |
| cit:pt-real-spectrum              | citedfact   | cited                | -                          |

| id                                       | kind        | derived status       | assumptions                |
|------------------------------------------|-------------|----------------------|----------------------------|
| obs:ringnorm-sign-correction             | observation | sketched-conditional | asm:explicit-formula-trace |
| asm:explicit-formula-trace               | assumption  | assumed              | -                          |
| asm:zero-count-location                  | assumption  | assumed              | -                          |
| asm:positive-distributions               | assumption  | assumed              | -                          |
| asm:landau-laplace                       | assumption  | assumed              | -                          |
| asm:lax-phillips-modes                   | assumption  | assumed              | -                          |
| asm:weil-curves                          | assumption  | assumed              | -                          |
| cit:lidskii-trace                        | citedfact   | cited                | -                          |
| cit:landau-dirichlet                     | citedfact   | cited                | -                          |
| cit:deninger-graded-lefschetz            | citedfact   | cited                | -                          |
| cit:connes-absorption                    | citedfact   | cited                | -                          |
| lem:supertrace-distribution              | lemma       | proved               | -                          |
| thm:supertrace-rigidity                  | theorem     | proved               | -                          |
| cor:forced-net-spectrum                  | corollary   | proved-conditional   | asm:explicit-formula-trace |
| thm:parity-forced-finite                 | theorem     | proved-conditional   | asm:explicit-formula-trace |
| prop:parity-infinite-dominated           | proposition | proved-conditional   | asm:explicit-formula-trace |
| obs:parity-infinite-open                 | observation | sketched             | -                          |
| thm:no-ungraded-trace                    | theorem     | proved               | -                          |
| prop:trace-supertrace-criterion          | proposition | proved               | -                          |
| cor:curve-counts-graded                  | corollary   | proved-conditional   | asm:weil-curves            |
| prop:riemann-graded-generator            | proposition | proved-conditional   | asm:explicit-formula-trace |
| prop:ladder-parity-free                  | proposition | proved-conditional   | asm:explicit-formula-trace |
| obs:phantasm-scope                       | observation | sketched             | -                          |
| cit:fmps-supertrace                      | citedfact   | cited                | -                          |
| cit:fcmps-graded-bond                    | citedfact   | cited                | -                          |
| cit:cmps-schmidt-fixed-point             | citedfact   | cited                | -                          |
| cit:bc-kms-unique                        | citedfact   | cited                | -                          |
| thm:vacuum-decay-finite                  | theorem     | proved               | -                          |
| cor:vacuum-decay-traces                  | corollary   | proved               | -                          |
| prop:absorbing-vacuum-channel            | proposition | proved-conditional   | asm:lax-phillips-modes     |
| obs:zero-coherences-populations          | observation | sketched             | -                          |
| prop:vacuum-renewal-trivial-entanglement | proposition | proved               | -                          |
| prop:normal-kms-gibbs                    | proposition | proved               | -                          |
| lem:mm1-spectrum                         | lemma       | proved               | -                          |
| prop:prime-chain-lindbladian             | proposition | proved               | -                          |
| prop:prime-chain-offdiagonal             | proposition | proved               | -                          |
| cor:gibbs-delocalisation                 | corollary   | proved               | -                          |
| num:riemann-graded-trace                 | numerical   | numerical            | -                          |
| num:bc-entropy                           | numerical   | numerical            | -                          |
| obs:gibbs-entanglement-hagedorn          | observation | sketched             | -                          |
| obs:character-sector-grading             | observation | sketched-conditional | asm:as-lpolynomial, asm:fr |
| obs:complementary-halves                 | observation | sketched             | -                          |
| obs:phantasm-established-scope           | observation | sketched             | -                          |
| conj:phantasm-both-halves                | conjecture  | open                 | -                          |
| conj:galois-graded-bond                  | conjecture  | open                 | -                          |
| asm:normal-basis                         | assumption  | assumed              | -                          |
| asm:as-lpolynomial                       | assumption  | assumed              | -                          |
| asm:weil-implementer-exact               | assumption  | assumed              | -                          |
| asm:frobenius-semisimple                 | assumption  | assumed              | -                          |
| cit:weil-character-formula               | citedfact   | cited                | -                          |
| cit:weil-egorov-uniqueness               | citedfact   | cited                | -                          |
| lem:gauss-quadratic-form                 | lemma       | proved               | -                          |

| id                              | kind        | derived status     | assumptions                |
|---------------------------------|-------------|--------------------|----------------------------|
| thm:as-sign-law                 | theorem     | proved-conditional | asm:normal-basis           |
| cor:as-periodic-sign            | corollary   | proved-conditional | asm:as-lpolynomial, asm:no |
| prop:as-sign-counterexamples    | proposition | proved-conditional | asm:normal-basis           |
| thm:as-frobenius-block          | theorem     | proved-conditional | asm:as-lpolynomial, asm:no |
| thm:as-super-transfer           | theorem     | proved-conditional | asm:as-lpolynomial, asm:fr |
| thm:as-symplectic-map           | theorem     | proved-conditional | asm:weil-implementer-exact |
| cor:as-weil-bound-fixed-space   | corollary   | proved-conditional | asm:normal-basis, asm:weil |
| num:artin-schreier-super        | numerical   | numerical          | -                          |
| cit:kl-identity                 | citedfact   | cited              | -                          |
| cit:kl-rh-four-way              | citedfact   | cited              | -                          |
| cit:kangyu-alternating          | citedfact   | cited              | -                          |
| cit:llp-geodesic-flow           | citedfact   | cited              | -                          |
| cit:llp-one-sided-rh            | citedfact   | cited              | -                          |
| cit:kamber-lp                   | citedfact   | cited              | -                          |
| cit:storm-reduction             | citedfact   | cited              | -                          |
| cit:bcds-triangulation          | citedfact   | cited              | -                          |
| cit:no-general-complex-zeta     | citedfact   | cited              | -                          |
| cit:deitmar-alternating         | citedfact   | cited              | -                          |
| cit:dz-order-chi                | citedfact   | cited              | -                          |
| cit:hashimoto-hprime            | citedfact   | cited              | -                          |
| cit:knill-sdet                  | citedfact   | cited              | -                          |
| cit:mo-dirac                    | citedfact   | cited              | -                          |
| asm:h-point                     | assumption  | assumed            | -                          |
| asm:h-llp                       | assumption  | assumed            | -                          |
| asm:h-kl                        | assumption  | assumed            | -                          |
| asm:h-ky                        | assumption  | assumed            | -                          |
| asm:h-ky-local                  | assumption  | assumed            | -                          |
| asm:h-kl-table                  | assumption  | assumed            | -                          |
| asm:h-lsv                       | assumption  | assumed            | -                          |
| asm:h-strong                    | assumption  | assumed            | -                          |
| asm:h-kamber                    | assumption  | assumed            | -                          |
| thm:ordered-vs-building         | theorem     | proved-conditional | asm:h-kl, asm:h-llp, asm:h |
| thm:total-zeta-vertex-l         | theorem     | proved-conditional | asm:h-kl                   |
| thm:cochain-cancellation        | theorem     | proved             | -                          |
| thm:pointed-euler-exponent      | theorem     | proved-conditional | asm:h-ky                   |
| thm:universal-bass-schur        | theorem     | proved             | -                          |
| cor:bass-every-graph            | corollary   | proved             | -                          |
| prop:tetrahedron-obstruction    | proposition | proved             | -                          |
| thm:graded-superdeterminant     | theorem     | proved             | -                          |
| prop:vertex-l-no-zeros          | proposition | proved-conditional | asm:h-kl                   |
| prop:parity-only                | proposition | proved             | -                          |
| prop:three-circles              | proposition | proved-conditional | asm:h-kl, asm:h-kl-table   |
| thm:incidence-two-schur         | theorem     | proved             | -                          |
| thm:berezinian-total            | theorem     | proved             | -                          |
| thm:twisted-bass-schur          | theorem     | proved             | -                          |
| thm:kraus-ring-positivity       | theorem     | proved             | -                          |
| prop:positivity-no-go-scope     | proposition | proved             | -                          |
| obs:graded-determinant-natural  | observation | proved             | -                          |
| num:small-complex-tests         | numerical   | numerical          | -                          |
| prop:pure-gauge-full-complex    | proposition | proved             | -                          |
| thm:artin-voltage-decomposition | theorem     | proved             | -                          |
| thm:euler-block-bookkeeping     | theorem     | proved-conditional | asm:h-ky, asm:h-ky-local   |

| id                                    | kind        | derived status       | assumptions                |
|---------------------------------------|-------------|----------------------|----------------------------|
| thm:quantum-vertex-transfer           | theorem     | proved-conditional   | asm:h-kl, asm:h-ky, asm:h- |
| prop:coverage-converse                | proposition | proved               | -                          |
| prop:one-sided-rh-variable            | proposition | proved-conditional   | asm:h-kamber, asm:h-llp    |
| prop:rank-three-dualities             | proposition | proved-conditional   | asm:h-kl-table, asm:h-lsv  |
| prop:weil-positivity-blocks           | proposition | proved               | -                          |
| num:a2-complex                        | numerical   | numerical            | -                          |
| num:a2-vertex-spectrum                | numerical   | numerical            | -                          |
| num:a2-identity                       | numerical   | numerical            | -                          |
| num:a2-flow-operators                 | numerical   | numerical            | -                          |
| num:a2-block-spectra                  | numerical   | numerical            | -                          |
| num:a2-quantum-twist                  | numerical   | numerical            | -                          |
| num:a2-voltage-euler                  | numerical   | numerical            | -                          |
| conj:vertex-collapse-characterisation | conjecture  | conjectured          | -                          |
| asm:graded-fock                       | assumption  | assumed              | -                          |
| cit:hcov-overlap-parity               | citedfact   | cited                | -                          |
| lem:ordered-contraction               | lemma       | sketched-conditional | asm:graded-fock            |
| lem:cmps-fock-isometry                | lemma       | sketched-conditional | asm:graded-fock            |
| lem:dyson-series                      | lemma       | sketched             | -                          |
| thm:cmps-overlap-parity               | theorem     | sketched-conditional | asm:graded-fock            |
| prop:cmps-graded-parity               | proposition | sketched-conditional | asm:graded-fock            |
| thm:cmps-twisted-supertrace           | theorem     | sketched-conditional | asm:graded-fock            |
| prop:cmps-twist-translation           | proposition | sketched-conditional | asm:graded-fock            |
| thm:cmps-sdet                         | theorem     | sketched-conditional | asm:graded-fock            |
| num:cmps-parity-supertrace            | numerical   | numerical            | -                          |
| obs:cmps-supertrace-reading           | observation | sketched-conditional | asm:graded-fock            |
| cit:frob-count-formula                | citedfact   | cited                | -                          |
| cit:hasse-degree-bound                | citedfact   | cited                | -                          |
| cit:serre-tate-lift                   | citedfact   | cited                | -                          |
| asm:deuring-lift                      | assumption  | assumed              | -                          |
| cit:degree-norm-index                 | citedfact   | cited                | -                          |
| cit:toral-lefschetz                   | citedfact   | cited                | -                          |
| cit:lenstra-structure                 | citedfact   | cited                | -                          |
| cit:latimer-macduffee                 | citedfact   | cited                | -                          |
| asm:supersingular-newton              | assumption  | assumed              | -                          |
| asm:kraus-hs-setting                  | assumption  | assumed              | -                          |
| thm:as-quadratic-supersingular        | theorem     | proved               | -                          |
| prop:as-finite-order                  | proposition | proved-conditional   | asm:as-lpolynomial, asm:fr |
| prop:ring-cut-rank                    | proposition | proved               | -                          |
| num:cut-rank-binary                   | numerical   | numerical            | -                          |
| conj:amplitude-locality               | conjecture  | conjectured          | -                          |
| thm:toral-frobenius-count             | theorem     | proved-conditional   | asm:deuring-lift           |
| prop:toral-zeta                       | proposition | proved-conditional   | asm:deuring-lift           |
| thm:hodge-doubling-tensor             | theorem     | proved-conditional   | asm:deuring-lift           |
| prop:fourier-shift-flat-trace         | proposition | proved-conditional   | asm:deuring-lift           |
| thm:elliptic-ph-unitary               | theorem     | proved-conditional   | asm:deuring-lift           |
| prop:integral-marking-gauge           | proposition | proved-conditional   | asm:deuring-lift           |
| prop:cyclic-digits-carries            | proposition | proved               | -                          |
| prop:carry-automaton-no-count         | proposition | proved               | -                          |
| obs:drafted-hs-bound-false            | observation | sketched             | -                          |
| thm:bosonic-blocks-hs                 | theorem     | proved               | -                          |
| lem:word-trace-cs                     | lemma       | proved               | -                          |
| thm:bosonic-genus-bound               | theorem     | proved-conditional   | asm:weil-curves            |

| id                                | kind        | derived status       | assumptions                 |
|-----------------------------------|-------------|----------------------|-----------------------------|
| prop:euler-bookkeeping            | proposition | proved               | -                           |
| prop:odd-species-cancellation     | proposition | proved               | -                           |
| thm:infinite-bond-bosonic-nogo    | theorem     | proved-conditional   | asm:kraus-hs-setting, asm:  |
| num:weil-ring-norm-check          | numerical   | numerical            | -                           |
| num:ring-norm-tensor              | numerical   | numerical            | -                           |
| cit:rosati-positive               | citedfact   | cited                | -                           |
| cit:cm-lift-frobenius             | citedfact   | cited                | -                           |
| cit:hodge-positive-forms          | citedfact   | cited                | -                           |
| asm:cm-hodge-type                 | assumption  | assumed              | -                           |
| asm:cm-polarised-lift             | assumption  | assumed              | -                           |
| lem:genus-two-blocks              | lemma       | proved               | -                           |
| prop:vacuum-ansatz-impossible     | proposition | proved-conditional   | asm:weil-curves             |
| thm:nine-letter-tensor            | theorem     | proved-conditional   | asm:weil-curves             |
| prop:f5-base-change               | proposition | proved-conditional   | asm:cm-polarised-lift, asm: |
| prop:three-letter-feasibility     | proposition | proved-conditional   | asm:cm-polarised-lift       |
| thm:f5-certified-tensor           | theorem     | proved-conditional   | asm:cm-polarised-lift       |
| conj:three-letter-universal       | conjecture  | conjectured          | -                           |
| num:ring-norm-certificate         | numerical   | numerical            | -                           |
| num:genus-two-cm-diagonal         | numerical   | numerical            | -                           |
| cor:genus-two-physical-norm       | corollary   | sketched-conditional | asm:cm-polarised-lift, asm: |
| prop:jacobian-curve-projector     | proposition | proved-conditional   | asm:cm-hodge-type, asm:cm-  |
| thm:rosati-ph-genus-two           | theorem     | proved-conditional   | asm:cm-hodge-type, asm:cm-  |
| prop:cm-type-gauge                | proposition | proved-conditional   | asm:cm-hodge-type           |
| obs:phantasm-bond-candidate       | observation | sketched-conditional | asm:zero-count-location     |
| obs:fermionic-jumps-necessary     | observation | sketched-conditional | asm:kraus-hs-setting, asm:  |
| obs:toral-contribution            | observation | sketched-conditional | asm:cm-hodge-type, asm:cm-  |
| obs:locality-dichotomy-restricted | observation | sketched-conditional | asm:deuring-lift, asm:weil  |
| prop:example-correlation-lengths  | proposition | sketched-conditional | asm:weil-curves             |
| prop:example-block-spectrum       | proposition | sketched             | -                           |
| obs:example-entanglement          | observation | sketched             | -                           |
| prop:example-parent-hamiltonian   | proposition | sketched             | -                           |
| num:ring-norm-examples            | numerical   | numerical            | -                           |

## 1.4 Lockstep rules

The gate (`make check`; `make ci` adds a fresh build) fails unless: every note under `notes/` is named by a shard header and every shard header names an existing note; every evidence script is named by a shard and has a captured output; every theorem-like environment has one label, one claims row, and one status tag equal to the derived status; every definition has one row and appears once; the notation macros are exactly the generated image of `db/notation.tsv` with no duplicate symbols; every provenance quote is found whole and contiguous at its stated locus; and the build has no undefined references or citations.

## 2 Conventions and Notation

Every symbol below is a macro generated from `db/notation.tsv`; shards may not define macros of their own, and the gate warns whenever a shard writes a reserved control sequence (such as the bare Greek letters) bare instead of through its macro. The table is regenerated by `make regen` and the gate fails if it is stale.

### 2.1 Standing conventions

1. **Channel normalisation.** A channel  $\Phi$  is normalised,  $\Phi(X) = \frac{1}{D} \sum_i U_i X U_i^\dagger$ , so that its leading eigenvalue is 1. The unnormalised sum is  $\Sigma = \sum_i \mathcal{E}_i = D\Phi$ . Every Ihara–Bass statement is written with  $\Sigma$  or  $D\Phi$ ; the notes that wrote  $\Phi$  without the  $1/D$  are read with this convention.
2. **The size parameter  $q$ .** One letter, three roles that never co-occur: the field size in Section 18, the LPS prime in Section 17, and  $D - 1$  for graphs and channels. In every role the critical modulus is  $\sqrt{q}$ . For a channel with  $D$  Kraus operators,  $q := D - 1$  is a standing identification.
3. **Ramanujan units.** Spectral bounds are reported in adjacency units: a channel eigenvalue is multiplied by  $D = q + 1$  and compared with  $2\sqrt{q}$ . Hastings’ bound  $\lambda_H = 2\sqrt{D - 1}/D$  is the same statement in channel units.
4. **The zeta variable is  $u$ ,** in  $\zeta_X(u)$ ,  $\zeta_T(u)$ ,  $Z_C(u)$  and  $\det(1 - uT)$ . The ring-length variable of the length flow is  $L$ , never  $u = t/2$ .
5. **Zeros.** A nontrivial zero is  $\rho = \sigma + i\gamma$  with  $\sigma$  its real part;  $\beta$  is reserved for the inverse temperature of the Bost–Connes system. The Riemann hypothesis reads  $\sigma = \frac{1}{2}$ .
6. **Indices.**  $n$  is the matrix size in  $M_n$  and, in the ring pictures, the number of sites, which for Artin–Schreier curves is the extension degree of  $\mathbb{F}_{q^n}$ . Walk, word and prime-cycle lengths are  $\ell$ . Counts are  $N_\ell$ .
7. **Column stacking.**  $\text{Ad}(A) = \bar{A} \otimes A$  as an  $n^2 \times n^2$  matrix, so  $\text{Tr Ad}(A) = |\text{Tr } A|^2$ .
8. **Non-backtracking operator.**  $T(v \otimes |i\rangle) = \sum_{j \neq \bar{i}} \mathcal{E}_i v \otimes |j\rangle$ : the label  $i$  names the superoperator applied, and the next label may not be its reversal. Word products are composed right to left,  $\mathcal{E}_{i_\ell} \cdots \mathcal{E}_{i_1}$ , with the cyclic condition  $i_{k+1} \neq \bar{i}_k$  for  $k \in \mathbb{Z}/\ell$ .
9. **Artin–Schreier sign.**  $\alpha_i = -\lambda_i(E)$ , so that  $\mathcal{S}_n = -\sum_i \alpha_i^n$  for all  $n$  and  $L = \prod_i (1 - \alpha_i u)$ , the standard normalisation of the  $L$ -function of an exponential sum. The curve count is  $N_n = q^n + 1 - \sum_i \alpha_i^n$ .
10. **Canonical form.** An MPS is in (left) canonical form when  $\sum_k A_k^\dagger A_k = 1$ ; then the transfer matrix has leading eigenvalue 1. The spectral normalisation used in the translation note (leading eigenvalue 1, rest inside the unit disc) is a consequence, not the definition.
11. **Status vocabulary.** The book uses only the tags of Section 3 (Definition 4.17). The two-value “established / not established” of the notes and the prior-art verdicts (**SAME**, **RELATED**, **NOT**) are read as **sketched/numerical** versus **open**, and as a provenance vocabulary confined to Section 32, respectively.
12. **Trace.**  $\text{Tr}$  is the operator trace; the field trace is  $\text{tr}_{q^n/q}$ .

## 2.2 Clashes found and resolved

The extraction of 2026-09-12 (`notes/extract/notation-and-definitions.md`) found 146 symbols across the seven notes, with these collisions. Each resolution below is what the table enforces.

- $\psi$  carried five meanings: Chebyshev function, additive character, Lévy exponent, digamma, MPS ring state. Now:  $\psi$  is Chebyshev;  $e_q$  the additive character;  $\kappa_\sigma$  the Lévy exponent;  $(\Gamma'/\Gamma)$  the digamma;  $\Psi_n$  the ring state.
- $T$  carried five: the non-backtracking operator, the length-flow generator, the curve-zeta variable, the Heisenberg–Weyl operator, the Hecke operator. Now:  $T$  is the non-backtracking operator;  $\mathcal{G}$  the generator;  $u$  the variable;  $T(u, v)$  the Heisenberg–Weyl operator (only with two arguments);  $A_q$  the Hecke adjacency operator.
- $\rho$  carried three: zero, density matrix, twist. Now:  $\rho$  is a zero;  $X$  a channel input;  $\varrho$  a twisting representation.
- $\beta$  carried three: real part of a zero, inverse temperature, Deligne weight. Now:  $\sigma$ ,  $\beta$ ,  $w$ .
- $\xi$  was both the completed zeta and the correlation length, within two pages. Now:  $\xi$  is the completed zeta;  $\ell_k^c$  the correlation length;  $\vartheta_k$  the phase.
- $S$  was the scattering symbol, the exponential sum and the start map. Now:  $S$ ,  $\mathcal{S}_n$ ,  $\mathbf{S}$ .
- $J$  was the Artin–Schreier range, the reversal operator and a splitting matrix. Now:  $J$  (italic),  $J$  (sans-serif),  $\mathbf{J}$  (bold).
- $D$  was the Kraus count, the dilation unitary, a diagonal degree matrix and the degree operator. Now:  $D$ ,  $\mathbf{D}$ , and  $\mathcal{D}(u)(u)$ ; the diagonal degree matrix of a general graph is never needed on a bouquet.
- $E$  was the MPS transfer matrix and the Artin–Schreier transfer matrix (the same concept, kept as  $E$ ) and the superoperators of a Kraus family (now  $\mathcal{E}_i$ , a different glyph).
- $W, V$ : the edge and vertex spaces of the non-backtracking construction are  $\mathcal{W}, \mathcal{V}$ ;  $W(g)$  and  $W_\pm$  are the Weil representation and its blocks;  $|V|$  is a vertex count.
- $A$ : the adjacency matrix  $A$ ; Kraus operators  $A_i$ ; the LPS Hecke adjacency  $A_q$ .
- $F$ : the Fourier transform  $\mathcal{F}$ ; Frobenius  $\text{Fr}$ ; the numerical window sum  $F$ ; Weil’s zero-side function  $F_W$ .
- $M$ : a generic matrix  $M$ ; the chirp  $\mathbf{M}$ .
- $\sigma$ : the real part of  $s$ ; the reversal involution is now  $\iota$ .  $\Sigma$  is only ever  $\Sigma = D\Phi$ .
- $\alpha$ : Frobenius eigenvalues carry a subscript,  $\alpha_i$ ; a quaternion is bare  $\alpha$ .
- $m$ : the number of Kraus pairs; walk length is  $\ell$ .  $N$ :  $\dim \mathcal{V}$ ; counts are  $N_\ell$ .
- $\mu$ : an edge eigenvalue; the Bost–Connes isometries are  $\mathfrak{m}_n$ .  $G$ : a group; the window width is  $G_w$ .  $\theta$ : the normal-basis generator  $\theta$ ; phases are  $\vartheta_k$ .  $s$ : the complex variable; a square root of  $q \bmod p$  is  $s_q$ .

## 2.3 Notation table

| symbol                 | meaning                                                                                              | domain                | where  |
|------------------------|------------------------------------------------------------------------------------------------------|-----------------------|--------|
| $\varrho_b$            | prescribed stationary density in a finite BC bond model                                              | finite BC states      | RC-02E |
| $\varrho^{\text{res}}$ | diagonal matrix extension of a finite BC phase marginal                                              | finite BC states      | RC-02E |
| $P$                    | local parity operator $x \rightarrow -x$ on $l\{2(F_p)$                                              | finite Weil model     | RC-02E |
| $\mathfrak{C}$         | cone of stationary covariant finite GKLS generators                                                  | finite channels       | RC-02E |
| $C$                    | unnormalised Choi matrix of a linear map                                                             | finite channels       | RC-02E |
| $\mathcal{D}$          | completely depolarizing channel $D_d(X) = \text{Tr}(X) I/d$                                          | finite channels       | RC-02E |
| $\text{Tr}$            | trace                                                                                                | general               | RC-01  |
| $\text{Ad}$            | adjoint action, $\text{Ad}(A): X \rightarrow A X A^\dagger$                                          | channels              | RC-01  |
| $\text{Re}$            | real part                                                                                            | general               | RC-01  |
| $\text{Im}$            | imaginary part                                                                                       | general               | RC-01  |
| $M_n$                  | $n \times n$ complex matrices with the Hilbert-Schmidt inner product                                 | channels              | RC-01  |
| $\mathcal{V}$          | vertex space of the non-backtracking construction ( $V = M_n$ in the channel case), $N = \dim V$     | quantum Ihara         | RC-08  |
| $\mathcal{W}$          | edge space $V(x) C\{D$                                                                               | quantum Ihara         | RC-08  |
| $T$                    | non-backtracking (Hashimoto) edge operator, classical on directed edges or quantum on the edge space | graphs, quantum Ihara | RC-02  |
| $\mathcal{G}$          | generator of the length flow whose trace is the prime measure, eigenvalues 1, $\rho$ , -2, -4, ...   | Riemann channel       | RC-03  |
| $\mathcal{E}_i$        | $i$ -th superoperator of a family ( $E_i = \text{Ad}(A_i)$ in the Kraus case)                        | quantum Ihara         | RC-08  |
| $A_i$                  | $i$ -th Kraus operator                                                                               | channels              | RC-01  |
| $U_i$                  | $i$ -th unitary Kraus operator                                                                       | channels              | RC-01  |
| $\Sigma$               | unnormalised transfer matrix $\sum_i E_i (= D \Phi$ for a unital channel)                            | channels, MPS         | RC-01  |
| $\Phi$                 | unital channel $(1/D) \sum_i U_i X U_i^\dagger$ , or a general channel                               | channels              | RC-01  |
| $D$                    | number of Kraus operators (= degree of the bouquet); $D = 2m$                                        | channels              | RC-01  |
| $m$                    | number of Kraus pairs, $D = 2m$                                                                      | quantum Ihara         | RC-08  |
| $\iota$                | fixed-point-free reversal involution on the index set $[D]$ , $\bar{i} = \iota(i)$                   | quantum Ihara         | RC-08  |
| $\bar{\cdot}$          | reversed index $\iota(i)$                                                                            | quantum Ihara         | RC-08  |
| $J$                    | backtracking operator $J(v(x)  i\rangle) = E_i v(x)  \bar{i}\rangle$                                 | quantum Ihara         | RC-08  |
| $R$                    | propagate-along-edge map $R: W \rightarrow V$                                                        | quantum Ihara         | RC-08  |
| $S$                    | start-any-edge map $S: V \rightarrow W$                                                              | quantum Ihara         | RC-08  |
| $\mathcal{A}(u)$       | deformed adjacency $u \sum_i E_i (1 - u\{2 E_{\{\bar{i}\}} E_i)\{-1\}$                               | quantum Ihara         | RC-08  |
| $\mathcal{D}(u)$       | deformed degree $u\{2 \sum_i E_{\{\bar{i}\}} E_i (1 - u\{2 E_{\{\bar{i}\}} E_i)\{-1\}$               | quantum Ihara         | RC-08  |
| $\zeta_T$              | zeta function $\det(1 - uT)\{-1\}$ of a non-backtracking operator                                    | quantum Ihara         | RC-08  |

| symbol       | meaning                                                                                                                    | domain                | where |
|--------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------|-------|
| $\zeta_\Phi$ | quantum Ihara zeta of a unital channel with unitary Kraus operators                                                        | quantum Ihara         | RC-02 |
| $\zeta_X$    | Ihara zeta of a graph X                                                                                                    | graphs                | RC-02 |
| $A$          | adjacency matrix of a graph                                                                                                | graphs                | RC-02 |
| $A.$         | adjacency operator of the LPS Cayley graph for norm-q quaternions (Hecke operator)                                         | Weil-LPS              | RC-05 |
| $q$          | size parameter: field size (Artin-Schreier), LPS prime (Weil-LPS), and $D - 1 = \text{degree} - 1$ for graphs and channels | all                   | RC-01 |
| $p$          | a rational prime; the modulus of the Weil representation in RC-05                                                          | all                   | RC-01 |
| $\lambda_2$  | second largest eigenvalue (or singular value) of a channel or graph, normalised to leading eigenvalue 1                    | expanders             | RC-02 |
| $\lambda_H$  | Hastings bound $2 \sqrt{(D-1)/D}$                                                                                          | expanders             | RC-02 |
| $\mu$        | eigenvalue of the edge operator T; pairs $\mu \mu' = q$ with an adjacency eigenvalue $\lambda$                             | graphs                | RC-02 |
| $\ell$       | length of a walk, word or prime cycle                                                                                      | graphs, quantum Ihara | RC-02 |
| $N.$         | closed non-backtracking walk count of length $\ell$ , or point count over $F_{q^\ell}$                                     | graphs, curves        | RC-02 |
| $P$          | prime cycle of a graph (primitive closed non-backtracking walk up to rotation)                                             | graphs                | RC-02 |
| $\text{Fr}$  | Frobenius                                                                                                                  | curves                | RC-06 |
| $\varrho$    | representation twisting a zeta (holonomy, Artin-Ihara twist, adjoint representation)                                       | graphs, Deligne       | RC-07 |
| $X$          | density matrix or general operator input of a channel                                                                      | channels              | RC-01 |
| $\rho$       | nontrivial zero of zeta, $\rho = \sigma + i \gamma$                                                                        | number theory         | RC-01 |
| $\sigma$     | real part of $s$ (or of a zero)                                                                                            | number theory         | RC-01 |
| $\gamma.$    | imaginary part of the $n$ -th zero                                                                                         | number theory         | RC-01 |
| $\xi$        | completed zeta $\xi(s) = (1/2) s (s-1) \pi^{\hat{s}/2} \Gamma(s/2) \zeta(s)$                                               | number theory         | RC-04 |
| $S$          | Lax-Phillips scattering symbol $S(\tau) = \xi(1 - 2i\tau)/\xi(1 + 2i\tau)$                                                 | Riemann channel       | RC-04 |
| $\tau$       | Mellin variable on the dilation group; dilation by $e^{\hat{t}}$ acts as $e^{\hat{t}}$                                     | Riemann channel       | RC-04 |
| $K_S$        | model space $H^{\hat{2}}_2 \text{ minus } S H^{\hat{2}}_2$                                                                 | Riemann channel       | RC-04 |
| $Z(t)$       | compressed dilation semigroup $P_{\{K_S\}} e^{\hat{t}}$ restricted to $K_S$                                                | Riemann channel       | RC-04 |
| $\Lambda$    | von Mangoldt function                                                                                                      | number theory         | RC-01 |
| $\psi$       | Chebyshev function $\psi(x) = \sum_{n \leq x} \Lambda(n)$                                                                  | number theory         | RC-03 |
| $e_q$        | additive character of $F_q$ composed with the trace                                                                        | curves                | RC-06 |
| $\kappa.$    | Bost-Connes Levy exponent $\kappa_\sigma(\tau) = \log \zeta(\sigma - i\tau) - \log \zeta(\sigma)$                          | Bost-Connes           | RC-04 |
| $\nu.$       | von Mangoldt jump measure $\nu_\sigma = \sum p^{\hat{k}} \{-k\sigma\} / k \delta_{\{k \log p\}}$                           | Bost-Connes           | RC-04 |
| $\beta$      | inverse temperature of the Bost-Connes system                                                                              | Bost-Connes           | RC-04 |
| $L$          | ring length variable of the length flow (prime powers sit at $L = \log n$ )                                                | Riemann channel       | RC-03 |

| symbol                        | meaning                                                                                                 | domain          | where      |
|-------------------------------|---------------------------------------------------------------------------------------------------------|-----------------|------------|
| $F$                           | windowed zero sum $F(u)$ of the ring-norm test                                                          | Riemann channel | RC-04      |
| $E$                           | transfer matrix of a matrix product state (the Artin-Schreier transfer matrix is the instance of RC-06) | MPS, curves     | RC-02      |
| $\Psi$                        | periodic matrix product state on $n$ sites                                                              | MPS             | RC-02      |
| $\langle \Psi   \Psi \rangle$ | ring norm of the periodic $n$ -site state, $= \text{Tr } E^{\hat{\cdot}}_n$                             | MPS             | RC-02      |
| $\ell^c$                      | $k$ -th correlation length of a transfer matrix                                                         | MPS             | RC-03      |
| $h$                           | entropy per site (log of the leading eigenvalue)                                                        | MPS             | RC-03      |
| $\mathbb{F}_q$                | finite field with $q$ elements                                                                          | curves          | RC-01      |
| $\mathbb{F}_p$                | finite field with $p$ elements                                                                          | curves, LPS     | Weil-RC-01 |
| $\mathbb{F}_{q^n}$            | extension of degree $n$                                                                                 | curves          | RC-06      |
| $\mathcal{S}$                 | exponential sum $S_n(g) = \sum_{x \in \mathbb{F}_{q^n}} e_{-q}(\text{Tr } g(x))$                        | curves          | RC-06      |
| $J$                           | range of the quadratic Artin-Schreier polynomial, $\dim E = q^{\frac{1}{2}} J$                          | curves          | RC-06      |
| $\alpha$                      | Frobenius eigenvalues (roots of the L-polynomial), $ \alpha  = \sqrt{q}$                                | curves          | RC-06      |
| $SL_2$                        | special linear group                                                                                    | Weil-LPS        | RC-05      |
| $PSL_2$                       | projective special linear group                                                                         | Weil-LPS        | RC-05      |
| $W$                           | Weil representation $W(g)$ of $SL_2(\mathbb{F}_p)$ on $\ell^2_2(\mathbb{F}_p)$                          | Weil-LPS        | RC-05      |
| $W_{\pm}$                     | even and odd blocks of the Weil representation, dimensions $(p \pm 1)/2$                                | Weil-LPS        | RC-05      |
| $\Phi$                        | Weil-LPS channel for the LPS prime $q$ (with superscript $pm$ for the block)                            | Weil-LPS        | RC-05      |
| $\alpha$                      | integer quaternion of norm $q$ in the LPS normalisation                                                 | Weil-LPS        | RC-05      |
| $g_{\alpha}$                  | image of the quaternion $\alpha$ in $SL_2(\mathbb{F}_p)$                                                | Weil-LPS        | RC-05      |
| $Z$                           | clock operator on $\ell^2_2(\mathbb{F}_p)$                                                              | Weil-LPS        | RC-05      |
| $X$                           | shift operator on $\ell^2_2(\mathbb{F}_p)$                                                              | Weil-LPS        | RC-05      |
| $\mathcal{F}$                 | Fourier transform on $\ell^2_2(\mathbb{F}_p)$                                                           | Weil-LPS        | RC-05      |
| $M$                           | chirp operator $\text{diag } e_p(x^2/2)$                                                                | Weil-LPS        | RC-05      |
| $D$                           | dilation unitary on $\ell^2_2(\mathbb{F}_p)$                                                            | Weil-LPS        | RC-05      |
| $I$                           | splitting matrix $I = ((x, y), (y, -x))$ with $x^2 + y^2 + 1 = 0 \pmod{p}$                              | Weil-LPS        | RC-05      |
| $J$                           | splitting matrix $J = ((0, 1), (-1, 0))$                                                                | Weil-LPS        | RC-05      |
| $T(u, v)$                     | Heisenberg-Weyl operator $X^u Z^v$                                                                      | Weil-LPS        | RC-05      |
| $c$                           | pointwise weight scalar of a twisted coefficient system ( $ \text{eigenvalue}  = c^{\frac{1}{2}}$ )     | Deligne         | RC-07      |
| $L$                           | L-function (of a curve with coefficients, or Artin-Ihara)                                               | graphs, curves  | RC-02      |
| $(\Gamma'/\Gamma)$            | digamma function in the archimedean term of the explicit formula                                        | number theory   | RC-04      |
| $w$                           | weight of a pure coefficient system ( $ \text{Frobenius eigenvalue}  = q^{\frac{w}{2}}$ )               | Deligne         | RC-07      |
| $\text{tr}_{q^n/q}$           | field trace from $\mathbb{F}_{q^n}$ to $\mathbb{F}_q$                                                   | curves          | RC-06      |
| $Z_C$                         | zeta function of a curve $C$ over $\mathbb{F}_q$ , $\exp \sum_n N_n u^n / n$                            | curves          | RC-06      |

| symbol                | meaning                                                                                                                                           | domain          | where  |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|
| $G_w$                 | Gaussian window width of the ring-norm test                                                                                                       | Riemann channel | RC-04  |
| $F_W$                 | zero-side function $\sum_{\rho} e^{\frac{1}{2}(\rho - 1/2)u}$ whose positive-definiteness is Weil's criterion                                     | number theory   | RC-03  |
| $\vartheta.$          | k-th oscillation frequency of a transfer matrix, $\lambda_k = \exp(-1/\ell \hat{c}_k + i \vartheta_k)$                                            | MPS             | RC-03  |
| $\theta$              | generator of a normal basis of $F_{\{q\}^n}$                                                                                                      | curves          | RC-06  |
| $s_q$                 | a square root of q modulo p                                                                                                                       | Weil-LPS        | RC-05  |
| $m.$                  | Bost-Connes isometry $\mu_n$                                                                                                                      | Bost-Connes     | RC-04  |
| $U.$                  | Shor's unitary $x \rightarrow ax \bmod N$ , the level-N Galois symmetry of Bost-Connes                                                            | Bost-Connes     | RC-04  |
| $\hat{g}$             | test window in the explicit formula (Fourier transform g)                                                                                         | number theory   | RC-04  |
| $t$                   | semigroup time in $Z(t)$ ; dilation by $e^{\hat{t}}$                                                                                              | Riemann channel | RC-04  |
| $u$                   | the zeta variable, in $\zeta_X(u)$ , $\zeta_T(u)$ , $Z_C(u)$ , $\det(1 - uT)$                                                                     | all             | RC-01  |
| $\ell^2(\cdot)$       | Hilbert space of square-summable functions on a set                                                                                               | general         | RC-01  |
| $\mathcal{O}_{Dw}$    | Dwork's p-adic transfer operator, multiplication by the splitting function then decimation                                                        | curves          | RC-06  |
| $\text{dec}_q$        | decimation $x^{\hat{m}} \rightarrow x^{\{m/q\}}$ on p-adic power series                                                                           | curves          | RC-06  |
| $\ell$ -adic          | ell-adic (cohomology); this ell is not a walk length                                                                                              | Deligne         | RC-06  |
| $\mathcal{X}$         | generator of the geodesic flow on $\Gamma \backslash G$ , $X = H/2$ through left-invariant fields; $(e^{\hat{t}}\{-tX\}f)(m) = f(\phi_{\{-t\}}m)$ | Selberg         | RC-09B |
| $H$                   | left-invariant field of $H = \text{diag}(1, -1)$                                                                                                  | Selberg         | RC-09B |
| $E$                   | left-invariant field of $E = ((0,1), (1,0))$                                                                                                      | Selberg         | RC-09B |
| $W$                   | left-invariant field of $W = ((0,1), (-1,0))$ , the compact direction                                                                             | Selberg         | RC-09B |
| $\Omega$              | Casimir $(1/4)(H^{\hat{2}} + E^{\hat{2}} - W^{\hat{2}})$                                                                                          | Selberg         | RC-09B |
| $\Delta$              | positive hyperbolic Laplacian $-y^{\hat{2}}(d_x^{\hat{2}} + d_y^{\hat{2}})$                                                                       | Selberg         | RC-09B |
| $\lambda.$            | j-th Laplace eigenvalue, $\lambda_j = 1/4 + r_j^{\hat{2}}$                                                                                        | Selberg         | RC-09B |
| $r.$                  | spectral parameter of $\lambda_j$ ; $r_0 = i/2$ for the constant mode, $r_j = i \nu_j$ for exceptional eigenvalues                                | Selberg         | RC-09B |
| $Z_{\text{Sel}}$      | Selberg zeta $\prod_{\gamma} \prod_{k \geq 0} (1 - e^{\hat{t}}\{-(s+k) \ell_{\gamma}\})$                                                          | Selberg         | RC-09B |
| $\zeta_R$             | Ruelle zeta $\prod_{\gamma} (1 - e^{\hat{t}}\{-s \ell_{\gamma}\}) = Z_{\text{Sel}}(s)/Z_{\text{Sel}}(s+1)$                                        | Selberg         | RC-09B |
| $P_{\gamma}$          | linearised Poincare map of the closed geodesic $\gamma$                                                                                           | Selberg         | RC-09B |
| $\text{Tr}^b$         | Guillemin flat trace                                                                                                                              | Selberg         | RC-09B |
| $\varsigma$           | Laplace variable of the flat trace (the resonance variable), never the Selberg variable s                                                         | Selberg         | RC-09B |
| $\Lambda_{\text{fl}}$ | Laplace transform of the flat trace, $\int_0^{\infty} e^{\hat{t}}\{-\text{varsigma } t\} \text{Tr}\{\}_{\text{flat}} e^{\hat{t}}\{-tX\} dt$       | Selberg         | RC-09B |
| $D_{\text{tow}}$      | the tower $\prod_{j \geq 1} Z_{\text{Sel}}(\text{varsigma} + j)$ , the flat dynamical determinant                                                 | Selberg         | RC-09B |

| symbol                     | meaning                                                                                                                                                            | domain          | where  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|
| $\mathcal{B}_0$            | nonconstant first band $\{-1/2 + i r_j : j \geq 1\}$                                                                                                               | Selberg         | RC-09B |
| $\varphi$                  | scattering determinant of $\text{PSL}(2, \mathbb{Z})$ , $\sqrt{\pi} \Gamma(s-1/2) \zeta(2s-1) / (\Gamma(s) \zeta(2s))$                                             | modular         | RC-09B |
| $m_{sc}$                   | scattering multiplier $-(1/2)(\text{varphi}'/\text{varphi})(1/2 + i r)$                                                                                            | modular         | RC-09B |
| $C_{\text{cusp}}$          | inverse Fourier transform of the scattering multiplier                                                                                                             | modular         | RC-09B |
| $P_+$                      | positive-time prime measure $\sum_n \Lambda(n)/n \delta(t - 2 \log n)$                                                                                             | modular         | RC-09B |
| $P_{\text{ch}}$            | prime measure $\sum_n \Lambda(n) n^{\hat{\cdot}} \{-1/2\} \delta(t - 2 \log n)$ of the Riemann-channel note                                                        | modular         | RC-09B |
| $A_{\text{arch}}$          | archimedean-plus-boundary distribution $1/2 - 1/(4 \sinh( t /2))$ away from zero                                                                                   | modular         | RC-09B |
| $S_{\text{triv}}$          | trivial sub-multiset of the spectrum, removed before rescaling                                                                                                     | Weil positivity | RC-08B |
| $r$                        | critical radius; $\sqrt{D-1}$ in the unitary Kraus case                                                                                                            | Weil positivity | RC-08B |
| $\mathcal{A}_{\text{ret}}$ | retained multiset $\text{spec}(X)$ minus $S_{\text{triv}}$ (algebraic multiplicities)                                                                              | Weil positivity | RC-08B |
| $n.$                       | rescaled trace sequence $n_l = r^{\hat{\cdot}} \{-1\} (\text{Tr } X^{\hat{\cdot}} l - \sum \{S_{\text{triv}}\} \mu^{\hat{\cdot}} l)$ , $n_{-1} = \text{conj}(n_l)$ | Weil positivity | RC-08B |
| $W$                        | Weil (Toeplitz) form $\sum_{l,m} c_l \text{conj}(c_m) \nu_{\{-l-m\}}$                                                                                              | Weil positivity | RC-08B |
| $M$                        | mode-pairing form $\sum_{\mu} f_c(\mu/r) \text{conj}(f_c(J(\mu)/r))$                                                                                               | Weil positivity | RC-08B |
| $J$                        | reflection in the critical circle, $J(\mu) = r^{\hat{\cdot}} \{2\} / \text{conj}(\mu)$ ; fixed points $ \mu  = r$                                                  | Weil positivity | RC-08B |
| $\mathcal{L}$              | Lindblad-type generator $K x + x K^{\hat{\cdot}} \dagger + \sum_j R_j x R_j^{\hat{\cdot}} \dagger$ on $M_n$                                                        | Weil positivity | RC-08B |
| $F$                        | rescaled continuous trace $F(t) = e^{\hat{\cdot}} \{-at\} (\text{Tr } e^{\hat{\cdot}} \{tL\} - \text{trivial part})$ , $F(-t) = \text{conj}(F(t))$                 | Weil positivity | RC-08B |
| $a$                        | critical real part, the line $\text{Re } \rho = a$ ; $a = -1/4$ for the Riemann channel                                                                            | Weil positivity | RC-08B |
| $f.$                       | polynomial $f_c(z) = \sum_l c_l z^{\hat{\cdot}} l$ of a finitely supported coefficient vector $c$                                                                  | Weil positivity | RC-08B |
| $R.$                       | reflection in the line $\text{Re } \rho = a$ , $R_a(\rho) = 2a - \text{conj}(\rho)$                                                                                | Weil positivity | RC-08B |
| $\varpi$                   | generic eigenvalue (mode) of a continuous transfer generator $L$                                                                                                   | Weil positivity | RC-08C |
| $\varkappa$                | eigenvalue of the unnormalised transfer matrix $\Sigma$ (the prover's $\alpha$ ); pairs with the two roots of $\mu^{\hat{\cdot}} \{2\} - \alpha \mu + q$           | Weil positivity | RC-08B |
| $\text{str}$               | supertrace: trace on the even part minus trace on the odd part of a $\mathbb{Z}_2$ -graded space                                                                   | graded          | RC-04B |
| $\text{comb}$              | prime comb $\sum_{n \geq 2} \Lambda(n) \delta(u - \log n)$ on $(0, \infty)$                                                                                        | number theory   | RC-04B |
| $\text{Tr}_{\text{dist}}$  | distributional trace of the compressed semigroup, $\sum_{\rho} e^{\hat{\cdot}} \{-\text{conj}(\rho) t/2\}$ , a definition by fiat                                  | Riemann channel | RC-04B |
| $\nu_{\text{net}}$         | net signed multiplicity $m(\lambda, +) - m(\lambda, -)$ of a graded spectral datum                                                                                 | graded          | RC-04B |
| $\mathfrak{B}$             | Riemann graded bond $C_+ (+) (K_S (+) l^{\hat{\cdot}} \{2(N_{\geq 1})\})_-$                                                                                        | Riemann channel | RC-04B |

| symbol                     | meaning                                                                                                                     | domain            | where  |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------|--------|
| $G$                        | graded generator 0 (+) B (+) $\text{diag}(-(k+1/2))$ on the Riemann graded bond                                             | Riemann channel   | RC-04B |
| $B$                        | generator of the compressed semigroup, $Z(t) = e^{\hat{\{tB\}}}$                                                            | Riemann channel   | RC-04B |
| $\mathcal{L}_{\text{vac}}$ | vacuum-decay Lindbladian on $B(C (+) H)$ : no-event part 0 (+) B, jump $ \text{vac}\rangle\langle j $                       | channels          | RC-04C |
| $\text{vac}$               | vacuum vector spanning the even line of the vacuum-decay model                                                              | channels          | RC-04C |
| $\Omega_{\text{vac}}$      | vacuum density $ \text{vac}\rangle\langle \text{vac} $ , the stationary state of the vacuum-decay Lindbladian               | channels          | RC-04C |
| $\mathcal{L}^{\text{pc}}$  | prime-chain detailed-balance Lindbladian at inverse temperature $\beta$ , jumps $\mu_p$ and $\mu_p^{\dagger}$               | Bost-Connes       | RC-04C |
| $\varrho^G$                | Bost-Connes Gibbs density $N^{\hat{\{}}\{-\beta\}}/\zeta(\beta)$ on $l^{\hat{\{}}2(N)$                                      | Bost-Connes       | RC-04C |
| $\mathcal{S}_{\text{ent}}$ | von Neumann entropy of a bond density (entanglement entropy of the half-line in the cMPS reading)                           | Bost-Connes       | RC-04C |
| TFD.                       | thermofield-double purification of the Gibbs density                                                                        | Bost-Connes       | RC-04C |
| $\mathbb{E}$               | super-transfer matrix of an Artin-Schreier curve: even block $E_0 (+) (1)$ , odd block the Frobenius blocks                 | curves            | RC-06B |
| $F$                        | corrected Frobenius block of the character sum $S_n(g)$ , $L(g, T) = \det(1 - T F_g)$                                       | curves            | RC-06B |
| $Q^{\circ}$                | ring quadratic form $\sum_j a_j \sum_i x_i x_{\{i+j\}}$ on $F_q^{\hat{\{}}n}$ (indices mod $n$ ), matrix $P(S)$             | curves            | RC-06B |
| $Q^{\text{tr}}$            | trace quadratic form $\text{tr}(g(x))$ in normal-basis coordinates, matrix $C P(S)$                                         | curves            | RC-06B |
| $P$                        | symmetrised coefficient Laurent polynomial $P_g(z) = \sum_j (a_j/2)(z^{\hat{\{}}j} + z^{\hat{\{}}\{-j\}})$                  | curves            | RC-06B |
| $S$                        | cyclic shift $(S x)_i = x_{\{i+1\}}$ on $F_q^{\hat{\{}}n}$ , the Frobenius in normal-basis coordinates                      | curves            | RC-06B |
| $\mathcal{R}$              | radical $\ker P_g(S)$ of the ring form on $F_q^{\hat{\{}}n}$                                                                | curves            | RC-06B |
| $\delta$                   | radical sign $\det(S \text{ restricted to the radical } R_n)$ in $\{+1, -1\}$                                               | curves            | RC-06B |
| $h$                        | period parameter: the smallest power of $q$ strictly greater than the order of vanishing of $z^{\hat{\{}}J} P_g(z)$ at $-1$ | curves            | RC-06B |
| $d$                        | rank deficiency of the ring form, $= \dim \ker(M_g^{\hat{\{}}n-1})$ , $= \log_q( S_n ^{\hat{\{}}2/q^{\hat{\{}}n})$          | curves            | RC-06B |
| $M$                        | symplectic matrix in $\text{Sp}(2J, F_q)$ implemented by the transfer unitary $E_g/\sqrt{q}$                                | curves            | RC-06B |
| $X$                        | centred Weyl operator $X_{\{u,v\}} s\rangle = e_q(v.(s + u/2)) s + u\rangle$ on $l^{\hat{\{}}2}(F_q^{\hat{\{}}J})$          | curves            | RC-06B |
| $\mathcal{U}$              | transfer unitary $E_g/\sqrt{q}$                                                                                             | curves            | RC-06B |
| $\text{Sp}$                | symplectic group                                                                                                            | curves            | RC-06B |
| $D$                        | transfer determinant $D_g(T) = \det(1 - T E_g)$                                                                             | curves            | RC-06B |
| $\mathcal{Z}$              | graded geodesic determinant (total $\zeta$ ) of a complex with specified geodesic data                                      | complexes         | RC-02D |
| $\text{sdet}$              | superdeterminant (Berezinian) of an even operator on a $\mathbb{Z}_2$ -graded space                                         | graded, complexes | RC-02D |

| symbol                  | meaning                                                                                                                                        | domain      | where  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------|
| $\mathfrak{S}$          | finite set of particle species of a boson-fermion cMPS                                                                                         | cMPS        | RC-02F |
| $\mathfrak{F}$          | graded Fock space over $L^{\hat{\{ \}}2}([0,L))$ for the species set                                                                           | cMPS        | RC-02F |
| $\Omega_{\mathfrak{F}}$ | Fock vacuum                                                                                                                                    | cMPS        | RC-02F |
| $(-1)^{\hat{F}}$        | fermion parity operator on the Fock space                                                                                                      | cMPS        | RC-02F |
| $\Delta_{\cdot}^{<}$    | open ordered simplex $\{0 < x_1 < \dots < x_n < L\}$                                                                                           | cMPS        | RC-02F |
| $\mathcal{K}_{\cdot}$   | ordered amplitude space $L^{\hat{\{ \}}2}(\text{ordered simplex } x \text{ species}^{\hat{\{ \}}n})$                                           | cMPS        | RC-02F |
| $\mathcal{J}$           | isometry from ordered amplitudes into the Fock space                                                                                           | cMPS        | RC-04E |
| $\Psi_{\cdot}^c$        | cMPS ring vector with boundary matrix B                                                                                                        | cMPS        | RC-02F |
| $c_{\cdot}$             | n-particle amplitude $c^{\hat{\{ \}}B_n}$ of a cMPS ring vector                                                                                | cMPS        | RC-02F |
| $\mathbb{T}$            | cMPS transfer generator $Q(x)1 + 1(x)\text{conj } Q + \sum_a s_a R_a(x) \text{conj } R_a$ on the doubled bond (subscript eta: fermionic signs) | cMPS        | RC-02F |
| $\Gamma_b$              | doubled-bond parity $P(x) \text{conj } P$                                                                                                      | cMPS        | RC-02F |
| $\Pi_{\cdot}$           | bond parity sector projections $(1 \pm P)/2$                                                                                                   | cMPS        | RC-02F |
| $\mathcal{E}_d$         | doubled transfer matrix $\sum_s A_s(x) \text{conj } A_s$ of a graded lattice tensor                                                            | MPS, graded | RC-02G |

### 3 Definitions

Each definition appears exactly once in the book. The column `merged_from` of `db/definitions.tsv` lists the notes and sections whose wordings were merged; where the wordings differed in substance, the difference is stated in the definition.

#### 3.1 The two sides

**Definition 3.1** (Side A and side B). For a zeta function written as  $\exp \sum_{\ell \geq 1} N_\ell u^\ell / \ell$  with nonnegative counts  $N_\ell$ , *side A* is the gas of rings: the counts, their Euler product over primitive closed orbits, and the product state they form. *Side B* is a transfer operator whose power sums reproduce the counts,  $N_\ell = \text{Tr } T^\ell$  (or a semigroup  $\text{Tr } e^{\text{L}\mathcal{G}}$  in continuous length); its eigenvalues are the poles and zeros.

**Definition 3.2** (The one identity). For any matrix  $M$  with eigenvalues  $\lambda_1, \dots, \lambda_N$ ,  $1/\det(1-uM) = \prod_i (1 - \lambda_i u)^{-1} = \exp \left( \sum_{\ell \geq 1} \text{Tr}(M^\ell) u^\ell / \ell \right)$ . The middle term is a bosonic partition function with one mode per eigenvector; the right term a sum over closed walks with the  $1/\ell$  of a ring's rotation symmetry.

**Definition 3.3** (Small matrix). For a shift or permutation with fixed-point counts  $\text{fix}(\ell)$  of its  $\ell$ -th power, a *small matrix* is any matrix  $M_{\text{small}}$  with  $\text{Tr } M_{\text{small}}^\ell = \text{fix}(\ell)$  for all  $\ell$ . It exists exactly when the zeta function is rational, and its eigenvalues are not those of the permutation matrix.

**Definition 3.4** (RH analogue, Ramanujan property). A side-B operator with leading eigenvalue  $\lambda_{\max}$  (or leading growth  $q$  per step) satisfies the *RH analogue* when every other nonzero eigenvalue has modulus  $\sqrt{\lambda_{\max}}$  (respectively  $\sqrt{q}$ ). In counting terms the fluctuation of the counts is the square root of the leading term. For graphs this is the Ramanujan property; for the Riemann zeta it is the Riemann hypothesis; for an MPS it is one correlation length equal to  $2/h$ .

#### 3.2 Graphs

**Definition 3.5** (Non-backtracking walk, prime cycle). A walk in a graph is non-backtracking if it never immediately retraces an edge; closed in the strong sense means the last step is not the reverse of the first. A *prime cycle*  $P$  is a closed non-backtracking walk that is not a repetition of a shorter one, taken up to choice of starting point;  $\ell(P)$  is its length. On a bouquet with  $D/2$  loops the directed edges are  $[D]$  and a closed non-backtracking walk is a cyclically reduced word.

**Definition 3.6** (Ihara zeta). For a finite graph  $X$  with  $N_\ell$  closed non-backtracking walks of length  $\ell$ ,  $\zeta_X(u) = \exp \left( \sum_\ell N_\ell u^\ell / \ell \right) = \prod_P (1 - u^{\ell(P)})^{-1}$ .

**Definition 3.7** (Hashimoto operator). The *Hashimoto* (non-backtracking edge) operator of a graph is the matrix  $T$  on directed edges with  $T_{ef} = 1$  when  $f$  can follow  $e$  without backtracking; then  $N_\ell = \text{Tr } T^\ell$  and  $\zeta_X(u) = \det(1 - uT)^{-1}$ .

**Definition 3.8** (Ihara–Bass formula). For a  $(q+1)$ -regular graph with adjacency matrix  $A$  on  $|V|$  vertices and  $|E|$  undirected edges,  $\det(1 - uT) = (1 - u^2)^{|E|-|V|} \det(1 - uA + qu^2)$ . Each adjacency eigenvalue  $\lambda$  produces two eigenvalues  $\mu, \mu'$  of  $T$  with  $\mu\mu' = q$ ,  $\mu + \mu' = \lambda$ ; the eigenvalue  $q+1$  gives  $\mu \in \{q, 1\}$ .

**Definition 3.9** (Ramanujan graph). A  $(q+1)$ -regular graph is *Ramanujan* when every adjacency eigenvalue other than  $\pm(q+1)$  satisfies  $|\lambda| \leq 2\sqrt{q}$ , the spectral radius of the universal covering tree. Equivalently all nontrivial  $\mu$  satisfy  $|\mu| = \sqrt{q}$ , which is the RH analogue for  $\zeta_X$ .

**Definition 3.10** (Artin–Ihara  $L$ -function). For a graph with a representation  $\varrho$  of its fundamental group assigned to closed walks (a holonomy),  $L(u, \varrho) = \prod_P \det(1 - \varrho(P)u^{\ell(P)})^{-1}$ , where the product is over prime cycles. For a bouquet the fundamental group is free and  $\varrho$  is a choice of one operator per loop.

### 3.3 Channels and matrix product states

**Definition 3.11** (Channel, Kraus operators). A *channel* on  $M_n$  is a completely positive trace-preserving map  $\Phi(X) = \sum_i A_i X A_i^\dagger$  with  $\sum_i A_i^\dagger A_i = 1$ ; the  $A_i$  are its Kraus operators. It is *unital* when  $\Phi(1) = 1$  and *mixed-unitary* when  $A_i = D^{-1/2} U_i$  with unitary  $U_i$ , in which case  $\Phi(X) = \frac{1}{D} \sum_i U_i X U_i^\dagger$  and the family is taken closed under adjoint.

**Definition 3.12** (Transfer matrix of an MPS). For a matrix product state with local tensors  $A_1, \dots, A_d \in M_n$  (physical dimension  $d$ , bond dimension  $n$ ), the *transfer matrix* is  $E = \sum_i \text{Ad}(A_i) = \Sigma$ , acting on  $M_n$ .

**Definition 3.13** (Ring norm). The *ring norm* of the periodic  $n$ -site state  $\Psi_n$  built from the tensors is  $\langle \Psi_n | \Psi_n \rangle = \text{Tr } E^n$ ; the zeta function of the state is the generating function  $\exp \sum_n \langle \Psi_n | \Psi_n \rangle u^n / n = \det(1 - uE)^{-1}$ .

**Definition 3.14** (Canonical form). An MPS is in (left) canonical form when  $\sum_k A_k^\dagger A_k = 1$ , i.e. when  $E$  is the transfer matrix of a channel; then  $E$  has leading eigenvalue 1 and the remaining eigenvalues lie in the closed unit disc.

**Definition 3.15** (Correlation length). In canonical form write the subleading eigenvalues as  $\lambda_k = e^{-1/\ell_k^c + i\vartheta_k}$ ; the  $\ell_k^c$  are the correlation lengths and the  $\vartheta_k$  the phases. The entropy per site is  $h = \log \lambda_{\max}$  of the unnormalised transfer matrix.

**Definition 3.16** (Quantum expander). A channel with  $D$  Kraus operators is a  $(n, D, \lambda)$  *quantum expander* when 1 is its unique fixed point (up to scale) and every other eigenvalue has modulus at most  $\lambda < 1$ .

**Definition 3.17** (Ramanujan quantum expander). A quantum expander is *Ramanujan* when  $\lambda_2 \leq \lambda_H$ , where, in Hastings' words,  $\lambda_H = \frac{2\sqrt{D-1}}{D}$ . (provenance `prov:hastings07-bound`); in adjacency units,  $D\lambda_2 \leq 2\sqrt{D-1}$ .

**Definition 3.18** (Quantum Hashimoto operator, quantum Ihara zeta). For superoperators  $\mathcal{E}_1, \dots, \mathcal{E}_D$  on  $\mathcal{V}$  and a fixed-point-free reversal  $\iota$  on  $[D]$ , the *quantum Hashimoto operator* is  $T$  on  $\mathcal{W} = \mathcal{V} \otimes \mathbb{C}^D$ ,  $T(v \otimes |i\rangle) = \sum_{j \neq \iota i} \mathcal{E}_i v \otimes |j\rangle$ , and  $\zeta_T(u) = \det(1 - uT)^{-1}$ . For a mixed-unitary channel,  $\mathcal{E}_i = \text{Ad}(U_i)$  with  $U_{\bar{i}} = U_i^\dagger$ , and the zeta is written  $\zeta_\Phi$ : the *quantum Ihara zeta* of the channel. It is the Artin–Ihara  $L$ -function of the bouquet twisted by  $\varrho = \text{Ad} \circ U$ .

## 4 Definitions, Continued

Continuation of Section 3; split for length only.

### 4.1 The Riemann channel

**Definition 4.1** (Scattering symbol). With  $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$  and  $\tau$  the Mellin variable on the dilation group, the *scattering symbol* of the modular surface is  $S(\tau) = \xi(1-2i\tau)/\xi(1+2i\tau)$ .

**Definition 4.2** (Inner function, model space, compressed semigroup). A function analytic and bounded by 1 in the lower half plane with unimodular boundary values is *inner*. For inner  $S$  the *model space* is  $K_S = H^2 \ominus SH^2$  and the *compressed semigroup* is  $Z(t) = P_{K_S} e^{it\tau}|_{K_S}$ ,  $t > 0$ .

**Definition 4.3** (Riemann channel). The *Riemann channel* is the completely positive semigroup  $X \mapsto Z(t)^* X Z(t)$  on  $B(K_S)$ , whose single Kraus operator is the compressed semigroup of the scattering symbol. Its modes are the zeros of  $\zeta$  and its distributional trace is the prime measure; the name is also used loosely for the whole side-A/side-B package of the Riemann zeta.

**Definition 4.4** (Weil explicit formula). For a test window  $\hat{g}$  with transform  $g$ ,  $\sum_{\gamma} \hat{g}(\gamma) = \hat{g}(\frac{i}{2}) + \hat{g}(-\frac{i}{2}) - g(0) \log \pi + \frac{1}{2\pi} \int \hat{g}(r) \operatorname{Re}(\Gamma'/\Gamma)(\frac{1}{4} + \frac{ir}{2}) dr - 2 \sum_n \frac{\Lambda(n)}{\sqrt{n}} g(\log n)$ , the sum on the left over the ordinates of the nontrivial zeros.

**Definition 4.5** (Bost–Connes system, Lévy exponent). The *Bost–Connes system* is the  $C^*$ -dynamical system generated by the phase operators  $e(a/b)$  and the isometries  $\mathbf{m}_n$  on  $\ell^2(\mathbb{N})$ , with time evolution scaling  $\mathbf{m}_n$  by  $n^{it}$ ; its  $\operatorname{KMS}_{\beta}$  states for  $\beta > 1$  have partition function  $\zeta(\beta)$  and  $\beta = 1$  is its phase transition. Its *Lévy exponent* at real part  $\sigma$  is  $\kappa_{\sigma}(\tau) = \log \zeta(\sigma - i\tau) - \log \zeta(\sigma)$ , the cumulant generating function of the zeta distribution, with jump measure  $\nu_{\sigma}$ .

**Definition 4.6** (Matrix product operator over the prime chain). Writing  $\ell^2(\mathbb{N}) = \bigotimes_p$  (oscillator at  $p$ ), an operator is a *matrix product operator over the prime chain* when it is a contraction of one local tensor per prime with a bond space carried between consecutive primes; for  $e(a/b)$  the bond space is  $\mathbb{Z}/b$  and the local transfer map is  $r \mapsto p^k r$ .

**Definition 4.7** (Hilbert–Pólya operator). On the subleading sector of the length flow write  $\mathcal{G} = -\frac{1}{2} + iH$ ;  $H$  is the *Hilbert–Pólya operator*, with eigenvalues  $\gamma_n - i(\sigma_n - \frac{1}{2})$ . It is not the entanglement Hamiltonian of any state.

**Definition 4.8** (Weil positivity). *Weil’s criterion*: the Riemann hypothesis holds iff  $F_W(u) = \sum_{\rho} e^{(\rho-1/2)u}$  is positive-definite on the dilation group, i.e. iff its spectral measure  $\sum_{\rho} \delta_{t_{\rho}}$  is a positive measure on the real line.

### 4.2 Weil–LPS channels

**Definition 4.9** (LPS generators). For a prime  $q \equiv 1 \pmod{4}$ , the *LPS generators* are the  $q+1$  integer quaternions  $\alpha = a_0 + a_1 i + a_2 j + a_3 k$  of norm  $q$  with  $a_0 > 0$  odd and  $a_1, a_2, a_3$  even; they generate a  $(q+1)$ -regular tree, and their images  $g_{\alpha}$  in  $\operatorname{PSL}_2(\mathbb{F}_p)$  (for  $q$  a square mod  $p$ , after division by  $s_q$ ) are the generators of the LPS Cayley graph.

**Definition 4.10** (Weil representation). The *Weil representation*  $W$  of  $\operatorname{SL}_2(\mathbb{F}_p)$  on  $\ell^2(\mathbb{F}_p)$  is the projective unitary representation characterised by  $W(g) T(u, v) W(g)^{\dagger} \propto T(g(u, v))$  for the Heisenberg–Weyl operators  $T(u, v) = X^u Z^v$ ; it is generated by  $\mathcal{F}$ ,  $\mathbf{M}$  and  $\mathbf{D}$  and splits into the blocks  $W_{\pm}$  of dimensions  $(p \pm 1)/2$  on which  $-1$  acts as a scalar.

**Definition 4.11** (Weil–LPS channel). For admissible  $(p; q)$  the *Weil–LPS channel* is  $\Phi_q^\pm(X) = \frac{1}{q+1} \sum_\alpha W_\pm(g_\alpha) X W_\pm(g_\alpha)^\dagger$  on  $M_{(p\pm 1)/2}$ : Harrow’s construction with the Weil representation and LPS generators.

**Definition 4.12** (Hecke relation). The adjacency operators  $A_q$  of the LPS Cayley graph satisfy  $A_q^2 - q1 = A_{q^2}$  and  $A_{q_1}A_{q_2} = A_{q_1q_2}$ , the tree relations “two steps without backtracking” and multiplicativity; the Weil–LPS channels for different  $q$  commute.

### 4.3 Curves

**Definition 4.13** (Artin–Schreier curve, exponential sum). For  $g \in \mathbb{F}_q[x]$  the *Artin–Schreier curve* is  $y^q - y = g(x)$ ; its affine point count over  $\mathbb{F}_{q^n}$  is  $N_n = q^n + \sum_{a \in \mathbb{F}_q^\times} \mathcal{S}_n(ag)$  with the *exponential sum*  $\mathcal{S}_n(g) = \sum_{x \in \mathbb{F}_{q^n}} e_q(\text{tr}_{q^n/q} g(x))$ . The quadratic family is  $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$ .

**Definition 4.14** (Self-dual normal basis). A basis  $\{\theta, \theta^q, \dots, \theta^{q^{n-1}}\}$  of  $\mathbb{F}_{q^n}$  is *normal*; Frobenius is then the cyclic shift of coordinates. It is *self-dual* when  $\text{tr}_{q^n/q}(ab) = \sum_i a_i b_i$ ; such bases exist for  $n$  odd.

**Definition 4.15** (Dwork transfer operator). *Dwork’s operator* is  $\mathcal{O}_{\text{Dw}} = \text{dec}_q \circ M_F$  on  $p$ -adic power series, multiplication by the splitting function  $M_F$  followed by the decimation  $\text{dec}_q : x^m \mapsto x^{m/q}$ ;  $\mathcal{S}_n^\times = (q^n - 1) \text{Tr } \mathcal{O}_{\text{Dw}}^n$  and  $L$  is its Fredholm determinant.

**Definition 4.16** (Pure of weight, big monodromy, slicing, the sign). A coefficient system on a curve over  $\mathbb{F}_q$  is *pure of weight*  $w$  when every eigenvalue of every holonomy of a prime cycle of length  $\ell$  has modulus  $q^{w\ell/2}$ ; it has *big monodromy* when the only invariants of its even tensor powers are the pairwise contractions by the given pairing. *Slicing* writes the middle block of a variety as the interesting block of a base curve with coefficients from hyperplane sections. *The sign* is the minus sign with which the interesting block enters the point count, so that its eigenvalues are zeros, not poles, of  $Z_C$ .

**Definition 4.17** (Status vocabulary). The tags **proved**, **sketched**, **numerical**, **cited**, **conjectured**, **open**, **refuted**, **assumed**, with the derived forms **unsupported** and **-conditional**, as fixed in Section 2 of the frontmatter; definitions are **stipulated** or **quoted**.

### 4.4 The continuous dictionary

**Definition 4.18** (Geodesic flow on the unit tangent bundle). For  $\Gamma < \text{PSL}_2(\mathbb{R})$  discrete, torsion-free and cocompact, the unit tangent bundle of  $\Gamma \backslash \mathbb{H}$  is  $\Gamma \backslash G$  with  $G = \text{PSL}_2(\mathbb{R})$ , and the *geodesic flow* is  $\phi_t(\Gamma g) = \Gamma g a_t$ ,  $a_t = \text{diag}(e^{t/2}, e^{-t/2})$ . Its generator is  $\mathcal{X} = H/2$  through the left-invariant fields  $H, E, W$  of the basis  $H, E, W$  of  $\mathfrak{sl}_2$ , and the Koopman operator is  $(e^{-t\mathcal{X}} f)(m) = f(\phi_{-t} m)$ . Primitive closed geodesics  $\gamma$  have lengths  $\ell_\gamma$  and opposite orientations are not identified.

**Definition 4.19** (Selberg zeta, Ruelle zeta).  $Z_{\text{Sel}}(s) = \prod_\gamma \prod_{k \geq 0} (1 - e^{-(s+k)\ell_\gamma})$ , in Marklof’s words  $Z(s) = \prod_{\{\gamma \in H_*\}} \prod_{m=0}^\infty (1 - e^{-(s+m)\ell_\gamma})$ , (provenance **prov:marklof04-selbergzeta**), and the *Ruelle zeta* in the direct-product convention is  $\zeta_R(s) = \prod_\gamma (1 - e^{-s\ell_\gamma})$ .

**Definition 4.20** (Guillemin flat trace). For an Anosov flow with closed orbits  $\gamma$ , primitive periods  $\ell_\gamma$  and linearised Poincaré maps  $P_\gamma$ , the *flat trace* of the Koopman operator is the distribution on  $t > 0$   $\text{Tr}^b e^{-t\mathcal{X}} = \sum_\gamma \sum_{k \geq 1} \ell_\gamma \delta(t - k\ell_\gamma) / |\det(1 - P_\gamma^k)|$  (Dyatlov–Zworski, provenance **prov:dz13-guillemin**). It is not a trace-class Hilbert-space trace.

**Definition 4.21** (Flat dynamical determinant, first band).  $A_{\text{fl}}(\varsigma) = \int_0^\infty e^{-\varsigma t} \text{Tr}^\flat e^{-t\mathcal{X}} dt$  for  $\text{Re } \varsigma > 0$ , and the *tower*  $D_{\text{tow}}(\varsigma) = \prod_{j \geq 1} Z_{\text{Sel}}(\varsigma + j)$ . *Trace resonances* are the poles of the continued  $A_{\text{fl}}$ . The *nonconstant first band* is  $\mathcal{B}_0 = \{-\frac{1}{2} \pm ir_j : j \geq 1\}$ , with  $\lambda_j = \frac{1}{4} + r_j^2$  the Laplace eigenvalues and  $r_0 = i/2$  for the constant mode excluded.

**Definition 4.22** (Scattering determinant of the modular group).  $\varphi(s) = \sqrt{\pi} \Gamma(s - \frac{1}{2}) \zeta(2s - 1) / (\Gamma(s) \zeta(2s))$ , in the source's words In the case when  $\Gamma$  is the modular group  $\text{PSL}(2, \mathbb{Z})$ , the scattering determinant is given by  $\phi(s) = \sqrt{\pi} \frac{\Gamma(s - 1/2)}{\Gamma(s)} \frac{\zeta(2s - 1)}{\zeta(2s)}$ , (provenance prov:fjs16-scatt) the *scattering multiplier* is  $m_{\text{sc}}(r) = -\frac{1}{2}(\varphi'/\varphi)(\frac{1}{2} + ir)$  and  $C_{\text{cusp}} = \mathcal{F}^{-1} m_{\text{sc}}$  with  $\mathcal{F}^{-1} b(t) = \frac{1}{2\pi} \int e^{irt} b(r) dr$ .

**Definition 4.23** (Prime comb measures).  $P_+(t) = \sum_{n \geq 2} \frac{\Lambda(n)}{n} \delta(t - 2 \log n)$  and  $P_{\text{ch}}(t) = \sum_{n \geq 2} \Lambda(n) n^{-1/2} \delta(t - 2 \log n)$  on  $(0, \infty)$ , locally finite measures;  $A_{\text{arch}}$  is the archimedean-plus-boundary distribution of Proposition 34.5, equal to  $\frac{1}{2} - 1/(4 \sinh(|t|/2))$  away from  $t = 0$ .

**Definition 4.24** (Trivial set, critical radius, rescaled trace sequence). For a linear operator  $X$  on a finite-dimensional complex space,  $\text{spec}(X)$  is its spectrum as a multiset with algebraic multiplicities, so  $\text{Tr } X^\ell = \sum_{\mu \in \text{spec}(X)} \mu^\ell$ . A *trivial set*  $S_{\text{triv}}$  is a sub-multiset of  $\text{spec}(X)$ ; the *retained multiset* is  $\mathcal{A}_{\text{ret}} = \text{spec}(X) \setminus S_{\text{triv}}$  (multiplicities subtracted). For a *critical radius*  $r > 0$  the *rescaled trace sequence* is  $\mathbf{n}_\ell = r^{-\ell} (\text{Tr } X^\ell - \sum_{\mu \in S_{\text{triv}}} \mu^\ell) = \sum_{\mu \in \mathcal{A}_{\text{ret}}} (\mu/r)^\ell$  for  $\ell \geq 0$  (with  $0^0 = 1$ ), extended by  $\mathbf{n}_{-\ell} = \overline{\mathbf{n}_\ell}$ . A sequence with this symmetry is *positive definite* if  $\sum_{\ell, k=0}^L c_\ell \bar{c}_k \mathbf{n}_{\ell-k} \geq 0$  for every  $L$  and every  $c \in \mathbb{C}^{L+1}$ .

**Definition 4.25** (Weil form, reflection, mode-pairing form). With the data of Definition 4.24, the *Weil form* of  $X$  is the Toeplitz form  $W(c) = \sum_{\ell, k} c_\ell \bar{c}_k \mathbf{n}_{\ell-k}$  on finitely supported coefficient vectors  $c$ , and  $f_c(z) = \sum_\ell c_\ell z^\ell$ . The *reflection in the critical circle* is  $J(\mu) = r^2/\bar{\mu}$  on  $\mathbb{C}^\times$ ; its fixed points are the circle  $|\mu| = r$ . When  $0 \notin \mathcal{A}_{\text{ret}}$  the *mode-pairing form* is  $M(c) = \sum_{\mu \in \mathcal{A}_{\text{ret}}} f_c(\mu/r) \overline{f_c(J(\mu)/r)}$ : each mode paired with its reflected partner.

**Definition 4.26** (Adjoint pairing and inverse pairing). A family  $\mathcal{E}_i = \text{Ad}(B_i)$ ,  $i \in [D]$ , with reversal  $\iota$  is *adjoint-paired* if  $B_{\bar{i}} = B_i^\dagger$  and *inverse-paired* if every  $B_i$  is invertible and  $B_{\bar{i}} = B_i^{-1}$  (so  $\mathcal{E}_{\bar{i}} = \mathcal{E}_i^{-1}$ , the hypothesis of Corollary 26.3). The Kraus families of Definition 3.18 are the adjoint-paired ones.

**Definition 4.27** (Rescaled continuous trace). For a linear operator  $\mathcal{L}$  on a finite-dimensional complex space, a trivial set  $S_{\text{triv}} \subseteq \text{spec}(\mathcal{L})$  and a real number  $a$ , the *rescaled continuous trace* is  $F(t) = e^{-at} (\text{Tr } e^{t\mathcal{L}} - \sum_{\varpi \in S_{\text{triv}}} e^{t\varpi}) = \sum_{\varpi \in \mathcal{A}_{\text{ret}}} e^{t(\varpi-a)}$  for  $t \geq 0$ , extended by  $F(-t) = \overline{F(t)}$ . A function with this symmetry is *positive definite* if  $\sum_{j, k} c_j \bar{c}_k F(t_j - t_k) \geq 0$  for all finite families of real times and complex coefficients. The *reflection in the line*  $\text{Re } \varpi = a$  is  $R_a(\varpi) = 2a - \bar{\varpi}$ .

## 5 Definitions, Continued: the Phantasm Campaign

Continuation of Section 4; split for length only.

**Definition 5.1** (Graded spectral datum, supertrace distribution, net multiplicity). A *graded spectral datum* is a countable multiset of pairs  $(\lambda, \epsilon)$ ,  $\lambda \in \mathbb{C}$ ,  $\epsilon \in \{+1, -1\}$  the *parity*, with multiplicities  $m(\lambda, \epsilon) \in \mathbb{N}$ , such that  $\operatorname{Re} \lambda \leq c$  for some real  $c$  and  $\sum m(\lambda, \epsilon)(1 + |\lambda|)^{-N} < \infty$  for some  $N$ . Its *supertrace distribution* is  $\operatorname{str} e^{tG} := \sum \epsilon m(\lambda, \epsilon) e^{\lambda t}$  on  $(0, \infty)$ , and its *net multiplicity* is  $\nu_{\text{net}}(\lambda) = m(\lambda, +1) - m(\lambda, -1)$ . Two data with the same  $\nu_{\text{net}}$  are *equivalent up to cancelling pairs*. For a  $\mathbb{Z}_2$ -graded operator the supertrace is the trace on the even part minus the trace on the odd part.

**Definition 5.2** (Distributional trace of the compressed semigroup, prime comb).  $\operatorname{Tr}_{\text{dist}} Z(t) := \sum_{\rho} e^{-\bar{\rho}t/2}$ , the sum over all nontrivial zeros with multiplicity, is a distribution on  $(0, \infty)$  (Lemma 12.3); this is a definition by fiat, not an application of a trace theorem to  $Z(t)$ . The *prime comb* is  $\operatorname{comb}(u) = \sum_{n \geq 2} \Lambda(n) \delta(u - \log n)$ , so that  $e^{-t/2} \operatorname{comb}(t/2) = 2P_+(t)$ .

**Definition 5.3** (Riemann graded bond and generator).  $\mathfrak{B} = \mathbb{C}_+ \oplus (K_S \oplus \ell^2(\mathbb{N}_{\geq 1}))_-$ , the even line spanned by the fixed vector and the odd part the model space together with an *archimedean ladder*;  $G = 0 \oplus B \oplus \operatorname{diag}(-(k + \frac{1}{2}))_{k \geq 1}$ , where  $Z(t) = e^{tB}$  is the compressed semigroup.

**Definition 5.4** (Vacuum-decay Lindbladian, absorbing-vacuum channel). For a Hilbert space  $H$ ,  $B$  on  $H$  and an exit vector  $j$  with  $-(B + B^\dagger) = |j\rangle\langle j|$ , the *vacuum-decay Lindbladian* on  $B(\mathbb{C}\text{vac} \oplus H)$  is  $\mathcal{L}_{\text{vac}}(x) = \bar{B}x + x\bar{B}^\dagger + JxJ^\dagger$  with  $\bar{B} = 0 \oplus B$  and  $J = |\text{vac}\rangle\langle j|$ ;  $\Omega_{\text{vac}} = |\text{vac}\rangle\langle \text{vac}|$ . The grading is  $\text{vac}$  even,  $H$  odd. The *absorbing-vacuum channel* of a strongly stable contraction semigroup  $Z_t$  is  $x \mapsto (1 \oplus Z_t)x(1 \oplus Z_t)^\dagger + \operatorname{Tr}[(1 - Z_t^\dagger Z_t)X] \Omega_{\text{vac}}$ ,  $X$  the  $H$ -block of  $x$ .

**Definition 5.5** (Prime-chain detailed-balance Lindbladian). With  $\mathfrak{m}_p|n\rangle = |pn\rangle$  the isometries of the Bost–Connes representation and rates  $r_p > 0$ ,

$$\mathcal{L}_\beta^{\text{pc}}(x) = \sum_p r_p (\mathfrak{m}_p x \mathfrak{m}_p^\dagger - \tfrac{1}{2} \{\mathfrak{m}_p^\dagger \mathfrak{m}_p, x\}) + \sum_p r_p p^\beta (\mathfrak{m}_p^\dagger x \mathfrak{m}_p - \tfrac{1}{2} \{\mathfrak{m}_p \mathfrak{m}_p^\dagger, x\}),$$

in the Schrödinger convention; its diagonal part is the multiplicative birth–death process  $n \rightarrow pn$  at rate  $r_p$ ,  $n \rightarrow n/p$  at rate  $r_p p^\beta$ . The Gibbs density is  $\varrho_\beta^G = N^{-\beta}/\zeta(\beta)$ .

**Definition 5.6** (Entanglement Hamiltonian and entropy of a bond density). For a bond density  $X$  (the fixed point of a cMPS transfer generator, which by cited fact 13.3 carries the squared Schmidt coefficients of a bipartition) the *entanglement Hamiltonian* is  $-\log X$ , its spectrum the *entanglement spectrum*, and the *entanglement entropy* is  $\mathcal{S}_{\text{ent}}(X) = -\operatorname{Tr} X \log X$ ;  $\operatorname{TFD}_\beta$  denotes the thermofield-double purification of  $\varrho_\beta^G$ .

**Definition 5.7** (Ring form, trace form, radical, radical sign, period parameter). For  $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$  over  $\mathbb{F}_q$ ,  $q$  an odd prime, on  $V = \mathbb{F}_q^n$  with cyclic indices let  $\mathbb{S}$  be the shift  $(\mathbb{S}x)_i = x_{i+1}$  and  $P_g(z) = \sum_j \frac{a_j}{2} (z^j + z^{-j})$ . The *ring form* is  $Q_g^\circ(x) = \sum_j a_j \sum_i x_i x_{i+j} = x^t P_g(\mathbb{S})x$ ; the *trace form* in the coordinates of a normal basis  $\theta$  is  $Q_g^{\text{tr}}(x) = \operatorname{tr}_{q^n/q} g(x) = x^t C P_g(\mathbb{S})x$  with  $C_{ik} = \operatorname{tr}_{q^n/q}(\theta^{q^i} \theta^{q^k})$  the (symmetric, invertible, circulant) Gram matrix. The *radical* is  $\mathcal{R}_n = \ker P_g(\mathbb{S})$ , the *radical sign*  $\delta_n = \det(\mathbb{S}|_{\mathcal{R}_n}) \in \{\pm 1\}$  ( $= 1$  if  $\mathcal{R}_n = 0$ ),  $d_n = \dim \mathcal{R}_n$ , and the *period parameter*  $h$  is the smallest power of  $q$  strictly greater than the order of vanishing of  $z^J P_g(z)$  at  $z = -1$ .

**Definition 5.8** (Super-transfer matrix of an Artin–Schreier curve).  $\mathbb{E} = (E_0 \oplus 1)_+ \oplus (\bigoplus_{a \in \mathbb{F}_q^\times} F_{ag})_-$ , where  $E_0$  is the transfer matrix of the zero polynomial (the de Bruijn adjacency matrix of memory

$J$ , or the scalar  $q$  when  $J = 0$ ), the summand 1 is the point at infinity, and  $F_g$  is the *Frobenius block* of the character sum: the diagonal  $q^J \times q^J$  matrix with  $L(g, u) = \det(1 - uF_g)$  (Theorem 19.7). The point count is its supertrace,  $N_n = \text{str } \mathbb{E}^n$ .

**Definition 5.9** (Centred Weyl operators, symplectic transfer map, Weil implementer). On  $\ell^2(\mathbb{F}_q^J)$  the *centred Weyl operators* are  $X_{u,v}|s\rangle = e_q(v \cdot (s + u/2))|s + u\rangle$ ,  $(u, v) \in \mathbb{F}_q^{2J}$ . The *transfer unitary* is  $\mathcal{U}_g = E_g/\sqrt{q}$ ; its *symplectic transfer map*  $M_g \in \text{Sp}(2J, \mathbb{F}_q)$  is defined by  $\mathcal{U}_g X_{u,v} \mathcal{U}_g^{-1} = X_{M_g(u,v)}$  (Theorem 19.13). A *Weil implementer* of  $M \in \text{Sp}(2J, \mathbb{F}_q)$  is a unitary  $W(M)$  with  $W(M)X_{u,v}W(M)^{-1} = X_{M(u,v)}$ ; it is determined up to a scalar (cited fact 19.10), and  $D_g(u) = \det(1 - uE_g)$ .

## 6 Definitions, Continued: Flows on Complexes and their Twists

Continuation of Section 5; split for length only. Throughout,  $X$  is a finite abstract simplicial complex of dimension  $d$  with face numbers  $f_0, \dots, f_d$  and Euler characteristic  $\chi(X) = \sum_i (-1)^i f_i$ , and  $s_k = (-1)^{k+1}$ .

**Definition 6.1** (Ordered  $k$ -cell, ordered geodesic flow). An *ordered  $k$ -cell* of  $X$  is a tuple  $[v_0, \dots, v_k]$  of distinct vertices whose underlying set is a  $k$ -simplex of  $X$ . Write  $\mathcal{O}_k$  for the set of ordered  $k$ -cells, so  $|\mathcal{O}_k| = (k+1)! f_k$ , and  $H_k = \mathbb{C}^{\mathcal{O}_k}$ . The *ordered geodesic flow* is the operator  $T_k$  on  $H_k$ ,  $1 \leq k \leq d$ ,

$$T_k[v_0, \dots, v_k] = \sum_{\substack{w \notin \{v_0, \dots, v_k\} \\ \{v_1, \dots, v_k, w\} \in X, \{v_0, \dots, v_k, w\} \notin X}} [v_1, \dots, v_k, w] :$$

drop the oldest vertex, append a new one keeping the ordered overlap, and forbid the step that a  $(k+1)$ -cell would short-cut. For  $d = 1$  this is exactly the Hashimoto operator of Definition 3.7, and the excluded vertex is the immediately previous one.

**Definition 6.2** (Pointed facet, opposition successor, algebraic length). Let  $X$  be a finite quotient of the Bruhat–Tits building  $\mathcal{B}$  of  $\mathrm{PGL}_n$  over a local field with residue order  $q$ , carrying the cyclic type structure of the affine diagram  $\tilde{A}_{n-1}$ . A *pointed  $k$ -facet* is a  $k$ -simplex with a distinguished vertex; ordering it cyclically in increasing type from that vertex gives its *ordered type*, a positive composition  $(\lambda_0, \dots, \lambda_k)$  of  $n$ , and its *algebraic step length* is  $\lambda_0$ . Write  $P_k$  for the set of pointed  $k$ -facets, so  $|P_k| = (k+1)f_k$ . A *successor* shifts the ordering and crosses to a vertex *opposite* in the spherical link of the common  $(k-1)$ -facet; the *pointed opposition flow* is

$$\mathsf{T}_k(u)[\sigma] = u^{\lambda_0(\sigma)} \sum_{\tau \in \mathrm{Succ}(\sigma)} [\tau] \quad \text{on } \mathbb{C}^{P_k}.$$

Opposition is strictly stronger than the non-cell condition of Definition 6.1, and the exponent  $u^{\lambda_0}$  is a *length*, never a matrix power or a multiplicity.

**Definition 6.3** (Graded geodesic determinant, total zeta of a complex). *Geodesic data*  $\mathfrak{g}$  on  $X$  is, in each degree  $1 \leq k \leq d$ : a finite state set  $\mathcal{S}_k$ , a successor relation with incidence multiplicities, a positive integer algebraic length  $\ell_k(\sigma)$ , and a step transport  $U_{\sigma\tau}$  on a finite-dimensional fibre  $V$ . Give the whole degree- $k$  space  $V \otimes \mathbb{C}^{\mathcal{S}_k}$  parity  $k+1$ , put  $\mathsf{T}_k^U(u)(v \otimes [\sigma]) = \sum_{\sigma \rightsquigarrow \tau} u^{\ell_k(\sigma)} U_{\sigma\tau} v \otimes [\tau]$ , and let  $W = \bigoplus_{k=1}^d V \otimes \mathbb{C}^{\mathcal{S}_k}$ . The *graded geodesic determinant* (the total zeta) is

$$\mathcal{Z}(X, \mathfrak{g}, U; u) = \mathrm{sdet}_W \left( I - \bigoplus_{k=1}^d s_k \mathsf{T}_k^U(u) \right)^{-1} = \prod_{k=1}^d \det(I - s_k \mathsf{T}_k^U(u))^{(-1)^k},$$

with the sign  $s_k = (-1)^{k+1}$  inserted once per successor arrow. Odd  $k$  (edges) sits in the denominator, even  $k$  (chambers) in the numerator. The intrinsic default for an arbitrary complex is  $\mathcal{S}_k = \mathcal{O}_k$ , unit length, the rule of Definition 6.1 and  $V = \mathbb{C}$ , written  $\mathcal{Z}_{\mathrm{ord}}(X, u)$ ; the building default is Definition 6.2, written  $\mathcal{Z}_{\mathrm{pt}}(X, u)$ .

**Definition 6.4** (Bruhat–Tits building of  $\mathrm{PGL}_n$ , colour-shift Hecke operators, Ramanujan complex). The vertices of  $\mathcal{B}$  carry a colour (type) function with values in  $\mathbb{Z}/n\mathbb{Z}$ , and the *colour-shift Hecke operators*  $A_k$ ,  $1 \leq k \leq n-1$ , sum a function over the neighbours of colour shift  $k$ : we may look at the  $d-1$  colored adjacency operators  $A_k$ ,  $k = 1, \dots, d-1$  on  $L^2(\mathcal{B}_0)$ , called the Hecke

operators. (math/0406208:main.tex:541-549). They commute, are normal, and satisfy  $A_k^* = A_{n-k}$ . A finite quotient is a *Ramanujan complex* when its nontrivial joint spectrum lies in the tempered set  $\text{Aspec}_n = \{(q^{k(n-k)/2} e_k(z))_k : |z_i| = 1, \prod_i z_i = 1\}$ : A finite quotient  $\mathbb{X}$  of  $\mathbb{B}$  is called a Ramanujan complex if the eigenvalues of every non-trivial simultaneous eigenvector  $v \in \bigwedge^2 \mathbb{X}$ ,  $A_k v = \lambda_k v$ , satisfy  $(\lambda_1, \dots, \lambda_{d-1}) \in \text{Aspec}_d$ . (math/0406208:main.tex:564-572). The condition is on the *joint* spectrum, which is strictly smaller than the product of the separate spectra.

**Definition 6.5** (Cayley complex). For a finite group  $G$  with generating set  $S$ , the *Cayley complex* is the clique complex of the Cayley graph: The Cayley complex of  $G$  with respect to a set of generators  $S$  is the simplicial complex whose 1-skeleton is the Cayley graph  $\text{Cayley}(G, S)$ , where a subset of  $i+1$  vertices is an  $i$ -cell iff every two vertices comprise an edge. (math/0406217: Lubotzky, Samuels and Vishne give explicit  $S$ , graded into colour classes  $S_1, \dots, S_{n-1}$  with  $|S_k| = \binom{n}{k}_q$  and  $S_k^{-1} = S_{n-k}$ , for which this complex is a Ramanujan complex covered by  $\mathcal{B}$ ).

**Definition 6.6** (Kraus connection on the 1-skeleton; flat, pure gauge, paired). A *connection* on the 1-skeleton of  $X$  with fibre  $V$  assigns to each directed edge  $(x, y)$  an arbitrary  $E_{(x,y)} \in \text{End}(V)$ ; it is a *Kraus connection* when  $V = M_n$  and each  $E_e$  is completely positive, for instance  $E_e = \text{Ad}(B_e)$ . Twist Definition 6.1 by anchoring the fibre at the *last* vertex,  $T_k^E(v \otimes [\sigma]) = \sum E_{(v_k, w)} v \otimes [\tau]$ . The connection is *flat* when  $E_{yx} = E_{xy}^{-1}$  and  $E_{yz} E_{xy} = E_{xz}$  on every 2-cell, and *pure gauge* when  $E_{(x,y)} = F_y F_x^{-1}$  for some invertible  $F$ . It is *adjoint-paired* when  $E_{\bar{e}} = E_e^\dagger$  and *inverse-paired* when  $E_{\bar{e}} = E_e^{-1}$  (Definition 4.26). No positivity, invertibility, pairing or flatness is assumed unless said.

**Definition 6.7** (Voltage quotient and Artin block). Let a finite group  $G$  act on  $X$  preserving the successor and length data and freely on the state set  $\mathcal{S}_k$ . Choose orbit representatives  $\sigma_a$  and encode each transition  $\sigma_a \rightarrow h\sigma_b$  by its *voltage*  $h \in G$ ; this is the *voltage quotient*, a digraph on the orbits with a group label on every arrow. For a unitary  $\varrho$  of  $G$ , the  $\varrho$ -block  $\mathsf{T}_{k,\varrho}(u)$  replaces that arrow by  $\varrho(h^{-1})$  on the fibre, keeping the length weight. For  $R = \pi \otimes \bar{\pi} = \bigoplus_{\varrho} a_{\varrho} \varrho$  the *Artin determinant* is  $D_{k,R}(u) = \prod_{\varrho} \det(I - s_k \mathsf{T}_{k,\varrho}(u))^{a_{\varrho}}$ . This is not the same object as the connection of Definition 6.6 on the full cover.

**Definition 6.8** (Ramanujan quantum expander of type  $\tilde{A}_{n-1}$ ; exceptional space). Let  $X = \text{Cay}(G, S)$  have the colour classes of Definition 6.5, let  $\pi: G \rightarrow U(N)$  be unitary and  $R = \text{Ad } \pi$  on  $M_n$ . Put  $\Phi_k(X) = |S_k|^{-1} \sum_{s \in S_k} \pi(s) X \pi(s)^\dagger$  and  $\hat{A}_k = |S_k| \Phi_k = \sum_{s \in S_k} R(s)$ . Let  $\mathcal{F}_R$  be the common fixed space and let  $\mathcal{E}_R \supseteq \mathcal{F}_R$  be the sum, inside  $R$ , of the type-character subrepresentations excluded by the convention of Definition 6.4. The tuple  $(\Phi_1, \dots, \Phi_{n-1})$  is *Ramanujan of type  $\tilde{A}_{n-1}$  relative to  $\mathcal{E}_R$*  when its joint spectrum on  $\mathcal{E}_R^\perp$  lies in  $\text{Aspec}_n$ . Only when  $\mathcal{E}_R = \mathcal{F}_R$  is this the fixed-point-complement condition; on a partite complex  $\mathcal{E}_R$  can be strictly larger.

## 7 Definitions, Continued: Finite BC States and Covariant Generators

**Definition 7.1** (Finite phase marginal and matrix extension). For an odd prime  $p$ , the phase algebra generated by  $e(k/p)$  is  $C(\mathbb{F}_p)$ , represented diagonally on  $H_p = \ell^2(\mathbb{F}_p)$ . Let  $w_\beta$  be the restriction of the unique BC state for  $0 < \beta \leq 1$ , or of the Galois average of extremal BC states for  $\beta > 1$ . Its *diagonal matrix extension* is  $\varrho_{p,\beta}^{\text{res}} = \text{diag}(w_\beta(x))$  in  $B(H_p)$ . A matrix extension is a density with these diagonal entries; its off-diagonal entries are additional data. This does not specify a representation of the full BC algebra on  $H_p$ .

**Definition 7.2** (Local arithmetic covariance and parity). The centred Weyl operators  $X_v$ ,  $v \in \mathbb{F}_p^2$ , are those of Definition 5.9 with one register. Three requirements on a superoperator are distinguished: commuting with  $\text{Ad}(U_a)$ ,  $U_a|x\rangle = |ax\rangle$ ,  $a \in \mathbb{F}_p^\times$  (*Galois torus*); commuting with  $\text{Ad}(W(g))$ ,  $g \in \text{Sp}(2, \mathbb{F}_p)$  (*linear symplectic covariance*); and commuting with all  $\text{Ad}(X_v)$  (*Weyl-translation covariance*). The first action sends  $(x, y)$  to  $(ax, a^{-1}y)$ , and induces the inverse-labelled BC action on the phase generators. Weyl-translation covariance is an extra condition, not a consequence of carrying Weyl operators. Parity is  $P|x\rangle = |-x\rangle$ ; operator parity is  $X \mapsto PXP$ .

**Definition 7.3** (Stationary covariant GKLS cone). For a density  $\varrho_b$  on  $\mathbb{C}^d$  and a unitary group action  $U$ , let  $\mathfrak{C}_U(\varrho_b)$  be the cone of Schrödinger generators of CPTP semigroups with  $\mathcal{L}(\varrho_b) = 0$  and  $[\mathcal{L}, \text{Ad}(U_g)] = 0$  for every  $g$ . Its dimension means the real dimension of its linear span, including the overall time scale. The identity is fixed by the dual Heisenberg semigroup; fixing  $I$  also in the Schrödinger picture is an additional condition, equivalent to stationarity of the tracial density  $I/d$ .

**Definition 7.4** (Conditional Choi matrix). For a complex-linear map  $L$  on  $M_d$ , set

$$\mathbf{C}_L = \sum_{i,j} |i\rangle\langle j| \otimes L(|i\rangle\langle j|), \quad b = d^{-1/2} \sum_i |i, i\rangle, \quad Q = I - |b\rangle\langle b|.$$

The *conditional Choi matrix* is  $Q\mathbf{C}_L Q$ . The completely depolarizing channel is  $\mathcal{D}_d(X) = \text{Tr}(X)I/d$ .

## 8 Definitions, Continued: Boson–Fermion cMPS Rings

Throughout,  $L > 0$  is the ring length,  $\mathfrak{S}$  a finite set of particle species with parity  $p: \mathfrak{S} \rightarrow \{0, 1\}$ ,  $\eta_a = (-1)^{p_a}$  and  $\eta_{ab} = (-1)^{p_a p_b}$ . For a word  $a = (a_1, \dots, a_n) \in \mathfrak{S}^n$  put  $\eta_a = \eta_{a_1} \cdots \eta_{a_n}$ . Bars denote entrywise complex conjugation of matrices,  $\otimes$  the Kronecker product, and  $\text{Tr}$  the trace of a matrix of the size at hand.

**Definition 8.1** (Graded Fock space, fermion parity, translations).  $\mathfrak{F}$  is the tensor product of the symmetric Fock space over  $\bigoplus_{p_a=0} L^2([0, L])$  and the antisymmetric Fock space over  $\bigoplus_{p_a=1} L^2([0, L])$ , with vacuum  $\Omega_{\mathfrak{F}}$ ,  $n$ -particle subspace  $\mathfrak{F}_n$  and finite-particle subspace  $\mathfrak{F}_{\text{fin}}$ . For  $f \in L^2([0, L])$ ,  $a_a^\dagger(f)$  creates a particle of species  $a$  in the mode  $f$  and  $a_a(f)$  is its adjoint on  $\mathfrak{F}_{\text{fin}}$ . For a family  $u = (u_a)_{a \in \mathfrak{S}}$  of unitaries on  $L^2([0, L])$ ,  $\Gamma(u)$  is its second quantisation. The *fermion parity* is  $(-1)^{\hat{F}} = \Gamma((\eta_a)_a)$ . The *periodic translation* by  $t \in [0, L]$  is  $U_t = \Gamma((u_t)_a)$  with  $(u_t f)(x) = f(x - t \bmod L)$ ; the *antiperiodic translation*  $U_t^{\text{ap}}$  uses  $u_t$  on bosonic species and, on fermionic species,  $(u_t^{\text{ap}} f)(x) = f(x - t)$  for  $x \geq t$  and  $-f(x - t + L)$  for  $x < t$ . The facts used about these objects are listed in assumption 15.1.

**Definition 8.2** (Boson–fermion cMPS ring vector, bond grading). *cMPS data* are a bond dimension  $D$ , matrices  $Q, R_a \in M_D(\mathbb{C})$  ( $a \in \mathfrak{S}$ ), constant in  $x$ , and a *boundary matrix*  $B \in M_D(\mathbb{C})$ . The open ordered simplex is  $\Delta_n^< = \{x \in (0, L)^n : x_1 < \cdots < x_n\}$ ,  $\mathcal{K}_n = L^2(\Delta_n^< \times \mathfrak{S}^n)$  (Lebesgue times counting measure), and the *n-particle amplitude* is

$$c_n^B(x; a) = \text{Tr}[B e^{x_1 Q} R_{a_1} e^{(x_2 - x_1) Q} R_{a_2} \cdots R_{a_n} e^{(L - x_n) Q}], \quad c_0^B = \text{Tr}[B e^{LQ}].$$

The *cMPS ring vector* is  $\Psi_B^c = \sum_{n \geq 0} \mathcal{J} c_n^B \in \mathfrak{F}$ , where  $\mathcal{J}$  is the isometry of Lemma 15.4 and the series converges by Theorem 15.6; this is the state of [33] with the positions increasing from left to right. A *bond grading* is  $P \in M_D(\mathbb{C})$  with  $P^2 = 1$ ,  $PQP = Q$  and  $PR_a P = \eta_a R_a$  for all  $a$ ; its sector projections are  $\Pi_\pm = \frac{1}{2}(1 \pm P)$ . The *untwisted* ring is  $\Psi_1^c$  and the *twisted* ring is  $\Psi_P^c$ . No regularity (supercommutation) condition and no normalisation of  $Q$  is imposed.

**Definition 8.3** (Sign-twisted transfer generators, doubled-bond parity, super-invariants). For two cMPS data  $(Q, R), (Q', R')$  with the same  $D$  and  $\mathfrak{S}$ , and signs  $s \in \{\pm 1\}^{\mathfrak{S}}$ , the *sign-twisted transfer generator* on  $\mathbb{C}^D \otimes \mathbb{C}^D$  is

$$\mathbb{T}_s = Q \otimes 1 + 1 \otimes \bar{Q}' + \sum_{a \in \mathfrak{S}} s_a R_a \otimes \bar{R}'_a.$$

$\mathbb{T}$  is  $\mathbb{T}_s$  with all  $s_a = 1$  and  $\mathbb{T}_\eta$  is  $\mathbb{T}_s$  with  $s_a = \eta_a$ ; when the primed data are not named they equal the unprimed ones. Given a bond grading  $P$ , the *doubled-bond parity* is  $\Gamma_b = P \otimes \bar{P}$ ; since  $\Gamma_b^2 = 1$ ,  $\mathbb{C}^D \otimes \mathbb{C}^D$  is the direct sum of its *even* (+1) and *odd* (−1) eigenspaces. Under  $|i\rangle \otimes |j\rangle \mapsto |i\rangle \langle j|$  with  $P = 1_{D_+} \oplus (-1)_{D_-}$  the even sector is the block-diagonal and the odd sector the block-off-diagonal matrices, of dimensions  $D_+^2 + D_-^2$  and  $2D_+ D_-$ . For an operator  $M$  commuting with  $\Gamma_b$ ,  $\text{str } M = \text{Tr}(\Gamma_b M)$ , which is the supertrace of Definition 5.1, and for  $z \in \mathbb{C}$  outside the spectrum of  $M$ ,  $\text{sdet}(z - M) = \det(z - M|_{\text{even}}) / \det(z - M|_{\text{odd}})$ .

## 9 Definitions, Continued: Graded Lattice Tensors and Lifted Frobenius

Continuation of Section 8; lattice (discrete-site) versions of the graded cMPS objects, and the arithmetic objects of the ring-norm-tensor campaign (Sections 20 and 23).

**Definition 9.1** (Graded lattice tensor, doubled transfer matrix,  $P$ -closed ring norm, ring zeta, gauge). A *graded bond* is  $V = V_+ \oplus V_- = \mathbb{C}^{m|k}$  with parity  $P = 1_m \oplus (-1_k)$ . A *graded lattice tensor* is a finite family  $A_s \in \text{End}(V)$ ,  $s$  in a finite alphabet (the *letters* or physical species), each homogeneous: *even* ( $PA_sP = A_s$ , block diagonal,  $A_s = a_s \oplus B_s$ ) or *odd* ( $PA_sP = -A_s$ , block off-diagonal,  $A_s = \begin{pmatrix} 0 & c_s \\ d_s & 0 \end{pmatrix}$ ). Its *doubled transfer matrix* is  $\mathcal{E}_d = \sum_s A_s \otimes \bar{A}_s$  on  $V \otimes \bar{V}$  (ordinary Kronecker product, no statistics sign), graded by the doubled parity  $\Gamma_b = P \otimes \bar{P}$  of Definition 8.3: the *even sector* is  $(+, +) \oplus (-, -)$  and the *odd sector*  $(+, -) \oplus (-, +)$ ;  $\mathcal{E}_d$  commutes with  $\Gamma_b$ . The  *$P$ -closed ring norm* of length  $n$  is  $\|\Psi_P(n)\|^2 := \text{str}_{\Gamma_b} \mathcal{E}_d^n = \sum_{s_1 \dots s_n} |\text{Tr}(PA_{s_1} \dots A_{s_n})|^2$  (the second equality is an unconditional identity, since  $\mathcal{E}_d^n = \sum_w A_w \otimes \bar{A}_w$  by Kronecker multiplicativity; its identification with the Fock norm of the physical fermion ring is Theorem 16.2), the *ring zeta* is  $\exp(\sum_n \text{str} \mathcal{E}_d^n u^n / n) = 1 / \text{sdet}(1 - u\mathcal{E}_d) = \det(1 - u\mathcal{E}_d|_{\text{odd}}) / \det(1 - u\mathcal{E}_d|_{\text{even}})$ , and an *even gauge* is  $A_s \mapsto GA_sG^{-1}$  with  $G$  even and invertible, which conjugates  $\mathcal{E}_d$  by  $G \otimes \bar{G}$  and fixes every ring norm. *Net spectrum* means after cancelling common even and odd eigenvalues; eigenvalue 0 and nilpotent blocks are invisible to all positive powers.

**Definition 9.2** (Lifted Frobenius of an ordinary elliptic curve; toral endomorphism; Hodge ket bond). Let  $E/\mathbb{F}_q$  be an ordinary elliptic curve with Frobenius endomorphism  $\pi$ , trace  $a = q + 1 - N_1$ , endomorphism order  $O = \text{End}(E)$  in the imaginary quadratic field  $K = \mathbb{Q}(\pi)$ . The *lifted Frobenius* is the pair  $(\Lambda, \pi)$  of assumption 20.4: a lattice  $\Lambda \subset \mathbb{C}$  that is a proper  $O$ -ideal, on which  $\pi$  acts by multiplication. Its *integral matrix*  $M \in M_2(\mathbb{Z})$  is the matrix of  $z \mapsto \pi z$  on a  $\mathbb{Z}$ -basis of  $\Lambda$ , and the *toral endomorphism* is the induced map  $M: \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2 \rightarrow \mathbb{T}^2$ . The *Hodge ket bond* is  $V = \Lambda^*(H^{1,0}) = \mathbb{C} \oplus \mathbb{C} dz = \mathbb{C}^{1|1}$  with  $P = (-1)^{\text{deg}}$ ; its double  $V \otimes \bar{V} = \Lambda^*(\mathbb{C} dz \oplus \mathbb{C} d\bar{z}) = H^*(\mathbb{T}^2, \mathbb{C})$  is graded by total form degree.

**Definition 9.3** (Elliptic norm tensor). For a unit vector  $\phi = \sum_s a_s |s\rangle$ ,  $\sum_s |a_s|^2 = 1$ , the *elliptic norm tensor* of  $E$  is the all-even graded lattice tensor  $A_s = a_s \text{diag}(1, \pi)$  on the Hodge ket bond, with  $P$ -closed ring vector  $\Psi_P(n) = (1 - \pi^n) \phi^{\otimes n}$ . The singleton alphabet  $a_0 = 1$  is allowed.

**Definition 9.4** (Genus-two bond and block notation). For a genus-two curve  $C/\mathbb{F}_q$  with simple ordinary Jacobian  $J$ , Frobenius  $\pi$ , quartic CM field  $K = \mathbb{Q}(\pi)$ , Frobenius eigenvalues  $\{\pi_1, \bar{\pi}_1, \pi_2, \bar{\pi}_2\}$  and CM type  $\Phi = \{\phi_1, \phi_2\}$  with  $\phi_j(\pi) = \pi_j$  (assumption 23.4), the *genus-two bond* is  $V = \mathbb{C} \oplus H^{1,0}(\tilde{J}) = \mathbb{C}^{1|2}$ ,  $P = (-1)^{\text{deg}}$ , with  $D = \text{diag}(\pi_1, \pi_2)$  on  $H^{1,0}$ . For a graded lattice tensor on  $\mathbb{C}^{1|2}$  with even letters  $a_s \oplus B_s$  and odd letters  $(c_f, d_f)$  ( $c_f$  a row,  $d_f$  a column) the *block notation* is  $M = \sum_s a_s \bar{B}_s$ ,  $N = \sum_f \bar{d}_f c_f$ ,  $t = \sum_s |a_s|^2$ ,  $u = \sum_f c_f \otimes \bar{c}_f$ ,  $v = \sum_f d_f \otimes \bar{d}_f$ ,  $W = \sum_s B_s \otimes \bar{B}_s$ , in the ordered odd basis  $(|0\rangle \otimes |\bar{j}\rangle)_j, (|i\rangle \otimes |\bar{0}\rangle)_i$  and row-major vectorisation of the  $(-, -)$  block (Lemma 23.6).

**Definition 9.5** (Nine-letter tensor). For  $q > 1$  and  $\pi_1, \pi_2$  with  $|\pi_j|^2 = q$  and  $q + 1 \geq 4\sqrt{q}$ , put  $t = (q + 1)/2$ ,  $w = (q + 1)/4$ ,  $h = (q - 1)/(2\sqrt{2})$ ,  $B_0 = D/\sqrt{t}$ ,  $b_0 = \text{vec } B_0$ ,  $\nu = b_0^* b_0 = 2q/t$ ,  $S = \sqrt{w} 1_4 + \nu^{-1}(\sqrt{w - \nu} - \sqrt{w}) b_0 b_0^*$  (the positive square root of  $w 1_4 - b_0 b_0^*$ ) and  $B_\ell = \text{unvec}(S e_\ell)$ ,  $\ell = 1, \dots, 4$ . The *nine-letter tensor* on  $\mathbb{C}^{1|2}$  has even letters  $A_0 = \sqrt{t} \oplus B_0$ ,  $A_\ell = 0 \oplus B_\ell$  and odd letters  $A_{\uparrow j} = \begin{pmatrix} 0 & \sqrt{h} e_j^T \\ 0 & 0 \end{pmatrix}$ ,  $A_{\downarrow j} = \begin{pmatrix} 0 & 0 \\ \sqrt{h} e_j & 0 \end{pmatrix}$ ,  $j = 1, 2$  (Theorem 23.8).

## 10 What the Riemann Hypothesis Has Become

Nothing here is new mathematics; it is a translation, written for TJO after the founding session from the ingredients built up there. Its point is one sentence, isolated in observation 10.3 and repeated in every later costume: the Riemann hypothesis says certain counts fluctuate no more than a random sequence would.

### 10.1 The one identity, and its two readings

*Observation 10.1* (The one identity). [sketched] For any matrix  $M$  with eigenvalues  $\lambda_1, \dots, \lambda_N$ ,

$$\frac{1}{\det(1 - uM)} = \prod_i \frac{1}{1 - \lambda_i u} = \exp\left(\sum_{\ell \geq 1} \frac{\text{Tr}(M^\ell)}{\ell} u^\ell\right).$$

The middle expression is a bosonic partition function, one mode per eigenvector with Boltzmann factor  $\lambda_i u$ ; the right expression is a sum over closed walks of length  $\ell$ , carrying the  $1/\ell$  of a ring's rotation symmetry. One object, two descriptions: a gas of modes and a gas of rings (Definitions 3.1 and 3.2).

The whole subject is the tension between the two sides. For a permutation matrix the ring side is concrete:  $\text{Tr}(M^\ell)$  counts the fixed points of the  $\ell$ -th power, the closed walks are the cycles and their repeats, and the identity collapses to  $\det(1 - uM)^{-1} = \prod_c (1 - u^{|c|})^{-1}$  over cycles  $c$ : a Bose gas with one mode per cycle, energy the cycle length. That product is the Euler product, and the primes will be the cycles.

### 10.2 A finite permutation: the analogue is free

*Observation 10.2* (RH for a finite permutation). [sketched]  
A permutation matrix permutes an orthonormal basis, hence is unitary, hence has every eigenvalue on the unit circle. The RH analogue at this level costs nothing.

### 10.3 An infinite permutation: a statement about a different matrix

For infinitely many cycles the permutation matrix is still unitary and still useless, its spectrum being roots of unity with infinite multiplicity. The count  $\text{fix}(\ell) = \sum_{d|\ell} d c_d$  of fixed points of the  $\ell$ -th power, with  $c_d$  the number of  $d$ -cycles, remains meaningful, and the interesting object is a *small matrix* (Definition 3.3) reproducing them.

*Observation 10.3* (RH for the small matrix). [sketched] A small matrix  $M_{\text{small}}$  with  $\text{Tr } M_{\text{small}}^\ell = \text{fix}(\ell)$  for all  $\ell$  exists exactly when the zeta function is rational, and then its eigenvalues are new numbers, off the unit circle. Writing  $\text{fix}(\ell) = \lambda_{\text{max}}^\ell + \sum_{i \geq 2} \lambda_i^\ell$ , the RH analogue is that every other nonzero  $\lambda_i$  has  $|\lambda_i| = \sqrt{\lambda_{\text{max}}}$  (Definition 3.4), so that the fluctuation has size  $\sqrt{\lambda_{\text{max}}^\ell}$ : the square root of the leading term, which is the fluctuation of a sum of  $\lambda_{\text{max}}^\ell$  independent coin tosses. In counting language, RH says the cycle counts fluctuate no more than a random sequence would.

So the RH analogue is here a property some systems have and others do not. The note records two shift examples, both perfectly good infinite permutations.

| shift                | eigenvalues of the small matrix | RH analogue |
|----------------------|---------------------------------|-------------|
| 4-symbol cyclic rule | 2, $1 + i$ , $1 - i$ , 0        | holds       |
| golden-mean shift    | $\phi$ , $-1/\phi$              | fails       |

## 10.4 A graph: Ramanujan

*Observation 10.4* (RH for a graph is the Ramanujan property).

[sketched]

For a  $(q+1)$ -regular graph the permutation is the shift on non-backtracking closed walks, the small matrix is the adjacency matrix  $A$ , and the Ihara–Bass formula (Definition 3.8) compresses one to the other: each eigenvalue  $\lambda$  of  $A$  produces a pair  $\mu, \mu'$  of eigenvalues of the edge operator  $T$  through

$$\mu^2 - \lambda\mu + q = 0, \quad \text{so} \quad \mu\mu' = q.$$

The leading  $\lambda = q+1$  gives  $\mu = q, \mu' = 1$ . For the rest,  $\lambda^2 < 4q$  forces a complex conjugate pair with  $|\mu| = |\mu'| = \sqrt{q}$ , on the critical circle, while  $\lambda^2 > 4q$  gives a real pair, one factor above  $\sqrt{q}$  and one below. Hence RH for  $\zeta_X$  holds if and only if  $|\lambda| \leq 2\sqrt{q}$  for every nontrivial  $\lambda$ , i.e. iff the graph is Ramanujan (Definition 3.9), and the closed non-backtracking walk counts are  $N_\ell = q^\ell + O(q^{\ell/2})$ .

Three remarks carry over to the primes. First, **the pairing comes from a symmetry, the hypothesis from a bound**:  $\mu\mu' = q$  is a functional equation pairing eigenvalues symmetrically about  $\sqrt{q}$  on a logarithmic scale, and RH is the extra statement that each pair is degenerate in modulus, which happens exactly when it is a conjugate pair. Second, **self-adjointness is not enough**:  $A$  is symmetric, so  $\lambda$  is real, which guarantees the pairing but not the bound; the prism graph  $C_{16} \times K_2$  is symmetric and fails it. Third, **the bound is geometric**:  $[-2\sqrt{q}, 2\sqrt{q}]$  is the spectrum of the infinite tree covering the graph, so RH says the graph has no eigenvalue outside the spectrum of its universal cover; the word is tempered.

## 10.5 A matrix product state: one correlation length

Let  $E = \Sigma$  be the transfer matrix of a matrix product state (Definition 3.12), so that  $\text{Tr } E^n = \langle \Psi_n | \Psi_n \rangle$  is the ring norm of the periodic state on  $n$  sites (Definition 3.13) and the zeta function of observation 10.1 is the generating function of ring norms. In canonical form (Definition 3.14) the leading eigenvalue is 1 and the rest lie inside the unit disc,  $\lambda_k = e^{-1/\ell_k^c + i\vartheta_k}$  (Definition 3.15), so  $\langle \Psi_n | \Psi_n \rangle = 1 + \sum_k e^{-n/\ell_k^c} e^{in\vartheta_k}$ .

*Observation 10.5* (RH for an MPS).

[sketched]

Translating observation 10.4, where the leading growth is  $q$  per step and the subleading modulus  $\sqrt{q}$ : in normalised units the subleading modulus is  $q^{-1/2}$ , so  $e^{-1/\ell_k^c} = q^{-1/2}$  and  $\ell_k^c = 2/\log q = 2/h$  for every  $k$ . The RH analogue for an MPS reads: all correlation lengths are equal, and equal to twice the inverse entropy rate. The phases  $\vartheta_k$  are unconstrained and there may be infinitely many of them; only the moduli are pinned.

The hierarchy this sits in has three levels, and each has a classical number-theoretic counterpart, anticipating observation 10.6 below.

| statement about $E$<br>(or $\mathcal{G}$ )                                        | name                           | counting meaning                                        | number theory                                                 |
|-----------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------|---------------------------------------------------------------|
| a unique eigenvalue on the unit circle                                            | primitivity (Perron–Frobenius) | ring norms follow the leading term                      | $\zeta(1+it) \neq 0$ , the prime number theorem               |
| all others inside a smaller circle                                                | spectral gap                   | fluctuation exponentially smaller than the leading term | the zero-free region, $\psi(x) = x + O(xe^{-c\sqrt{\log x}})$ |
| all others on <i>one</i> circle of radius $e^{-1/\ell_k^c}$ with $\ell_k^c = 2/h$ | Ramanujan / RH                 | fluctuation is the square root of the leading term      | all zeros on $\text{Re } s = \frac{1}{2}$                     |

## 10.6 The primes: continuous length, and the sign

For the primes the cycle lengths are  $\log p$ , rationally independent, so the matrix powers become a semigroup  $e^{\mathbf{L}\mathcal{G}}$  in a continuous length  $\mathbf{L}$  and the ring norms become a measure in  $\mathbf{L}$ . Start from the Riemann–von Mangoldt formula for  $\psi$ , valid for  $x$  not a prime power, put  $x = e^{\mathbf{L}}$  and differentiate in  $\mathbf{L}$ : the left side becomes a sum of delta functions, one per prime power, of weight  $\log p$ , and the right side a sum of exponentials.

*Observation 10.6* (The explicit formula as a trace, and the pairing).

[sketched] With

$$\psi(x) = x - \sum_{\rho} x^{\rho}/\rho - \log 2\pi - \frac{1}{2} \log(1 - x^{-2}),$$

$$\sum_n \Lambda(n) \delta(\mathbf{L} - \log n) = e^{\mathbf{L}} - \sum_{\rho} e^{\rho \mathbf{L}} - (e^{-2\mathbf{L}} + e^{-4\mathbf{L}} + \dots),$$

which is  $\text{Tr } e^{\mathbf{L}\mathcal{G}}$  for a generator  $\mathcal{G}$  whose eigenvalues are 1 (the pole of  $\zeta$ , the leading term, entropy 1), every zero  $\rho$ , and the trivial zeros  $-2, -4, \dots$ : the trace identity of observation 10.1 with  $M^{\ell}$  replaced by  $e^{\mathbf{L}\mathcal{G}}$ , ring side the prime powers at lengths  $\log n$ , mode side the zeros. The functional equation  $\xi(s) = \xi(1-s)$  makes  $\rho$  and  $1-\rho$  both zeros, so after dividing by the leading  $e^{\mathbf{L}}$  the normalised decay rates come in pairs  $(1-\sigma, \sigma)$  summing to 1, exactly as  $\mu\mu' = q$  paired the graph eigenvalues around  $\sqrt{q}$ . That pairing is a theorem; RH is the extra statement that every pair is degenerate,  $\sigma = 1-\sigma = \frac{1}{2}$ .

Two features of this trace identity are visible immediately. The first is *the sign*: the zeros enter with a minus sign relative to the leading term, the same minus sign that marks the fermionic  $H^1$  block in the elliptic-curve case (Section 25) and the absorption spectrum in Connes’s language. The total measure stays positive because  $\Lambda(n) \geq 0$ , but the subleading sector of the bond space contributes negatively, so whatever state this is, its bond space is graded, not plainly bosonic. The second is the pairing just stated. The numerical check against several thousand zeros is recorded in Section 11. Collecting the costumes:

| language                  | the Riemann hypothesis says                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| counting small matrix MPS | $\psi(x) = x + O(x^{1/2+\epsilon})$ : the prime powers fluctuate like coin tosses<br>every subleading mode has modulus the square root of the leading one<br>the state whose rings are the primes has <i>one</i> correlation length, $\ell_k^c = 2$ since $h = 1$ , and an infinite tower of frequencies $\gamma_n$ |
| channel                   | the transfer semigroup $e^{\mathbf{L}\mathcal{G}}$ has a single uniform decay rate $\frac{1}{2}$                                                                                                                                                                                                                    |

## 10.7 The Hilbert space where this is literally true

The channel version is not only an analogy. It is a theorem on a concrete Hilbert space: the model space  $K_S$  of the scattering symbol  $S$  of the modular surface, on which the compressed dilation semigroup  $Z(t)$  has one mode per zero, with decay rate and frequency read off from that zero. This is Lax–Phillips [56], and “RH iff every mode of  $Z(t)$  decays at the same rate” is Faddeev–Pavlov [26]. What the founding session added is where  $S$  comes from: its phase is the imaginary part of the Bost–Connes Lévy exponent at the critical temperature  $\beta = 1$  [10], so the ring gas and the transfer semigroup are one object seen from its two ends, bridged by the critical phase. Construction, normalisations (the factor  $\frac{1}{2}$  from the doubled variable) and numerics are in Section 11; Definitions 4.1 to 4.3 fix the objects.

## 10.8 Why it is still hard, in this language

The graph case is the honest guide. There side B exists, the trace identity holds and the small matrix is symmetric for every graph, and most graphs fail the bound. Being Ramanujan is never a consequence of the structure; it is an extra fact, and in every known case it comes from arithmetic input of Deligne’s type [18]. For Selberg’s surfaces the same statement, no Laplace eigenvalue below  $\frac{1}{4}$ , is open.

What the translation settles: RH is a statement about the *sizes* of the subleading modes of a transfer semigroup, not about its existence. The semigroup exists, the zeros are its modes, the primes are its rings, the functional equation is the pairing of its rates. What it does not settle: a mechanism forcing the pairs to be degenerate. In positivity form (Definition 4.8), RH holds iff  $F_W$  is positive-definite on the dilation group, iff its spectral measure is positive on the real line, which by Bochner is the same as all its atoms being real.

*Observation 10.7* (Hermiticity of the Hilbert–Pólya operator).

[sketched]

Write the length-flow generator on the subleading sector as  $\mathcal{G} = -\frac{1}{2} + iH$  (Definition 4.7); then  $H$  has eigenvalues  $\gamma_n - i(\sigma_n - \frac{1}{2})$ , and RH says exactly that  $H$  is Hermitian, i.e. that the transfer semigroup is a uniform damping times an oscillation with real frequencies.  $H$  is the Hilbert–Pólya operator. The non-Hermitian generator carries the complex pairs and the Hermitian operator their symmetric combination, here  $\rho(1 - \rho) = \sigma(1 - \sigma) + \gamma^2 + i\gamma(1 - 2\sigma)$ , which is real iff  $\sigma = \frac{1}{2}$ ;  $\mathcal{G}$  is to  $H$  what the Hashimoto edge operator  $T$  is to the adjacency matrix  $A$  of a graph. For graphs the Hermitian object is given and its reality is a theorem; for the primes it is the conjecture.

Proving it would mean exhibiting that state, or a Stinespring form of  $Z(t)$  with the prime dilations as Kraus operators, in a way that makes the positivity manifest; that is conjecture 11.10. The graph lesson says the Kraus structure alone will not do it, and that some arithmetic rigidity has to enter, playing the role that weights play for curves (Section 25).

## 10.9 Erratum on words

An earlier draft, and the founding session, called  $-\log E$  the entanglement Hamiltonian. That is wrong: the entanglement Hamiltonian of an MPS is  $-\log$  of a reduced density matrix built from the fixed points of  $E$  alone, is Hermitian by construction, and does not see the subleading spectrum. The operator whose spectrum contains the zeros is the transfer generator, non-Hermitian in general, whose eigenvalue phases are the oscillation frequencies of the ring norms. observation 10.7 states the correct relation; the route is listed among the dead ones in Section 35.

## 10.10 Summary

A zeta function is one partition function with two Fock-space descriptions: a gas of rings (cycles, closed walks, prime powers) and a gas of modes (eigenvalues of a small transfer matrix); the trace identity links them. For the primes the rings have lengths  $\log n$  and weights  $\Lambda(n)$ , the modes are the zeros of  $\zeta$  plus the pole, and the functional equation pairs the modes' decay rates as  $(\sigma, 1 - \sigma)$ . The Riemann hypothesis says every pair is degenerate at  $\frac{1}{2}$ : the transfer semigroup has one correlation length,  $\ell_k^c = 2$ , so the ring counts fluctuate by no more than the square root of their size, so the primes are as regular as coin tosses. That semigroup exists explicitly, its scattering symbol is the phase of the Bost–Connes system at its critical temperature, and RH is the statement that this channel is Ramanujan.

## 11 The Riemann Channel: Bost–Connes at Critical Temperature, Compressed

This section is the lab-book version of `notes/riemann-channel-note.md`, a scratch note of the founding session. It assembles one object out of three standard pieces: Lax–Phillips scattering on the modular surface, the Sz.-Nagy–Foias functional model of an inner function, and the Bost–Connes system at its phase transition. Nothing here is new mathematics; the content is the identification of the three, and the numerical checks that it is stated correctly. Everything below is *equivalent to*, never a proof of, the Riemann hypothesis. Several statements are theorem-level standard theory (Faddeev–Pavlov, Lax–Phillips, Adamyan–Arov, Khinchin) for which this repo holds no local source and no written-out argument; they carry the tag `sketched` for that reason, not because their truth is in doubt.

### 11.1 Setting

Let  $\tau$  be the Mellin variable on the dilation group  $\mathbb{R}_+^\times$ , so that dilation by  $e^t$  acts on  $L^2(\mathbb{R}_+^\times)$  as the multiplier  $e^{it\tau}$ . With the completed zeta

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

the scattering symbol of the modular surface in this variable (Definition 4.1) is

$$S(\tau) = \frac{\xi(1-2i\tau)}{\xi(1+2i\tau)}.$$

Write a nontrivial zero as  $\rho_n = \sigma_n + i\gamma_n$ , so that the Riemann hypothesis is  $\sigma_n = \frac{1}{2}$  for all  $n$  (Section 2).

### 11.2 The symbol is inner

**Proposition 11.1** (The scattering symbol is inner, with the zeros as its zeros). *[sketched]  $S$  is unimodular on the real line, analytic and contractive in the lower half plane  $\text{Im } \tau < 0$ , and its zeros there are exactly*

$$\tau_n = \frac{\gamma_n}{2} - i \frac{\sigma_n}{2}, \quad \rho_n = \sigma_n + i\gamma_n \text{ a nontrivial zero of } \zeta.$$

The note’s argument:  $\xi$  is real on the real axis and satisfies  $\xi(s) = \xi(1-s)$ , which gives  $|S| = 1$  for real  $\tau$ ; for  $\text{Im } \tau < 0$  the denominator’s argument  $1+2i\tau$  has real part greater than 1, inside the Euler product, so  $S$  is analytic there. Contractivity in the interior is the Phragmén–Lindelöf step, which the note does not write out (step 3 of `HANDOFF.md`); for the modular surface it is classical [26, 56]. The zeros are a substitution:  $\xi(1-2i\tau) = 0$  iff  $1-2i\tau$  is a nontrivial zero.

### 11.3 The compressed semigroup

Since  $S$  is inner, the model space  $K_S = H^2 \ominus SH^2$  (Definition 4.2) carries the compressed dilation semigroup  $Z(t) = P_{K_S} e^{it\tau}|_{K_S}$ ,  $t > 0$ . By Adamyan–Arov this is unitarily equivalent to the Lax–Phillips semigroup of the modular surface with the cusp forms removed; the note states this without a written argument, and no local source exists for it here.

**Proposition 11.2** (One mode per zero; RH as a single decay rate).

[sketched]

For each zero  $\lambda$  of  $S$  in the lower half plane the reproducing kernel  $k_\lambda(\tau) = (\tau - \bar{\lambda})^{-1}$  lies in  $K_S$ , and  $Z(t)^* k_\lambda = e^{-it\bar{\lambda}} k_\lambda$ . With  $\lambda = \gamma_n/2 - i\sigma_n/2$  the eigenvalue is  $\exp(-\sigma_n t/2 - i\gamma_n t/2)$ : one mode per zero, decay rate  $\sigma_n/2$ , frequency  $\gamma_n/2$ . Hence the Riemann hypothesis holds if and only if every mode of  $Z(t)$  decays at the single rate  $1/4$ .

The kernel is orthogonal to  $SH^2$  precisely because  $S(\lambda) = 0$ . The last sentence is the Faddeev–Pavlov / Lax–Phillips reading of RH [26, 56]; it is the “one correlation length” criterion of Section 10 (observation 10.5) on an explicit Hilbert space, with  $K_S$  in the role of the bond space. The completely positive lift is immediate and uninformative:  $\Phi_t(X) = Z(t)^* X Z(t)$  on  $B(K_S)$  is the Riemann channel of Definition 4.3, a sub-unital semigroup with one Kraus operator, whose Stinespring dilation is the automorphic wave group and whose environment is the pair of incoming and outgoing subspaces, that is, the cusp. Whether a less trivial Kraus decomposition exists is conjecture 11.10.

## 11.4 Where the primes are

Two placements of the primes appear, and the note is explicit that they are different objects. The first is a dilation-covariant Markov semigroup on  $L^2(\mathbb{R}_+^\times)$  in Holevo form, with the von Mangoldt jump measure  $\nu_\sigma = \sum_{p,k} \frac{p^{-k\sigma}}{k} \delta_{k \log p}$  and symbol

$$\left( \frac{\zeta(\sigma - i\tau)}{\zeta(\sigma)} \right)^t = \exp \left( t \sum_{p,k} \frac{p^{-k\sigma}}{k} (e^{ik\tau \log p} - 1) \right),$$

a genuine Markov semigroup for  $\sigma > 1$  by Khinchin’s theorem that the zeta distribution is compound Poisson (standard; no local source). It has no zeros for  $\sigma > 1$ , and at  $\sigma = \frac{1}{2}$  it is not a semigroup at all, the symbol being unbounded. The Bost–Connes phase transition  $\beta = 1$  is the finite- versus infinite-activity threshold of  $\nu_\sigma$  [10].

The second is  $Z(t)$  on  $K_S$ : the zeros as resonances, no jump structure, the primes entering only through  $S$ . The bridge between the two is the boundary phase.

**Proposition 11.3** (The symbol is the phase of the critical Lévy exponent). [sketched] For real  $\tau$ ,  $S(\tau) = \exp(-2i \arg \zeta(1 + 2i\tau))$ , and

$$-\arg \zeta(1 + 2i\tau) = \sum_p \sum_{k \geq 1} \frac{p^{-k}}{k} \sin(2k\tau \log p),$$

which is the imaginary part of the Bost–Connes Lévy exponent  $\kappa_\sigma(\tau') = \log \zeta(\sigma - i\tau') - \log \zeta(\sigma)$  at the critical temperature  $\sigma = 1$ ,  $\tau' = 2\tau$ . Hence  $S$  is the phase of the critical Lévy exponent times the Gamma phase.

The real part of  $\kappa_1$  diverges, since  $\sum_p p^{-1} = \infty$ ; that divergence is the Bost–Connes phase transition. The imaginary part converges conditionally, at prime-number-theorem strength, i.e. by  $\zeta(1 + it) \neq 0$ . The bridge the note proposes: take the Bost–Connes process at  $\beta = 1$ , discard the divergent real part of its Lévy exponent, keep the phase, and use that unimodular function as the characteristic function of a contraction semigroup. Symbolically, Lax–Phillips = (Bost–Connes at  $\beta = 1$ ) + (Sz.-Nagy–Foias), at the level of boundary phases.

## 11.5 Numerical check of the symbol

*Numerical result* 11.4 (Unimodularity, zeros, innerness and the prime phase). [numerical]  
`scripts/scat.py` (`outputs/scat.txt`, mpmath at 20 digits) reports  $||S| - 1| = 0$  to machine precision at  $\tau = 0.3, 2.0, 7.5, 21.3$ ; a root search from  $\gamma_n/2 - \frac{1}{4}i$  converging for  $n = 1, \dots, 6$  to  $\gamma_n/2 - 0.250000i$  with  $|S| \leq 1.5 \cdot 10^{-20}$ ;  $|S| = 0.9348, 0.5393, 0.8344, 0.2718$  at the interior points  $\tau = 3 - 0.6i, 10 - i, 0.5 - 2i, 30 - 0.3i$ ; and Proposition 11.3 to  $1.3 \cdot 10^{-3}$  or better at five points:

| $\tau$ | exact    | prime sum + tail | residual             |
|--------|----------|------------------|----------------------|
| 0.7    | +0.80572 | +0.80469         | $-1.0 \cdot 10^{-3}$ |
| 1.9    | -0.03879 | -0.03854         | $+2.5 \cdot 10^{-4}$ |
| 4.2    | -0.17233 | -0.17306         | $-7.3 \cdot 10^{-4}$ |
| 9.0    | +0.06468 | +0.06594         | $+1.3 \cdot 10^{-3}$ |
| 15.5   | -0.06432 | -0.06409         | $+2.3 \cdot 10^{-4}$ |

The middle column is the prime sum over  $p < 3 \cdot 10^6$  truncated at  $k \leq 6$  plus the prime-number-theorem tail  $\pi/2 - \text{Si}(2\tau \log X)$ , which is what makes the residual  $10^{-3}$  rather than  $10^{-2}$ ; `outputs/scat.txt` archives the bare prime sums, before the tail, at three of the five points (0.82380, -0.05607, -0.18043). The innerness numbers are a check at four points, not a proof.

## 11.6 Ring norms: the trace of the semigroup is the prime measure

Distributionally  $\text{Tr } Z(t) = \sum_n e^{-t\sigma_n/2 + it\gamma_n/2}$ , under RH  $e^{-t/4} \sum_n e^{it\gamma_n/2}$ . The Weil explicit formula (Definition 4.4) turns that zero sum into a prime sum: in the ring-length variable  $L = t/2$ , with window  $\hat{g}$  and transform  $g$ ,

$$\sum_{\gamma} \hat{g}(\gamma) = \hat{g}\left(\frac{i}{2}\right) + \hat{g}\left(-\frac{i}{2}\right) - g(0) \log \pi + \frac{1}{2\pi} \int \hat{g}(r) \text{Re}(\Gamma'/\Gamma)\left(\frac{1}{4} + \frac{ir}{2}\right) dr - 2 \sum_n \frac{\Lambda(n)}{\sqrt{n}} g(\log n).$$

**Proposition 11.5** (The primes are the ring spectrum of the Riemann channel). [proved-conditional]  
For  $t > 0$  the distributional trace of Definition 5.2 is

$$\text{Tr}_{\text{dist}} Z(t) = 1 - 2P_+(t) - \frac{e^{-t/2}}{e^t - 1} = 1 - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{n} \delta(t - 2 \log n) - \sum_{k \geq 1} e^{-(k+1/2)t} :$$

the primes are dips of weight  $-2\Lambda(n)/n$  at the ring lengths  $t = 2 \log n$ , the constant 1 is the pole, the smooth term the archimedean ladder.

Restated 2026-09-12 night from the prover's T4.1; the founding version had weight  $+\Lambda(n)n^{-1/2}$  and no pole term (see Proposition 12.20). In the MPS dictionary of Section 10: rings of length  $\log n$  with weight  $\Lambda(n)$ , a transfer generator with eigenvalues  $-\frac{1}{2} \pm i\gamma_n$ , and a single correlation length iff RH. The primes are the ring spectrum, the zeros the transfer spectrum, and one identity, the explicit formula, links them (Definition 3.2).

*Observation* 11.6 (Sign and factor of the prime weight). [sketched-conditional] In the convention of the explicit formula displayed above, the prime part of the centred trace of  $Z(t)$  carries weight  $-2\Lambda(n)n^{-1/2}$  at  $t = \pm 2 \log n$  (dips), and the positive-time uncentred prime part is  $-2\Lambda(n)/n$ . Proposition 11.5 was right about the support and off by the factor  $-2$  in the weight; it has since been restated in the form printed above.

Recorded 2026-09-12 from the Selberg review’s action item (b) and, independently, from the codex prover’s T6 in `notes/weil-positivity/astra-proofs.md`; the Weil positivity review checks it as its item X3 (Section 27).

*Numerical result* 11.7 (Windowed zero sum against the full explicit formula). [numerical]  
`scripts/ringnorm.py` (`outputs/ringnorm.txt`) sums the first 3000 zeros ( $\gamma_{\max} = 3533.3$ ) against the Gaussian window  $\hat{g}(r) = \cos(rL)e^{-r^2/2G_w^2}$  with  $G_w = 1009.5$ , for which the truncation weight at  $\gamma_{\max}$  is  $2.2 \cdot 10^{-3}$ . The numeric side is  $F(L) = \sum_{\gamma>0} e^{-\gamma^2/2G_w^2} \cos(\gamma L)$  on  $L \in [0.3, 3.3]$ ; the prediction is every term of the explicit formula above, archimedean digamma integral included. The prime terms are Gaussian dips of depth  $\frac{G_w}{2\sqrt{2\pi}} \Lambda(n)n^{-1/2}$  at  $L = \log n$ :

| $n$ | $\log n$ | $F(L_n)$ | prediction | ratio | $\Lambda(n)/\sqrt{n}$ |
|-----|----------|----------|------------|-------|-----------------------|
| 2   | 0.6931   | −97.95   | −98.13     | 0.998 | 0.490                 |
| 3   | 1.0986   | −126.83  | −127.02    | 0.999 | 0.634                 |
| 4   | 1.3863   | −68.77   | −68.94     | 0.998 | 0.347                 |
| 5   | 1.6094   | −143.66  | −143.86    | 0.999 | 0.720                 |
| 7   | 1.9459   | −146.71  | −146.92    | 0.999 | 0.736                 |
| 8   | 2.0794   | −47.87   | −48.03     | 0.997 | 0.245                 |
| 9   | 2.1972   | −72.19   | −72.36     | 0.998 | 0.366                 |
| 11  | 2.3979   | −143.86  | −144.07    | 0.999 | 0.723                 |
| 13  | 2.5649   | −141.21  | −141.41    | 0.999 | 0.711                 |
| 16  | 2.7726   | −32.88   | −33.03     | 0.995 | 0.173                 |
| 27  | 3.2958   | −39.93   | −40.09     | 0.996 | 0.211                 |

The maximum of  $|F - \text{prediction}|$  over the whole interval is 0.20, against dip depths up to 147; away from the dips it is 0.18, so the residual is flat and is the truncation of the zero sum, not a mismatch of terms. Depths scale as  $\Lambda(n)/\sqrt{n}$ , prime powers included. The archived output carries four further rows ( $n = 17, 19, 23, 25$ ) with the same ratios; the table above is the note’s selection.

## 11.7 The Bost–Connes phase operators as an MPO over the prime chain

Item 2 of `HANDOFF.md` concerns the Bost–Connes system itself rather than its compression. Its Hilbert space factorises over the primes, one oscillator per prime (Definition 4.6); the Gibbs state at inverse temperature  $\beta$  is a product state with local weight  $(1 - p^{-\beta})p^{-\beta k}$  at occupation  $k$ , and the isometries  $m_n$  act locally.

**Proposition 11.8** (The Bost–Connes phase operator is a prime-chain MPO). [sketched]  
*The Bost–Connes phase operator  $e(a/b)$  is a matrix product operator over the prime chain with bond space  $\mathbb{Z}/b$ , the local transfer map being  $r \mapsto p^k r$ , that is, Shor’s map  $U_{p^k}$  at the site  $p$ . In the Dirichlet character basis the transfer eigenvalues are  $L(\beta, \bar{\chi})/\zeta(\beta)$ .*

Extremal  $\text{KMS}_\beta$  states differ only by the boundary residue of the chain, and  $\beta \rightarrow 1^+$  kills every nontrivial character sector: the symmetry breaking of [10] read as a boundary effect. Shor’s unitary  $U_a$  is the level- $N$  Galois symmetry of the system and a snapshot of the dilation flow.

*Numerical result* 11.9 (MPO contraction against Hurwitz zeta and Dirichlet  $L$ -values). [numerical]  
`scripts/bcmpto.py` (`outputs/bcmpto.txt`) contracts the chain over all primes below  $2 \cdot 10^5$  (occupations  $k \leq 60$ ) and compares with the Hurwitz-zeta closed form  $\zeta(\beta)^{-1} b^{-\beta} \sum_r e^{2\pi i ar/b} \zeta(\beta, r/b)$ :

| $a/b, \beta$ | MPO                     | closed form             | diff                |
|--------------|-------------------------|-------------------------|---------------------|
| 1/3, 2.0     | $-0.333333 + 0.411340i$ | $-0.333333 + 0.411340i$ | $1.6 \cdot 10^{-7}$ |
| 1/4, 2.0     | $-0.125000 + 0.556841i$ | $-0.125000 + 0.556840i$ | $2.1 \cdot 10^{-7}$ |
| 2/5, 1.5     | $-0.263990 + 0.143625i$ | $-0.263949 + 0.143579i$ | $6.1 \cdot 10^{-5}$ |
| 1/6, 1.7     | $+0.103130 + 0.502210i$ | $+0.103130 + 0.502200i$ | $1.0 \cdot 10^{-5}$ |

For  $b = 5$ ,  $\beta = 1.5$  the contracted transfer operator on the units is diagonal in the character basis to a spread of  $2.6 \cdot 10^{-14}$ , with eigenvalues  $0.910557$ ,  $0.354022 \mp 0.067781i$ ,  $0.225025$  against  $\text{mpmath } L(\beta, \chi)/\zeta(\beta)$  values  $0.910557$ ,  $0.353909 \pm 0.067760i$ ,  $0.224953$ : agreement to  $1.1 \cdot 10^{-4}$ , with the expected conjugation between the two lists.

The residuals are the truncation of the Euler product at  $2 \cdot 10^5$  and grow as  $\beta$  approaches 1, as they must. The relevance here is structural: the bond space of the phase operators is  $\mathbb{Z}/b$ , finite and abelian, whereas that of the compressed semigroup is  $K_S$ , and the note offers no map between them.

## 11.8 What is not established

RH itself is not touched; every statement above is an equivalent form of the uniform-rate statement. The innerness argument is not written out (Proposition 11.1), the Adamyan–Arov identification is quoted without a local source, and the bridge is a statement about boundary phases, not about the operators. No identification with the  $\beta > 1$  constructions of the parent repository is claimed: those sit above the phase transition, the object here is the  $\beta = 1$  phase compressed to a co-invariant subspace.

**Conjecture 11.10** (Stinespring form with prime jumps). *[open] There is a Stinespring form of the compressed semigroup  $Z(t)$  on  $K_S$  whose Kraus (jump) operators are the prime dilations.*

The single-Kraus lift exists trivially and says nothing; the Holevo-form covariant Lindbladian with jump measure  $\nu_\sigma$  lives on the full regular representation and is not the compressed object. Whether a jump-type generator on  $K_S$  reproduces  $Z(t)$  is the open question this note isolates, and its physical index would be the Stinespring environment. A negative answer with a reason is a result; the route is listed in Section 35.

## 12 The Phantasm I: What the Prime Measure Forces

*Phantasm* is the name used in this book for the object the Bost–Connes reframe chases (HANDOFF.md, central priority of 2026-09-12): a continuous matrix product state whose one-dimensional space is dilation time, whose bond carries a “Riemann Lindbladian” with the pole of  $\zeta$  at  $s = 1$  as fixed point and the zeros as relaxation modes, and whose ring norm is the prime measure. Because the ring norms of an ordinary cMPS are sums of squares (Theorem 28.3) while the zeros enter the explicit formula with a minus sign, the bond was expected to be  $\mathbb{Z}_2$ -graded, the zeros odd, the pole even, and the ring norm a supertrace. This shard and the next record what can be proved about that object from the data at hand: what is forced, what cannot exist, and what exists. The proofs are the codex prover’s (notes/riemann-cmps/astra-proofs.md, Lamport style, labels T1–T8), reviewed adversarially by an Opus reviewer (notes/reviews/riemann-cmps-2026-09-12.md); the brief that drafted the statements, several of which the prover corrected, is notes/riemann-cmps/astra-brief.md.

### 12.1 Prior art: the grading is Deninger’s, the sign is Connes’s

*Cited fact* 12.1 (Deninger’s graded dynamical cohomology). [cited] In Deninger’s conjectural picture [19] the completed zeta is an alternating product of regularised determinants over  $H^0$ ,  $H^1$ ,  $H^2$  of a flow  $\Theta$ :  $H^0 = \mathbb{R}$  with  $\Theta = 0$ ;  $H^1$  infinite dimensional, “the spectrum of  $\Theta$  consisting of the non-trivial zeroes  $\rho$  of  $\zeta_K(s)$  with their multiplicities”;  $H^2 \cong \mathbb{R}$  “but with  $\Theta = \text{id}$ ”; and on  $\mathbb{R}^{>0}$  the Lefschetz formula reads  $\sum_i (-1)^i \text{Tr}(\phi^* | H^i) = 1 - \sum_{\rho} e^{t\rho} + e^t$ , equal to the prime comb plus archimedean terms. Verbatim:  $H^1_{\text{dyn}}(\overline{\text{spec } \Theta}, \mathbb{R})$  is infinite dimensional, the spectrum of  $\Theta$  consisting of the non-trivial zeroes  $\rho$  of  $\zeta_K(s)$  with their multiplicities, (math/0505354:main.tex:323);  $H^2_{\text{dyn}}(\overline{\text{spec } \Theta}, \mathbb{R}) \cong \mathbb{R}$  but with  $\Theta = \text{id}$ . (math/0505354:main.tex:323)  $\left( \sum_{i=0}^2 (-1)^i \text{Tr}(\phi^* | H^i_{\text{dyn}}(\overline{\text{spec } \Theta}, \mathbb{R})) = 1 - \sum_{\rho} e^{t\rho} + e^t \right)$  (math/0505354:main.tex:878)

*Cited fact* 12.2 (Connes: the zeros as an absorption spectrum). [cited] Connes [12]: “the spectral interpretation of the zeros of the Riemann zeta function should be as an absorption spectrum rather than as an emission spectrum, to borrow the language of spectroscopy”; the minus sign is that of  $H^1$  in the Lefschetz formula. Verbatim: the spectral interpretation of the zeros of the Riemann zeta function should be as an absorption spectrum rather than as an emission spectrum, to borrow the language of spectroscopy. (math/9811068:main.tex:579–581).

So the forced part of the Phantasm, a graded operator with the pole even and the zeros odd whose Lefschetz trace is the explicit formula, is Deninger’s programme, and the sign is Connes’s. What this book adds is the reading through ring norms and transfer operators: which data are forced by the ring norms alone, which realisations cannot exist, and which exist concretely.

### 12.2 Rigidity: the supertrace determines the graded spectrum

**Lemma 12.3** (Existence of the supertrace distribution). [proved] *For a graded spectral datum (Definition 5.1) the series  $\sum \epsilon m e^{\lambda t}$  converges absolutely after pairing with every test function in  $C_c^\infty(0, \infty)$  and defines a distribution there, independently of the order of summation.*

The estimate integrates by parts in the full complex parameter  $\lambda$ , not only in its imaginary part (the draft’s version was corrected there).

**Theorem 12.4** (Rigidity up to cancelling pairs). *[proved]* Two graded spectral data with the same supertrace distribution on  $(0, \infty)$  have the same net multiplicity  $\nu_{\text{net}}(\lambda)$  at every  $\lambda \in \mathbb{C}$ .

The proof pairs the supertrace with cut-off versions of  $t^k e^{-st}$ ,  $k$  large, the cutoffs nonnegative and monotone in both parameters (which is what part (i) of Proposition 12.12 later needs), shows the limit is  $k! \sum_{\lambda} \nu_{\text{net}}(\lambda)(s - \lambda)^{-k-1}$ , meromorphic on  $\mathbb{C}$  with principal part  $k! \nu_{\text{net}}(\lambda)(s - \lambda)^{-k-1}$  at each  $\lambda$ , and compares principal parts. Ring norms therefore see exactly the net graded spectrum and nothing else: not Jordan blocks, not the Hilbert-space geometry, not the operator domain.

**Assumption 12.5** (Explicit formula in trace form). *[assumed]* For  $u > 0$ , as distributions on  $(0, \infty)$ ,  $\sum_{\rho} e^{\rho u} = e^u - \text{comb}(u) - 1/(e^{2u} - 1)$ , the zero sum in symmetric order; equivalently the  $x$ -derivative of  $\psi_0(x) = x - \sum_{\rho} x^{\rho}/\rho - \log 2\pi - \frac{1}{2} \log(1 - x^{-2})$  at  $x = e^u$ .

**Assumption 12.6** (Zero count and location). *[assumed]*  $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T)$ ; every nontrivial zero has  $0 < \text{Re } \rho < 1$  and  $\text{Im } \rho \neq 0$ ; the zero multiset is locally finite and invariant, with multiplicity, under conjugation and  $\rho \mapsto 1 - \rho$ . No RH input.

**Corollary 12.7** (The forced net spectrum). *[proved-conditional]* A graded datum whose supertrace on  $(0, \infty)$  is  $\text{comb}(u)$  has  $\nu_{\text{net}}(1) = 1$ ,  $\nu_{\text{net}}(\rho) = -m_{\rho}$  at each nontrivial zero,  $\nu_{\text{net}}(-2k) = -1$  for  $k \geq 1$ , and  $\nu_{\text{net}} = 0$  elsewhere. One whose supertrace is  $2P_+(t)$  has  $\nu_{\text{net}}(0) = 1$ ,  $\nu_{\text{net}}(-\bar{\rho}/2) = -m_{\rho}$ ,  $\nu_{\text{net}}(-(k + \frac{1}{2})) = -1$ , and 0 elsewhere.

The second datum is the first transported by  $\lambda \mapsto (\lambda - 1)/2$  and the symmetry  $\rho \mapsto 1 - \rho$ , with the half-time Jacobian  $\text{comb}(t/2) = 2 \sum \Lambda(n) \delta(t - 2 \log n)$ ; the archimedean term  $1/(e^{2u} - 1) = \sum_{k \geq 1} e^{-2ku}$  becomes the odd ladder  $-(k + \frac{1}{2})$ .

### 12.3 Positivity forces the parities

**Assumption 12.8** (Positive distributions). *[assumed]* A distribution on  $(0, \infty)$  that is nonnegative on nonnegative test functions is integration against a unique positive Radon measure, and conversely.

**Theorem 12.9** (Finite parity changes). *[proved-conditional]* Assign one parity per value of  $\{1\} \cup \{\rho\} \cup \{-2k\}$  and let  $F_Z$  be the nontrivial zeros whose parity differs from the reference (pole even, everything else odd),  $F_T$  the flipped trivial zeros,  $b$  the pole flag. If  $F_Z$  is finite, the supertrace is a positive distribution iff  $b = 0$  and  $F_Z = \emptyset$ , while  $F_T$  may be any set, even infinite. The supertrace is real iff the flipped set is conjugation-invariant.

The flipped part is then a real-analytic function; positivity of  $\text{comb}$  plus a continuous function forces the function nonnegative; a flipped pole gives  $-2e^u$ , and a flipped set of zeros gives a leading trigonometric polynomial with zero mean and positive mean square, which is negative on an unbounded set. The draft's “a cosine is negative somewhere” was replaced by this argument on the whole leading-frequency sum.

**Assumption 12.10** (Landau's theorem, Laplace form). *[assumed]* The Laplace transform of a positive measure on  $(0, \infty)$  with finite abscissa of convergence  $\sigma_c$  has a singularity at  $s = \sigma_c$ .

*Cited fact 12.11* (Landau's theorem, Dirichlet form). *[cited]* A Dirichlet series with nonnegative coefficients does not extend holomorphically to a neighbourhood of its abscissa of absolute convergence [68]. Verbatim: Suppose that  $f(s) = \sum a_n n^{-s}$  has abscissa of absolute convergence equal to 0. If  $a_n \in \mathbb{R}$ ,  $a_n \geq 0$  for all  $n$  then  $f$  does not extend holomorphically to a neighborhood of  $s = 0$ . (1009.0228:main.tex:177).

**Proposition 12.12** (Infinite flips: the pole, and the zeros under domination). *[proved-conditional]* With  $F_Z$  possibly infinite: (i) positivity of the supertrace alone forces  $b = 0$ , the pole even, by monotone convergence of the cutoff pairings, with no domination premise and no Landau input; (ii) if moreover the supertrace dominates the comb as measures then  $F_Z = \emptyset$ . In both parts  $F_T$  remains arbitrary.

Part (i) is the reviewer’s strengthening of the prover’s statement; part (ii) cannot be freed of its premise by the same route, because for infinite  $F_Z$  the flipped subseries is only a distribution and can carry negative mass at the prime-power atoms.

*Observation 12.13* (The unrestricted infinite case is open). *[sketched]* Under positivity alone, with infinitely many zero parities flipped, the difference between the supertrace and the comb is a signed measure whose atoms at the prime powers can be negative, so the Laplace-transform argument does not close; whether positivity alone forces every parity is open here. Exact equality to the prime measure is rigid by Theorem 12.4 regardless.

## 12.4 The no-go: numerators are odd

*Cited fact 12.14* (Lidskii’s theorem). *[cited]* The trace of a trace-class operator on a Hilbert space coincides with its spectral trace, the sum of its eigenvalues counted with algebraic multiplicity [72, 74]. Verbatim: V.B. Lidskii [2], in 1959, proved his famous theorem on the coincidence of the trace of the  $S_1$ -operator in an (infinite dimensional) Hilbert space with its spectral trace  $\sum_{k=1}^{\infty} \mu_k(T)$ . (1105.2914:main.tex:119-121).

**Theorem 12.15** (No ungraded realisation of a numerator). *[proved]* Let  $N_n = \sum_i b_i^n - \sum_j \alpha_j^n$  with nonzero  $b_i, \alpha_j$ , the two multisets disjoint after cancellation and at least one  $\alpha_j$ . Then no finite matrix and no trace-class operator on a separable Hilbert space has  $\text{Tr } E^n = N_n$  for all  $n \geq 1$ .

At  $u = 1/\alpha_j$  the generating function  $\sum_n N_n u^n$  has residue  $+m_j/\alpha_j$ , whereas  $\sum_\lambda m_E(\lambda) \lambda u / (1 - \lambda u)$  has residue  $-m_E(\alpha_j)/\alpha_j$  with  $m_E \geq 0$ ; a numerator root needs a negative multiplicity.

**Proposition 12.16** (Trace and supertrace criteria). *[proved]*  $(N_n)$  is the trace sequence of a trace-class operator iff  $\exp(\sum_n N_n u^n / n) = \prod_i (1 - \lambda_i u)^{-1}$  with  $\sum |\lambda_i| < \infty$ ; it is the supertrace sequence of a finite graded matrix iff that germ is a rational function with value 1 at 0, the even net spectrum being the reciprocal poles and the odd net spectrum the reciprocal zeros after cancellation. Eigenvalue 0 and nilpotent blocks are invisible to all positive powers.

**Assumption 12.17** (Weil’s theorem for curves). *[assumed]* For a smooth projective geometrically connected curve  $C$  of genus  $g$  over  $\mathbb{F}_q$  there are  $2g$  nonzero algebraic numbers  $\alpha_j$ ,  $|\alpha_j| = \sqrt{q}$ , the Frobenius eigenvalues on  $H^1$ , with  $\#C(\mathbb{F}_{q^n}) = 1 + q^n - \sum_j \alpha_j^n$ .

**Corollary 12.18** (Curves need the grading; graphs do not). *[proved-conditional]* For a curve of genus  $g \geq 1$  the counts  $N_n$  are not the trace sequence of any finite matrix or trace-class operator, hence not the ring norms  $\text{Tr}(\sum_s A_s \otimes \overline{A_s})^n$  of any translation-invariant bosonic MPS with a fixed local tensor; they are the supertrace sequence of  $\text{diag}(q, 1)_+ \oplus (\text{Fr}|H^1)_-$ , and the odd net spectrum is forced to be the Frobenius eigenvalues. Non-backtracking graph counts  $\text{Tr } T^\ell$  are ordinary trace sequences, and the Ihara zeta  $\det(1 - uT)^{-1}$  has no finite zeros.

This is the precise content of “the sign” of Section 25: a zeta function with a numerator has its interesting eigenvalues as negative net multiplicities, and the grading is the only way to write them as a trace. It is also why the quadratic Artin–Schreier transfer matrix of Section 18, whose trace is an exponential sum, is a complex partition-function transfer and not the norm transfer of a state (a point the prover insisted on): the honest state, the uniform superposition over the points of the curve, has no fixed-tensor MPS with those norms.

## 12.5 What exists: the compressed semigroup is the odd sector

**Assumption 12.19** (Lax–Phillips modes of the compressed semigroup). [assumed]  $Z(t) = e^{tB}$  is a strongly continuous contraction semigroup on  $K_S$  whose generator has pure point spectrum  $\{-\bar{\rho}/2\}$ , one eigenvector per zero with multiplicity  $m_\rho$ , and  $Z(t) \rightarrow 0$  strongly. (The book’s own sketch of this standard input is Proposition 11.2.)

**Proposition 12.20** (The Riemann graded generator). [proved-conditional] On  $\mathfrak{B}$  (Definition 5.3),  $e^{tG} = 1 \oplus Z(t) \oplus \text{diag}(e^{-(k+1/2)t})$  is a strongly continuous contraction semigroup whose fixed vectors are exactly the even line, and for  $t > 0$

$$\text{str } e^{tG} = 1 - \text{Tr}_{\text{dist}} Z(t) - \sum_{k \geq 1} e^{-(k+1/2)t} = 2P_+(t) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{n} \delta(t - 2 \log n).$$

The single-particle graded spectrum is thus forced (Corollary 12.7) and realised. Its odd  $K_S$  part is the Riemann channel of Section 11; the missing even fixed line is the pole; the ladder is the archimedean place. The uncentred trace of  $Z(t)$  alone is restated as Proposition 11.5.

**Proposition 12.21** (The ladder parity is free; all-even is impossible). [proved-conditional] Over the even line plus odd  $K_S$  alone the supertrace is  $2P_+ + e^{-t/2}/(e^t - 1)$ ; with the ladder counted even it is  $2P_+ + 2e^{-t/2}/(e^t - 1)$ ; both are positive, so positivity does not fix the ladder parity. Exact equality to  $2P_+$  forces the net spectrum of  $G$ , and no all-even graded datum has supertrace  $2P_+$  or comb.

*Observation 12.22* (Scope of the interpretation). [sketched] RH is the statement that every odd  $K_S$  mode of  $G$  has real part  $-\frac{1}{4}$ ; the even line represents the pole of  $\zeta$  (the constant function on the modular surface, the residue of the Eisenstein series at  $s = 1$ ), an identification not constructed here;  $Z(t)$  is not trace class for  $t > 0$ ; and the identity above is a spectral identity, not a cMPS tensor whose Hilbert norm is the prime measure.

The numerical check of the supertrace identity, with 3000 zeros, is numerical result 13.19 in the next shard.

## 13 The Phantasm II: Vacuum Decay, the Prime Chain, and the Two Halves

Section 12 realised the forced graded spectrum by a Hilbert-space semigroup. The reframe asks for more: a Lindbladian on operators, with a fixed point that is a state and the zeros as relaxation modes. This shard records the two Lindbladians that can actually be built, and why each is only half of what is wanted.

### 13.1 Prior art on graded bonds

*Cited fact* 13.1 (Fermionic matrix product states contract by a supertrace). [cited] Bultinck, Williamson, Haegeman and Verstraete [11] contract fermionic MPS with the graded contraction  $\mathcal{C}(|i\rangle \otimes |j\rangle) = (-1)^{|i|} \delta_{ij}$ , so a closed ring of a  $\mathbb{Z}_2$ -graded bond is a supertrace, with a parity insertion in the odd sector. Verbatim: `\mathcal{C}(\ket{i} \otimes \bra{j}) = (-1)^{|i||j|} \mathcal{C}(\bra{j} \otimes \ket{i}) = (-1)^{|i|} \delta_{i,j}`. (1610.07849:FermionicMPS.tex)

*Cited fact* 13.2 (The fermionic cMPS already has the graded bond). [cited] Haegeman, Cirac, Osborne and Verstraete [33] give the fermionic cMPS: a  $\mathbb{Z}_2$ -graded bond with block-diagonal  $Q$ , block-off-diagonal  $R$  for fermionic species, and parity-graded boundary matrices. Verbatim: `R_{\alpha} = \begin{bmatrix} 0_{D^{(+)}} \times D^{(+)} & R_{\alpha}^{(+)} \\ R_{\alpha}^{(-)} & 0_{D^{(-)}} \times D^{(-)} \end{bmatrix} \quad \text{(particle species } \alpha \text{ is fermionic)}`. (1211.3935:calculus.tex:327).

*Cited fact* 13.3 (The transfer fixed point is the half-line reduced state). [cited] Verstraete and Cirac [83]: the eigenvalues of the fixed point of the cMPS transfer Lindbladian are the squared Schmidt coefficients of a bipartition. Verbatim: the eigenvalues of the matrix  $\rho(x)$  would exactly correspond to the squares of the Schmidt coefficients when considering a bipartition at site  $x$  (1002.1824:mpsQFT4.tex:81).

So the graded bond with a supertrace ring contraction is not new as a tensor-network construction; the arithmetic reading is what this book proposes.

### 13.2 Vacuum decay: the zeros as odd coherences, the vacuum as fixed point

**Theorem 13.4** (The finite vacuum-decay model). [proved] For finite-dimensional  $H$ ,  $B$  with  $-(B + B^\dagger) = |j\rangle\langle j|$  and  $e^{tB} \rightarrow 0$ , the vacuum-decay Lindbladian  $\mathcal{L}_{\text{vac}}$  of Definition 5.4 is a trace-preserving GKLS generator with unique stationary density  $\Omega_{\text{vac}}$ ; in the block decomposition of  $B(\mathbb{C} \oplus H)$  it is triangular, the coherence blocks evolve exactly by  $e^{tB}$ , and

$$\text{spec } \mathcal{L}_{\text{vac}} = \{0\} \uplus \text{spec } B \uplus \overline{\text{spec } B} \uplus \{b_i + \bar{b}_j\},$$

ordered pairs with multiplicities; the coherences are odd and the populations even.

Explicitly  $e^{t\mathcal{L}_{\text{vac}}}$  is conjugation by  $1 \oplus e^{tB}$  plus the completely positive feed  $X \mapsto \text{Tr}[(1 - e^{tB^\dagger} e^{tB})X] \Omega_{\text{vac}}$ .

**Corollary 13.5** (Traces of the vacuum-decay semigroup). [proved]  $\text{Tr } e^{t\mathcal{L}_{\text{vac}}} = |1 + \text{Tr } e^{tB}|^2$  and  $\text{str } e^{t\mathcal{L}_{\text{vac}}} = |1 - \text{Tr } e^{tB}|^2$ : the vacuum-decay transfer carries both coherence lists and all pair sums, and is not the single-particle generator  $\mathbf{G}$ .

**Proposition 13.6** (The absorbing-vacuum channel). [proved-conditional] For any strongly continuous contraction semigroup  $Z_t \rightarrow 0$  (strongly) on a separable Hilbert space, the

absorbing-vacuum channel of Definition 5.4 is a strongly continuous CPTP semigroup on trace-class operators on  $\mathbb{C} \oplus H$  with unique stationary density  $\Omega_{\text{vac}}$  and exact odd coherence evolution by  $Z_t$ . For  $H = K_S$  it realises the Riemann channel as its odd coherence sector.

*Observation 13.7* (Zeros in the coherences, pair sums in the populations). [sketched] For  $H = K_S$  each zero appears once as an odd coherence mode  $-\bar{\rho}/2$  and once as its conjugate (odd multiplicity  $2m_\rho$ ), the even populations carry the pair sums  $-(\bar{\rho}_n + \rho_m)/2$ , and their formal trace  $|\text{Tr}_{\text{dist}} Z(t)|^2$  is not a defined product of distributions; no pair-correlation statement follows.

**Proposition 13.8** (The vacuum is absorbing, and the entanglement is trivial). [proved]  
*The jump reinserts into the vacuum, but the renewal integral  $\int_0^T e^{t\bar{B}} \Omega_{\text{vac}} e^{tB^\dagger} dt = T \Omega_{\text{vac}}$  diverges: the vacuum is absorbing, not a rebound state of finite holding time, so this model does not solve the renewal equation of the SHW picture with a rebound inside  $K_S$ . The canonical cMPS selected by the stationary bond density has a single nonzero Schmidt weight: its half-line entanglement is trivial.*

This is the first half: side B without side A. The construction has the right relaxation modes and no entanglement.

### 13.3 The prime chain: the Gibbs fixed point, and no zeros

*Cited fact 13.9* (Bost–Connes KMS classification). [cited]

For  $0 < \beta \leq 1$  the Bost–Connes system has a unique  $\text{KMS}_\beta$  state; for  $\beta > 1$  the extremal KMS states are the Gibbs states with partition function  $\zeta(\beta)$ , labelled by embeddings of  $\mathbb{Q}^{\text{ab}}$  [14, 10]. Verbatim: For  $0 < \beta \leq 1$  there exists a unique  $\{\text{rm KMS}\}_{\beta}$  state  $\varphi_{\beta}$  for the above system. (math/0404128:houcheschapter1final7.tex:1820–182

**Proposition 13.10** (Normal KMS states are the Gibbs densities, only above the transition). [proved]  
*For the dynamics  $\text{Ad}(N^{\text{it}})$  on  $B(\ell^2(\mathbb{N}))$  a normal  $\text{KMS}_\beta$  state exists iff  $\beta > 1$ , and then it is  $\varrho_\beta^{\text{G}}$ ; the spectrum of  $-\log \varrho_\beta^{\text{G}}$  is  $\{\beta \log n + \log \zeta(\beta)\}$ ; and  $\text{TFD}_\beta$  factorises over the prime sites, with bond dimension one along the prime chain. The unique critical KMS state of the Bost–Connes algebra is not a normal density in this representation.*

**Lemma 13.11** (M/M/1 spectrum). [proved] *The birth–death generator on  $\mathbb{N}_0$  with constant up-rate  $\lambda$  and down-rate  $\lambda' > \lambda$ , on  $L^2$  of its geometric stationary law, has spectrum  $\{0\} \cup [-(\sqrt{\lambda} + \sqrt{\lambda'})^2, -(\sqrt{\lambda} - \sqrt{\lambda'})^2]$ , the zero simple.*

**Proposition 13.12** (The classical prime-chain dynamics). [proved] *The diagonal part of  $\mathcal{L}_\beta^{\text{pc}}$  ( $\sum_p r_p < \infty$ ,  $\beta > 1$ ) is a nonexplosive product of independent birth–death chains, one per prime, with unique stationary law  $n^{-\beta}/\zeta(\beta)$ ; its self-adjoint generator on  $L^2$  of that law has spectrum the closure of  $\{0\}$  and the finite Minkowski sums of the bands  $[-r_p(p^{\beta/2} + 1)^2, -r_p(p^{\beta/2} - 1)^2]$ . It is real, and contains no zero of  $\zeta$ .*

**Proposition 13.13** (Off-diagonal sectors). [proved] *If  $\sum_p r_p p^\beta < \infty$ ,  $\mathcal{L}_\beta^{\text{pc}}$  generates a bounded CPTP semigroup with  $\varrho_\beta^{\text{G}}$  stationary, and on the weighted Hilbert space with norm  $\|\varrho_\beta^{\text{G}-1/4} x \varrho_\beta^{\text{G}-1/4}\|_{\text{HS}}$  it is bounded, self-adjoint and nonpositive, off-diagonal sectors included: its spectrum is real.*

**Corollary 13.14** (Delocalisation at the transition). [proved] *As  $\beta \downarrow 1$  the Gibbs density has no trace-norm limit and its mass on every finite-rank projection tends to zero; for  $0 < \beta \leq 1$  the prime occupation process has no stationary probability on finite-support configurations. This is the precise content available for the Bost–Connes transition; it is not a proof of ergodicity breaking of a Riemann Lindbladian.*

This is the second half: side A without side B. The stationary state is right and the spectrum is blind to the zeros, as every prime-by-prime construction is (observation 35.5).

### 13.4 Entropy of the bond state and its cutoffs

*Numerical result* 13.15 (Entropy of the Gibbs density under cutoffs). [numerical]  
`scripts/bc_entropy.py` (`outputs/bc_entropy.txt`):  $\mathcal{S}_{\text{ent}}(\varrho_\beta^G) = \log \zeta(\beta) - \beta \zeta'(\beta)/\zeta(\beta)$  against the prime-product sum; with the prime cutoff  $p \leq P$  at  $\beta = 1$  the entropy follows  $\log P + \log \log P$  to  $10^{-3}$  at  $P = 10^6$ , grows like  $\beta P^{1-\beta}/(1-\beta)$  for  $\beta < 1$  and saturates for  $\beta > 1$ ; with the bond cutoff  $n \leq N$  at  $\beta = 1$  the ratio  $\mathcal{S}_{\text{ent}}/\log N$  falls towards  $\frac{1}{2}$  (0.657 at  $N = 10^7$ ) against 1 for  $\beta < 1$  and 0 for  $\beta > 1$ ; the number of entanglement levels below  $E$  is  $\lfloor e^E \rfloor$ . Constants fixed by the review:  $\mathcal{S}_{\text{ent}} = 1/(\beta - 1) + \log(1/(\beta - 1)) + (1 - \gamma_E) + o(1)$ ; the prime-cutoff constant is exactly 0, the two Mertens constants cancelling; the bond-cutoff constant is  $-\gamma_E/2$ .

*Observation* 13.16 (The entanglement Hamiltonian is free and Hagedorn). [sketched]  
 If the Phantasm's bond fixed point is  $\varrho_\beta^G$ , then by cited fact 13.3 its entanglement Hamiltonian is the Bost–Connes Hamiltonian  $\beta \log N$ , free over the primes with single-particle levels  $\log p$ ; the entanglement entropy is finite for  $\beta > 1$ , grows like  $\log P$  (prime cutoff) or  $\frac{1}{2} \log N$  (bond cutoff) at  $\beta = 1$ , and like a power of the cutoff for  $\beta < 1$ : gapped, critical, and not a cMPS at all. The level density  $e^E$  is Hagedorn growth with  $\beta_H = 1$  (Julia's reading of the transition [43]), not Cardy growth, so the finite-entanglement scaling law of critical CFTs does not apply to this state.

### 13.5 The two halves, and the guiding observation

*Observation* 13.17 (Complementary halves). [sketched] The vacuum-decay channel has the zeros as odd relaxation modes (each zero twice, as a coherence and its conjugate, plus an even pair-sum sector) but a pure stationary state, hence trivial entanglement. The prime-chain Gibbs Lindbladian, for  $\beta > 1$ , has  $\varrho_\beta^G$  as stationary state, with entanglement spectrum  $\beta \log n + \log \zeta(\beta)$ , proportional to the ring lengths  $2 \log n$ , but a real spectrum with no zeros: unconditionally for its classical part, and on the weighted space of Proposition 13.13 for the full generator when  $\sum r_p p^\beta < \infty$ . No construction in this book has both, and the Phantasm would need both (conjecture 35.3).

*Observation* 13.18 (The grading is trivial versus nontrivial characters). [sketched-conditional]  
 In the Artin–Schreier super-transfer matrix (Theorem 19.9) the even block is the trivial additive-character sector, the de Bruijn count  $q^n$  (the pole at  $u = 1/q$ ), together with the point at infinity (the pole at  $u = 1$ , not a character sector), and the odd block is the direct sum of the corrected Frobenius blocks  $F_{ag}$  over the nontrivial characters, which carries the zeros as its net spectrum. By analogy, a Galois-symmetric Riemann bond decomposed under  $\hat{\mathbb{Z}}^\times$  would have as even sector the pole of  $\zeta$ , the trivial Dirichlet character being the only one with a pole, and as odd sector the zeros of every  $L(s, \chi)$ : the graded object would compute the zeta of the cover, the Dedekind zeta of the cyclotomic field at level  $N$ , not  $\zeta$  alone. This is consistent with the character-basis transfer eigenvalues  $L(\beta, \bar{\chi})/\zeta(\beta)$  of Proposition 11.8, which by Proposition 13.12 is nevertheless not that object, and it is where Connes's trace formula lands (all  $L$ -functions with Grössencharakter). It is an analogy, not a theorem; its finite-level test is conjecture 35.4.

*Numerical result* 13.19 (Supertrace, vacuum decay and prime-chain bands, checked). [numerical]  
`scripts/riemann_graded_trace.py` (`outputs/riemann_graded_trace.txt`, 42/42 checks): the windowed supertrace of  $G$  with 3000 zeros against  $2P_+$  has ratio  $1 - 5 \cdot 10^{-4}$  at twelve prime powers and is positive to within truncation (minimum  $-0.041$  against peaks of 147); both ladder alternatives match and are positive; flipping the first zero pair to even gives  $-2.88$ ; the finite

vacuum-decay spectral multiset matches Theorem 13.4 to  $7 \cdot 10^{-14}$  for  $D = 3, 5, 8$  with CPTP and coherence checks; single-prime birth–death spectra lie in the M/M/1 bands and the product gap equals the  $p = 2$  gap.

*Observation 13.20* (Established scope). [sketched] Established: the forced graded net spectrum and its realisation by a graded contraction semigroup with one even fixed line; the no-go for ungraded trace-class realisations; a genuine absorbing-vacuum channel with the Riemann channel as its odd coherences; the normal-Gibbs classification and the prime-chain spectra. Not established: unrestricted infinite parity rigidity, the analytic realisation of the Lax–Phillips input, a cMPS whose Hilbert norm is the prime measure, and a stationary entangled bond with the critical arithmetic state.

## 14 The Phantasm III: Stationarity, Symmetry, and Generator Freedom

TJO's specification is an unknown Lindbladian with the BC state stationary, Galois symmetry, and compatible adelic symplectic structure. The no-event generator and the jumps are both unknown. This section completes a finite constraint calculation, not the construction of the adelic bond. Detailed arguments B0–B6 are in `notes/bc-symmetry-generators.md`; source loci are in `notes/extract/bc-symmetry-sources.md`. These arguments have not had an independent review and therefore carry **sketched**, including the statements with complete written derivations. Numerical evidence is separately tagged. No claim of novelty is made for the general GKLS machinery.

### 14.1 The state on the finite phase algebra

**Proposition 14.1** (Residue probabilities).

*[sketched] For the state of Definition 7.1,*

$$w_\beta(0) = p^{-\beta}, \quad w_\beta(x) = \frac{1 - p^{-\beta}}{p - 1} \quad (x \neq 0).$$

*At criticality the phase marginal is uniform at every integer level  $N$ , compatible under reduction, and  $\varrho_{p,1}^{\text{res}} = I/p$ .*

This is finite Fourier inversion of the BC KMS formula [14]; for  $\beta > 1$  it also follows by averaging the residue-class Gibbs sums over the Galois action. A single extremal state is generally not invariant. Uniform phase probabilities fix the diagonal of a matrix extension, not its off-diagonal entries. The full critical BC state is not a normal density in the standard representation (Proposition 13.10).

**Proposition 14.2** (Finite representations collapse the BC phases).

*[sketched]*

*Every finite-dimensional unital  $*$ -representation of the full BC algebra sends all  $e(r)$ ,  $r \in \mathbb{Q}/\mathbb{Z}$ , to  $I$ . Consequently it cannot realise the nontrivial critical finite phase marginal.*

Indeed,  $\mathbf{m}_n^* \mathbf{m}_n = I$  makes  $\mathbf{m}_n$  unitary in finite dimension. The BC relation gives  $I = \mathbf{m}_n \mathbf{m}_n^* = n^{-1} \sum_{k=0}^{n-1} e(k/n)$ , the spectral projection onto the eigenvalue 1 of  $e(1/n)$ . Thus  $e(1/n) = I$  for every  $n$ . This excludes an exact finite representation; it does not exclude restrictions, CP compressions, or asymptotic models.

**Proposition 14.3** (Invariant state extensions).

*[sketched] A unique stationary density of a*

*covariant semigroup is invariant under its symmetry group. The diagonal extension  $\varrho_{p,\beta}^{\text{res}}$  is Galois invariant, and is Weil invariant iff  $\beta = 1$ . The same phase marginal has a unique Weil-invariant matrix extension, when it is positive:*

$$\varrho_b = aI + (r - a)P, \quad r = p^{-\beta}, \quad a = \frac{1 - r}{p - 1}.$$

*Its parity-block eigenvalues are  $r$  and  $2a - r$ , so positivity holds iff  $r \leq 2/(p+1)$ . Full Weyl invariance forces  $I/p$ .*

The Weil-invariant operator space is  $\text{span}\{I, P\}$ : the symplectic group is transitive on the nonzero Weyl labels and  $\sum_v X_v = pP$ . The diagonal fixes the two coefficients. These coherent extensions are explicit state-extension examples, not an identification with the full BC bond. Above the transition, covariance can instead act on a family of stationary extremal states; uniqueness is the premise of the first assertion.

## 14.2 The complete finite feasibility problem

**Theorem 14.4** (A linear section of the conditional Choi cone). *[sketched]* The cone  $\mathfrak{C}_U(\varrho_b)$   
*consists exactly of complex-linear maps  $L$  with*

$$\begin{aligned} L(X^*) &= L(X)^*, \quad \text{Tr } L(X) = 0, \quad L(\varrho_b) = 0, \\ [L, \text{Ad}(U_g)] &= 0 \quad \text{for every } g, \quad QC_LQ \geq 0. \end{aligned}$$

*Equivalently, in an orthonormal traceless operator basis  $F_i$ ,*

$$L(X) = -i[H, X] + \sum_{i,j} A_{ij} (F_i X F_j^* - \tfrac{1}{2} \{F_j^* F_i, X\}), \quad A \geq 0, \quad H = H^*.$$

*With  $\text{Tr } H = 0$ , covariance requires  $H$  invariant and  $A$  commuting with the induced representation on traceless operators. If that representation is  $\bigoplus_\lambda V_\lambda \otimes \mathbb{C}^{m_\lambda}$ , then  $A = \bigoplus_\lambda I_{V_\lambda} \otimes A_\lambda$ ,  $A_\lambda \geq 0$ ; stationarity supplies further linear equations. Every invariant faithful  $\varrho_b$  has an interior feasible generator  $X \mapsto \varrho_b \text{Tr } X - X$ .*

Necessity follows by differentiating a positive channel Choi matrix on the complement of the maximally entangled vector. For sufficiency, the positive matrix  $QC_LQ$  supplies the traceless Kraus part; the remainder is the Choi matrix of  $KX + XK^*$ , and trace annihilation fixes its Hermitian part. This yields the bounded GKLS form recalled in [73]. The reset generator has conditional Choi matrix  $Q(I \otimes \varrho_b)Q$ , strictly positive on the range of  $Q$ .

**Theorem 14.5** (Dimensions at the critical tracial extension). *[sketched]* For odd prime  $p$ , at stationary density  $I/p$ , the real dimensions of the generator cones' linear spans are

| <i>commuting actions</i>                      | <i>dimension, including time scale</i> |
|-----------------------------------------------|----------------------------------------|
| <i>Galois torus</i>                           | $(p-1)(p+1)^2$                         |
| <i>full linear Weil action</i>                | $2p-2$                                 |
| <i>Weyl translations and Galois torus</i>     | $p+1$                                  |
| <i>Weyl translations and full Weil action</i> | 1.                                     |

In Weyl coordinates, trace annihilation and stationarity remove the identity row and column. The first two dimensions count simultaneous group orbits on ordered pairs of nonzero vectors. The torus acts freely; the symplectic orbits are  $w = av$  ( $a \neq 0$ ) and  $\det(v, w) = b$  ( $b \neq 0$ ), giving  $2(p-1)$ . Since  $-I$  belongs to both groups, Hermiticity makes the orbit coefficients real. The depolarizing generator is an interior point of the conditional Choi cone, so positivity does not reduce these dimensions. Orbit coefficients are not independent nonnegative rates in the first two rows.

**Proposition 14.6** (The cost of extra Weyl covariance). *[sketched]*  
*All Weyl-covariant generators are  $L(X) = \sum_{v \neq 0} c_v (X_v X X_v^* - X)$ ,  $c_v \geq 0$ . Galois covariance makes the rates constant on the two punctured axes and the  $p-1$  sets  $xy = b \neq 0$ . In particular  $c_v = c_{-v}$ , making every Weyl eigenvalue real and nonpositive. Full symplectic covariance makes all the rates equal and gives exactly  $L = \gamma(\mathcal{D}_p - \text{id})$ ,  $\gamma \geq 0$ .*

This extra commuting symmetry removes the oscillatory zero-like modes in this model. It is not a consequence of merely requiring a symplectic representation or an adelic structure.

### 14.3 Odd spectral freedom and coupling freedom

**Proposition 14.7** (A symplectic odd-sector counterexample). *[sketched] Let  $a$  be a primitive root modulo  $p$  and set  $R_a(I) = 0$ ,  $R_a(X_v) = X_{av}$  for  $v \neq 0$ . Then*

$$L = \mathcal{D}_p - \text{id} + \varepsilon R_a, \quad \varepsilon = \frac{1}{4p^2}$$

*is a CPTP generator with unique stationary state  $I/p$  and full Weil, Galois and parity covariance. For  $p = 7$  its odd eigenvalues are  $-1 - \varepsilon$  and  $-1 + \varepsilon/2 \pm i\sqrt{3}\varepsilon/2$ , each with multiplicity 8. The odd decay rates are unequal.*

The map  $R_a$  is Hermiticity preserving and equivariant, but need not itself be positive. Its Choi operator norm is bounded by its Frobenius norm  $\sqrt{p^2 - 1} < p$ , whereas the depolarizer's conditional Choi matrix is  $Q/p$ ; the specified perturbation therefore remains strictly positive on the range of  $Q$ . At  $p = 7$ , scalar multiplication has eight cycles of length six, and the odd sector selects the roots whose cube is  $-1$ . Thus the counterexample does not fit any input zero positions.

**Proposition 14.8** (Couplings invisible to the one-prime restrictions). *[sketched] For distinct primes  $p, q$ , fixed positive rates  $a, b$ , and  $0 \leq c \leq \min(a, b)$ , let*

$$L_c = (a - c)(\mathcal{D}_p - \text{id}) \otimes \text{id} + (b - c)\text{id} \otimes (\mathcal{D}_q - \text{id}) \\ + c(\mathcal{D}_p \otimes \mathcal{D}_q - \text{id}).$$

*These CPTP generators have the unique stationary state  $I/(pq)$  and full local Weyl/Weil covariance. Their restrictions on the one-prime observable algebras are independent of  $c$ , with rates  $a$  and  $b$ , while the rate on the joint traceless sector is  $a + b - c$ .*

Each summand is a nonnegative multiple of a channel minus the identity. The four tensor sectors give the eigenvalues  $0, -a, -b, -a - b + c$  directly. This supplies compatible two-prime residue dynamics under CRT with a variable coupling; it is not an infinite adelic construction.

*Numerical result 14.9* (Finite arithmetic symmetry checks). *[numerical] The script `scripts/bc_symmetry_generators.py` passes 427 checks; its output is `outputs/bc_symmetry_generators.txt`. The checks cover  $p = 3, 5, 7$ : BC residue probabilities against Hurwitz sums, invariant matrix extensions, orbit dimensions against independent Burnside counts, conditional and finite-time Choi positivity, trace preservation, stationarity and uniqueness, covariance, Egorov, odd-sector spectra, and two-prime restrictions. The four cone dimensions are respectively  $(32, 4, 4, 1)$ ,  $(144, 8, 6, 1)$ , and  $(384, 12, 8, 1)$ .*

### 14.4 What is still missing

The finite inverse problem is now a concrete positive-semidefinite cone. The missing input is a representation or CP comparison map connecting the full BC algebra, including its prime isometries, to the scattering bond, with compatible local Weil data. It must select the permitted commuting symmetries and constrain the generator blocks and the inter-prime couplings. The  $\Gamma_0(N)$  character/scattering computation of conjecture 35.4 remains open. Prime-power levels, the real place, metaplectic compatibility and the critical operator-algebra limit have not been constructed. A chosen graded transfer spectrum alone does not identify a physical cMPS norm.

The earlier reset test  $-(B\rho_b + \rho_b B^*) \geq 0$  is a criterion for a fixed finite no-event generator  $B$  and scalar reset, when its exit flux is positive. It is not a no-go criterion for the full cone in Theorem 14.4, where  $B$  and all jumps may vary.

## 15 The Fermion Parity of a Boson–Fermion cMPS Ring

*Question (TJO, 2026-09-13).* The Phantasm programme reads the ring norm of a graded cMPS as a supertrace (HANDOFF, “forced consequence”). Had that been proved, and what does a direct calculation of the expectation value of  $(-1)^F$  in a general boson–fermion cMPS give? *Answer on arrival.* It had not been proved. The book contained the graded contraction of lattice fermionic MPS as a citation (cited fact 13.1), the graded cMPS matrix structure as a citation (cited fact 13.2), and theorems about abstract graded spectral data (Section 12); no cMPS calculation. This section and the next supply it, with the definitions of Section 8.

*Result in one paragraph.* For arbitrary data,  $\langle \Psi_{B'}^c | (-1)^{\hat{F}} | \Psi_B^c \rangle = \text{Tr}[(B \otimes \bar{B}') e^{L\mathbb{T}\eta}]$ , a signed sum of squares (Theorem 15.6). With a bond grading this collapses: the physical parity only moves  $B$  to  $PBP$ , so  $\langle (-1)^{\hat{F}} \rangle = \pm 1$  for a parity-homogeneous boundary (Proposition 16.1). The supertrace appears instead as the norm of the ring closed with the bond parity,  $\|\Psi_P^c\|^2 = \text{str } e^{L\mathbb{T}}$ , graded by  $\Gamma_b = P \otimes \bar{P}$ , whose ring-length transform is  $1/\text{sdet}(z - \mathbb{T})$  (Theorems 16.2 and 16.4). So the HANDOFF sentence is correct as stated for finite bonds, but the sign is the bond parity of the closure, not the expectation value of the physical  $(-1)^F$ .

*Status.* The author of every argument here is `claud:opus-5`, working alone at TJO’s instruction; no independent review has been performed, so the rows are **sketched**. The numerical row is independent evidence at the level of explicit Jordan–Wigner operators.

### 15.1 Inputs

**Assumption 15.1** (Graded canonical relations). `[assumed]` In  $\mathfrak{F}$  (Definition 8.1), for  $f, g \in L^2([0, L])$  and  $a, b \in \mathfrak{S}$ , on  $\mathfrak{F}_{\text{fin}}$ : (F1)  $a_b(g)a_a^\dagger(f) - \eta_{ab}a_a^\dagger(f)a_b(g) = \delta_{ab}\langle g, f \rangle$  and  $a_a^\dagger(f)a_b^\dagger(g) = \eta_{ab}a_b^\dagger(g)a_a^\dagger(f)$ ; (F2)  $a_b(g)\Omega_{\mathfrak{F}} = 0$ ; (F3)  $\langle \Upsilon, a_a^\dagger(f)\Upsilon' \rangle = \langle a_a(f)\Upsilon, \Upsilon' \rangle$  for  $\Upsilon, \Upsilon' \in \mathfrak{F}_{\text{fin}}$ ,  $a_a^\dagger(f)$  is linear in  $f$  and maps  $\mathfrak{F}_n$  into  $\mathfrak{F}_{n+1}$ ; (F4) the  $\mathfrak{F}_n$  are mutually orthogonal,  $\|\Omega_{\mathfrak{F}}\| = 1$  and  $\mathfrak{F}_0 = \mathbb{C}\Omega_{\mathfrak{F}}$ ; (F5)  $\Gamma(u)$  is unitary,  $\Gamma(u)\Omega_{\mathfrak{F}} = \Omega_{\mathfrak{F}}$  and  $\Gamma(u)a_a^\dagger(f) = a_a^\dagger(u_a f)\Gamma(u)$ . These are the standard properties of the boson and fermion Fock spaces.

*Cited fact 15.2* (Overlap, transfer matrix and parity action of cMPS). `[cited]` Haegeman, Cirac, Osborne and Verstraete [33] compute the overlap of two boson–fermion cMPS as  $\text{Tr}[(B \otimes \bar{B}) \mathcal{P} \exp \int \mathbb{T}]$  with the transfer matrix carrying a plus sign for every species, using the ordered contraction of creation operators formally with delta functions, and state that the parity operator replaces  $R_\alpha$  by  $\eta_\alpha R_\alpha$ . Verbatim: `\voperator {T}(x)=Q(x)\otimes \text{one}_{\{D\}}+\text{one}_{\{D\}}\otimes \overline{\{Q(x)\}} + \sum_{\alpha=1}^N R_{\alpha}(x)\otimes \overline{R_{\alpha}(x)}`. (1211.3935:calculus) and `Since \operator {P}\ket {\Psi [Q,\{R_{\alpha}\}]}=\ket {\Psi [Q,\{\eta_{\alpha} R_{\alpha}\}]}` (1211.3935:calculus.tex:319).

So the overlap formula and the parity action are prior art at a formal level, and Theorem 15.6(ii) below is their combination. What this section adds is a proof from (F1)–(F5) in which the absence of a statistics sign in the norm is a checked step (Lemma 15.3, step (1)3), not a convention.

### 15.2 Ordered contraction and the Fock embedding

A *grid* is a partition of  $[0, L]$  into finitely many nonempty half-open intervals  $C_1, \dots, C_M$  listed from left to right,  $\chi_i$  the indicator of  $C_i$ . For a strictly increasing  $i = (i_1 < \dots < i_n)$  and  $a \in \mathfrak{S}^n$  put  $e(i, a) = a_{a_1}^\dagger(\chi_{i_1}) \cdots a_{a_n}^\dagger(\chi_{i_n})\Omega_{\mathfrak{F}}$  ( $e(\emptyset) = \Omega_{\mathfrak{F}}$ ), and let  $\text{box}(i, a) \in \mathcal{K}_n$  be  $(x; b) \mapsto \delta_{ab} \prod_k \chi_{i_k}(x_k)$ , which is supported in  $\Delta_n^< \times \mathfrak{S}^n$  because the cells are ordered.

**Lemma 15.3** (Ordered creation vectors contract without sign).

[sketched-conditional]

For strictly increasing  $i$  of length  $n$ ,  $j$  of length  $m$  on one grid, and words  $a, b$ ,  $\langle e(j, b), e(i, a) \rangle = \delta_{mn} \prod_{k=1}^n \delta_{i_k j_k} \delta_{a_k b_k} |C_{i_k}|$ .

*Proof.*  $\langle 1 \rangle 1$ . For  $g, h_1, \dots, h_n \in L^2$  and  $b, a_1, \dots, a_n \in \mathfrak{S}$ :  $a_b(g) a_{a_1}^\dagger(h_1) \cdots a_{a_n}^\dagger(h_n) \Omega_{\mathfrak{F}} = \sum_{k=1}^n \left( \prod_{l < k} \eta_{b a_l} \right) \delta_{b a_k} \langle g, h_k \rangle$

*Proof.* Induction on  $n$ . For  $n = 0$  both sides vanish by (F2). For  $n \geq 1$  write the vector as  $a_{a_1}^\dagger(h_1)X$  and use (F1):  $a_b(g) a_{a_1}^\dagger(h_1)X = \delta_{b a_1} \langle g, h_1 \rangle X + \eta_{b a_1} a_{a_1}^\dagger(h_1) a_b(g)X$ ; expand  $a_b(g)X$  by the induction hypothesis.  $\square$

$\langle 1 \rangle 2$ . For  $m \geq 1$  and  $\Upsilon \in \mathfrak{F}_{\text{fin}}$ ,  $\langle e(j, b), \Upsilon \rangle = \langle e(j', b'), a_{b_1}(\chi_{j_1}) \Upsilon \rangle$  with  $j' = (j_2, \dots, j_m)$ ,  $b' = (b_2, \dots, b_m)$ . *Proof.*  $e(j, b) = a_{b_1}^\dagger(\chi_{j_1}) e(j', b')$  and (F3).  $\square$

$\langle 1 \rangle 3$ . **The lemma, by induction on  $m$ .** If  $m = 0$  the claim is  $\langle \Omega_{\mathfrak{F}}, e(i, a) \rangle = \delta_{n0}$ , true by (F3), (F4). If  $m \geq 1$  and  $n = 0$ , the inner product is  $\langle e(j', b'), a_{b_1}(\chi_{j_1}) \Omega_{\mathfrak{F}} \rangle = 0$  by  $\langle 1 \rangle 2$  and (F2). Let  $m, n \geq 1$ . By  $\langle 1 \rangle 2$  and  $\langle 1 \rangle 1$ ,

$$\langle e(j, b), e(i, a) \rangle = \sum_{k=1}^n \left( \prod_{l < k} \eta_{b_1 a_l} \right) \delta_{b_1 a_k} \langle \chi_{j_1}, \chi_{i_k} \rangle \langle e(j', b'), e(i^{(k)}, a^{(k)}) \rangle,$$

$i^{(k)}, a^{(k)}$  being  $i, a$  with the  $k$ -th entry deleted. Since distinct cells are disjoint,  $\langle \chi_{j_1}, \chi_{i_k} \rangle = |C_{i_k}| \delta_{j_1 i_k}$ . Let  $k \geq 2$ . If  $m = 1$ , then  $j'$  is empty and  $i^{(k)}$  is not, so the last factor is 0 by the induction hypothesis. If  $m \geq 2$ , the induction hypothesis makes the last factor vanish unless  $j' = i^{(k)}$ , whose first entry is  $i_1$ ; that would give  $j_2 = i_1 < i_k = j_1$ , contradicting  $j_1 < j_2$ . Hence only  $k = 1$  survives; its sign product is empty, so  $\langle e(j, b), e(i, a) \rangle = \delta_{b_1 a_1} \delta_{j_1 i_1} |C_{i_1}| \langle e(j', b'), e(i^{(1)}, a^{(1)}) \rangle$ , and the induction hypothesis finishes.  $\square$

The statistics factors  $\prod_{l < k} \eta_{b_1 a_l}$  ride only on the terms  $k \geq 2$ , which vanish: this is the whole reason no sign enters the norm of an ordered state.  $\square$

**Lemma 15.4** (Isometric embedding of ordered amplitudes).

[sketched-conditional]

There is a unique bounded linear  $\mathcal{J}: \bigoplus_n \mathcal{K}_n \rightarrow \mathfrak{F}$  with  $\mathcal{J} \text{box}(i, a) = e(i, a)$  for every grid, strictly increasing  $i$  and word  $a$ . It is isometric, maps  $\mathcal{K}_n$  into  $\overline{\mathfrak{F}_n}$ , and satisfies  $(-1)^{\hat{F}} \mathcal{J} = \mathcal{J} \hat{\eta}$  where  $(\hat{\eta}c)(x; a) = \eta_a c(x; a)$ .

*Proof.*  $\langle 1 \rangle 1$ . **One grid.** The box functions of a grid  $G$  are pairwise orthogonal in  $\mathcal{K}_n$  with  $\|\text{box}(i, a)\|^2 = \prod_k |C_{i_k}|$ . By Lemma 15.3 the vectors  $e(i, a)$  have the same Gram matrix, so the linear extension  $\mathcal{J}_G$  to their span  $S_G$  is well defined and isometric.  $\square$

$\langle 1 \rangle 2$ . **Refinement.** If  $G'$  refines  $G$  then  $S_G \subseteq S_{G'}$  and  $\mathcal{J}_{G'}|_{S_G} = \mathcal{J}_G$ . *Proof.* Each  $C_{i_k}$  is a disjoint union of  $G'$ -cells, all to the left of those in  $C_{i_{k+1}}$ , so  $\text{box}(i, a) = \sum \text{box}'(i', a)$  over the strictly increasing  $i'$  with  $C'_{i'_k} \subseteq C_{i_k}$ , and  $e(i, a) = \sum e'(i', a)$  by linearity of  $a^\dagger$  in its argument (F3).  $\square$

$\langle 1 \rangle 3$ . Any two grids have a common refinement, so  $S = \bigcup_G S_G$  is a linear subspace on which  $\mathcal{J}$  is well defined and isometric.  $\square$

$\langle 1 \rangle 4$ . **Density.**  $S$  is dense in  $\mathcal{K}_n$ . *Proof.* Continuous  $c$  with compact support in  $\Delta_n^< \times \mathfrak{S}^n$  are dense. On the uniform grid of mesh  $h$  let  $c_h$  equal  $c$  at the left corner of each strictly increasing box and 0 elsewhere. On those boxes  $|c - c_h| \leq \text{osc}_c(\sqrt{n}h)$ ; every other point of  $\Delta_n^<$  has two consecutive coordinates in one cell, a set of measure at most  $(n-1)hL^{n-1}$ . So  $\|c - c_h\|^2 \leq |\mathfrak{S}|^n (\text{osc}_c(\sqrt{n}h)^2 L^n + (n-1)hL^{n-1} \sup |c|^2) \rightarrow 0$  by uniform continuity.  $\square$

(1)5. **Extension.**  $\mathcal{J}$  extends uniquely by continuity to an isometry on each  $\mathcal{K}_n$  with range in  $\overline{\mathfrak{F}_n}$  (F3); ranges for different  $n$  are orthogonal (F4), so the sum over  $n$  is isometric. Uniqueness: a bounded map is fixed by its values on the dense set  $S$ .  $\square$

(1)6. **Parity.** By (F5),  $(-1)^{\hat{F}}e(i, a) = \eta_a e(i, a)$ . Both  $(-1)^{\hat{F}}\mathcal{J}$  and  $\mathcal{J}\hat{\eta}$  are bounded and agree on  $S$ .  $\square$

$\square$

For box amplitudes  $\mathcal{J}$  is literally the smeared form of the formal integral  $\int c a_{a_1}^\dagger(x_1) \cdots a_{a_n}^\dagger(x_n) \Omega_{\mathfrak{F}}$  of [33]; Lemma 15.4 is its rigorous meaning.

### 15.3 The overlap and parity theorem

**Lemma 15.5** (Dyson series). *[sketched] Let  $A, C \in M_N(\mathbb{C})$ ,  $L \geq 0$ ,  $I_0(L) = e^{LA}$  and, for  $n \geq 1$ ,  $I_n(L) = \int_{0 < x_1 < \cdots < x_n < L} e^{x_1 A} C e^{(x_2 - x_1)A} C \cdots C e^{(L - x_n)A} dx$ . Then  $\|I_n(L)\| \leq e^{L\|A\|} \|C\|^n L^n / n!$  and  $\sum_{n \geq 0} I_n(L) = e^{L(A+C)}$ .*

*Proof.* (1)1. **Bound.**  $\|e^{yA}\| \leq e^{y\|A\|}$  and the exponents in each integrand add up to  $L$ , so the integrand has norm at most  $e^{L\|A\|} \|C\|^n$ ; the simplex has volume  $L^n / n!$ .  $\square$

(1)2. **Recursion.**  $I_n(L) = \int_0^L I_{n-1}(y) C e^{(L-y)A} dy$  for  $n \geq 1$ . *Proof.* Fubini, integrating first over  $0 < x_1 < \cdots < x_{n-1} < y = x_n$ .  $\square$

(1)3. On  $[0, L_0]$  the series  $F = \sum_n I_n$  converges uniformly by (1)1,  $F$  is continuous, and summing (1)2 termwise gives  $F(L) = e^{LA} + \int_0^L F(y) C e^{(L-y)A} dy$ .  $\square$

(1)4.  $G(L) = e^{L(A+C)}$  satisfies the same equation. *Proof.* Put  $H(L) = G(L)e^{-LA}$ . Then  $H' = G(A+C)e^{-LA} - GAe^{-LA} = GCe^{-LA}$  and  $H(0) = 1$ , so  $G(L)e^{-LA} = 1 + \int_0^L G(y) C e^{-yA} dy$ ; multiply by  $e^{LA}$  on the right.  $\square$

(1)5. **QED.**  $Z = F - G$  is continuous with  $Z(L) = \int_0^L Z(y) C e^{(L-y)A} dy$ , so  $\|Z(L)\| \leq K \int_0^L \|Z(y)\| dy$  with  $K = \|C\| e^{L_0\|A\|}$ ; iterating,  $\|Z(L)\| \leq \sup_{[0, L_0]} \|Z\| (KL)^k / k!$  for every  $k$ , hence  $Z = 0$ .  $\square$

$\square$

**Theorem 15.6** (Overlap and fermion parity of cMPS rings).

*[sketched-conditional]*

Let  $(Q, R, B)$  and  $(Q', R', B')$  be cMPS data with the same  $D$  and  $\mathfrak{S}$ , and let  $\Psi = \Psi_B^c$ ,  $\Psi'$  the ring vector of  $(Q', R', B')$ . The series defining them converge in  $\mathfrak{F}$ , and

$$(i) \quad \langle \Psi', \Psi \rangle = \text{Tr}[(B \otimes \bar{B}') e^{L\mathbb{T}}];$$

$$(ii) \quad \langle \Psi', (-1)^{\hat{F}} \Psi \rangle = \text{Tr}[(B \otimes \bar{B}') e^{L\mathbb{T}_\eta}];$$

$$(iii) \quad \langle \Psi, (-1)^{\hat{F}} \Psi \rangle = \sum_{n \geq 0} \sum_{a \in \mathfrak{S}^n} \eta_a \int_{\Delta_n^c} |c_n^B(x; a)|^2 dx, \text{ and } \|\Psi\|^2 \text{ is the same sum without the signs } \eta_a.$$

No bond grading, regularity condition or normalisation is assumed.

*Proof.* Write  $\epsilon \in \{0, 1\}$ , with  $\epsilon = 1$  for the parity insertion, and  $A = Q \otimes 1 + 1 \otimes \bar{Q}'$ ,  $C_\epsilon = \sum_a \eta_a^\epsilon R_a \otimes \bar{R}'_a$ , so that  $A + C_0 = \mathbb{T}$  and  $A + C_1 = \mathbb{T}_\eta$ .

- $\langle 1 \rangle 1$ . **Convergence.** With  $r = \max_a \|R_a\|$  and the trace norm  $\|B\|_1$ ,  $|c_n^B(x; a)| \leq \|B\|_1 e^{L\|Q\|} r^n$ , so  $\|c_n^B\|_{\mathcal{K}_n}^2 \leq \|B\|_1^2 e^{2L\|Q\|} (|\mathfrak{S}| r^2 L)^n / n!$ . The vectors  $\mathcal{J} c_n^B$  are mutually orthogonal (Lemma 15.4) with summable squared norms, so  $\sum_n \mathcal{J} c_n^B$  converges.  $\square$
- $\langle 1 \rangle 2$ .  $\langle \Psi', ((-1)^{\hat{F}})^\epsilon \Psi \rangle = \sum_n \sum_a \eta_a^\epsilon \int_{\Delta_n^<} \overline{c'_n(x; a)} c_n^B(x; a) dx$ ,  $c'_n$  the amplitude of the primed data. *Proof.* Continuity of the inner product and of  $(-1)^{\hat{F}}$ , then  $(-1)^{\hat{F}} \mathcal{J} = \mathcal{J} \hat{\eta}$ , isometry of  $\mathcal{J}$  and orthogonality for different  $n$  (Lemma 15.4).  $\square$
- $\langle 1 \rangle 3$ . **Pointwise factorisation.**  $\overline{c'_n(x; a)} c_n^B(x; a) = \text{Tr}[(B \otimes \bar{B}') e^{x_1 A} (R_{a_1} \otimes \bar{R}'_{a_1}) e^{(x_2 - x_1) A} \dots (R_{a_n} \otimes \bar{R}'_{a_n}) e^{(L - x_n) A}]$ . *Proof.*  $\overline{\text{Tr } Y} = \text{Tr } \bar{Y}$  and  $\overline{e^{yQ'}} = e^{y\bar{Q}'}$ ;  $\text{Tr } X \text{Tr } Y = \text{Tr}(X \otimes Y)$ ;  $(X_1 \otimes Y_1)(X_2 \otimes Y_2) = X_1 X_2 \otimes Y_1 Y_2$ ; and  $e^{yQ} \otimes e^{y\bar{Q}'} = e^{yA}$  because  $Q \otimes 1$  and  $1 \otimes \bar{Q}'$  commute.  $\square$
- $\langle 1 \rangle 4$ .  $\sum_a \eta_a^\epsilon$  of the matrix in  $\langle 1 \rangle 3$  is  $e^{x_1 A} C_\epsilon e^{(x_2 - x_1) A} \dots C_\epsilon e^{(L - x_n) A}$ . *Proof.*  $\eta_a^\epsilon = \prod_k \eta_{a_k}^\epsilon$ , and the sum over each  $a_k$  acts on the  $k$ -th factor alone.  $\square$
- $\langle 1 \rangle 5$ . **QED.** Integrating  $\langle 1 \rangle 4$  over  $\Delta_n^<$  gives the Dyson term  $I_n(L)$  of  $(A, C_\epsilon)$ ; the trace is linear and continuous and the sum over  $n$  converges absolutely (Lemma 15.5), so  $\langle 1 \rangle 2$  equals  $\text{Tr}[(B \otimes \bar{B}') \sum_n I_n(L)] = \text{Tr}[(B \otimes \bar{B}') e^{L(A + C_\epsilon)}]$ . This is (i) for  $\epsilon = 0$  and (ii) for  $\epsilon = 1$ ; (iii) is  $\langle 1 \rangle 2$  with primed data equal to unprimed.  $\square$

$\square$

So, with no structure at all, the parity expectation is  $\langle (-1)^{\hat{F}} \rangle_B = \text{Tr}[(B \otimes \bar{B}) e^{L\mathbb{T}_\eta}] / \text{Tr}[(B \otimes \bar{B}) e^{L\mathbb{T}}]$ : the sign sits in the generator, every  $n$ -particle term is weighted by  $\eta_a$ , and  $\mathbb{T}_\eta$  is in general not the generator of a positive semigroup. It is a trace, not a supertrace. The next section shows what a bond grading does to it.

## 16 Graded cMPS Rings: Supertrace and Superdeterminant

Throughout this section  $(Q, R)$  carry a bond grading  $P$  (Definition 8.2),  $\mathbb{T}$ ,  $\mathbb{T}_\eta$  and  $\Gamma_b = P \otimes \bar{P}$  are as in Definition 8.3, and  $\Psi_B^c$  is the ring vector of length  $L$  with boundary matrix  $B$ .

**Proposition 16.1** (With a bond grading the physical parity moves the boundary). *[sketched-conditional]*

(i)  $\Gamma_b^2 = 1$  and  $\Gamma_b \mathbb{T} = \mathbb{T} \Gamma_b$ . (ii)  $(P \otimes 1) \mathbb{T} (P \otimes 1) = \mathbb{T}_\eta = (1 \otimes \bar{P}) \mathbb{T} (1 \otimes \bar{P})$ . (iii)  $(-1)^{\hat{F}} \Psi_B^c = \Psi_{PBP}^c$  for every  $B$ . (iv) If  $PB = BP$  then  $(-1)^{\hat{F}} \Psi_B^c = \Psi_B^c$ , and if  $PB = -BP$  then  $(-1)^{\hat{F}} \Psi_B^c = -\Psi_B^c$ ; in particular  $\langle \Psi_B^c, (-1)^{\hat{F}} \Psi_B^c \rangle = \pm \|\Psi_B^c\|^2$  for  $B \in \{1, P, \Pi_+, \Pi_-\}$  (sign +).

*Proof.*  $\langle 1 \rangle 1$ . (i):  $\bar{P}^2 = \overline{P^2} = 1$ . Conjugating term by term,  $(P \otimes \bar{P})(Q \otimes 1)(P \otimes \bar{P}) = PQP \otimes 1 = Q \otimes 1$ , likewise for  $1 \otimes \bar{Q}$ , and  $(P \otimes \bar{P})(R_a \otimes \bar{R}_a)(P \otimes \bar{P}) = PR_a P \otimes \bar{P} R_a \bar{P} = \eta_a^2 R_a \otimes \bar{R}_a$ .  $\square$

$\langle 1 \rangle 2$ . (ii):  $(P \otimes 1)$  fixes  $Q \otimes 1$  and  $1 \otimes \bar{Q}$  and sends  $R_a \otimes \bar{R}_a$  to  $\eta_a R_a \otimes \bar{R}_a$ ; the same for  $1 \otimes \bar{P}$ .  $\square$

$\langle 1 \rangle 3$ . (iii):  $c_n^{PBP}(x; a) = \eta_a c_n^B(x; a)$ . *Proof.* By cyclicity  $c_n^{PBP} = \text{Tr}[B P X P]$  with  $X = e^{x_1 Q} R_{a_1} \cdots R_{a_n} e^{(L-x_n) Q}$ . Insert  $P^2 = 1$  between all factors:  $P e^{y Q} P = e^{y P Q P} = e^{y Q}$  and  $P R_a P = \eta_a R_a$ , so  $P X P = \eta_a X$ . Hence  $\Psi_{PBP}^c = \sum_n \mathcal{J} \hat{\eta} c_n^B = (-1)^{\hat{F}} \sum_n \mathcal{J} c_n^B$  by Lemma 15.4 and continuity of  $(-1)^{\hat{F}}$ .  $\square$

$\langle 1 \rangle 4$ . (iv):  $PB = \pm BP$  gives  $PBP = \pm B$ ; apply  $\langle 1 \rangle 3$ . The four listed matrices commute with  $P$ .  $\square$

As a consistency check, Theorem 15.6(ii) and (ii) above give  $\text{Tr}[(B \otimes \bar{B}) e^{L \mathbb{T}_\eta}] = \text{Tr}[(PBP \otimes \bar{B}) e^{L \mathbb{T}}] = \langle \Psi_B^c, \Psi_{PBP}^c \rangle$ , the same statement as (iii). The direct answer to TJO's question is therefore: in a graded cMPS the expectation value of  $(-1)^F$  is identically  $\pm 1$  for the natural boundaries and carries no spectral information.

**Theorem 16.2** (The twisted ring norm is a supertrace). *[sketched-conditional]* (i)  $\|\Psi_1^c\|^2 = \text{Tr} e^{L \mathbb{T}}$  and  $\|\Psi_P^c\|^2 = \text{str} e^{L \mathbb{T}} = \text{Tr}(e^{L \mathbb{T}}|_{\text{even}}) - \text{Tr}(e^{L \mathbb{T}}|_{\text{odd}})$ . (ii)  $\langle \Psi_P^c, (-1)^{\hat{F}} \Psi_P^c \rangle = \text{str} e^{L \mathbb{T}} = \text{Tr}[\Gamma_b e^{L \mathbb{T}_\eta}]$  and  $\langle \Psi_1^c, (-1)^{\hat{F}} \Psi_1^c \rangle = \text{Tr} e^{L \mathbb{T}}$ . (iii)  $\text{Tr}(e^{L \mathbb{T}}|_{\text{even}}) = \|\Psi_{\Pi_+}^c\|^2 + \|\Psi_{\Pi_-}^c\|^2$  and  $\text{Tr}(e^{L \mathbb{T}}|_{\text{odd}}) = 2 \text{Re} \langle \Psi_{\Pi_-}^c, \Psi_{\Pi_+}^c \rangle$ . (iv)  $\text{str} e^{L \mathbb{T}} = \sum_n \sum_a \int_{\Delta_n^<} |\text{Tr}[P e^{x_1 Q} R_{a_1} \cdots R_{a_n} e^{(L-x_n) Q}]|^2 dx \geq 0$ , with equality iff  $\Psi_P^c = 0$ .

*Proof.*  $\langle 1 \rangle 1$ . For  $M$  commuting with  $\Gamma_b$  and  $E_\pm = \frac{1}{2}(1 \pm \Gamma_b)$ :  $E_\pm$  are complementary idempotents commuting with  $M$ , so  $M$  is block diagonal for  $\text{ran} E_+ \oplus \text{ran} E_-$ ,  $\text{Tr}(E_\pm M) = \text{Tr}(M|_{\text{ran} E_\pm})$  and  $\text{Tr}(\Gamma_b M) = \text{Tr}(E_+ M) - \text{Tr}(E_- M)$ .  $e^{L \mathbb{T}}$  commutes with  $\Gamma_b$  by Proposition 16.1(i).  $\square$

$\langle 1 \rangle 2$ . (i): Theorem 15.6(i) with  $B = B' = 1$ , and with  $B = B' = P$ , where  $P \otimes \bar{P} = \Gamma_b$ ; then  $\langle 1 \rangle 1$ .  $\square$

$\langle 1 \rangle 3$ . (ii): the first equality is Proposition 16.1(iv) and (i); the second is Theorem 15.6(ii) with  $B = B' = P$ ; the untwisted case likewise.  $\square$

$\langle 1 \rangle 4$ . (iii): expanding  $\Pi_\pm = \frac{1}{2}(1 \pm P)$ ,  $\Pi_+ \otimes \bar{\Pi}_+ + \Pi_- \otimes \bar{\Pi}_- = \frac{1}{2}(1 + \Gamma_b) = E_+$  and  $\Pi_+ \otimes \bar{\Pi}_- + \Pi_- \otimes \bar{\Pi}_+ = E_-$ . By Theorem 15.6(i),  $\|\Psi_{\Pi_\pm}^c\|^2 = \text{Tr}[(\Pi_\pm \otimes \bar{\Pi}_\pm) e^{L \mathbb{T}}]$  and  $\langle \Psi_{\Pi_+}^c, \Psi_{\Pi_-}^c \rangle = \text{Tr}[(\Pi_+ \otimes \bar{\Pi}_+) e^{L \mathbb{T}}]$ ; these two cross terms are complex conjugates. Add and use  $\langle 1 \rangle 1$ .  $\square$

$\langle 1 \rangle 5$ . (iv): Theorem 15.6(iii) without signs for  $B = P$ , and (i).  $\square$

Two elementary readings. First, the minus sign of the supertrace is the minus sign in  $\Psi_P^c = \Psi_{\Pi_+}^c - \Psi_{\Pi_-}^c$ : the odd sector of the doubled bond is the interference term between the two parity-sector closures, and positivity of the norm is automatic. Second, physical fermions are not needed to have an odd sector: if every species is bosonic and  $P$  commutes with  $Q$  and all  $R_a$ , the bond splits into two decoupled cMPS on the  $\pm 1$  eigenspaces and  $\text{str } e^{L\mathbb{T}} = \|\Psi_+ - \Psi_-\|^2$ , the odd sector being their cross-transfer. Fermionic  $R_a$  are what couple the two blocks.

**Proposition 16.3** (The twisted ring is the periodic fermion ring). *[sketched-conditional]* For every  $t \in [0, L)$ ,  $U_t \Psi_P^c = \Psi_P^c$  and  $U_t^{\text{ap}} \Psi_1^c = \Psi_1^c$  (Definition 8.1).

*Proof.* **(1)1. Action on amplitudes.** For  $(y; b) \in \Delta_n^< \times \mathfrak{S}^n$  let  $w = \#\{k : y_k < t\}$  and define  $(T_t c)(y; b) = \varepsilon_w(b) c(y_{w+1}-t, \dots, y_n-t, y_1-t+L, \dots, y_w-t+L; b_{w+1}, \dots, b_n, b_1, \dots, b_w)$  with  $\varepsilon_w(b) = \prod_{k \leq w < l} \eta_{b_k b_l}$ , and  $T_t^{\text{ap}} = (-1)^{w_f} T_t$ ,  $w_f$  the number of fermionic  $b_1, \dots, b_w$ . These are unitary on  $\mathcal{K}_n$ , and  $U_t \mathcal{J} = \mathcal{J} T_t$ ,  $U_t^{\text{ap}} \mathcal{J} = \mathcal{J} T_t^{\text{ap}}$ . *Proof.* On a grid having  $L-t$  as a cell boundary,  $u_t$  maps cells to the cells of the shifted grid, the  $w$  cells beyond  $L-t$  landing first; by (F5)  $U_t e(i, a)$  is the ordered product with the last  $w$  creation operators moved to the front, which by (F1) costs exactly  $\varepsilon_w$ ;  $u_t^{\text{ap}}$  adds  $-1$  per wrapped fermion. So the identities hold on box functions of such grids, which are dense (Lemma 15.4, steps (1)2 and (1)4, refining by the point  $L-t$ ); both sides are bounded.  $\square$

**(1)2. Cyclic shift of the amplitude.** With  $x$  the argument on the right of (1)1, split  $X = e^{x_1 Q} R_{a_1} \dots e^{(L-x_n)Q}$  at the position  $L-t$  as  $X = X_1 X_2$ ,  $X_2$  containing the last  $w$  factors  $R$ . Then  $X_2 X_1 = e^{y_1 Q} R_{b_1} e^{(y_2-y_1)Q} \dots R_{b_n} e^{(L-y_n)Q}$ , because  $e^{(L-x_n)Q} e^{x_1 Q} = e^{(y_{w+1}-y_w)Q}$  and the end pieces become  $e^{y_1 Q}$  and  $e^{(L-y_n)Q}$ . Hence  $c_n^1(x; a) = c_n^1(y; b)$  and, moving  $X_2$  past  $P$  with  $P X_2 = \eta_{b_1} \dots \eta_{b_w} X_2 P$ ,  $c_n^P(x; a) = (-1)^{w_f} c_n^P(y; b)$ .  $\square$

**(1)3. Parity of nonzero amplitudes.** If  $\eta_b = -1$  then  $c_n^1(y; b) = c_n^P(y; b) = 0$ . *Proof.*  $PXP = -X$  and  $PB = BP$  give  $\text{Tr}[BX] = \text{Tr}[PBXP] = -\text{Tr}[BX]$ .  $\square$

**(1)4. QED.** Let  $n_f$  be the (even, by (1)3) number of fermionic letters of  $b$ . Then  $\varepsilon_w(b) = (-1)^{w_f(n_f-w_f)} = (-1)^{w_f}$ . By (1)2,  $T_t c_n^P = (-1)^{w_f} (-1)^{w_f} c_n^P = c_n^P$  and  $T_t^{\text{ap}} c_n^1 = (-1)^{w_f} (-1)^{w_f} c_n^1 = c_n^1$ . Apply (1)1 and sum over  $n$ .  $\square$

So  $\|\Psi_P^c\|^2$  is the Ramond (periodic) ring norm and  $\|\Psi_1^c\|^2$  the Neveu–Schwarz (antiperiodic) one: the supertrace of the transfer semigroup in the dilation direction is the ordinary norm of the physically periodic fermionic ring, the familiar exchange of a parity insertion in one direction with a boundary condition in the other.

**Theorem 16.4** (Ring-length transforms: supertrace resolvent and superdeterminant). *[sketched-conditional]* Let  $N_P(L) = \|\Psi_P^c\|^2$  and  $N_1(L) = \|\Psi_1^c\|^2$  as functions of the ring length, the data  $(Q, R, P)$  fixed; let  $m_{\pm}(\lambda)$  be the algebraic multiplicity of  $\lambda$  in  $\mathbb{T}|_{\text{even}}$ ,  $\mathbb{T}|_{\text{odd}}$ ,  $\nu(\lambda) = m_+(\lambda) - m_-(\lambda)$ , and  $\omega = \max \text{Re spec } \mathbb{T}$ . (i)  $N_P(L) = \sum_{\lambda} \nu(\lambda) e^{\lambda L}$ , and  $N_P$  determines  $\nu$ . (ii) For  $\text{Re } z > \omega$ ,  $\int_0^{\infty} e^{-zL} N_P(L) dL = \text{str}(z - \mathbb{T})^{-1} = \frac{d}{dz} \log \text{sdet}(z - \mathbb{T}) = \sum_{\lambda} \frac{\nu(\lambda)}{z - \lambda}$ . (iii) For  $\text{Re } z, \text{Re } z_0 > \omega$ ,

$$\exp\left(-\int_0^{\infty} (e^{-zL} - e^{-z_0 L}) N_P(L) \frac{dL}{L}\right) = \frac{\text{sdet}(z - \mathbb{T})}{\text{sdet}(z_0 - \mathbb{T})}.$$

(iv) The same holds for  $N_1$  with  $\text{Tr}$ ,  $\det$  and  $m_+ + m_-$ . Hence the ring zeta of the twisted rings is  $1/\text{sdet}(z - \mathbb{T}) = \det(z - \mathbb{T}|_{\text{odd}})/\det(z - \mathbb{T}|_{\text{even}}) = \prod_{\lambda} (z - \lambda)^{-\nu(\lambda)}$ : poles at net-even and zeros at net-odd eigenvalues.

*Proof.*  $\langle 1 \rangle 1$ . For  $M \in M_k(\mathbb{C})$  with algebraic multiplicities  $m$ :  $\text{Tr } e^{LM} = \sum m(\lambda) e^{\lambda L}$ ,  $\text{Tr}(z - M)^{-1} = \sum m(\lambda)/(z - \lambda)$ ,  $\det(z - M) = \prod (z - \lambda)^{m(\lambda)}$ . *Proof.* Schur:  $M = UTU^\dagger$  with  $T$  upper triangular carrying each eigenvalue  $m(\lambda)$  times on the diagonal;  $e^{LT}$  and  $(z - T)^{-1}$  are upper triangular with diagonals  $e^{\lambda L}$  and  $(z - \lambda)^{-1}$ .  $\square$

$\langle 1 \rangle 2$ . (i): Theorem 16.2(i) and  $\langle 1 \rangle 1$  on the invariant sectors (Proposition 16.1(i)). Distinct exponentials are linearly independent, so the coefficients  $\nu(\lambda)$  are determined.  $\square$

$\langle 1 \rangle 3$ . (ii):  $\int_0^\infty e^{-(z-\lambda)L} dL = (z - \lambda)^{-1}$  for  $\text{Re}(z - \lambda) > 0$ ; sum the finitely many terms and use  $\langle 1 \rangle 1$  on each sector;  $\text{sdet}(z - \mathbb{T}) = \prod (z - \lambda)^{\nu(\lambda)}$  has logarithmic derivative  $\sum \nu(\lambda)/(z - \lambda)$ .  $\square$

$\langle 1 \rangle 4$ . **Frullani.** For  $\text{Re } \kappa_1, \text{Re } \kappa_2 > 0$ ,  $\int_0^\infty (e^{-\kappa_1 L} - e^{-\kappa_2 L}) \frac{dL}{L} = \text{Log } \kappa_2 - \text{Log } \kappa_1$  (principal branch). *Proof.*  $|e^{-\kappa_1 L} - e^{-\kappa_2 L}| \leq |\kappa_1 - \kappa_2| L$  for  $L \geq 0$ , so the integral converges and, by dominated differentiation on compact subsets of the right half plane, is holomorphic in  $\kappa_1$  with derivative  $-\int_0^\infty e^{-\kappa_1 L} dL = -1/\kappa_1$ ; it vanishes at  $\kappa_1 = \kappa_2$ , as does  $\text{Log } \kappa_2 - \text{Log } \kappa_1$ , which has the same derivative on the connected half plane.  $\square$

$\langle 1 \rangle 5$ . (iii): by (i) and  $\langle 1 \rangle 4$  with  $\kappa_1 = z - \lambda$ ,  $\kappa_2 = z_0 - \lambda$ , the exponent equals  $-\sum \nu(\lambda)[\text{Log}(z_0 - \lambda) - \text{Log}(z - \lambda)]$ ; exponentiating the integer multiples gives  $\prod (z - \lambda)^{\nu(\lambda)} / \prod (z_0 - \lambda)^{\nu(\lambda)}$ .  $\square$

$\langle 1 \rangle 6$ . (iv): identical, with  $m_+ + m_-$  in place of  $\nu$ ; the zeta statement is  $\langle 1 \rangle 3$ .  $\square$

*Numerical result 16.5* (Parity, overlap, supertrace and sdet against explicit Fock vectors). [numerical] `scripts/cms_parity_supertrace.py` (`outputs/cms_parity_supertrace.txt`, 89 checks). Lattice, exact at fixed spacing: 6 sites with one hard-core boson and two fermion species, Jordan–Wigner creation operators whose graded relations and parity action are checked first, bond  $2 + 2$ , random data. For ungraded data the Fock inner products  $\langle \Psi' | \Psi \rangle$  and  $\langle \Psi' | (-1)^{\hat{F}} | \Psi \rangle$  match  $\text{Tr}[(B \otimes \bar{B}') E^N]$  and  $\text{Tr}[(B \otimes \bar{B}') E_\eta^N]$  to  $10^{-10}$ , and  $\langle (-1)^{\hat{F}} \rangle = +0.0107$  is nontrivial. For graded data:  $(-1)^{\hat{F}} \Psi_B^c = \Psi_{PBP}^c$  as vectors,  $\langle (-1)^{\hat{F}} \rangle = 1$  for  $B = 1, P$ ,  $\|\Psi_P^c\|^2 = \text{str } E^N$ , the odd trace equals  $2 \text{Re} \langle \Psi_{\Pi_-}^c, \Psi_{\Pi_+}^c \rangle$ ,  $\Psi_P^c$  has only even fermion number, and  $\Psi_P^c$  ( $\Psi_1^c$ ) is invariant under the periodic (antiperiodic) lattice translation and not under the other. Continuum, bond  $2 + 1$ : the Dyson series against `expm`; lattice  $\text{str } E^N \rightarrow \text{str } e^{L\mathbb{T}}$  at first order; the Laplace transform of  $N_P$  equals  $\text{str}(z - \mathbb{T})^{-1}$  to  $10^{-8}$  and the Frullani exponential equals the sdet ratio (0.6002951682 both).

*Observation 16.6* (What this does and does not give the Phantasm). [sketched-conditional] For a finite graded bond the HANDOFF reading “the ring norm is a supertrace, the ordinary norm of a fermionic ring with its parity insertion” is a theorem, with three sharpenings. (a) The sign is not the expectation value of the physical  $(-1)^{\hat{F}}$ , which is  $\pm 1$  (Proposition 16.1); it is the bond parity inserted in the closure, i.e. the periodic fermion boundary condition (Proposition 16.3). (b) The grading is  $\Gamma_b = P \otimes \bar{P}$  on the doubled bond, whose odd sector is the block-off-diagonal operators: if zeros are to appear as the odd part of such a ring zeta, they are relaxation rates of  $\mathbb{T}$  on coherences between the two parity blocks, which agrees with the vacuum-decay picture of observation 13.7; the supertrace over the bond itself in the graded realisation of Section 12 is a different operator and its relation to a  $\mathbb{T}$  is not established. (c) Positivity  $\text{str } e^{L\mathbb{T}} = \|\Psi_P^c\|^2 \geq 0$  is automatic, and the ring zeta is exactly  $1/\text{sdet}$ , consistent with the rigidity of Theorem 12.4. Not established: anything for an infinite-dimensional bond (trace class, regularisation of the Laplace transforms), and any identification of  $\mathbb{T}$  with a Riemann Lindbladian.

## 17 Weil–LPS Channels: Exactly Ramanujan Quantum Expanders

This is the finite model of the Riemann channel of Section 11 with every ingredient present and every claim checkable: commuting dilations indexed by primes, an arithmetic quotient supplying the expansion, and the Weil representation as the Hilbert space. The construction was proposed in the founding session’s handoff and computed on 2026-09-11; everything below was computed, and every construction step carries its own numerical check. The object itself is Definition 4.11, its ingredients Definition 4.9 and Definition 4.10.

### 17.1 The object

Fix an odd prime  $p$  and an LPS parameter  $q \equiv 1 \pmod{4}$  with  $q$  a square mod  $p$ . The Weil intertwiner exists only for symplectic (determinant one) matrices, and  $q$  being a square is what lets the norm- $q$  quaternions land in  $\mathrm{SL}_2(\mathbb{F}_p)$  after scaling. The admissible pairs computed are these.

| $p$ | $q$    | $ \mathrm{PSL}_2(\mathbb{F}_p) $ | Weil blocks $M_{(p+1)/2}, M_{(p-1)/2}$ |
|-----|--------|----------------------------------|----------------------------------------|
| 5   | 29     | 60                               | $M_3, M_2$                             |
| 7   | 29     | 168                              | $M_4, M_3$                             |
| 11  | 5      | 660                              | $M_6, M_5$                             |
| 13  | 17, 29 | 1092                             | $M_7, M_6$                             |
| 17  | 13     | 2448                             | $M_9, M_8$                             |
| 19  | 5, 17  | 3420                             | $M_{10}, M_9$                          |
| 29  | 5, 13  | 12180                            | $M_{15}, M_{14}$                       |

**Step 1, generators.** The  $q+1$  integer quaternions  $\alpha = a_0 + a_1i + a_2j + a_3k$  of norm  $q$  with  $a_0 > 0$  odd and  $a_1, a_2, a_3$  even. Jacobi’s four-square theorem gives exactly  $q+1$  of them; the enumeration was checked against the divisor sum  $\sum_{d|n} d$  for the norms  $q, q^2$  and  $q_1q_2$  used below. The same  $p=5$  quaternions appear in Appendix C of [41].

**Step 2, images in  $\mathrm{SL}_2(\mathbb{F}_p)$ .** Split the Hamilton quaternions over  $\mathbb{F}_p$  by

$$\mathbf{I} = \begin{pmatrix} x & y \\ y & -x \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad x^2 + y^2 + 1 \equiv 0 \pmod{p}.$$

Then  $\alpha \mapsto a_0 + a_1\mathbf{I} + a_2\mathbf{J} + a_3\mathbf{IJ}$  is a ring homomorphism whose determinant is the quaternion norm. Dividing by a square root  $s_q$  of  $q \bmod p$  gives  $g_\alpha \in \mathrm{SL}_2(\mathbb{F}_p)$ . Checked:  $\det g_\alpha = 1$ ,  $g_{\bar{\alpha}} = g_\alpha^{-1}$ , and all  $q+1$  images distinct.

**Step 3, the Weil representation, built and checked numerically.** On  $\ell^2(\mathbb{F}_p)$  take the clock and shift operators  $Z, X$  and the Heisenberg–Weyl operators  $T(u, v) = X^u Z^v$ . The Fourier transform  $\mathcal{F}$ , the chirp  $M$  and a dilation  $D$  each satisfy an intertwining relation of the form

$$W(g)T(u, v)W(g)^\dagger \propto T(g(u, v)),$$

with a symplectic  $g$  that the code *reads off* rather than assumes. A breadth-first search over  $\mathrm{SL}_2(\mathbb{F}_p)$  in these three generators assigns a unitary  $W(g)$  to every group element, projectively; for  $p=13$  the table has the expected  $p(p^2-1) = 2184$  entries. The relation above was then verified directly for each LPS generator  $g_\alpha$ , with error at most  $4 \cdot 10^{-14}$  in every case ( $7.2 \cdot 10^{-15}$  for  $(p, q) = (13, 17)$ ,  $7.0 \cdot 10^{-15}$  for  $(13, 29)$ ). Since only  $\mathrm{Ad} W$  enters the channel, the projective phases are irrelevant.

**Step 4, the two irreducible blocks.**  $W(-1)$  is the parity operator, so  $W$  splits into even and odd functions, of dimensions  $(p \pm 1)/2$ . Each block  $W_{\pm}$  is irreducible and  $-1$  acts on it as a scalar, so it is a representation of  $\text{PSL}_2(\mathbb{F}_p)$  up to phase. This matters: on the full  $M_p$  the parity operator would be a second fixed point of the channel, and Harrow’s theorem needs irreducibility. The channels are

$$\Phi_q^{\pm}(X) = \frac{1}{q+1} \sum_{\alpha} W_{\pm}(g_{\alpha}) X W_{\pm}(g_{\alpha})^{\dagger} \quad \text{on } M_{(p \pm 1)/2},$$

that is,  $D = q + 1$  unitary Kraus operators in the normalisation of Section 2. Block unitarity and Hilbert–Schmidt self-adjointness of the superoperator were checked at  $10^{-14}$  (for (13, 17): deviations  $2.7 \cdot 10^{-15}$  and  $1.8 \cdot 10^{-15}$ ).

## 17.2 Exact Ramanujan behaviour

**Proposition 17.1** (Weil–LPS channels are exactly Ramanujan). *[sketched]* For every admissible pair  $(p; q)$ , each block channel  $\Phi_q^{\pm}$  is a Ramanujan quantum expander in the sense of Definition 3.17:  $\lambda_2(\Phi_q^{\pm}) \leq \lambda_H$ , equivalently  $(q+1)\lambda_2 \leq 2\sqrt{q}$  in adjacency units.

The argument is two lines. Harrow’s transfer inequality (cited fact 32.5, [34]) says that for the channel built from an irreducible representation of a group and a symmetric generating set, the second singular value of the channel is at most the second singular value of the transition matrix of the Cayley graph on that generating set. Here the group is  $\text{PSL}_2(\mathbb{F}_p)$ , the generating set is the  $q+1$  images  $g_{\alpha}$  of Definition 4.9, and the representation is  $W_{\pm}$ , irreducible by Step 4; so  $(q+1)\lambda_2(\Phi_q^{\pm})$  is bounded by the largest modulus of a nontrivial eigenvalue of the LPS Cayley graph  $A_q$ . That graph is Ramanujan,  $|\lambda| \leq 2\sqrt{q}$ , by Lubotzky, Phillips and Sarnak [61], and  $2\sqrt{q} = 2\sqrt{D-1}$  is exactly Hastings’ bound  $\lambda_H$  in adjacency units [36]. The status is *sketched* rather than *proved* because the Lubotzky–Phillips–Sarnak theorem has no byte-verified local source in **refs**/, and because the irreducibility of  $W_{\pm}$  and the projective-phase bookkeeping of Step 4 are verified numerically here rather than cited. Iyer, Jain, Jordan and Somma [41] state the  $\text{SU}(2)$  version of the same mechanism and call explicit Ramanujan quantum expanders a longstanding open problem; see Section 32.

*Numerical result 17.2* (Ramanujan, Harrow containment, Hecke, commutation). *[numerical]*  
For  $(p; q)$  in  $\{(5; 29), (7; 29), (11; 5), (13; 17, 29), (17; 13), (19; 5, 17), (29; 5, 13)\}$  the channels  $\Phi_q^{\pm}$  on the even and odd Weil blocks satisfy the Ramanujan bound of Definition 3.17 and have a unique fixed point; Harrow containment, the Hecke relation of Definition 4.12 and commutation across  $q$  hold to  $10^{-14}$ .

Every entry of the table below is on the correct side of the bound  $2\sqrt{q}$ , in adjacency units.

| $p$ | $q$ | Cayley $A_q$ |                  | $\Phi_q^{+}$     |                       | $\Phi_q^{-}$     |                       | $2\sqrt{q}$ |
|-----|-----|--------------|------------------|------------------|-----------------------|------------------|-----------------------|-------------|
|     |     | $\lambda_2$  | $\lambda_{\min}$ | $(q+1)\lambda_2$ | $(q+1)\lambda_{\min}$ | $(q+1)\lambda_2$ | $(q+1)\lambda_{\min}$ |             |
| 5   | 29  | 6.000        | −8.000           | 6.000            | −2.000                | 2.000            | 2.000                 | 10.770      |
| 7   | 29  | 8.000        | −6.000           | 6.000            | −6.000                | 2.000            | −6.000                | 10.770      |
| 11  | 5   | 4.372        | −4.024           | 3.562            | −4.024                | 3.406            | −4.024                | 4.472       |
| 13  | 17  | 6.928        | −7.851           | 6.000            | −6.685                | 5.555            | −5.555                | 8.246       |
| 13  | 29  | 9.932        | −9.208           | 9.000            | −9.000                | 9.000            | −9.000                | 10.770      |
| 17  | 13  | 6.106        | −7.090           | 6.000            | −6.828                | 6.000            | −6.828                | 7.211       |
| 19  | 5   | 4.275        | −4.274           | 4.243            | −4.000                | 4.243            | −4.000                | 4.472       |
| 19  | 17  | 7.671        | −7.638           | 7.128            | −7.638                | 7.128            | −7.638                | 8.246       |
| 29  | 5   | 4.442        | −4.411           | 4.442            | −4.049                | 4.442            | −4.049                | 4.472       |
| 29  | 13  | 6.765        | −6.949           | 6.765            | −6.912                | 6.579            | −6.912                | 7.211       |

The further checks, all passed in every case, are these.

- **Harrow containment.** Every channel eigenvalue, times  $q + 1$ , lies in the spectrum of the  $\mathrm{PSL}_2(\mathbb{F}_p)$  Cayley graph to  $5 \cdot 10^{-7}$  (for  $(13, 17)$ : maximum distance  $4.4 \cdot 10^{-7}$  on the even block,  $3.1 \cdot 10^{-7}$  on the odd). The graph spectrum was computed in double precision on up to 3420 vertices; for  $p = 29$  only the extreme eigenvalues were computed, by Lanczos, which is why this is a containment check and not an equality check.
- **Unique fixed point.** The eigenvalue 1 of each block channel has multiplicity one, so the channels are quantum expanders in the sense of Definition 3.16 and not merely bounded.
- **Hecke relation.**  $A_q^2 - q1 = A_{q^2}$  on  $\mathrm{PSL}_2(\mathbb{F}_p)$ , with  $A_{q^2}$  built from the  $q^2 + q + 1$  normalised quaternions of norm  $q^2$  (307 for  $q = 17$ , 871 for  $q = 29$ ). Residual exactly 0. This is the tree relation “two steps without backtracking”.
- **Commutation across  $q$ .** For  $(p; q_1, q_2) = (13; 17, 29), (19; 5, 17), (29; 5, 13)$  the commutator  $\|[\Phi_{q_1}^\pm, \Phi_{q_2}^\pm]\|$  is at most  $3 \cdot 10^{-15}$  (for  $p = 13$ :  $5.4 \cdot 10^{-16}$  even,  $4.5 \cdot 10^{-16}$  odd), and  $A_{q_1} A_{q_2} = A_{q_1 q_2}$  exactly, with the  $(q_1 + 1)(q_2 + 1) = 540$  quaternions of norm  $q_1 q_2$ .

### 17.3 The quantum Ihara zeta of these channels

*Numerical result 17.3* (Edge spectrum and Ihara–Bass for  $(p, q) = (13, 17)$ ). [numerical] For  $(13, 17)$ , odd block, the non-backtracking edge superoperator  $T$  on  $M_6 \otimes \mathbb{C}^{18}$ , of dimension 648, has  $|\mu|$  in  $\{17, \sqrt{17}, 1\}$  with multiplicities 1, 70, 577, and satisfies the Ihara–Bass identity to  $10^{-14}$ .

The edge superoperator of Definition 3.18 was built directly, and the Ihara–Bass factorisation of Definition 3.8,

$$\det(1 - uT) = (1 - u^2)^{n^2(D-2)/2} \det(1 - u D \Phi_q^- + q u^2), \quad n = 6, D = 18,$$

holds with relative residuals  $2.8 \cdot 10^{-15}$  at  $u = 0.11$ ,  $2.3 \cdot 10^{-15}$  at  $u = 0.2 + 0.07i$  and  $1.5 \cdot 10^{-14}$  at  $u = -0.23$ . The multiplicities  $1 + 70 + 577 = 648$  exhaust the edge space, and  $\sqrt{q} = 4.12311$ . So the quantum Ihara zeta  $\zeta_\Phi$  of this channel satisfies the RH analogue of Definition 3.4 on the nose: every nontrivial zero sits on the critical circle. For every other case the same conclusion follows from the channel spectrum through the quadratic relation  $\mu^2 - \lambda\mu + q = 0$  that links an edge eigenvalue to an adjacency eigenvalue  $\lambda$ , without building  $T$ .

### 17.4 What the joint spectrum looks like

For  $(13; 17, 29)$  the even block has 19 distinct joint eigenvalue pairs  $(a_{17}, a_{29})$  of the commuting pair  $\Phi_{17}^+, \Phi_{29}^+$ . Examples, in adjacency units.

| $(a_{17}, a_{29})$                                         | multiplicity | remark                                   |
|------------------------------------------------------------|--------------|------------------------------------------|
| $(18, 30)$                                                 | 1            | trivial: $q + 1$                         |
| $(6, 2)$                                                   | 2            | integers                                 |
| $(3, -1), (3, 3), (3, 9)$                                  | 3, 3, 2      | integers                                 |
| $(-5, -5)$                                                 | 6            | integers                                 |
| $(\frac{3+\sqrt{17}}{2}, -1 - \sqrt{17})$                  | 3            | quadratic, field $\mathbb{Q}(\sqrt{17})$ |
| $(5.6847, -2), (5.5549, -0.3344), (1.7730, 5.1119), \dots$ | 3 each       | higher degree                            |

That the eigenvalues are algebraic integers is not a test: the Cayley adjacency matrix  $A_q$  is an integer matrix by construction. What is visible is the structure a Hecke identification would predict, namely joint eigenvalue pairs, small-degree algebraic integers, multiplicities equal to the multiplicity of the corresponding irreducible representation of  $\mathrm{PSL}_2(\mathbb{F}_p)$  in  $W_\pm \otimes \overline{W}_\pm$ , and every pair inside the box  $[-2\sqrt{q_1}, 2\sqrt{q_1}] \times [-2\sqrt{q_2}, 2\sqrt{q_2}]$ .

**Conjecture 17.4** (Hecke identification of the Weil–LPS spectra). *[open] The joint spectra of the commuting Weil–LPS channels  $\Phi_q^\pm$  are Hecke eigenvalues  $a_q(f)$  of weight-2 forms attached to the quaternion algebra ramified at  $\{2, \infty\}$ , with level determined by  $p$ , via Jacquet–Langlands.*

The identification was *not* done here; it needs LMFDB and the Jacquet–Langlands bookkeeping.

## 17.5 Reading

| Riemann channel                         | Weil–LPS channel at $p$                                                 |
|-----------------------------------------|-------------------------------------------------------------------------|
| commuting dilations by primes           | commuting Hecke channels $\Phi_q$ , one per LPS prime $q$               |
| arithmetic quotient supplying expansion | $\mathrm{PSL}_2(\mathbb{F}_p)$ as quotient of the $(q+1)$ -regular tree |
| Weil representation                     | $W_\pm$ on $\ell^2(\mathbb{F}_p)$ , even and odd                        |
| joint spectrum = zeros                  | joint spectrum = Hecke eigenvalues                                      |
| RH: all modes at $\mathrm{Re} = 1/2$    | Ramanujan: all modes at $ \mu  = \sqrt{q}$                              |
| source of the bound: unknown            | source of the bound: Deligne, via LPS                                   |

The last row is the honest one. These channels are exactly Ramanujan because Deligne proved the Weil conjectures (Section 25), not because they are channels. What the computation establishes is that the object proposed in the founding session’s handoff exists, is cheap to build, is self-checking at every step, and lands on the parent campaign’s Hilbert space with the parent campaign’s representation.

## 17.6 Not established

- The identification of the joint spectrum with specific modular forms in LMFDB (conjecture 17.4).
- Any statement about  $p \rightarrow \infty$ , or about assembling the  $\Phi_q$  over all  $p$  into an object whose rings are the primes; that is the parent repository’s global question in a new guise, recorded as conjecture 35.2 in Section 35.
- Anything about the Riemann hypothesis itself.

## 18 A Matrix-Product Formulation of the Weil Conjectures for Quadratic Artin–Schreier Curves

The question the founding session put to this family is narrow and answerable: is there an honest matrix product state behind the Weil conjectures for curves, meaning a *fixed* local tensor whose  $n$ -site ring contraction is the point count  $N_n$ ? In the vocabulary of this book the ingredients line up:  $N_n = \#C(\mathbb{F}_{q^n})$  is a ring norm, the Frobenius eigenvalues  $\alpha_i$  appearing in  $Z_C(u)$  are a transfer spectrum, and the Riemann hypothesis for the curve is the RH analogue of Definition 3.4. What is at issue is only whether the defining equation of the curve is a *local* constraint on the ring. For the quadratic Artin–Schreier family the answer is yes, and the answer stops exactly at degree three.

### 18.1 The shift picture

By Definition 4.14 the field  $\mathbb{F}_{q^n}$  carries a normal basis  $\{\theta, \theta^q, \dots, \theta^{q^{n-1}}\}$  in which Frobenius  $x \mapsto x^q$  is the cyclic shift of coordinates. An element of  $\mathbb{F}_{q^n}$  is then a ring of  $n$  sites with alphabet  $\mathbb{F}_q$ , Frobenius is translation by one site, and points over  $\mathbb{F}_{q^n}$  are ring configurations. This is the matrix-product setting of Definition 3.12 with no further choices to make.

For the Artin–Schreier curve  $y^q - y = g(x)$  of Definition 4.13, Hilbert 90 reduces the affine count to exponential sums,

$$N_n = q^n + \sum_{a \in \mathbb{F}_q^\times} \mathcal{S}_n(ag), \quad \mathcal{S}_n(g) = \sum_{x \in \mathbb{F}_{q^n}} e_q(\text{tr}_{q^n/q} g(x)), \quad (1)$$

so everything is carried by the  $\mathcal{S}_n$ . Writing the exponent  $m$  of a monomial  $x^m$  in base  $q$ ,  $m = \sum_j m_j q^j$ , one has  $x^m = \prod_j (x^{q^j})^{m_j}$ : a product of shifted copies of  $x$ , whose range is the number of base- $q$  digits. In a self-dual normal basis, which exists for  $n$  odd, the trace form is the dot product, and so for the quadratic family  $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$ ,

$$\text{tr}_{q^n/q} g(x) = \sum_{j=0}^J a_j \sum_i x_i x_{i+j}, \quad (2)$$

a translation-invariant quadratic spin model of range  $J$  on the ring. Its partition function is  $\text{Tr } E^n$  for the transfer matrix

$$E_{(s_1, \dots, s_J) \rightarrow (s_2, \dots, s_J, x)} = e_q \left( a_0 x^2 + \sum_{j=1}^J a_j s_{J+1-j} x \right), \quad (3)$$

of size  $q^J$ , with physical index  $\mathbb{F}_q$  and bond dimension  $q^J$ . The spins  $s_i$  are the  $J$  most recent site values; the map reads in the newest value  $x$  and forgets the oldest spin  $s_1$ .

**Proposition 18.1** (Transfer matrix for the quadratic family: the two forms). *[proved-conditional]* For  $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$  and every  $n \geq 1$ , including  $n \leq J$ :  $\text{Tr } E_g^n = \sum_{x \in \mathbb{F}_q^n} e_q(Q_g^\circ(x))$  with  $Q_g^\circ$  the ring form of matrix  $P_g(\mathbb{S})$ , and  $\mathcal{S}_n(g) = \sum_x e_q(Q_g^{\text{tr}}(x))$  with  $Q_g^{\text{tr}}$  the trace form of matrix  $CP_g(\mathbb{S})$  in normal-basis coordinates (Definition 5.7). The two forms have the same radical  $\mathcal{R}_n$ , so  $|\mathcal{S}_n(g)| = |\text{Tr } E_g^n|$  and  $\log_q(|\mathcal{S}_n|^2/q^n) = d_n$  is a nonnegative integer.

Two remarks on the statement, both corrected on 2026-09-12 night. First, the founding session’s reading of the even- $n$  sign flip as the normalisation  $\alpha_i = -\lambda_i(E)$ , “uniformly in  $n$ ”, is false in general: it holds exactly when  $P_g(-1) \neq 0$ , and the exact law  $\mathcal{S}_n = (-1)^{n-1} \delta_n \text{Tr } E_g^n$  with the radical sign

$\delta_n$  is Theorem 19.3 of Section 19, where the corrected Frobenius block is built. Second, a self-dual normal basis exists only for odd  $n$ ; in a general normal basis the trace form is  $C P_g(\mathbb{S})$ , not  $P_g(\mathbb{S})$ , and the discrepancy between the two Gauss sums is precisely the discriminant of  $C$  on the complement of the radical, which is Stickelberger's sign  $(-1)^{n-1}$  corrected on the radical.

## 18.2 What was computed

The script brute-forces  $\mathcal{S}_n$  in  $\mathbb{F}_{q^n}$  in a polynomial basis, with the modulus verified irreducible, and compares with  $\text{Tr } E^n$ .

| $q$ | $a = (a_0, \dots, a_J)$ | $\dim E$ | $n$ tested | $\mathcal{S}_n$ against $\text{Tr } E^n$   |
|-----|-------------------------|----------|------------|--------------------------------------------|
| 3   | (0, 1)                  | 3        | 2–7        | equal for odd $n$ , sign flip for even $n$ |
| 3   | (1, 1)                  | 3        | 3, 5, 7    | equal                                      |
| 5   | (0, 1)                  | 5        | 2–5        | equal for odd $n$ , sign flip for even $n$ |
| 3   | (0, 1, 1)               | 9        | 3, 5, 7    | equal                                      |
| 3   | (0, 0, 1)               | 9        | 3, 5, 7    | equal                                      |

*Numerical result 18.2* (Transfer identity and unitarity, checked). [numerical] For the five  $(q, a)$  of the table above and all  $n$  listed, the brute-force exponential sum agrees with  $\text{Tr } E^n$  up to the sign convention  $\alpha_i = -\lambda_i(E)$ . For eight  $(q, a)$  with  $q \in \{3, 5, 7\}$  and  $\dim E$  up to 49, every eigenvalue of  $E$  has modulus  $\sqrt{q}$  and  $\|EE^\dagger - qI\| \leq 5.2 \times 10^{-15}$ .

The ratios  $|\mathcal{S}_n|/q^{n/2}$  are 1 at every tested  $n$  except  $n = 4$ , where they are 3 for  $q = 3$  and 5 for  $q = 5$ : the sums saturate the Weil bound  $|\mathcal{S}_n| \leq (\deg g - 1) q^{n/2}$ , the maximal-curve phenomenon. The degree of  $L$  is  $q^J = \deg g - 1$ , matching Weil's count.

## 18.3 RH is unitarity

**Proposition 18.3** (Scaled unitarity of the transfer matrix). [proved] *If  $a_J \neq 0$  then  $EE^\dagger = qI$  on the  $q^J$ -dimensional bond (the scalar quadratic Gauss sum when  $J = 0$ ). Hence  $E/\sqrt{q}$  is unitary, every eigenvalue of  $E$  is nonzero, semisimple and of modulus  $\sqrt{q}$ , and, through the corrected Frobenius block of Theorem 19.7, the Weil bound  $|\alpha_i| = \sqrt{q}$  holds for the whole quadratic Artin–Schreier family.*

The note's argument is one line. In  $(EE^\dagger)_{s's''}$  the oldest spin  $s_1$ , which is the index summed over, appears only through the linear term  $a_J s_1(x' - x'')$  of (3); provided  $a_J \neq 0$  the sum over  $s_1$  is therefore  $q \delta_{x'x''}$ . The hypothesis  $a_J \neq 0$  is not cosmetic: it is what makes  $J$  the true range, and without it the same argument applies at the largest  $j$  with  $a_j \neq 0$  and a smaller transfer matrix. This is the mechanism already seen for a finite permutation in observation 10.2: the RH analogue is manifest because the transfer matrix is a scaled unitary. In the same terms, rationality of  $Z_C(u)$  holds because  $E$  is finite, and the note records that the functional equation is the involution  $E \mapsto q E^{-\dagger}$ , which scaled unitarity makes trivial. The proof with the output-row index bookkeeping of (3) made explicit is the prover's T7.6 (Section 19); the even- $n$  sign is settled there as well.

## 18.4 Where it stops, and why that is the right place

Take a non-quadratic exponent, say  $g(x) = x^3$  over  $\mathbb{F}_q$  with  $q \geq 5$ . The base- $q$  expansion of 3 has a single digit, so  $x^3$  is an on-site cubic; but in normal-basis coordinates  $\text{tr}_{q^n/q}(x^3) = \sum_{ijk} c_{ijk} x_i x_j x_k$ , with the multiplication structure constants  $c_{ijk}$  of the basis, which are non-local and depend on  $n$ . A self-dual normal basis makes the *bilinear* trace form local; it says nothing about the trilinear one.

| $g$                               | locality in the shift basis | transfer matrix                        | RH             |
|-----------------------------------|-----------------------------|----------------------------------------|----------------|
| quadratic, $\sum_j a_j x^{1+q^j}$ | local, range $J$            | $q^J \times q^J$ , $\sqrt{q}$ -unitary | manifest       |
| degree $\geq 3$ in any digit      | non-local, $n$ -dependent   | none of fixed size in this basis       | Weil / Deligne |

*Observation 18.4* (The non-quadratic case and Dwork’s operator).

[sketched]

For non-quadratic  $g$  the trace form in the shift basis is non-local and depends on  $n$ , so no fixed local tensor represents the ring model. The uniform transfer operator that does exist is Dwork’s  $p$ -adic operator (Definition 4.15, [21]), which delivers rationality and, through Dwork’s duality, the functional equation, but which sees the  $p$ -adic sizes of its eigenvalues and not their complex absolute values.

Dwork’s operator has exactly the form of a coarse-graining tensor-network step, “local weight, then decimate by  $q$ ”: multiplication by a splitting function followed by  $x^m \mapsto x^{m/q}$ , with infinite but nuclear bond dimension, and  $L$  as a Fredholm determinant. What it cannot deliver is the RH analogue, because the Newton polygon it computes is a statement about  $p$ -adic valuations. That is the structural reason Deligne’s proof [18] lives in  $\ell$ -adic cohomology, and the Weil/Deligne divide is taken up in Section 25. The sums that fall on the far side of the table, cubic Gauss sums and Kloosterman-type sums, are precisely the ones that needed algebraic geometry.

## 18.5 Not established

The note is explicit about what was not shown. No matrix product state with a fixed local tensor is known here for a non-quadratic Artin–Schreier curve, or for any curve of genus  $\geq 1$  that is not of this maximal type. Whether a *different* basis of  $\mathbb{F}_{q^n}$ , uniform in  $n$ , makes cubic traces local was neither settled nor searched: no candidate was found. And the unitarity argument of Proposition 18.3 is now proved, and the transfer unitary is identified as a Weil-representation operator in Section 19.

The boundary itself is the result worth keeping, and it is the same boundary met for graphs and for the Riemann channel: the Ramanujan bound is either manifest, because the transfer matrix is a scaled unitary, or it comes from arithmetic input that sits outside the transfer structure altogether.

## 19 Artin–Schreier II: the Sign Law, the Super-Transfer Matrix, and the Weil Representation

Section 18 recorded that  $\mathcal{S}_n(g) = \text{Tr } E_g^n$  for odd  $n$  and guessed that the even- $n$  sign flip was the normalisation  $\alpha_i = -\lambda_i(E_g)$ . That guess is false in general. The numerics lane found the failures, the orchestrator’s replacement (a conjugated law) failed in two further cases, and the codex prover found and proved the exact law; the numerics lane then confirmed it in 624 further checks. This shard is the corrected account, with the prototype now carrying, explicitly, every feature the Phantasm is supposed to have: a graded bond, a supertrace count, a fermionic sign that is a permutation sign, and a transfer unitary from a Weil representation. Throughout  $q$  is an odd prime and the notation is that of Definition 5.7.

### 19.1 The two quadratic forms and the exact sign law

**Lemma 19.1** (Gauss sums of quadratic forms). *[proved] For a quadratic form  $Q$  on  $\mathbb{F}_q^n$  of rank  $r$  with nondegenerate part  $Q_{\text{nd}}$ ,  $\sum_x e_q(Q(x)) = q^{n-r} \eta(\det Q_{\text{nd}}) \mathfrak{g}^r$  with  $\mathfrak{g} = \sum_c e_q(c^2)$ ,  $\mathfrak{g}^2 = \eta(-1)q$  and  $\bar{\mathfrak{g}} = \eta(-1)\mathfrak{g}$ ; every such sum is nonzero, has  $|\cdot|^2 = q^{n+(n-r)}$ , and its conjugate is  $\eta(-1)^r$  times itself.*

**Assumption 19.2** (Normal bases). *[assumed]  $\mathbb{F}_{q^n}$  has a normal basis over  $\mathbb{F}_q$ ; for  $n$  odd it can be chosen self-dual [59, 57].*

**Theorem 19.3** (The sign law). *[proved-conditional] For every  $n \geq 1$ , including  $q \mid n$ ,*

$$\mathcal{S}_n(g) = (-1)^{n-1} \delta_n \text{Tr } E_g^n, \quad \delta_n = \det(\mathbb{S}|_{\mathcal{R}_n}) \in \{\pm 1\},$$

*equivalently  $\mathcal{S}_n(g) = (-1)^{n-1} \delta_n \eta(-1)^{n-d_n} \text{Tr } E_{-g}^n$ .*

The proof builds a square root of the Gram matrix over  $\overline{\mathbb{F}_q}$ : the conjugate-embedding matrix  $A_{ri} = \theta^{q^{r+i}}$  satisfies  $A^t = A$ ,  $A^2 = C$ ,  $A^{[q]} = \mathbb{S}A$ , so  $\eta(\det(C|_R)) = \det(\mathbb{S}|_R)$  for every  $\mathbb{S}$ -invariant subspace  $R$  on which the identities restrict. Applied to the whole space this is Stickelberger’s theorem,  $\eta(\text{disc}) = (-1)^{n-1}$ , the sign of the  $n$ -cycle by which Frobenius permutes the conjugates; applied to the radical it gives the correction  $\delta_n$ . Both Gauss sums have the same rank and the same power of  $\mathfrak{g}$ ; only the discriminants of their nondegenerate parts differ, by  $\det C / \det(C|_{\mathcal{R}_n})$ .

**Assumption 19.4** (Polynomiality of the character  $L$ -function). *[assumed] For  $\deg g = q^J + 1$  and every  $a \in \mathbb{F}_q^\times$ ,  $L(ag, u) = \exp(\sum_n \mathcal{S}_n(ag) u^n / n)$  is a polynomial of degree exactly  $q^J$  with constant term 1. Root sizes are not assumed.*

**Corollary 19.5** (The periodic sign and the endpoint criteria). *[proved-conditional] With  $f(z) = z^J \mathcal{P}_g(z)$ ,  $v_-$  its order of vanishing at  $-1$  (always even) and  $\mathfrak{h}$  the period parameter:  $\delta_n = 1$  for  $n$  odd and  $(-1)^{\min(v_-, q^{v_q(n)})}$  for  $n$  even, so*

$$\mathcal{S}_n(g) = (1 - 2 \cdot \mathbf{1}_{2\mathfrak{h}|n}) \text{Tr } E_g^n.$$

*The claim  $\alpha_i = -\lambda_i(E_g)$  of Section 18 holds for all  $n$  iff  $\mathcal{P}_g(-1) \neq 0$ ; the conjugated law  $\mathcal{S}_n = -(-\eta(-1))^n \overline{\text{Tr } E_g^n}$  holds for all  $n$  iff  $\mathcal{P}_g(-\eta(-1)) \neq 0$ .*

**Proposition 19.6** (Both drafted laws fail). *[proved-conditional] For  $q = 3$ ,  $g = 2x^2 + x^4$ ,  $n = 1$ :  $\mathcal{S}_1 = \text{Tr } E_g = 3$ , while the conjugated law predicts  $-3$  ( $\mathcal{P}_g(1) = 0$ ). For  $q = 5$ ,  $g = x^2 + x^6$ ,  $n = 2$ :  $\mathcal{S}_2 = \text{Tr } E_g^2 = 5\sqrt{5}$ , while both laws predict  $-5\sqrt{5}$  ( $\mathcal{P}_g(-1) = 0$ ). The failures occur at  $n$  prime to  $q$ .*

| $q$ | $a$       | $P_g(1)$ | $P_g(-1)$ | $v_-$ | $h$ | $\alpha = -\lambda(E)$ | conjugated law |
|-----|-----------|----------|-----------|-------|-----|------------------------|----------------|
| 3   | (0, 1)    | 1        | 2         | 0     | 1   | holds                  | holds          |
| 3   | (1, 1)    | 2        | 0         | 2     | 3   | fails                  | holds          |
| 3   | (0, 1, 1) | 2        | 0         | 4     | 9   | fails                  | holds          |
| 3   | (2, 1)    | 0        | 1         | 0     | 1   | holds                  | fails          |
| 5   | (1, 2)    | 3        | 4         | 0     | 1   | holds                  | holds          |
| 5   | (0, 1, 1) | 2        | 0         | 2     | 5   | fails                  | fails          |
| 7   | (1, 1)    | 2        | 0         | 2     | 7   | fails                  | holds          |

## 19.2 The Frobenius block and the super-transfer matrix

**Theorem 19.7** (The corrected Frobenius block).

[proved-conditional]

$L(g, u)^h = \prod_{\omega^{2h}=1} D_g(\omega u) / D_g(u)^h$ ; the reciprocal-root multiplicity of  $L(g, u)$  at  $z$  is  $m_F(z) = \frac{1}{h} \sum_{\omega^{2h}=1} m_E(\omega^{-1}z) - m_E(z)$ , a nonnegative integer, and the diagonal  $q^J \times q^J$  matrix  $F_g$  with these multiplicities satisfies  $F_g F_g^\dagger = q$ ,  $L(g, u) = \det(1 - u F_g)$  and  $\mathcal{S}_n(g) = -\text{Tr } F_g^n$  for all  $n$ . If  $P_g(-1) \neq 0$  one may take  $F_g = -E_g$ ; if  $P_g(-\eta(-1)) \neq 0$ ,  $F_g = -\eta(-1)E_{-g}$ .

**Assumption 19.8** (Frobenius semisimplicity).

[assumed] The Frobenius action on  $H^1$  of a

smooth projective curve over a finite field is semisimple over an algebraic closure of the coefficient field. Used only for operator similarity, never for net spectra.

**Theorem 19.9** (The point count is a supertrace).

[proved-conditional]

The projective curve  $y^q - y = g(x)$  has one point at infinity;  $N_n = 1 + q^n + \sum_{a \neq 0} \mathcal{S}_n(ag)$  (additive Hilbert 90);  $\text{Tr } E_0^n = q^n$ ; and the super-transfer matrix  $\mathbb{E}$  of Definition 5.8 satisfies  $N_n = \text{str } \mathbb{E}^n$  for all  $n \geq 1$ . Its odd block  $F$  has  $FF^\dagger = q$ , and its multiset is the  $2g = (q-1)q^J$  Frobenius eigenvalues with no even-odd cancellation; under assumption 19.8 it is similar to Frobenius on  $H^1$ . The simpler odd block  $-\eta(-1) \bigoplus_a E_{ag}$  is correct iff  $P_g(-\eta(-1)) \neq 0$ : for  $q = 3$ ,  $g = 2x^2 + x^4$  it gives  $-2$  against  $N_1 = 10$ .

The even block is the trivial-character sector and the odd block the nontrivial characters: this is the grading observation of observation 13.18. The count is a supertrace because, by Corollary 12.18, it cannot be a trace.

## 19.3 The transfer unitary is a Weil-representation operator

*Cited fact* 19.10 (Uniqueness of Weil implementers).

[cited] Gurevich and Hadani [32]: the

intertwiner of the Heisenberg representation with its twist by  $g \in \text{Sp}$  “is determined up to scalar” by the Egorov relation. Verbatim: satisfies and in fact is determined up to scalar by the following equation (math/0610818:main.tex:422);  $\rho(g)\pi(h)\rho(g)^{-1} = \pi(g \cdot h)$ , \label {Egorov2} (math/0610818:main.tex:425).

*Cited fact* 19.11 (Character of the Weil representation).

[cited] Thomas [82]: for a finite

field  $F$  the trace of the Weil representation at  $g \in \text{Sp}(V)$  is  $|F|^{\frac{1}{2} \dim \ker(g-1)}$  times a product of Weil-index phases; in particular its modulus squared is  $|F|^{\dim \ker(g-1)}$  (Howe [38]). Verbatim:  $\text{Tr } \rho(g) = |F|^{\frac{1}{2} \dim \ker(g-1)} \gamma(1)^{\dim V - \dim \ker(g-1)} \gamma(\det \sigma_g)$ . (math/0610644:ArxivThomasWeil3.tex:196).

**Assumption 19.12** (Exact centred Weil implementers exist).

[assumed] There is a genuine

unitary representation  $W$  of  $\text{Sp}(2J, \mathbb{F}_q)$ ,  $q$  odd, with  $W(M)X_{u,v}W(M)^{-1} = X_{M(u,v)}$  exactly for the centred Weyl operators.

**Theorem 19.13** (The symplectic transfer map). *[proved-conditional]  $\mathcal{U}_g = E_g/\sqrt{q}$  factors as the Fourier transform on the retiring register (scaled by  $a_J$ ), then the quadratic phases, then the cyclic relabelling, and  $\mathcal{U}_g X_{u,v} \mathcal{U}_g^{-1} = X_{M_g(u,v)}$  exactly, where*

$$u'_k = u_{k+1}, \quad u'_J = -v_1/a_J, \quad v'_k = v_{k+1} + a_{J-k}u'_J \quad (k < J), \quad v'_J = a_J u_1 + \sum_{k \geq 2} a_{J+1-k} u_k + 2a_0 u'_J,$$

a symplectic map. Hence  $\mathcal{U}_g = e^{i\vartheta_g} W(M_g)$ .

**Corollary 19.14** (Fixed spaces, the Weil bound, and the permutation sign). *[proved-conditional]  $|\text{Tr } E_g^n|^2 = q^{n+\dim \ker(M_g^n - 1)}$  (from  $\text{Tr Ad}(\mathcal{U}_g^n) = \text{the number of fixed Weyl labels}$ ), so  $d_n = \dim \ker(M_g^n - 1) \leq 2J$ ;  $|\mathcal{S}_n(g)| \leq q^J q^{n/2}$  with equality iff  $M_g^n = 1$ ; and when  $d_n = 0$ ,  $\mathcal{S}_n(g) = (-1)^{n-1} \text{Tr } E_g^n$ , the sign of the Frobenius  $n$ -cycle. For  $q = 3$ ,  $g = x^4$ :  $M_g = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  of order 4,  $d_2 = 0$ ,  $d_4 = 2$ .*

So the maximal-curve phenomenon of Section 18 ( $|\mathcal{S}_4|/q^2 = q$  for  $g = x^4$ ) is the statement  $M_g^4 = 1$ , and the Weil bound for the whole quadratic family is Howe's character bound for the Weil representation. The fermionic sign of the count on a ring of  $n$  sites is the sign of the Frobenius permutation, corrected on the fixed space of  $M_g^n$  by the radical sign.

*Numerical result 19.15* (The corrected account, checked). [numerical]  
`scripts/artin_schreier_super.py` (`outputs/artin_schreier_super.txt`; 1969 checks, of which the 24 failures are exactly the refutation of the two drafted laws): eleven  $(q, g)$  over  $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$ ,  $n \leq 9$ ; the exact sign law and its periodic form and endpoint criteria (624/624); the corrected Frobenius blocks and  $\text{str } \mathbb{E}^n = N_n$  against direct point counts in every case; the Clifford property of  $E_g/\sqrt{q}$  with symplectic matrices matching the explicit formula matrix for matrix (orders 4, 3, 30); the character formula in 258/258 cases; and Prony on the genus-3 counts recovering the Frobenius eigenvalues with coefficient  $-1$ .

## 20 Curve Counts as Twisted Ring Norms I: Inputs and the Supersingular Boundary

*Question (TJO, 2026-09-14).* The previous session realised the elliptic-curve count  $N_n = 1 + q^n - \pi^n - \bar{\pi}^n$  as the  $P$ -closed ring norm of a bosonic tensor on  $\mathbb{C}^{1|1}$  with  $\pi$  put in by hand. Wanted: for the absolute simplest variety without an *obvious* Polya–Hilbert (PH) Hamiltonian, (1) a natural tensor determined by the variety, (2) the graded decomposition of the bond and why, (3) a theorem that the ring norms are the counts, (4) the PH operator, (5) the Ramanujan property, all modulo gauge. In this notebook the obvious PH Hamiltonian is the quadratic Artin–Schreier transfer matrix (Section 19); Theorem 20.11 shows that family is supersingular, so the first ordinary variety is an elliptic curve. Genus one is Section 21, the bosonic no-go Section 22, genus two Section 23. Proofs by the codex prover (notes/ring-norm-tensor/astra-proofs.md, sections T0–T5), Opus REFUTE review (notes/reviews/ring-norm-tensor-2026-09-14.md: 0 INVALID, 3 MINOR, 2 MAJOR, all applied in these shards; the prover’s file is kept as written). The finite-bond input “the  $P$ -closed ring norm is  $\text{str} \mathcal{E}_d^n$ ” is used in two halves (review MAJ-2): the algebraic identity  $\text{str}_{\Gamma_b} \mathcal{E}_d^n = \sum_w |\text{Tr}(PA_w)|^2$ , which is a two-line theorem for any letters (Kronecker multiplicativity) and is built into Definition 9.1, and the identification with the Fock norm of the physical fermion ring, which is Theorem 16.2 (sketched) and enters only Corollary 23.15; independent numerics in two scripts; sources byte-cited in notes/ring-norm-tensor/sources.md. Definitions in Section 9.

### 20.1 Inputs

*Cited fact 20.1* (Point counts and the Frobenius quadratic, without the root sizes). [cited] Milne [69]: for a curve of genus  $g$ ,  $N_n = 1 + q^n - \sum_{j \leq 2g} \alpha_j^n$  with the  $\alpha_j$  nonzero algebraic; Sutherland [79]: for an elliptic curve the Frobenius satisfies  $\pi^2 - t\pi + q = 0$  with  $t = q + 1 - \#E(\mathbb{F}_q)$ . Verbatim:  $N_{\{n\}} = 1 + q^{\{n\}} - (a_{\{1\}}^{\{n\}} + \dots + a_{\{2g\}}^{\{n\}})$ . (1509.00797:prh.tex:176); and  $\pi_E$  satisfies the characteristic equation  $\pi_E^2 - t\pi_E + q = 0$ . (1208.5370:IsogenyVolcanoes.tex:35).

*Cited fact 20.2* (Degree is a nonnegative quadratic form;  $\deg \pi = q$ ; Hasse’s bound). [cited] Milne [69]: the degree of  $m - n\alpha$  is nonnegative, so the discriminant of the Frobenius quadratic is  $\leq 0$ ;  $d = \deg \pi = q$ ;  $|\#E(k) - q - 1| \leq 2q^{1/2}$ . Verbatim: The right hand side is the degree of the map  $m - n\alpha$ , which is always nonnegative, and so the discriminant  $c^2 - 4d \leq 0$ , i.e.,  $c^2 \leq 4d$ . (1509.00797:prh.tex:241–242); Then  $d = \deg(\pi) = q$ , and  $c^2 \leq 4d = 4q$ . (1509.00797:prh.tex:248–249).

*Cited fact 20.3* (Serre–Tate canonical lift and the lifted Frobenius on  $H_1$ ). [cited] Goresky and Tai [31]: an ordinary abelian variety over  $\mathbb{F}_q$  has a canonical lift over  $W(k)$ , hence a complex variety, and the Frobenius lifts to an endomorphism of  $T = H_1(A_{\mathbb{C}}, \mathbb{Z})$ . Verbatim: The geometric action of  $F$  on  $A$  lifts to an automorphism  $F_A$  on  $[T = T_A = H_1(A_{\mathbb{C}}, \mathbb{Z})]$  (1701.07742:main).

**Assumption 20.4** (Deuring lifting). [assumed] For  $E/\mathbb{F}_q$  ordinary with Frobenius  $\pi$  and  $\text{End}(E) = O$  there is an elliptic curve over a number field with good reduction to  $E$  at a prime above  $p$ , endomorphism ring  $O$ , reduction map an isomorphism on endomorphisms taking  $\tilde{\pi}$  to  $\pi$ ; over  $\mathbb{C}$  it is  $\mathbb{C}/\Lambda$  with  $\Lambda$  a proper  $O$ -ideal and  $\tilde{\pi}$  multiplication by the complex number  $\pi$  (Deuring 1941; the Serre–Tate form is cited fact 20.3).

*Cited fact 20.5* (Degree of an ideal isogeny is the norm, i.e. the lattice index). [cited] Sutherland [79]. Verbatim: we have  $\deg \varphi = N(\mathfrak{f}) = [\mathbb{Q}(\mathfrak{f}) : \mathbb{Q}]$ . (1208.5370:IsogenyVolcanoes.tex:2).

*Cited fact 20.6* (Toral Lefschetz count and dynamical zeta). [cited] Baake, Lau and Paskunas [2]: for  $M \in \text{Mat}(d, \mathbb{Z})$  the number of isolated fixed points of  $M^m$  on  $\mathbb{T}^d$  is  $|\det(1 - M^m)|$  and the dynamical

$\zeta$  is the alternating product of  $\det(1 - z\Lambda^k M)$ . Verbatim:  $a^{\{-\}_{\{m\}} = \lvert \det (\one - M^m) \rvert$ . (0810.1855:main.tex:98);  $\widetilde{\zeta}^{\{-\}_{\{M\}}(z) = \prod_{k=0}^d \det \bigl( \one - z \ep^k(M) \bigr)^{(-1)^{k+1}}$ . (0810.1855:main.tex:199–200).

*Cited fact* 20.7 (Lenstra’s structure theorem). [cited] Springer [75], quoting Lenstra 1996, Theorem 1(a): for  $\pi \notin \mathbb{Z}$ ,  $R = \text{End}_{\mathbb{F}_q}(E)$  has rank 2 and  $E(\mathbb{F}_{q^n}) \cong R/(\pi^n - 1)R$  as  $R$ -modules. Verbatim:  $E(\mathbb{F}_{q^n}) \cong R/(\pi^n - 1)R$ . (2006.00637:main.tex:103).

*Cited fact* 20.8 (Latimer–MacDuffee; Waterhouse’s torsor). [cited] Knight and Stasinski [50]:  $\text{GL}_n(\mathbb{Z})$ -similarity classes of integer matrices with irreducible characteristic polynomial  $f$  are in bijection with the ideal classes of  $\mathbb{Z}[x]/(f)$ . Sutherland [79]: the  $j$ -invariants of curves with  $\text{End} \cong O$  form a torsor for  $\text{cl}(O)$ . Verbatim: and the ideal classes of the order  $\mathbb{Z}[x]/(f(x))$ . (2205.02094:main.tex:97); Provided it is non-empty, the set  $\text{Ell}_{\emptyset}(k)$  is thus a principal homogeneous space, a *torsor*, for the group  $\text{cl}(\emptyset)$ . (1208.5370:IsogenyVolc).

**Assumption 20.9** (Spectral versus Newton supersingularity). [assumed] An abelian variety over a finite field has all Newton slopes  $\frac{1}{2}$  iff all its Frobenius eigenvalues are  $\sqrt{q}$  times roots of unity; a curve is supersingular when its Jacobian is.

**Assumption 20.10** (Analytic setting for the infinite-bond obstruction). [assumed]  $H_{\pm}$  separable Hilbert spaces; at each  $t > 0$  a normal completely positive map on  $B(H_+ \oplus H_-)$  with a countable bounded block-diagonal Kraus family  $K_s = a_s \oplus B_s$  and weakly convergent expansion, whose blocks on Hilbert–Schmidt operators extend to bounded  $S_{\pm}(t)$ ,  $M(t)$  with  $S_{\pm}$  trace class; compression to finite bond subspaces agrees with compression of the Kraus expansion; the odd semigroup has the zeros as eigenmodes with multiplicity. Not established for any Riemann cMPS.

## 20.2 Where the obvious Polya–Hilbert Hamiltonian lives, and why it stops

**Theorem 20.11** (The quadratic Artin–Schreier family is supersingular). [proved] For  $q = p^r$ ,  $J \geq 0$ ,  $a_J \neq 0$ ,  $g = \sum_{j \leq J} a_j x^{1+q^j}$ , every Frobenius eigenvalue of the smooth model of  $y^q - y = g(x)$  is  $\sqrt{q}$  times a root of unity, for every  $p$  including  $p = 2$ ; for  $J = 0$  likewise in odd characteristic, while in characteristic two the  $J = 0$  curve has genus zero.

*Proof.* Sketch (T0.2).  $\text{Tr}_{\mathbb{F}_{q^n}/\mathbb{F}_p}(bg(x))$  is a quadratic form on  $\mathbb{F}_p^{rn}$  whose polar radical has dimension  $\leq 2rJ$  (its kernel is the root set of a fixed  $q$ -linearised polynomial of degree  $q^{2J}$ , using that  $b, a_j \in \mathbb{F}_q$  are fixed by  $x \mapsto x^q$ ; review items m2, m3). Quadratic Gauss sums are 0 or of modulus  $p^{(m+d)/2}$  with a phase in  $\{\pm 1, \pm i\}$ , so  $S_{b,n}/q^{n/2}$  ranges over a finite set independent of  $n$ ; by Hilbert 90 so do the normalised Frobenius power sums, which then satisfy a linear recurrence with finitely many state vectors, hence are eventually periodic; a Vandermonde argument gives  $(\alpha_j/\sqrt{q})^h = 1$ . No root size is assumed.  $\square$

**Proposition 20.12** (Finite order of the quadratic transfer unitary). [proved-conditional] For  $q$  an odd prime,  $E_g/\sqrt{q}$  has finite order: its Fourier factor has determinant a fourth root of unity, the phases and the register permutation have root-of-unity determinants,  $W(M_g)^h = 1$  for  $M_g$  in the finite group  $\text{Sp}(2J, \mathbb{F}_q)$ , so the scalar  $c_g$  in  $E_g/\sqrt{q} = c_g W(M_g)$  is a root of unity. Hence all eigenvalues of  $E_g$  are  $\sqrt{q}$  times roots of unity (the orchestrator’s drafted argument “the entries are roots of unity” was insufficient: zero entries and the normalisation).

**Proposition 20.13** (Cut rank of a ring amplitude). [proved] For  $A_s \in M_D(\mathbb{C})$  the flattening rank of  $\text{Tr}(A_{s_1} \cdots A_{s_n})$  across two contiguous arcs is at most  $D^2$ , sharply (matrix units at  $n = 2$ ), and at most  $D^b$  across a bipartition cutting  $b$  virtual edges; the drafted bound  $D$  is the open-chain bound.

*Numerical result 20.14* (Binary cut ranks; supersingularity of the tested family; basis dependence).

[numerical] `scripts/ring_norm_tensor.py` (`outputs/ring_norm_tensor.txt`, N4; the prover's kernel is reproduced in `notes/ring-norm-tensor/scratch/t04_cutrank.py`). For all eleven  $(q, a)$  cases of Section 19 every eigenvalue of  $E_g/\sqrt{q}$  is a root of unity. Over  $\mathbb{F}_2$ ,  $x^3 = x^{1+2}$  is a quadratic Frobenius form: in a self-dual normal basis ( $n$  odd) its cut rank stays  $\leq 4$  while the cubic  $\text{Tr}(x^{1+2+4})$  has ranks 2, 4, 8, 16 at  $n = 3, 5, 7, 9$  (over  $\mathbb{F}_3$ :  $\leq 9$  versus 1, 9, 27); in a plain normal basis even the quadratic amplitude grows (16 at  $n = 10$ ). The prover's table with an  $n$ -dependent basis prescription reaches 16 at  $n = 8$  for the quadratic form and left boundedness open; the reviewer closed it (MAJ-1): in a self-dual normal basis  $\text{Tr}(x^3) = \text{Tr}(x \cdot x^2) = \sum_i x_i x_{i-1}$  is a cyclic nearest-neighbour form, so the contiguous-cut rank is at most  $4 = D^2$  for every  $n$  (checked for  $n = 3, 5, 6, 7, 9, 10, 11$ ; no self-dual normal basis exists over  $\mathbb{F}_2$  for  $n = 4, 8$ ). Cut rank is basis dependent.

**Conjecture 20.15** (Restricted amplitude locality). *[conjectured] With a self-dual normal-basis prescription for every  $n$  and trace-invisible Artin–Schreier coboundaries quotiented out, a fixed finite lattice tensor with amplitudes  $\psi(\text{Tr } g(x))$  for all  $n$  exists only when the trace function has quadratic degree over the prime field. This is a necessity conjecture about equation-local amplitudes; it says nothing against the finite ordinary norm tensors below, and the broad slogan “finite tensors give only supersingular zetas” is false (observation 23.22).*

## 21 Curve Counts as Twisted Ring Norms II: the Ordinary Elliptic Curve

Tier A of the campaign of Section 20; notation of Definitions 9.2 and 9.3; proofs in notes/ring-norm-tensor/astra-T1–T2.

### 21.1 Tier A: the ordinary elliptic curve

**Theorem 21.1** (Frobenius on the lifted torus; the count is a ring norm). [proved-conditional]  
*With  $M$  the integral matrix of Definition 9.2,  $\chi_M(X) = X^2 - aX + q$  and for every  $n \geq 1$  the total endomorphism  $M^n$  has exactly  $\det(1 - M^n) = |1 - \pi^n|^2 = N_{K/\mathbb{Q}}(1 - \pi^n) = N_n$  fixed points, all nondegenerate of index +1, so that  $N_n = 1 - \text{Tr } M^n + \det M^n$  is the Lefschetz number; moreover  $N_n = |O/(\pi^n - 1)O|$ , the index of the principal ideal  $(1 - \pi^n)$  in the ring  $O$ , which under cited fact 20.7 is the point group  $E(\mathbb{F}_{q^n})$  itself as an  $O$ -module.*

*Proof.* Sketch (T1.1).  $\pi$  preserves  $\Lambda$ , so  $M$  is integral with eigenvalues  $\pi, \bar{\pi}$ , determinant  $|\pi|^2 = q$  by cited facts 20.2 and 20.5 and trace  $a$  by cited fact 20.1. The fixed points of  $M^n$  form  $\mathbb{Z}^2/(M^n - 1)\mathbb{Z}^2$  of order  $|\det(M^n - 1)|$ ;  $\det(1 - M^n) = (1 - \pi^n)(1 - \bar{\pi}^n) > 0$  gives the index. Lefschetz (cited fact 20.6) with the actions  $1, M^T, q$  in degrees 0, 1, 2 gives the count.  $\square$

**Proposition 21.2** (Dynamical zeta equals Hasse–Weil zeta). [proved-conditional]  
 $\exp(\sum_n \# \text{Fix}(M^n) u^n / n) = (1 - au + qu^2) / ((1 - u)(1 - qu)) = Z(E, u)$ ; by Möbius inversion the primitive periodic orbits of  $M$  on  $\mathbb{T}^2$  of each length are equinumerous with the closed points of  $E$  of each degree, so the gas side and the transfer side are both carried by the dynamical system  $(\mathbb{T}^2, M)$ ; the bijection is noncanonical.

**Theorem 21.3** (Hodge doubling and the elliptic norm tensor). [proved-conditional] *On  $\mathbb{C}/\Lambda$ ,  $M^*dz = \pi dz$  and  $M^*d\bar{z} = \bar{\pi}d\bar{z}$ . The double of the Hodge ket bond is  $H^*(\mathbb{T}^2, \mathbb{C})$  graded by form degree, with transfer  $\mathcal{E}_d = \text{diag}(1, \bar{\pi}, \pi, q)$  in the basis  $(1, d\bar{z}, dz, dz \wedge d\bar{z})$  and  $\Gamma_b = \text{diag}(1, -1, -1, 1)$ :  $(+, +) = H^0$ , the mixed lines are  $H^{0,1}, H^{1,0}$ , and  $(-, -) = H^2$  with eigenvalue  $\pi\bar{\pi} = q$ , the degree. The elliptic norm tensor (Definition 9.3) is all-even,  $\Psi_P(n) = (1 - \pi^n)\phi^{\otimes n}$ ,  $\|\Psi_P(n)\|^2 = N_n$ , the untwisted norm is  $|1 + \pi^n|^2$ , and the ring zeta is  $Z(E, u)$ . The fermionic modes are the odd-degree forms; the physical letters are bosonic; the physical state has Schmidt rank one and its configurations do not enumerate points.*

**Proposition 21.4** (Fourier pullback is an injection with one finite orbit; the flat supertrace). [proved-conditional]  
*The pullback  $U = M^*$  on  $L^2(\mathbb{T}^2) \otimes \Lambda^*(\mathbb{C}^2)$  sends  $e_k \otimes \omega$  to  $e_{M^T k} \otimes \Lambda^*(M^T)\omega$ ;  $k \mapsto M^T k$  is injective of index  $q$  on  $\mathbb{Z}^2$  (not a permutation), its only finite orbit is  $\{0\}$ , and the Fourier-diagonal flat supertrace  $\sum_{M^n T k = k} \text{str } \Lambda^*(M^n) = \det(1 - M^n) = N_n$  is contributed by  $k = 0$  alone. The infinite-label tensor  $A_k = |M^T k\rangle\langle k| \otimes \text{diag}(1, \pi)$  has  $P$ -closed ring vector  $(1 - \pi^n)|0, \dots, 0\rangle$ : an honest but redundant physical index; every closed ring except the vacuum mode vanishes.*

**Theorem 21.5** (The Polya–Hilbert unitary and the ordinary Hasse bound). [proved-conditional]  
*On the flat torus the Hodge ( $L^2$ ) metric on harmonic one-forms is scaled by  $|\pi|^2$  under  $M^*$  on both  $H^{1,0}$  and  $H^{0,1}$ , so  $U_{\text{PH}} = q^{-1/2} M^*|_{H^1}$  is unitary iff  $|\pi|^2 = q$ , which is derived, not assumed: the lifted Frobenius is a conformal similarity whose oriented area ratio is its covering degree  $[\Lambda : \pi\Lambda] = |\pi|^2$  (cited fact 20.5), and degree is preserved under good reduction with  $\deg \text{Fr} = q$  (the clause  $d = \deg(\pi) = q$  of cited fact 20.2, not its inequality; review m1). Hence the two nontrivial eigenvalues of  $\mathcal{E}_d$  have modulus  $\sqrt{q}$  (the Ramanujan property here) and  $a^2 < 4q$ , from the positive*

definite degree form  $\deg(m + n\pi) = m^2 + amn + qn^2$ ; no root-size input is used. The real PH model  $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{C}$  with  $\operatorname{Re}(x\bar{y})$  (the Rosati trace form up to the factor 2) is the real form of  $H^1(\mathbb{T}^2, \mathbb{R})$ , not the complex  $H^1$ ; a self-adjoint logarithm needs a phase branch the curve does not specify.

**Proposition 21.6** (Integral markings versus MPS gauge). *[proved-conditional] The integral matrix  $M$  changes by  $\operatorname{GL}_2(\mathbb{Z})$  conjugacy under a change of lattice basis; its class is an ideal-lattice class of  $\mathbb{Z}[\pi]$  (cited fact 20.8, all full ideal lattices, not only invertible ones) whose multiplier order is the endomorphism order, and the ordinary classification is recovered as Waterhouse’s torsor (indexed by  $j$ -invariants, up to quadratic twist; review m4). Complex spectral data forget these classes: all matrices in the isogeny class are  $\operatorname{GL}_2(\mathbb{C})$ -conjugate, the cohomology matrix is  $M^T$  on the dual lattice, and the unitary gauge is the Hodge basis. The even ket gauge  $G \in \operatorname{GL}(1|1)$  acts on  $\mathcal{E}_d$  by  $G \otimes \bar{G}$  and fixes all ring norms; exchanging the two Hodge embeddings is not an even gauge.*

## 21.2 Digits in base $\pi$

**Proposition 21.7** (Ordinary digit expansions, cyclic fibres, bounded carries). *[proved] With  $D$  a residue system of  $O/\pi O$  ( $|D| = q$ ), every class of  $O/\pi^n O$  has a unique expansion  $\sum_{i < n} d_i \pi^i$ . The cyclic map  $D^n \rightarrow O/(\pi^n - 1)O$  need not be onto (for  $O = \mathbb{Z}[i]$ ,  $\pi = 1 + 2i$ ,  $D = \{0, \pm 1, \pm 2\}$  the target  $O/(2i)$  has four classes and the digits hit two;  $|1 - \pi^2|^2 = 32 > 25$  excludes surjectivity at  $n = 2$  for any five digits; this is  $y^2 = x^3 + x$  over  $\mathbb{F}_5$ ). Two words have the same image iff they admit integral cyclic carries  $d_i - e_i = \pi c_{i+1} - c_i$ ,  $c_n = c_0$ , which are unique and bounded by  $|c_i| \leq \max_{d,e} |d - e|/(\sqrt{q} - 1)$ : the cyclic equivalence relation has a finite carry automaton.*

**Proposition 21.8** (What finite carries do not count). *[proved] No finite weighted adjacency matrix and no trace-class operator has power traces  $N_n$  (Theorem 12.15), so no automaton with nonnegative transition counts counts the classes of  $O/(\pi^n - 1)$  by its unweighted rings; the carry automaton recognises equality of represented words and does not count classes with weight one. A graded signed transfer (diag(1,  $q$ ) even,  $M$  odd) has supertrace  $N_n$  but is not by itself a doubled-Kraus factorisation; the elliptic norm tensor supplies that. The drafted “carries must be unbounded” was wrong: cyclic carries are bounded here.*

## 22 Curve Counts as Twisted Ring Norms III: the Bosonic No-Go

Notation of Definition 9.1; proofs in notes/ring-norm-tensor/astra-proofs.md T3.

### 22.1 The bosonic no-go and its bookkeeping

*Observation 22.1* (The orchestrator's drafted bound was false). [sketched] The 2026-09-14 morning draft claimed  $\sum_{\text{odd}} |\mu|^2 \leq 2 \text{Tr } \mathcal{E}_{d++} \text{Tr } \mathcal{E}_{d--}$  for bosonic tensors, and its script computed  $\sum_s \|a_s\|_{\text{HS}}^2$  while calling it a trace. Since  $\text{Tr}(A \otimes \bar{A}) = |\text{Tr } A|^2$ , one even letter  $a = B = \text{diag}(1, -1)$  on  $\mathbb{C}^{2|2}$  has both traces zero and odd squared eigenvalue sum 8; the prover and the numerics lane found the error independently (173 of 400 random instances violate it). The genus bound survives by Theorem 22.4.

**Theorem 22.2** (Block structure and the correct Hilbert–Schmidt bounds). [proved] For all-even tensors  $A_s = a_s \oplus B_s$  on  $\mathbb{C}^{m|k}$ :  $\mathcal{E}_{\text{deven}} = S_+ \oplus S_-$  with  $S_+ = \sum a_s \otimes \bar{a}_s$ ,  $S_- = \sum B_s \otimes \bar{B}_s$ , and  $\mathcal{E}_{\text{dodd}} = M \oplus \bar{M}$  with  $M = \sum_s a_s \otimes \bar{B}_s$ . Then  $\sum_{\text{odd}} |\mu|^2 = 2 \sum_{\text{spec } M} |\mu|^2 \leq 2 \|M\|_{\text{HS}}^2 \leq 2(\sum_s \|a_s\|_{\text{HS}}^2)(\sum_s \|B_s\|_{\text{HS}}^2)$  and  $\|M\|_{\text{HS}}^2 \leq \|S_+\|_{\text{HS}} \|S_-\|_{\text{HS}}$  (Gram-matrix Cauchy–Schwarz);  $\text{Tr } S_{\pm} = \sum_s |\text{Tr } a_s|^2, \sum_s |\text{Tr } B_s|^2$ . Schur's inequality is proved from triangularisation.

**Lemma 22.3** (All-power word-trace inequality). [proved] With  $s_{\pm}(n) = \text{Tr } S_{\pm}^n = \sum_w |\text{Tr } a_w|^2, \sum_w |\text{Tr } B_w|^2$  over words  $w$  of length  $n$  and  $\text{Tr } M^n = \sum_w \text{Tr } a_w \text{Tr } \bar{B}_w$ :  $s_{\pm}(n) \geq 0$  and  $|\text{Tr } M^n|^2 \leq s_+(n)s_-(n)$  for every  $n$ , with equality iff the two word-trace vectors are proportional.

**Theorem 22.4** (Bosonic species cannot pass genus one). [proved-conditional] If the nonzero even spectrum is exactly  $\{1, q\}$  with no even–odd cancellation and the odd spectrum is  $2g$  values of modulus  $\sqrt{q}$ ,  $g \geq 1$ , then  $g \leq 1$  and  $\{\text{Tr } S_+, \text{Tr } S_-\} = \{1, q\}$ . For  $m = k = 1$  and  $g = 1$  the two coefficient vectors are proportional, giving Definition 9.3; for larger  $m, k$  literal matrix proportionality and normality are false (nilpotent transients, nonunitary gauges).

*Proof.* Sketch (T3.3). A block with no nonzero eigenvalue has  $s_{\pm}(n) = 0$  for all  $n$ , so Lemma 22.3 makes  $\text{Tr } M^n = 0$  for all  $n$  and  $M$  nilpotent, contradicting the nonempty odd spectrum; so each block carries one of  $1, q$  and  $s_+(n)s_-(n) = q^n$ . Writing the  $g$  eigenvalues of  $M$  as  $\sqrt{q}\beta_j$ ,  $|\beta_j| = 1$ , the inequality reads  $|\sum_j \beta_j^n|^2 \leq 1$ ; the Cesàro mean of the left side over  $n \leq N$  tends to  $\sum_i m_i^2 \geq g$ , so  $g \leq 1$ .  $\square$

**Proposition 22.5** (Euler characteristic bookkeeping and the minimal bond). [proved] If a graded tensor on  $\mathbb{C}^{m|k}$  has  $\text{str } \mathcal{E}_{\text{d}}^n = N_n$  for all  $n$  with net spectrum  $\{1, q\}_+ - \{\alpha_{1..2g}\}_-$ , and  $z_e, z_o$  count zero eigenvalues, then  $(m - k)^2 - (z_e - z_o) = 2 - 2g$ ; necessarily  $mk \geq g$ ; if the odd dimension is exactly  $2g$  then  $mk = g$ , no odd zero modes, no cancellation, and  $m^2 + k^2 - 2$  nilpotent even modes ( $g^2 - 1$  on  $\mathbb{C}^{1|g}$ ). The minimal bond dimension is  $\min\{m + k : mk \geq g\}$ , not generally  $g + 1$ ; at genus two it is three.

**Proposition 22.6** (Odd species and cancellation evade the bound). [proved] Odd letters add blocks  $\sum_f c_f \otimes \bar{c}_f, \sum_f d_f \otimes \bar{d}_f$  between  $(+, +)$  and  $(-, -)$  (zero trace at length one, nonzero in higher powers) and  $\sum_f c_f \otimes \bar{d}_f$  between the mixed sectors, so  $\mathcal{E}_{\text{dodd}}$  is no longer  $M \oplus \bar{M}$ . With all-even letters and a cancelling multiset  $X$ :  $s_+(n) + s_-(n) = 1 + q^n + \sum_X x^n$  and  $|\sum_j \alpha_j^n + \sum_X x^n| \leq 2\sqrt{s_+(n)s_-(n)}$ , which bounds nothing independent of  $X$ . Fermionic letters are therefore necessary on the minimal genus-two bond  $\mathbb{C}^{1|2}$  (no room to cancel), not on every larger bond; the failed bosonic  $\mathbb{C}^{2|2}$  search is no certificate.

**Theorem 22.7** (Infinite bond: bosonic species cannot carry the zeros). [proved-conditional]  
*Under assumption 20.10 and assumption 12.6, if the odd eigenmodes at time  $t$  include  $e^{t\mu_\rho}$  over the nontrivial zeros (with  $\mu_\rho = \rho$  or  $-\bar{\rho}/2$ ), then  $\sum_\rho e^{2t\operatorname{Re}\mu_\rho} \leq \|\mathcal{E}_{\text{odd}}(t)\|_{\text{HS}}^2 \leq 2\|S_+(t)\|_{\text{HS}}\|S_-(t)\|_{\text{HS}} < \infty$ , which is impossible because the left side diverges; hence such a purely bosonic Kraus realisation does not exist and fermionic jumps are necessary in that setting. The trace version  $2\operatorname{Tr} S_+ \operatorname{Tr} S_-$  needs  $S_\pm$  positive on the Hilbert–Schmidt spaces, which complete positivity alone does not give.*

*Numerical result 22.8* (Ring norms of real curves; the false bound; the repaired chain). [numerical]  
**scripts/weil\_ring\_norm\_check.py** (**outputs/weil\_ring\_norm\_check.txt**, 135 checks, author claude:fable-5.1): brute-force counts of four elliptic curves over  $\mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$  against  $|1 - \pi^n|^2$ ,  $\operatorname{str} \mathcal{E}_d^n$ , the explicit vector  $\Psi_P = \Psi_+ - \Psi_-$  with  $\Psi_- = \pi^n \Psi_+$  (Schmidt rank one), the untwisted  $|1 + \pi^n|^2$ ; the decoupled-block Cauchy–Schwarz statement; the Hilbert–Schmidt chain of Theorem 22.2 on random  $\mathbb{C}^{m|k}$  (its printed labels are the Hilbert–Schmidt sums, see the docstring); the genus-two searches of Section 23; the  $y^3 - y = x^2$  indicator state matching the count only at odd  $n$ .

*Numerical result 22.9* (Toral Frobenius, Lenstra structure, gauge and dichotomy). [numerical]  
**scripts/ring\_norm\_tensor.py** (**outputs/ring\_norm\_tensor.txt**, 193 checks, author claude:opus-5, 562 s): for four ordinary curves,  $N_n = |\det(M^n - 1)| = 1 - \operatorname{Tr} M^n + \det M^n$ , the fixed points enumerated by Smith normal form all of index +1,  $k \mapsto M^T k$  with no periodic point but 0 in  $[-50, 50]^2$  up to period 12, the dynamical zeta to order  $u^8$ ,  $|\pi|^2 = q$  and  $M^\dagger G M = qG$  in the Hodge basis; the group structure of  $E(\mathbb{F}_{q^n})$  by brute force against the invariant factors of  $O_K/(\pi^n - 1)$  for  $n \leq 3$ ; the refutation of the drafted trace bound and the validity of the Hilbert–Schmidt chain on 400 random instances; the genus-two items of numerical result 23.14; the supersingularity and cut-rank items of numerical result 20.14.

## 23 Curve Counts as Twisted Ring Norms IV: Genus Two

Tier B of the campaign of Section 20: the first case where fermionic letters are forced (Proposition 22.6). Notation of Definitions 9.4 and 9.5; proofs in notes/ring-norm-tensor/astra-proofs.md T4–T5.

### 23.1 Inputs

*Cited fact* 23.1 (Rosati positivity and  $\pi\pi^\dagger = q$ ). [cited]  
Milne [69], Theorem: the Rosati involution on  $\text{End}^0(A)$  is positive,  $(\alpha, \beta) \mapsto \text{Tr}_{E/\mathbb{Q}}(\alpha \circ \beta^\dagger)$  positive definite; and  $\pi \circ \pi^\dagger = q_A$ . Verbatim: `\={\rm End}^{\sim 0}(A)` is positive, i.e., the pairing (1509.00797:prh.tex:1559); This will follow from (\ref {r39}) once we show that  $\pi \circ \pi^\dagger = q_A$ . This can be proved by a direct calculation, but it is more (1509.00797:prh.tex:1647–1648).

*Cited fact* 23.2 (CM lift: eigenvalues of modulus  $\sqrt{q}$ , CM type with positivity;  $\mathbb{Q}(\pi)$  is CM). [cited] Goresky and

Tai [31]: the lifted Frobenius is semisimple with eigenvalues of magnitude  $\sqrt{q}$ , and the lift carries a CM type  $\Phi$  on  $\mathbb{Q}[F]$  with a positivity condition; Dupuy, Kedlaya, Roe and Vincent [20]:  $\mathbb{Q}(\pi)$  is a CM field. Verbatim: `\item` The mapping  $F$  is semisimple and all of its eigenvalues in  $\mathbb{C}$  have magnitude (1701.07742:main.tex:565); and where  $\Phi$  is a CM type on  $\mathbb{Q}[F]$  such that the following `\em` positivity condition holds: (1701.07742:main.tex:604–605), `\item`  $\mathbb{Q}(\pi)$  is a CM-field with maximal totally real subfield  $\mathbb{Q}(\beta)$ , and  $\pi$  has no real embeddings; (2003.05380:weil.tex:332).

*Cited fact* 23.3 (Positivity of the Hodge form on holomorphic one-forms). [cited] Lesfari [58]: for a holomorphic one-form  $\omega = f dz$  on a compact Riemann surface,  $\omega \wedge \bar{\omega} = -2i|f|^2 dx \wedge dy$  and  $i \int \omega \wedge \bar{\omega} > 0$  for  $\omega \neq 0$ . Verbatim:  $\omega \wedge \overline{\omega} = -2i|f|^2 dx \wedge dy$ . (0903.2156:main.tex:1254);  $\int_X |f|^2 dx \wedge dy > 0$ ,  $\omega \neq 0$ , (0903.2156:main.tex:1262).

**Assumption 23.4** (CM type and the Hodge decomposition of the canonical lift). [assumed]

For a simple ordinary abelian variety  $A/\mathbb{F}_q$  of dimension  $g$  with Frobenius  $\pi$ ,  $K = \mathbb{Q}(\pi)$  is a CM field of degree  $2g$ ,  $\text{End}(A) \otimes \mathbb{Q} = K$ , and  $H^1(\tilde{A}(\mathbb{C}), \mathbb{C}) = \bigoplus_{\phi: K \rightarrow \mathbb{C}} \mathbb{C}_\phi$  with  $H^{1,0} = \bigoplus_{\phi \in \Phi} \mathbb{C}_\phi$  for a CM type  $\Phi$ ,  $\tilde{\pi}$  acting on  $\mathbb{C}_\phi$  by  $\phi(\pi)$  (Shimura–Taniyama; the eigenspace statement is not quotable from the fetched TeX, cf. cited fact 23.2).

**Assumption 23.5** (Polarised lift, Hodge adjunction, Jacobian comparison; abelian cohomology; real closed transfer; functoriality). [assumed]

(i) A polarisation lifts along the canonical lift and identifies Rosati adjunction with adjunction for the positive Hodge form on  $H^{1,0}$ ; for  $i: C \hookrightarrow J$ ,  $i^*: H^1(J) \rightarrow H^1(C)$  is an isomorphism,  $[C]$  is the principal polarisation in dimension two,  $\int_C i^* \xi = \int_J \xi \smile [C]$  and  $\int_J \theta^2 = 2$ . (ii)  $H^*(A) = \Lambda^* H^1(A)$  with Frobenius acting by exterior powers and  $\#A(\mathbb{F}_{q^n}) = \det(1 - F^n | H^1)$ . (iii) A finite system of real polynomial equalities and inequalities over the real algebraic numbers with a real solution has a real algebraic solution. (iv) Frobenius characteristic polynomials are multiplicative in exact sequences of abelian varieties; ordinarity is preserved by finite base extension.

### 23.2 The block equations and two impossibilities

**Lemma 23.6** (Block equations on  $\mathbb{C}^{1|2}$ ). [proved] In the notation of Definition 9.4,  $\mathcal{E}_{\text{dodd}} = \begin{pmatrix} M & N \\ N & M \end{pmatrix}$  and  $\mathcal{E}_{\text{even}} = \begin{pmatrix} t & u \\ v & w \end{pmatrix}$ ; equivalently the complete positive

map is  $x \oplus X \mapsto (tx + \sum_f c_f X c_f^*) \oplus (\sum_f d_f x d_f^* + \sum_s B_s X B_s^*)$  on even operators, with no Koszul sign.

**Proposition 23.7** (The literal placement and the vacuum ansatz are impossible). *[proved-conditional]* If the odd spectrum is the four Frobenius values and the even spectrum  $\{1, q, 0, 0, 0\}$ , then the pure trace line  $0 \oplus 1_2$  is not a  $q$ -eigenvector and the vacuum line is not invariant (either would force all  $c_f = 0$  or all  $d_f = 0$ , delete the odd letters without changing the spectra, and contradict Theorem 22.4); both coupling directions must occur. The drafted ansatz  $a_1 = 1, B_1 = 0, a_2 = 0, B_2 = B$  makes  $M = 0$ , so one odd letter gives  $\text{rank} \leq 2$  and any number gives a spectrum symmetric under  $z \mapsto -z$  with zero trace; it fails for every curve with  $e_1 \neq 0$ .

### 23.3 Existence

**Theorem 23.8** (The nine-letter tensor for  $q + 1 \geq 4\sqrt{q}$ ). *[proved-conditional]* For  $|\pi_j|^2 = q$  and  $q + 1 \geq 4\sqrt{q}$  (every prime power  $q \geq 16$ ) the nine-letter tensor of Definition 9.5 has  $\text{spec } \mathcal{E}_{\text{even}} = \{1, q, 0, 0, 0\}$  with the three zero modes the traceless  $2 \times 2$  operators killed in one step,  $\mathcal{E}_{\text{odd}} = \text{diag}(\bar{D}, D)$ , a normal transfer in an orthonormal doubled basis, and  $\text{str } \mathcal{E}_{\text{d}}^n = 1 + q^n - \sum_j (\pi_j^n + \bar{\pi}_j^n) = N_n(C)$  for every  $n$ , ring zeta  $Z(C, u)$ . The even letters realise the depolarising map  $\sum_\ell B_\ell X B_\ell^* = w \text{Tr}(X) 1_2$ , the odd letters couple vacuum and trace, and on  $\text{span}(\text{vac}, 1_2/\sqrt{2})$  the even block is  $\begin{pmatrix} (q+1)/2 & (q-1)/2 \\ (q-1)/2 & (q+1)/2 \end{pmatrix}$  with eigenvalues  $q, 1$  and eigenvectors  $(\pm\sqrt{2}, 1_2)$ : neither is the pure vacuum nor the pure polarisation line. On  $\mathbb{C}^{1|g}$  the same works with  $w = (q + 1)/(2g)$ ,  $h = (q - 1)/(2\sqrt{g})$  whenever  $q + 1 \geq 2g\sqrt{q}$ . Entries are algebraic over  $\mathbb{Q}(\pi_1, \pi_2)$ ; the letters have no equation-local meaning and their number is not claimed minimal.

**Proposition 23.9** (Base change of the  $\mathbb{F}_5$  curve to  $\mathbb{F}_{25}$ ). *[proved-conditional]* For  $C: y^2 = x^5 + x^3 + x^2 - 2$  over  $\mathbb{F}_5$ ,  $\chi_F = z^4 - 3z^3 + 7z^2 - 15z + 25$  (matching the counts 3, 31, 117, 619); over  $\mathbb{F}_{25}$ ,  $\chi_{F^2} = z^4 + 5z^3 + 9z^2 + 125z + 625$ , the nine-letter tensor applies, and the first six norms are 31, 619, 15991, 390739, 9759526, 244128859. Both quartics are irreducible over  $\mathbb{Q}$  (reduction mod 2), so both Jacobians are simple over their fields; ordinarity is preserved.

**Proposition 23.10** (The three-letter problem is a finite real feasibility problem). *[proved-conditional]* Two even and one odd letter on  $\mathbb{C}^{1|2}$  realise  $N_n(C)$  for all  $n$  iff  $\det(z - \mathcal{E}_{\text{even}}) = z^3(z - 1)(z - q)$  and  $\det(z - \mathcal{E}_{\text{odd}}) = \chi_F(z)$  (28 real unknowns; Proposition 22.5 forbids cancellation on this bond), and then an algebraic solution exists. Least-squares residuals and four-digit printouts certify nothing; only Theorem 23.11 does.

**Theorem 23.11** (A certified two-even, one-odd tensor for the  $\mathbb{F}_5$  curve). *[proved-conditional]* There are two even and one odd letter on  $\mathbb{C}^{1|2}$  with  $\chi_e = z^3(z - 1)(z - 5)$  and  $\chi_o = z^4 - 3z^3 + 7z^2 - 15z + 25$ , hence  $P$ -closed ring norm  $N_n(C)$  for every  $n \geq 1$ . The entries are the unique zero, in the box  $\|y - y_0\|_\infty \leq 10^{-30}$  around a rational centre  $y_0$  (nine free real coordinates, the other nineteen frozen rationals), of the nine power-trace equations  $f(y) = (\text{Tr } \mathcal{E}_{\text{even}}^n - (1 + 5^n))_{n \leq 5}, (\text{Tr } \mathcal{E}_{\text{odd}}^n - p_n)_{n \leq 4}$ ; a rational approximate inverse  $B$  of  $Df(y_0)$  satisfies, by exact rational comparison,  $\|B\|_\infty < 2$ ,  $\|Bf(y_0)\|_\infty < 10^{-70}$ ,  $\|1 - BDf(y_0)\|_\infty < 10^{-50}$ , and with the uniform Hessian bound  $10^{16}$  the map  $y \mapsto y - Bf(y)$  contracts the box into itself. Newton's identities turn the nine traces into the two characteristic polynomials. The entries are algebraic (unique real point of a rational semialgebraic set). This is a computer-assisted existence proof for one curve, not a formula.

**Conjecture 23.12** (Universal three-letter tensor). *[conjectured]* For every ordinary simple genus-two curve the feasibility problem of Proposition 23.10 has a solution;

and some solution is selected naturally by the equation or the polarised CM data. Both parts are open; failed smaller-alphabet searches prove no lower bound on the number of letters.

*Numerical result 23.13* (Certificate, nine-letter tensor, Fock-vector norms). [numerical]  
`scripts/ring_norm_certificate.py` (`outputs/ring_norm_certificate.txt`, 65 checks): the rational certificate of Theorem 23.11 re-run in exact arithmetic ( $\|B\|_\infty = 1.499$ , residual  $1.0 \times 10^{-90}$ , inverse defect  $3.5 \times 10^{-58}$ ,  $\kappa = 1.2 \times 10^{-12}$ ); the certified centre in floating point gives  $\chi_e, \chi_o$  to  $10^{-9}$ ,  $\text{str } \mathcal{E}_d^n = N_n$  for  $n \leq 12$ , and the explicit Jordan–Wigner Fock vector has norm  $N_n$  and periodic-translation invariance for  $n = 2, \dots, 5$ ; the nine-letter tensor at  $q = 25$  (the base change), 17, 16 has even spectrum  $\{1, q, 0, 0, 0\}$ , odd spectrum the four inputs, normality  $10^{-15}$ ,  $\text{str } \mathcal{E}_d^n = N_n$  to  $10^{-15}$  for  $n \leq 8$ , and fails at  $q = 13$  exactly as  $q + 1 < 4\sqrt{q}$ .

*Numerical result 23.14* (Structured genus-two solutions: the CM-diagonal ansatz). [numerical]  
`scripts/ring_norm_tensor.py`, N3: on the  $\mathbb{F}_5$  curve, two even and one odd letter reach residual  $10^{-24}$  (the seed of Theorem 23.11); a vacuum letter with one odd letter never converges (every search stalls at  $t \rightarrow q$ ,  $\text{spec } M \rightarrow \{\bar{\pi}_j\}$ ); a vacuum letter with two odd letters, and the CM-diagonal ansatz  $A_s = \text{diag}(a_s, b_{s,1}, b_{s,2})$  (all even letters simultaneously diagonal, the direct lift of Definition 9.3) with two odd letters, both solve it to  $10^{-25}$  with even spectrum  $\{1, 5, 0, 0, 0\}$ ; in every solution  $M \neq \text{diag}(\bar{\pi}_j)$  and  $|m_i|^2 \neq q$ , and the solution set is positive dimensional ( $t = \sum |a_s|^2$  varies between restarts; only  $t + \text{Tr } W = 1 + q$  and  $2 \text{Re Tr } M = e_1$  are pinned). The three nilpotent modes appear numerically at  $|z| \sim 10^{-4}$ , the cube root of the residual. The Jacobian identity and the curve subspace of Proposition 23.16 check for  $n \leq 6$ .

**Corollary 23.15** (The genus-two ring norms are Hilbert norms of fermionic ring states). [sketched-conditional]  
*By Theorem 16.2 on the lattice, the supertraces of Theorems 23.8 and 23.11 are the squared norms of the physical periodic-fermion rings built from the letters; the explicit Jordan–Wigner check of numerical result 23.13 confirms this for  $n \leq 5$ . The count of a genus-two curve is therefore literally the norm of a graded matrix product state with a three-dimensional bond and a fermionic letter.*

## 23.4 Jacobian, Rosati metric, gauge

**Proposition 23.16** (Jacobian product states and the curve projector). [proved-conditional] *The tensor product of two elliptic norm tensors  $\text{diag}(1, \pi_j)$  on the ket  $V_J = \Lambda^* H^{1,0} = \mathbb{C}^{2|2}$  has  $P$ -closed ring norm  $\prod_j |1 - \pi_j^n| = \#J(\mathbb{F}_{q^n})$  and double  $H^*(J) = \Lambda^* H^1(J)$ . Inside it the Frobenius-invariant graded subspace  $\mathcal{S} = \mathbb{C}1 \oplus H^1(J) \oplus \mathbb{C}\omega$ ,  $\omega = e_1 \wedge f_1 + e_2 \wedge f_2$  a polarisation class with  $F\omega = q\omega$ , has  $\text{str } F^n|_{\mathcal{S}} = N_n(C)$ ; the principal class restricts to  $2[\text{pt}]$  on  $C$ . The projector onto  $\mathcal{S}$  (rank six) is not of the form  $B \otimes \bar{B}'$  (a product range would have dimension  $\geq 9$ ): extracting the curve from the Jacobian ring needs a non-product boundary, which the single fermionic rings above avoid. This does not say every boundary with the same scalar counts is that projector.*

**Theorem 23.17** (Rosati positivity gives the moduli and the Polya–Hilbert metric). [proved-conditional]  
*For a simple ordinary polarised abelian variety, the Rosati involution acts as complex conjugation on each factor of  $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \prod_{j \leq g} \mathbb{C}$  (a primitive idempotent sent elsewhere or  $ie_j$  fixed would contradict  $\text{Tr}(xx^\dagger) > 0$ ), so  $\pi\pi^\dagger = q$  gives  $|\phi_j(\pi)|^2 = q$  at every place and  $q^{-1/2}\bar{\pi}^*$  is unitary on  $H^{1,0}$  and on  $H^1(\tilde{J}, \mathbb{C})$  with their Hodge metrics: the ordinary simple instance of the root sizes from positivity, with no use of assumption 12.17. The PH models  $H^1(\tilde{J}, \mathbb{R})$ ,  $K \otimes \mathbb{R}$  with the Rosati trace form, and the realification of  $H^{1,0}$  agree up to positive weights and the CM type. In Theorem 23.8 the odd block is  $\sqrt{q}$  times a unitary by construction, so RH is manifest only after the moduli are supplied; the arithmetic proof of those moduli is Rosati (Hodge index on  $C \times C$ ), not complete positivity of a transfer. For the certified tensor the distinct odd roots give complex similarity to  $\sqrt{5}$  times a unitary;*

whether that similarity is an even ket gauge carrying the arithmetic Hodge metric is not determined by the count equations.

**Proposition 23.18** (Gauge and the CM-type marking). [proved-conditional] *The even gauge  $G = \text{diag}(g_0, H)$ ,  $H \in \text{GL}_2(\mathbb{C})$ , acts by  $B_s \mapsto HB_sH^{-1}$ ,  $c_f \mapsto g_0c_fH^{-1}$ ,  $d_f \mapsto Hd_fg_0^{-1}$  and fixes all ring norms; same-parity unitary rotations of the letters fix  $\mathcal{E}_d$  itself. The CM type is a marking: replacing a selected  $\pi_j$  by  $\bar{\pi}_j$  changes the mixed-sector eigenvalues of Theorem 23.8 and is not an even gauge; a coupled solution with  $N \neq 0$  mixes the two coherences and its counts do not recover a marked half. The unitary gauge is an orthonormal Hodge basis.*

### 23.5 Assessment for the Phantasm (observations, not theorems)

*Observation 23.19* (Candidate bond; conjugation is not the functional equation). [sketched-conditional]

A plausible ket is  $\mathbb{C} \oplus H_{>0}$  with one odd mode per positive-height zero, the bra supplying conjugates; its  $(-, -)$  sector is far larger than the pole sector and what kills or regularises those modes is unspecified. Ket/bra exchange is complex conjugation; the functional equation also involves  $\rho \mapsto 1 - \rho$ , which coincides with conjugation on each pair only on the critical line, so reading the functional equation as ket/bra exchange would assume RH.

*Observation 23.20* (Fermionic jumps are necessary in the stated setting; sufficiency open). [sketched-conditional]

Theorem 22.7 excludes the all-even Kraus class under assumption 20.10. It does not follow that odd jumps suffice, that finitely many do, or that a distributional regularisation is a Hilbert norm; Theorem 16.2 is a finite-bond statement.

*Observation 23.21* (What the toral example contributes and where the analogy breaks). [sketched-conditional]

For an ordinary elliptic curve the lifted Frobenius is an injective Fourier shift whose only finite cycle is the zero label and the flat Lefschetz supertrace sits in that fibre; the physical state is one configuration. This is no principle that every variety's tensor is a permutation: the shift is not surjective, and for an abelian surface the torus counts the Jacobian, not the curve. No lifted Riemann dynamics, positive Hodge/Rosati metric or physical-label construction is established; that missing dynamics and positivity is the substantive break.

*Observation 23.22* (The locality dichotomy must be restricted). [sketched-conditional]

“Fixed finite lattice tensors give only supersingular zetas” is false: Theorem 21.3 and Theorem 23.8 are finite ordinary norm tensors. Only the equation-local amplitude question survives as conjecture 20.15. Infinitely many zeros and dilation time motivate an infinite-bond cMPS, but supply none of the domains, regularised trace, tensor factorisation or positivity it would need.

## 24 Curve Counts as Twisted Ring Norms V: Explicit Examples, Entanglement, Parent Hamiltonians

*Question (TJO, 2026-09-14, evening).* How hard is it to describe a specific example of each genus and write down the actual matrices of the corresponding MPS; and what are their usual entanglement properties and parent Hamiltonians? *Answer.* Genus 0 and 1 are one-liners, genus 2 is the closed form of Theorem 23.8 for  $q \geq 16$  and the certified decimals of Theorem 23.11 below that, and every genus  $g$  is the same closed form on  $\mathbb{C}^{1|g}$  once  $q + 1 \geq 2g\sqrt{q}$ ; the only input at every genus is the  $L$ -polynomial (brute-force counts  $N_1, \dots, N_g$ , Newton's identities, the functional equation). Everything below is the notebook's own calculation (`scripts/ring_norm_examples.py`, author claudes:fable-5.1, unreviewed); the contraction is the Kronecker one of Definition 9.1, with the fermionic reading checked against explicit Jordan–Wigner vectors.

### 24.1 The five states

**Definition 24.1** (The example states). All rings are  $P$ -closed unless stated. (0)  $\mathbb{P}^1/\mathbb{F}_5$ : bond  $\mathbb{C}^2$  ungraded, letters  $\{\infty\} \cup \mathbb{F}_5$ ,  $A_\infty = e_{11}$ ,  $A_s = e_{22}$  for  $s \in \mathbb{F}_5$ ;  $\Psi = |\infty\rangle^{\otimes n} + \sum_{s \in \mathbb{F}_5^n} |s\rangle$ ,  $\|\Psi\|^2 = 1 + 5^n$ ; the configurations are the points. (1)  $y^2 = x^3 + x + 1/\mathbb{F}_5$ :  $N_1 = 9$ ,  $a = -3$ ,  $\pi = (-3 + i\sqrt{11})/2$ ; bond  $\mathbb{C}^{1|1}$ , letters  $A_s = \phi_s \text{diag}(1, \pi)$  with  $\phi = (1, 1)/\sqrt{2}$ ,  $P = \text{diag}(1, -1)$ ; the same Frobenius as the toral endomorphism  $M = \begin{pmatrix} 0 & -5 \\ 1 & -3 \end{pmatrix}$ . (2a)  $y^2 = x^5 + x^3 + x^2 - 2/\mathbb{F}_5$ : the certified letters of Theorem 23.11, two even, one odd, bond  $\mathbb{C}^{1|2}$ . (2b) the same curve over  $\mathbb{F}_{25}$  (eigenvalues  $\pi_j^2$ ): the nine-letter tensor with  $t = 13$ ,  $w = 13/2$ ,  $h = 6\sqrt{2}$ . (3)  $y^2 = x^7 + x + 1/\mathbb{F}_{37}$ : counts 41, 1455, 50915, six Frobenius eigenvalues of modulus  $\sqrt{37}$ ; the closed form on  $\mathbb{C}^{1|3}$  with ten even and six odd  $4 \times 4$  letters,  $t = 19$ ,  $w = 19/3$ ,  $h = 6\sqrt{3}$ ;  $\mathbb{F}_{37}$  is the smallest prime field with  $q + 1 \geq 6\sqrt{q}$ .

**Proposition 24.2** (Correlation lengths: RH as a single odd correlation length). *[sketched-conditional]* For every graded tensor whose doubled transfer has net even spectrum  $\{1, q\}$  and odd spectrum the Frobenius eigenvalues, the dominant eigenvalue is  $q$ , the even correlation length is  $1/\log q$ , and the correlation length of an odd mode  $\alpha_j$  is  $1/\log(q/|\alpha_j|)$ ; hence RH for the curve is the statement that every odd (fermionic) mode has the same correlation length  $2/\log q$ , twice the even one. In the five examples this holds to  $10^{-6}$  (numerical result 24.6).

*Proof.* Immediate from the spectra; the modulus  $|\alpha_j| = \sqrt{q}$  is assumption 12.17.  $\square$

### 24.2 Entanglement

**Proposition 24.3** (Block Schmidt spectrum of a twisted ring from the doubled bond). *[sketched]* For a graded tensor on a bond of dimension  $D$  with Kronecker contraction, the reduced state of a contiguous block of  $l$  sites in the  $P$ -closed ring of length  $n$  has at most  $D^2$  nonzero Schmidt weights, equal to the spectrum of the  $D^2 \times D^2$  matrix  $K G_X^T$  with  $G_X[(ab), (a'b')] = (\mathcal{E}_d^l)_{(aa'), (bb')}$  and  $K[(ab), (a'b')] = (\mathcal{E}_d^{n-l} \Gamma_b)_{(bb'), (aa')}$ , normalised by the ring norm; hence the block entanglement entropy is at most  $2 \log D$ . For a contiguous block starting at the first site the fermionic (Jordan–Wigner) reduced state has the same spectrum, because the string acting on the block is a parity sign, a unitary on the block.

*Proof.* Write  $\Psi_P = \sum_{ab} X_{ab} \otimes Y_{ba}$  with  $X_{ab} = \sum_w (A_w)_{ab} |w\rangle$  over the block and  $Y_{ba} = \sum_{w'} (A_{w'} P)_{ba} |w'\rangle$  over the rest; then  $\rho_{\text{block}} = \sum K_{\alpha\beta} |X_\alpha\rangle \langle X_\beta|$  with  $K$  the Gram matrix of the  $Y$ , and on coefficient vectors  $\rho$  acts as  $K G_X^T$ . The Gram matrices are the stated entries of  $\mathcal{E}_d^l$  and  $\mathcal{E}_d^{n-l}(P \otimes \bar{P})$ . Checked against explicit vectors in numerical result 24.6.  $\square$

*Observation 24.4* (What the entanglement looks like at each genus).

[sketched]

Genus 0: the state is a cat state of two orthogonal product states with weights 1 and  $5^n$ , so every block has Schmidt rank two and entropy  $H(1/(1+5^n))$ , which vanishes exponentially; the parent Hamiltonian is two-fold degenerate (both branches), as for a GHZ state. Genus 1:  $\Psi_P = (1-\pi^n)\phi^{\otimes n}$  is a product state; the two bond lines carry the same physical vector, so the count is entirely in the normalisation and the entanglement is zero. Genus 2 (certified letters over  $\mathbb{F}_5$ ): Schmidt rank  $9 = D^2$  for every half-ring, entropy saturating at 1.6636 for  $n \geq 20$  (bound  $2 \log 3 = 2.197$ ), largest weights 0.380, 0.199, 0.199, 0.105. Genus 2 (nine letters over  $\mathbb{F}_{25}$ ): the half-ring spectrum is exactly  $\{\frac{1}{4}, \frac{1}{8}^{\times 4}, \frac{1}{16}^{\times 4}\}$  already at  $n = 10$ , entropy  $\log 8 = 2.0794$ ; genus 3 (sixteen letters over  $\mathbb{F}_{37}$ ): weights  $\{\frac{1}{4}, \frac{1}{12}^{\times 6}, \frac{1}{36}^{\times 9}\}$ , entropy  $\log 12 = 2.4849$ , in both cases  $\frac{1}{4}$  on the vacuum line and a flat spectrum on the rest, the signature of the depolarising even letters of Theorem 23.8. The odd sector contributes to the entanglement only through the doubled bond: the physical letters of the certified tensor carry one fermion, those of the closed form  $2g$ .

### 24.3 Parent Hamiltonians

**Proposition 24.5** (Parent Hamiltonians and the two boundary conditions).

[sketched] *Let*

*$S_k$  be the span of the  $k$ -site block vectors  $X_{ab}$  and  $h = 1 - \Pi_{S_k}$ ; the parent Hamiltonian on the ring is  $H = \sum_i h_i$  (translates, wrapping around). Every ring  $\Psi_B$  with  $B$  commuting with the letters is annihilated by all non-wrapping terms; the term wrapping the closure sees  $B$ . For an all-even tensor both  $\Psi_1$  and  $\Psi_P$  are ground states. When odd letters are present,  $A_s P A_t = \eta_s P A_s A_t$ , so  $\Psi_P$  is a ground state exactly of the Hamiltonian whose wrapping terms are conjugated by the site parities of the wrapped sites (the Jordan–Wigner image of the periodic fermion ring), and  $\Psi_1$  (antiperiodic) of the untwisted one; the fermionic parent Hamiltonian is one local Hamiltonian with these two boundary conditions.*

*Proof.* Sketch. The block at sites  $i..i+k-1$  of  $\Psi_B$  has components  $\sum_{ab} (A_w)_{ab} (\cdots B \cdots)_{ba}$ , in  $S_k$ ; for the wrapped block the word is  $A_{w_1} B A_{w_2}$ , and for  $B = P$  moving  $P$  to the left multiplies the wrapped letters by their parities, i.e. applies  $\prod Z$  on the wrapped sites, which is undone by conjugating the term. Checked numerically in numerical result 24.6: for the fermionic examples  $\Psi_P$  lies in the kernel of the twisted and not of the untwisted Hamiltonian, and conversely for  $\Psi_1$ .  $\square$

*Numerical result 24.6* (The five examples: spectra, entanglement, injectivity, parent Hamiltonians).

[numerical] `scripts/ring_norm_examples.py` (`outputs/ring_norm_examples.txt`). For each of the five states: the counts  $\text{str } \mathcal{E}_d^n = N_n$  for  $n \leq 4$ ; the even and odd spectra and the correlation lengths ( $1/\log q$  even,  $2/\log q$  for every odd mode); the injectivity ranks  $\dim \text{span}\{A_w : |w| = k\}$  for  $k \leq 4$ : genus 0 gives  $2 < 4$  (two blocks), genus 1 gives 1 (product state), the certified genus-two letters reach  $9 = D^2$  at  $k = 2$ , the closed forms are injective at  $k = 1$ ; the block Schmidt spectra from Proposition 24.3 against explicit vectors for  $n \leq 7$  (genus 2) and against explicit Jordan–Wigner Fock vectors for  $n \leq 5$ ; the half-ring entropies of observation 24.4; the parent Hamiltonians ( $k = 2$  for genus 0,  $k = 1$  for genus 1,  $k = 3$  for the certified letters,  $k = 2$  for the nine letters) with ring ground-space dimensions, gaps, and the membership of  $\Psi_1$  and  $\Psi_P$  in the kernels of the untwisted and parity-twisted Hamiltonians as in Proposition 24.5; genus 3 has sixteen letters and its parent Hamiltonian is not diagonalised. Results: genus 0, ring  $n = 5$ , ground space of dimension 2 (the two branches), gap 2; genus 1,  $n = 6$ , unique ground state, gap 1; certified genus-two letters,  $n = 6$ , unique ground state for each boundary condition, gaps 0.239 (untwisted,  $\Psi_1$ ) and 0.232 (twisted,  $\Psi_P$ ); nine letters over  $\mathbb{F}_{25}$ ,  $n = 4$ , unique ground state for each, gaps 0.639 and 0.636. 80 checks.

## 25 Deligne's Proof, Told Through Regular Graphs

Expository. Nothing here is new; the point is to isolate what is actually hard in the proof of the Riemann hypothesis over finite fields [18], using the Ihara zeta [40] of a regular graph as the running model, and to read off which ingredients this project has and which it does not.

### 25.1 One graph, two sides

Fix a finite connected  $(q+1)$ -regular graph  $X$ . The objects are those of Section 3: non-backtracking walks and prime cycles (Definition 3.5), the count  $N_\ell \geq 0$  of closed non-backtracking walks of length  $\ell$ , the Ihara zeta with its Euler product (Definition 3.6), the Hashimoto operator with  $N_\ell = \text{Tr } T^\ell$  and  $\zeta_X(u) = \det(1 - uT)^{-1}$  (Definition 3.7), and its compression to the adjacency matrix  $A$  by Ihara–Bass (Definition 3.8, [3]). Side A is the orbit side, side B the small matrix (Definition 3.1); passing between them is bookkeeping.

Call  $\pm 1$  and the pair  $\mu \in \{q, 1\}$  from the adjacency eigenvalue  $\lambda = q+1$  the *trivial* eigenvalues of  $T$ , everything else nontrivial. Then

$$N_\ell = q^\ell + 1 + \sum_{\mu \text{ nontrivial}} \mu^\ell + (|E| - |V|)(1 + (-1)^\ell),$$

with extra terms if  $X$  is bipartite: the leading term is the growth rate of the  $(q+1)$ -regular tree, and the nontrivial eigenvalues are the error. Since  $\mu\mu' = q$ , a pair either has both members of modulus  $\sqrt{q}$  (when  $|\lambda| \leq 2\sqrt{q}$ , where they are complex conjugates) or is real with one member larger. So all nontrivial  $|\mu| = \sqrt{q}$  iff all nontrivial  $|\lambda| \leq 2\sqrt{q}$  iff the error in  $N_\ell$  is  $O(q^{\ell/2})$ : the Ramanujan property (Definition 3.9) and the RH analogue (Definition 3.4) are the same statement, that closed walks spread as evenly as they possibly could.

Two examples from the founding session.  $K_4$  has  $q = 2$ , adjacency spectrum  $3, -1, -1, -1$ , hence  $\mu = (-1 \pm i\sqrt{7})/2$  with  $|\mu|^2 = 2$ : Ramanujan, with  $N_3 = 8 + 1 + 6 \cdot \frac{5}{2} = 24$ , the oriented pointed triangles. The prism  $C_{16} \times K_2$  has an eigenvalue  $2 \cos(\pi/8) + 1 \approx 2.848 > 2\sqrt{2}$ : a real pole inside the circle and an error growing faster than  $2^{\ell/2}$ . A long thin graph always fails: a bottleneck gives a real  $\mu$  near  $q$ , and walks pile up on one side of it.

### 25.2 The same statement for a curve, with one sign changed

A curve  $C$  over  $\mathbb{F}_q$  has  $N_\ell$  points over  $\mathbb{F}_{q^\ell}$ : again a nonnegative count, again a generating function  $Z_C(u)$  with an Euler product, now over closed points, the orbits of Frobenius  $\text{Fr}$  on the points over the algebraic closure. Side A is identical in form; side B gives a  $2g \times 2g$  matrix ( $g$  the genus) with

$$N_\ell = q^\ell + 1 - \text{Tr } \text{Fr}^\ell = q^\ell + 1 - \sum_{i=1}^{2g} \alpha_i^\ell, \quad Z_C(u) = \frac{\det(1 - u\text{Fr})}{(1 - u)(1 - qu)}.$$

|                             | graph                         | curve                                 |
|-----------------------------|-------------------------------|---------------------------------------|
| count                       | closed non-backtracking walks | points over $\mathbb{F}_{q^\ell}$     |
| leading term                | $q^\ell + 1$                  | $q^\ell + 1$                          |
| error term                  | $+\sum \mu^\ell$              | $-\sum \alpha_i^\ell$                 |
| pairing                     | $\mu \leftrightarrow q/\mu$   | $\alpha_i \leftrightarrow q/\alpha_i$ |
| interesting eigenvalues are | poles of $\zeta_X$            | zeros of $Z_C$                        |
| RH                          | $ \mu  = \sqrt{q}$            | $ \alpha_i  = \sqrt{q}$               |

Everything matches except the sign of the error term, hence whether the interesting eigenvalues are poles or zeros. For the graph  $\zeta_X$  is the reciprocal of a polynomial and every eigenvalue is a pole; for the curve the trivial eigenvalues 1 and  $q$  are poles and the interesting ones are zeros. This is the sign fixed in Section 2,  $\mathcal{S}_n = -\sum_i \alpha_i^n$ , and the sign of the explicit formula. For a variety of dimension  $n$  the pattern repeats as an alternating sum over  $n + 1$  blocks, block  $i$  of modulus  $q^{i/2}$ ; the one that matters is the middle,  $i = n$ .

### 25.3 What is cheap and what is not

| ingredient           | graph                         | variety                      |
|----------------------|-------------------------------|------------------------------|
| side B exists        | $T$ plus Ihara–Bass           | Grothendieck’s trace formula |
| the pairing          | $\mu\mu' = q$ from Ihara–Bass | Poincaré duality             |
| trivial bound        | Perron: $ \mu  \leq q$        | $ \alpha_i  \leq q$          |
| the bound $\sqrt{q}$ | <i>false in general</i>       | Deligne’s theorem            |

The first three rows are cheap. Rationality of the zeta function is the first row and nothing more; the trace formula is heavy technology but the same move, that the count is the trace of a power of a finite matrix. The pairing turns the one-sided bound into equality, so the whole problem is one-sided; and the trivial bound is just convergence of the Euler product for  $|u| < 1/q$ . The gap between  $q$  and  $\sqrt{q}$  is the entire content.

The last row is where graph and variety part. There is no theorem that a  $(q + 1)$ -regular graph is Ramanujan; most are not, the prism is a counterexample, and each Ramanujan construction carries its own proof: arithmetic (Lubotzky–Phillips–Sarnak [61], through RH for a modular curve), probabilistic (Friedman), or interlacing (Marcus–Spielman–Srivastava). So the right question is not how one proves the bound but what varieties have that graphs lack, which makes every one of them Ramanujan. Three things: varieties can be *multiplied*, *sliced* over a curve, and their interesting eigenvalues enter the count with a *minus sign*. A graph has none of the three.

### 25.4 The engine: power up, dominate, take roots

The trick the proof is built on, in its purest graph form: why the Ramanujan bound is  $2\sqrt{q}$ . On the infinite  $(q + 1)$ -regular tree let  $c_{2k}$  count closed ordinary walks of length  $2k$  from a root, so the spectral radius of the tree’s adjacency operator is  $\lim_k c_{2k}^{1/2k}$ . Such a walk is a Dyck path (about  $4^k$ , up to a polynomial factor) together with a fresh neighbour at each of the  $k$  outward steps ( $q$  choices,  $q + 1$  at the root), so  $c_{2k} = 4^k q^k$  times a bounded factor times a polynomial in  $k$ , and  $c_{2k}^{1/2k} \rightarrow 2\sqrt{q}$ . Three moves: a nonnegative count dominates a power of an eigenvalue,  $c_{2k} = \langle \delta_{\text{root}}, A^{2k} \delta_{\text{root}} \rangle$ ; the count is known up to a loss that does not grow exponentially in  $k$ ; and the  $2k$ -th root as  $k \rightarrow \infty$  kills every fixed loss. This computation is the tree side of the Ramanujan comparison. Deligne makes the same three moves twice; the nontrivial content is entirely in the second.

### 25.5 The pointwise theorem, in twisted-graph language

Deligne’s core theorem is not about a variety but about a curve carrying matrix-valued coefficients. A *twist* on a base graph  $B$  assigns to each directed edge  $e$  an invertible  $d \times d$  matrix  $\varrho(e)$  with  $\varrho(\bar{e}) = \varrho(e)^{-1}$ ; a closed walk  $w$  has holonomy  $\varrho(w)$ , the ordered product along it, and the twisted zeta is the Artin–Ihara  $L$ -function of Definition 3.10,

$$L(u, \varrho) = \prod_P \det(1 - u^{\ell(P)} \varrho(P))^{-1} = \exp\left(\sum_w \frac{1}{\ell(w)} \text{Tr } \varrho(w) u^{\ell(w)}\right) = \det(1 - uT_\varrho)^{-1}.$$

The quantum Ihara zeta of Section 26 is this with  $B$  a bouquet of  $D$  loops and  $\varrho = \text{Ad}(U_i)$  on the  $i$ -th loop; a quantum expander is a Ramanujan twist of a bouquet. Two kinds of eigenvalue now live on side B: the *global* ones, the eigenvalues of  $T_\varrho$  and poles of  $L$ , analogues of the  $\mu$ ; and the *pointwise* ones, the eigenvalues of each  $\varrho(P)$ . The twist is *pure with constant*  $c$  when every eigenvalue of every  $\varrho(P)$  has modulus  $c^{\ell(P)}$ : a unitary twist is pure with  $c = 1$ , Deligne's coefficient systems with  $c = q^{w/2}$  for an integer weight  $w$  (Definition 4.16).

Deligne's pointwise theorem (his Theorem 3.2, in this language): if (a) all traces  $\text{Tr } \varrho(w)$  are real, in fact rational; (b)  $\varrho$  preserves an alternating pairing up to the scalar  $c^{2\ell(w)}$ , so eigenvalues of  $\varrho(P)$  come in pairs  $\alpha_i$  and  $c^{2\ell(P)}/\alpha_i$ ; and (c) the holonomies generate a group as large as (b) permits; then  $\varrho$  is pure with constant  $c$ .

The note argues in four steps, for a fixed prime cycle  $P$  and eigenvalue  $\alpha_1$  of  $\varrho(P)$ , the target being  $|\alpha_1| \leq c^{\ell(P)}$  with (b) supplying equality. *Power up*: replace  $\varrho$  by  $\varrho^{\otimes 2k}$ , so traces become  $(\text{Tr } \varrho(w))^{2k}$  and  $\alpha_1^{2k}$  is an eigenvalue at  $P$ . *Dominate*: by (a) those traces are nonnegative, so every Euler factor of  $L_{2k} := L(u, \varrho^{\otimes 2k})$  has nonnegative coefficients and constant term 1, the product dominates any single factor coefficientwise, and the factor at  $P$  has a pole at  $|u| = |\alpha_1|^{-2k/\ell(P)}$  which a dominated series cannot outrun. *The known pole*: suppose the only poles of  $L_{2k}$  were the trivial ones, at  $|u| = 1/(q c^{2k})$ ; then

$$|\alpha_1|^{-2k/\ell(P)} \geq 1/(q c^{2k}) \iff |\alpha_1| \leq c^{\ell(P)} q^{\ell(P)/2k}.$$

*Take roots*: as  $k \rightarrow \infty$  the loss  $q^{\ell(P)/2k}$  vanishes. The fixed loss is the single factor  $q$  in the trivial pole, paid once by the base curve however large  $k$  is.

A graph breaks the third step, and must, since the theorem is false for graphs. Take  $B$  a bouquet of two loops with  $\varrho(a) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\varrho(b) = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ : traces are integers, so (a); determinants are 1, so (b) with  $c = 1$ ; and the two generate a group as large as  $\text{SL}_2(\mathbb{Z})$  permits, so (c). Yet  $\varrho(a)$  has eigenvalue  $(3+\sqrt{5})/2 > 1$ , so the twist is not pure:  $L_{2k} = \det(1 - uT_{\varrho^{\otimes 2k}})^{-1}$  has a pole at  $|u| = ((3+\sqrt{5})/2)^{-2k}$ , far inside  $1/q$ , from a nontrivial eigenvalue of the twisted edge operator. For a graph every side-B eigenvalue is a pole, so nothing forces the poles to the trivial place. What makes the third step true for a curve is one hypothesis and one structural fact.

## 25.6 Big monodromy, and the abelian failure

The trivial poles of  $L_{2k}$  come from invariant vectors: a vector in the  $d^{2k}$ -dimensional coefficient space fixed by every holonomy. On such a vector the twist acts like no twist, so  $L_{2k}$  acquires a copy of the untwisted zeta with its pole at  $1/q$ , rescaled by the scalar the twist puts on the vector; if that scalar is known, so is the pole. The pairing (b) supplies invariants for free, by contracting the  $2k$  tensor slots in pairs, each contraction giving a pole at exactly  $1/(q c^{2k})$ . Hypothesis (c), *big monodromy*, says these are the only invariants: if the holonomies fill out the whole group preserving the pairing, invariant theory makes every invariant of  $\varrho^{\otimes 2k}$  a combination of pairwise contractions, so no other trivial pole appears.

*Observation 25.1* (The abelian obstruction is inside Deligne's proof). [sketched] Commuting holonomies produce uncontrolled invariants in tensor powers, exactly as commuting Kraus unitaries never give an expander: commuting  $\varrho(w)$  have common eigenvectors, and a common eigenvector with eigenvalue  $\chi(w)$  becomes invariant in  $\varrho^{\otimes N}$  as soon as  $\chi^N = 1$ , carrying an unknown  $\chi$ -dependent scalar, so the trivial poles sit where the argument cannot control them. Any Ramanujan lift of the prime dilations must break the commutativity by a quotient, as [61] does with the arithmetic lattice and as Deligne does with the shears of a generic slicing.

In Deligne's setting the commuting case is a coefficient system with abelian monodromy, which can have arbitrary pointwise weights; (c) excludes it. In this project it is the observation of the

founding session that commuting Kraus unitaries never give an expander, the adjoint action of an abelian group having as many invariants as it has joint eigenprojectors: the same obstruction on both sides of the dictionary. Geometrically, big monodromy is verified through shears. The holonomy around each bad slice is the identity plus a rank-one nilpotent along one vector, those vectors are carried to one another by the holonomy group, and a theorem of Kazhdan and Margulis says a group generated by one conjugacy class of shears and acting without invariant subspaces is automatically as large as the pairing allows. Big monodromy is the statement that the bad slices are all alike, which holds because they are generic.

## 25.7 The sign

For a graph, side B is the one matrix  $T_\varrho$  and every eigenvalue is a pole. For a curve with coefficients the trace formula has two blocks of opposite sign,

$$\sum_{w: \ell(w)=\ell} \text{Tr } \varrho(w) = \text{Tr}(\text{Fr}^\ell \mid \text{invariant}) - \text{Tr}(\text{Fr}^\ell \mid \text{interesting}), \quad L(u, \varrho) = \frac{\det(1 - u\text{Fr} \mid \text{interesting})}{\det(1 - u\text{Fr} \mid \text{invariant})}.$$

The invariant block is the one big monodromy computes, and it produces the poles; the interesting block produces zeros, and zeros do not obstruct convergence, so the radius of convergence of  $L_{2k}$  is set by the invariant block alone. That is the third step, and it is the minus sign of the table above: positivity of counts is available on both sides, but on the graph side it constrains the sum of everything, which is the Perron bound and no more, while on the curve side the interesting eigenvalues are subtracted, so positivity bounds them above by the trivial ones and even tensor powers squeeze the bound to equality. The two-loop bouquet is not a curve because its walk counts cannot be written with the interesting part subtracted.

## 25.8 From pointwise to global: slice, bound, multiply

*Slice.* Take a one-parameter family of hyperplane sections of a variety  $X$  of dimension  $n$ ; the parameter runs over a curve and all but finitely many slices are smooth of dimension  $n - 1$ . Up to pieces already understood, the middle block of  $X$  is the interesting block of that base curve with coefficients, the coefficient at a parameter value being the middle block of the slice there. In graph terms this is the Artin–Ihara factorisation (Definition 3.10) used in Section 26, with the fibre replaced by a variety of one dimension less. The coefficients satisfy (a), (b), (c): the pairing is Poincaré duality on the slice, big monodromy is the shear argument, and the traces are rational because the zeta of a slice is rational with rational coefficients, off which the blocks known by induction are peeled. So they are pure with  $c = q^{(n-1)/2}$ .

*Bound.* With pure coefficients the Euler product over the base converges absolutely for  $|u| < 1/(qc)$ , there being about  $q^\ell/\ell$  prime cycles of length  $\ell$  each with holonomy of norm  $c^\ell$ . So the interesting block has no eigenvalue larger than  $qc = q^{(n+1)/2}$ , where RH wants  $q^{n/2}$ : a loss of one factor  $\sqrt{q}$ , paid by the base being a curve, independent of  $n$ .

*Multiply.* The product  $X^k$  has dimension  $nk$  and its middle block contains the  $k$ -fold tensor products of the middle block of  $X$ , so the products  $\alpha_1 \cdots \alpha_k$  are among its eigenvalues. The bound applied to  $X^k$  still loses one factor  $\sqrt{q}$ , because the base of a slicing is always a curve, whence

$$|\alpha_i|^k \leq q^{(nk+1)/2} \implies |\alpha_i| \leq q^{n/2} q^{1/2k} \xrightarrow{k \rightarrow \infty} q^{n/2},$$

with duality giving the reverse inequality: the engine a second time, count, fixed loss, roots. The pointwise theorem is what makes the last move legitimate, since the slices of  $X^k$  have dimension

$nk - 1$ , far above anything the induction on dimension knows, so their purity must come from rationality and big monodromy alone. That is why the core theorem is stated for coefficients on a curve: it certifies coefficients in any dimension at the price of one curve. Even for a curve  $C$  the argument passes through  $C^k$  for all  $k$ ; the surface  $C \times C$  is where Weil’s proof lived, and the products are the mechanism. The blocks  $i \neq n$  follow by induction from lower-dimensional sections and from duality.

## 25.9 What the proof does not do

It never exhibits a Hermitian structure. Weil’s proof for curves does: by the Hodge index theorem the intersection form on  $C \times C$  is definite on the relevant part, which makes  $\text{Fr}/\sqrt{q}$  unitary for an explicit positive inner product on the middle block and produces a Hilbert–Pólya operator (Definition 4.7). In graph terms, on the nontrivial part  $T$  is  $\sqrt{q}$  times a unitary in some inner product, which for the block  $\begin{pmatrix} \lambda & -q \\ 1 & 0 \end{pmatrix}$  holds exactly when  $|\lambda| < 2\sqrt{q}$ . Grothendieck’s plan was to prove a definite form in all dimensions, the standard conjectures, and deduce RH as Weil had; that plan is still open. Deligne bypassed it: no form, no operator, only positivity and a limit. RH over finite fields is therefore a theorem, while in dimension above one the inner product that would make Frobenius normal is not known to exist.

The Artin–Schreier computation of Section 18 sits on the Weil side of this divide:  $EE^\dagger = qI$  is manifest from Parseval for the additive characters (Proposition 18.3), an explicit inner product. Deligne’s proof is what one has when no such structure is in sight. Two honesty notes the note records: the Ramanujan property of the LPS graphs of Section 17 needs only the curve case, since Eichler–Shimura transports Hecke eigenvalues to Frobenius eigenvalues of a modular curve, so Weil’s theorem suffices there and Deligne’s method is what extends it to higher weight; and the technology behind “side B exists” is the largest single piece of the proof by page count, though not by idea.

## 25.10 What this says for the project

*Observation 25.2* (The three ingredients, and no Hermitian form). [sketched] Deligne’s proof uses three ingredients that a Hilbert–Pólya approach would have to replace: products, under which eigenvalues multiply while the trivial loss stays fixed; slicing over a curve, which keeps that loss at one factor  $\sqrt{q}$  in every dimension; and the sign, which makes the interesting eigenvalues zeros rather than poles, so that positivity of the counts bounds them. It exhibits no Hermitian form and no operator.

- **Which ingredients  $\zeta$  has.** The zeros are subtracted from the prime count, so  $\zeta$  is on the curve side of the table, not the graph side, and the positivity  $\Lambda(n) \geq 0$  is the right kind. The sign and the positivity are present; the product is absent, since there is no  $X \times X$  with  $\zeta$  as its slice, so the multiply step has nothing to act on. This is the standard diagnosis, stated in the terms of this section.
- **The number-field shadow of the pointwise theorem is Rankin–Selberg.** For a modular form the Euler product of  $L(f \times \bar{f})$  has nonnegative coefficients and a known pole, and dominating one local factor gives  $|a_p| \leq p^{(k-1)/2} \cdot p^{1/2}$ , the same fixed loss; each symmetric-power  $L$ -function whose poles are known shaves it (Kim–Sarnak’s 7/64 is the current state). Knowing the poles of all the tensor-power  $L$ -functions is the analogue of big monodromy; over number fields that is Langlands functoriality, unproved.
- **The abelian obstruction is inside the proof** (observation 25.1), on both sides of the dictionary.

- **Ramanujan without Hermiticity is possible.** Deligne’s route and the interlacing route both reach the spectral bound with no Hilbert–Pólya operator. For the steering question of where Hermiticity enters, this is the cleanest evidence that it need not enter at all; what must enter is a positivity plus an operation under which eigenvalues multiply at fixed loss.

## 26 The Ihara–Bass Formula for an Arbitrary Kraus Family

This is the load-bearing section of the book. The classical Ihara–Bass formula (Definition 3.8) compresses a determinant on the edge space of a regular graph to one on the vertex space, and Definition 3.18 lifts the edge operator to superoperators. The question the founding session left open, and which TJO steered towards, is whether the compression survives when the Kraus operators are not unitary. It does, at the price of two  $u$ -dependent operators in place of the adjacency matrix and the degree.

*Status.* Theorem 26.1 and its three corollaries are tagged **proved**. The author is **claudé:fable-5.1**; the reviewers are of a different model, **claudé:opus**; and their reports in **notes/reviews/** — a round-1 algebra report, a round-1 provenance report, and a round-2 re-verification — return **VALID** on every verdict. Only one model family was declared for this campaign, so **proved** here means exactly that: author  $\neq$  reviewer, inside one family. The two **numerical** rows are independent evidence.

### 26.1 Setting and conventions

- $\mathcal{V}$  is a finite-dimensional complex vector space,  $N = \dim \mathcal{V}$ . In the application  $\mathcal{V} = M_n$  with the Hilbert–Schmidt inner product, so  $N = n^2$ .
- $D = 2m$  is even, and  $\iota$  is a fixed-point-free involution on  $[D] = \{1, \dots, D\}$ , the *reversal*; we write  $\bar{i} = \iota(i)$ , so that  $\bar{\bar{i}} = i$ .
- $\mathcal{E}_1, \dots, \mathcal{E}_D \in \text{End}(\mathcal{V})$  are arbitrary linear maps. No invertibility, positivity or unitarity is assumed.
- The edge space is  $\mathcal{W} = \mathcal{V} \otimes \mathbb{C}^D$ , with  $\mathbb{C}^D$  spanned by  $|1\rangle, \dots, |D\rangle$ ; an element is  $x = \sum_i x_i \otimes |i\rangle$  with  $x_i \in \mathcal{V}$  its *component* at  $i$ .
- $T \in \text{End}(\mathcal{W})$  is the quantum Hashimoto operator of Definition 3.18,  $T(v \otimes |i\rangle) = \sum_{j \neq \bar{i}} \mathcal{E}_i v \otimes |j\rangle$ , and  $\zeta_T(u) = \det_{\mathcal{W}}(1 - uT)^{-1}$ , a rational function of  $u \in \mathbb{C}$ .
- $\|\cdot\|$  is the operator norm; any submultiplicative norm on  $\text{End}(\mathcal{V})$  would do. Determinants carry the space they are taken over,  $\det_{\mathcal{V}}$  and  $\det_{\mathcal{W}}$ .
- Sylvester’s identity  $\det(1 - XY) = \det(1 - YX)$ , for  $X: \mathcal{V} \rightarrow \mathcal{W}$  and  $Y: \mathcal{W} \rightarrow \mathcal{V}$ , is used without further comment; so is the block determinant with identity diagonal blocks,  $\det \begin{pmatrix} 1 & B \\ C & 1 \end{pmatrix} = \det(1 - CB) = \det(1 - BC)$  (Schur complement, then Sylvester).

**The Kraus case.**  $\mathcal{V} = M_n$ ;  $A_1, \dots, A_m \in M_n$  are arbitrary,  $A_{m+k} = A_k^\dagger$ ,  $\iota(k) = m + k$ , and  $\mathcal{E}_i = \text{Ad}(A_i): X \mapsto A_i X A_i^\dagger$ . With column stacking (Section 2),  $\text{Ad}(A) = \bar{A} \otimes A$  as an  $n^2 \times n^2$  matrix, so  $\text{Tr}_{\mathcal{V}} \text{Ad}(A) = |\text{Tr } A|^2$  and  $\text{Ad}(A) \text{Ad}(B) = \text{Ad}(AB)$ .

**Auxiliary maps.** Three maps do all the work: **R** propagates along an edge, **S** starts along every edge, **J** is the backtracking term.

$$\mathbf{R}: \mathcal{W} \rightarrow \mathcal{V}, \quad \mathbf{R}(v \otimes |i\rangle) = \mathcal{E}_i v; \quad \mathbf{S}: \mathcal{V} \rightarrow \mathcal{W}, \quad \mathbf{S}v = \sum_{j=1}^D v \otimes |j\rangle; \quad \mathbf{J}(v \otimes |i\rangle) = \mathcal{E}_i v \otimes |\bar{i}\rangle.$$

The two  $u$ -dependent operators on  $\mathcal{V}$  are the *deformed adjacency* and the *deformed degree*,

$$\mathcal{A}(u) = u \sum_{i=1}^D \mathcal{E}_i (1 - u^2 \mathcal{E}_i \mathcal{E}_i)^{-1}, \quad \mathcal{D}(u) = u^2 \sum_{i=1}^D \mathcal{E}_i \mathcal{E}_i (1 - u^2 \mathcal{E}_i \mathcal{E}_i)^{-1}.$$

The zeta variable is  $u$  throughout (Section 2); the channel normalisation  $\Sigma = \sum_i \mathcal{E}_i = D\Phi$  is the one fixed there, and nothing below depends on it.

## 26.2 Statement

**Theorem 26.1** (Ihara–Bass for an arbitrary family of superoperators). *[proved] Let  $u \in \mathbb{C}$  be such that  $1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i$  is invertible for every  $i \in [D]$ . Then*

$$\det_{\mathcal{W}}(1 - uT) = \prod_{\{i, \bar{i}\}} \det_{\mathcal{V}}(1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i) \cdot \det_{\mathcal{V}}(1 + \mathcal{D}(u) - \mathcal{A}(u)),$$

the product being over the  $m$  unordered pairs  $\{i, \bar{i}\}$ , which is well defined because  $\det(1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i) = \det(1 - u^2 \mathcal{E}_i \mathcal{E}_{\bar{i}})$ . The hypothesis holds for all  $|u| < (\max_i \|\mathcal{E}_{\bar{i}} \mathcal{E}_i\|)^{-1/2}$ , and for all  $u$  if every  $\mathcal{E}_{\bar{i}} \mathcal{E}_i = 0$ ; the identity holds pointwise at every such  $u$ , and since the left side is a polynomial of degree  $\leq ND$  and the right side is rational, the two sides agree as rational functions.

**Corollary 26.2** (Kraus families and positivity of side A). *[proved] For arbitrary  $A_1, \dots, A_m \in M_n$  with the adjoint pairing above,  $\det_{\mathcal{W}}(1 - uT) = \prod_{k=1}^m \det(1 - u^2 \text{Ad}(A_k^\dagger A_k)) \cdot \det(1 + \mathcal{D}(u) - \mathcal{A}(u))$ , valid for  $|u| < \min_k \|A_k\|^{-2}$ ; here  $\mathcal{E}_{\bar{i}} \mathcal{E}_i$  is  $\text{Ad}(A_k^\dagger A_k)$  for  $i = k \leq m$  and  $\text{Ad}(A_k A_k^\dagger)$  for  $i = m + k$ , both positive semidefinite. Moreover, for every  $\ell \geq 1$ ,*

$$\text{Tr}_{\mathcal{W}} T^\ell = \sum_{\substack{(i_1, \dots, i_\ell) \in [D]^\ell \\ i_{k+1} \neq i_k \text{ (} k \in \mathbb{Z}/\ell \text{)}}} |\text{Tr}(A_{i_\ell} \cdots A_{i_1})|^2 \geq 0,$$

so  $\zeta_T(u) = \exp(\sum_\ell u^\ell \text{Tr} T^\ell / \ell)$  has nonnegative Taylor coefficients, and  $\zeta_T(u) = \prod_P \det_{\mathcal{V}}(1 - u^{\ell(P)} \text{Ad}(A_P))^{-1}$  over primitive cyclically non-backtracking classes  $P$  of length  $\ell(P)$ , with  $A_P$  the ordered product along  $P$ .

**Corollary 26.3** (Inverse pairing: the unitary case). *[proved] If  $\mathcal{E}_{\bar{i}} = \mathcal{E}_i^{-1}$  for all  $i$  — in particular for unitary Kraus operators with  $U_{\bar{i}} = U_i^\dagger = U_i^{-1}$  — then, with  $\Sigma = \sum_i \mathcal{E}_i$ ,*

$$\det_{\mathcal{W}}(1 - uT) = (1 - u^2)^{N(D-2)/2} \det_{\mathcal{V}}(1 - u\Sigma + (D-1)u^2).$$

For  $\mathcal{V} = M_n$  and  $\Sigma = D\Phi$  this is the formula recorded as item 1 of `HANDOFF.md` and, on the bouquet, equation (2.11) of Matsuura–Ohta [65].

**Corollary 26.4** (Which factor carries the spectrum). *[proved] The poles of  $\zeta_T$  lie among the roots of  $\det_{\mathcal{V}}(1 + \mathcal{D}(u) - \mathcal{A}(u)) = 0$  together with the points  $u^2 = 1/\lambda$  for  $\lambda \in \text{spec}(\mathcal{E}_{\bar{i}} \mathcal{E}_i) \setminus \{0\}$ ; the latter may cancel against poles of the second factor, as they do in Corollary 26.3 (exponent  $ND/2$  against  $-N$ , a total cancellation when  $D = 2$ ). For  $D = 2$  in general,  $T = \mathcal{E}_1 \oplus \mathcal{E}_2$  on  $\mathcal{W} = \mathcal{V} \oplus \mathcal{V}$ , since the only non-backtracking successor of  $i$  is  $i$  itself, so  $\det_{\mathcal{W}}(1 - uT) = \det_{\mathcal{V}}(1 - u\mathcal{E}_1) \det_{\mathcal{V}}(1 - u\mathcal{E}_2)$ . Conversely a pole of the second factor, which can only sit where some  $1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i$  is singular, is not a pole of  $\zeta_T$  unless it survives multiplication by the first.*

## 26.3 Proof of the theorem

The proof is written in Lamport structure: each step is a claim followed by its own proof, and the last step assembles them.

*Proof of Theorem 26.1.* ASSUME:  $u \in \mathbb{C}$  with  $1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i$  invertible for all  $i \in [D]$ . PROVE: the displayed identity.

- (1)1. **The decomposition  $T = \text{SR} - \text{J}$ .** *Proof.* On  $v \otimes |i\rangle$ ,  $\text{SR}(v \otimes |i\rangle) = \sum_j \mathcal{E}_i v \otimes |j\rangle$  and  $\text{J}(v \otimes |i\rangle) = \mathcal{E}_i v \otimes |\bar{i}\rangle$ , so  $(\text{SR} - \text{J})(v \otimes |i\rangle) = \sum_{j \neq \bar{i}} \mathcal{E}_i v \otimes |j\rangle = T(v \otimes |i\rangle)$ . Both sides are linear and these vectors span  $\mathcal{W}$ .  $\square$
- (1)2. **Block structure of  $\text{J}$ .** For each pair  $p = \{i, \bar{i}\}$  put  $\mathcal{W}_p = \mathcal{V} \otimes \text{span}\{|i\rangle, |\bar{i}\rangle\}$ . Then  $\mathcal{W} = \bigoplus_p \mathcal{W}_p$ ,  $\text{J}\mathcal{W}_p \subseteq \mathcal{W}_p$ , and in the ordered decomposition  $\mathcal{W}_p = \mathcal{V} \oplus \mathcal{V}$  (component at  $i$ , component at  $\bar{i}$ ),  $\text{J}|_{\mathcal{W}_p} = \begin{pmatrix} 0 & \mathcal{E}_{\bar{i}} \\ \mathcal{E}_i & 0 \end{pmatrix}$ . *Proof.*  $\text{J}$  sends the component at  $i$  to the component at  $\bar{i}$  through  $\mathcal{E}_i$ , and the component at  $\bar{i}$  to the component at  $i$  through  $\mathcal{E}_{\bar{i}}$ , using  $\bar{\bar{i}} = i$ . The direct sum is the partition of  $[D]$  into  $\iota$ -orbits, which have size two because  $\iota$  is fixed-point-free.  $\square$
- (1)3.  **$1+u\text{J}$  is invertible, with  $\det_{\mathcal{W}}(1+u\text{J}) = \prod_p \det_{\mathcal{V}}(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)$ .** *Proof.* By (1)2,  $\det_{\mathcal{W}}(1+u\text{J}) = \prod_p \det \begin{pmatrix} 1 & u\mathcal{E}_{\bar{i}} \\ u\mathcal{E}_i & 1 \end{pmatrix} = \prod_p \det_{\mathcal{V}}(1-u^2\mathcal{E}_i\mathcal{E}_{\bar{i}}) = \prod_p \det_{\mathcal{V}}(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)$ , by the block determinant and then Sylvester. Each factor is nonzero by ASSUME.  $\square$
- (1)4. **Explicit inverse on a pair.** Write  $B = u\mathcal{E}_{\bar{i}}$ ,  $C = u\mathcal{E}_i$ ,  $P = (1-BC)^{-1} = (1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1}$  and  $Q = (1-CB)^{-1} = (1-u^2\mathcal{E}_i\mathcal{E}_{\bar{i}})^{-1}$ , both existing by ASSUME (the second is the hypothesis at the index  $\bar{i}$ ). Then  $(1+u\text{J})^{-1}|_{\mathcal{W}_p} = \begin{pmatrix} P & -BQ \\ -CP & Q \end{pmatrix}$ . *Proof.* Multiply out:  $\begin{pmatrix} 1 & B \\ C & 1 \end{pmatrix} \begin{pmatrix} P & -BQ \\ -CP & Q \end{pmatrix} = \begin{pmatrix} (1-BC)P & -BQ+BPQ \\ CP-CP & (1-CB)Q \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ . A right inverse of a square operator on a finite-dimensional space is the inverse.  $\square$
- (1)5. **Sylvester step.**  $\det_{\mathcal{W}}(1-uT) = \det_{\mathcal{W}}(1+u\text{J}) \cdot \det_{\mathcal{V}}(1-u\text{R}(1+u\text{J})^{-1}\text{S})$ . *Proof.* By (1)1,  $1-uT = 1+u\text{J} - u\text{SR} = (1+u\text{J})(1-u(1+u\text{J})^{-1}\text{SR})$ , the inverse existing by (1)3. Take determinants and apply Sylvester with  $X = u(1+u\text{J})^{-1}\text{S}: \mathcal{V} \rightarrow \mathcal{W}$  and  $Y = \text{R}: \mathcal{W} \rightarrow \mathcal{V}$ .  $\square$
- (1)6. **Computation of  $M(u) := \text{R}(1+u\text{J})^{-1}\text{S}$ .** It equals  $\sum_{i=1}^D (\mathcal{E}_i - u\mathcal{E}_{\bar{i}}\mathcal{E}_i)(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1}$ . *Proof.* For  $v \in \mathcal{V}$ ,  $\text{S}v$  has component  $v$  at every index. By (1)4, on the pair  $p = \{i, \bar{i}\}$  the vector  $(1+u\text{J})^{-1}\text{S}v$  has component  $(P-BQ)v$  at  $i$  and  $(-CP+Q)v$  at  $\bar{i}$ . Applying  $\text{R}$ , which applies  $\mathcal{E}_i$  to the component at  $i$ ,  $\mathcal{E}_{\bar{i}}$  to the component at  $\bar{i}$ , and sums over all indices, and substituting  $B, C, P, Q$ , the pair  $p$  contributes

$$\mathcal{E}_i P - u\mathcal{E}_i \mathcal{E}_{\bar{i}} Q + \mathcal{E}_{\bar{i}} Q - u\mathcal{E}_{\bar{i}} \mathcal{E}_i P = (\mathcal{E}_i - u\mathcal{E}_{\bar{i}}\mathcal{E}_i)P + (\mathcal{E}_{\bar{i}} - u\mathcal{E}_i\mathcal{E}_{\bar{i}})Q.$$

The first summand is the  $i$ -th term of the claimed sum and the second is the  $\bar{i}$ -th, since  $Q = (1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1}$ . Summing over the  $m$  pairs gives the sum over all  $D$  indices.  $\square$

- (1)7.  **$1-uM(u) = 1+\mathcal{D}(u)-\mathcal{A}(u)$ .** *Proof.* Expand (1)6 linearly:  $uM(u) = u \sum_i \mathcal{E}_i(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1} - u^2 \sum_i \mathcal{E}_{\bar{i}}\mathcal{E}_i(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1} = \mathcal{A}(u) - \mathcal{D}(u)$ .  $\square$
- (1)8. **QED.** *Proof.* Combine (1)5, (1)3 and (1)7. Steps (1)1–(1)7 hold pointwise at every  $u$  satisfying ASSUME, with no analyticity used. The bound  $|u| < (\max_i \|\mathcal{E}_{\bar{i}}\mathcal{E}_i\|)^{-1/2}$  makes  $\|u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i\| < 1$ , so the Neumann series supplies the inverses and ASSUME holds there. The set of admissible  $u$  is cofinite: its complement is the zero set of the polynomial  $\prod_i \det_{\mathcal{V}}(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)$ , which equals 1 at  $u = 0$  and so is not identically zero. Two rational functions agreeing on a cofinite set are equal.  $\square$

$\square$

## 26.4 Proofs of the corollaries

*Proof of Corollary 26.2.*  $\mathcal{E}_i \mathcal{E}_{\bar{i}} = \text{Ad}(A_{\bar{i}}) \text{Ad}(A_i) = \text{Ad}(A_{\bar{i}} A_i)$ , which is  $\text{Ad}(A_k^\dagger A_k)$  or  $\text{Ad}(A_k A_k^\dagger)$ ; for  $P \geq 0$  with eigenvalues  $p_a$ ,  $\text{Ad}(P) = \bar{P} \otimes P$  has spectrum  $\{\bar{p}_a p_b\} \subseteq [0, \infty)$ , and  $\|\text{Ad}(A_k^\dagger A_k)\| = \|A_k\|^4$ , so the hypothesis of Theorem 26.1 holds for  $|u| < \min_k \|A_k\|^{-2}$ ; the two orderings of a pair have equal determinant by Sylvester, leaving one factor per  $k$ . For the trace formula, induction on  $\ell$  gives  $T^\ell(v \otimes |i_1\rangle) = \sum \mathcal{E}_{i_\ell} \cdots \mathcal{E}_{i_1} v \otimes |i_{\ell+1}\rangle$ , summed over  $(i_2, \dots, i_{\ell+1})$  with  $i_{k+1} \neq \bar{i}_k$  for  $k = 1, \dots, \ell$ . Tracing over  $\mathcal{W} = \mathcal{V} \otimes \mathbb{C}^D$  sets  $i_{\ell+1} = i_1$ , which adds the cyclic condition  $i_1 \neq \bar{i}_\ell$ , and leaves  $\text{Tr}_{\mathcal{V}} \text{Ad}(A_{i_\ell} \cdots A_{i_1}) = |\text{Tr}(A_{i_\ell} \cdots A_{i_1})|^2$ . Nonnegativity of the coefficients of  $\log \zeta_T = \sum_\ell u^\ell \text{Tr } T^\ell / \ell$  implies nonnegativity for  $\zeta_T = \exp(\cdot)$ . The Euler product is the standard regrouping of cyclic words into powers of primitive classes, using  $\sum_{k \geq 1} \frac{u^{k\ell(P)}}{k} \text{Tr } \text{Ad}(A_P)^k = -\log \det(1 - u^{\ell(P)} \text{Ad}(A_P))$ , valid for  $|u|$  small.  $\square$

*Proof of Corollary 26.3.* With  $\mathcal{E}_{\bar{i}} \mathcal{E}_i = 1$  each pair factor is  $\det_{\mathcal{V}}(1 - u^2) = (1 - u^2)^N$ , so the product is  $(1 - u^2)^{ND/2}$ ; and  $\mathcal{A}(u) = \frac{u}{1-u^2} \Sigma$ ,  $\mathcal{D}(u) = \frac{Du^2}{1-u^2}$ . Hence  $1 + \mathcal{D}(u) - \mathcal{A}(u) = \frac{1}{1-u^2} (1 - u\Sigma + (D-1)u^2)$ , whose determinant is  $(1 - u^2)^{-N} \det_{\mathcal{V}}(1 - u\Sigma + (D-1)u^2)$ . The total exponent is  $ND/2 - N = N(D-2)/2$ . The computation divides by  $1 - u^2$  and so is carried out for  $u^2 \neq 1$ ; both sides are polynomials in  $u$ , so the identity extends to  $u^2 = 1$ .  $\square$

*Proof of Corollary 26.4.* Immediate from the factorisation: a zero of the polynomial  $\det_{\mathcal{W}}(1 - uT)$  is a zero of the product, each factor is analytic away from the points named, and a polynomial has no poles, so every pole of the second factor is cancelled by a zero of the first.  $\square$

## 26.5 Provenance and relation to the literature

Watanabe and Fukumizu [85] prove an Ihara–Bass formula for graph zeta functions with arbitrary matrix weights, in the shape  $\det(I + \hat{\mathcal{D}} - \hat{\mathcal{A}}) \prod_{[e]} \det(I - u_e u_{\bar{e}})$ , with  $\hat{\mathcal{D}}$  and  $\hat{\mathcal{A}}$  built from  $(I - u_e u_{\bar{e}})^{-1}$  exactly as  $\mathcal{D}(u)$  and  $\mathcal{A}(u)$  are; the byte-verified quotes are in Section 32. On a one-vertex graph whose  $2m$  directed edges are  $[D]$  with reversal  $\iota$ , and with  $u_e = u \mathcal{E}_e$ , their right-hand side becomes the right-hand side of Theorem 26.1 after the push-through identity  $\mathcal{E}_i (1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i)^{-1} = (1 - u^2 \mathcal{E}_i \mathcal{E}_{\bar{i}})^{-1} \mathcal{E}_i$  and the relabelling  $i \leftrightarrow \bar{i}$  in the degree term. Two further conventions differ and wash out of  $\det(1 - u \cdot)$ : they compose a prime cycle in the order opposite to our  $\mathcal{E}_{i_\ell} \cdots \mathcal{E}_{i_1}$ , and their edge operator weights the target edge where  $T$  weights the source. Reversal of words is a bijection of cyclically non-backtracking words because  $\iota$  is an involution, and  $\det(1 - XY) = \det(1 - YX)$ .

The framing matters, and was corrected by the round-1 provenance review: their corollary is stated for loopless graphs (their graphs are hypergraphs with two-element hyperedges), so a bouquet is not one of their graphs and their corollary does not apply to it. Theorem 26.1 is therefore the loop-admitting bouquet analogue of their arbitrary-weight corollary, not an instance of it; their proof goes through a hypergraph theorem, the proof above is direct. The shape-matching itself is checked numerically only (`notes/reviews/scratch_wf_mo_conventions.py`).

Matsuura and Ohta [65] treat invertible edge weights on a general graph, which is Corollary 26.3 in that generality; on a bouquet with  $X_e = U_e \otimes U_e^\dagger$  their equation (2.11) is the formula of `HANDOFF.md` item 1. No local source covers arbitrary weights on a bouquet.

## 26.6 Numerical checks

*Numerical result 26.5* (Ihara–Bass for random unitary Kraus sets). `[numerical]` For Haar-random unitary Kraus sets closed under adjoint, Corollary 26.3 holds at  $u \in \{0.13, 0.29 + 0.11i, -0.4\}$  with relative error at most  $2.1 \cdot 10^{-15}$  for  $(n, D) = (3, 4)$  and at most  $3.9 \cdot 10^{-15}$  for  $(n, D) = (4, 6)$ . The four

largest eigenvalues of  $\Sigma = D\Phi$  are 1.465, 1.850, 3.622, 4.000 for (3, 4) against the Ramanujan bound  $2\sqrt{q} = 3.464$ , and 2.782, 3.128, 3.594, 6.000 for (4, 6) against 4.472: the first instance violates the bound of Definition 3.17 and the second satisfies it. (`scripts/qihara.py`, `outputs/qihara.txt`.)

*Numerical result 26.6* (The general formula in five configurations). [numerical]  
Theorem 26.1, in both the  $1 - uM(u)$  and the displayed  $1 + \mathcal{D}(u) - \mathcal{A}(u)$  form, at three values of  $u$  (one complex), with maximal relative error per configuration as in the table; the trace formula of Corollary 26.2 for  $\ell \leq 4$  to  $5.5 \cdot 10^{-12}$ ; and exact verification in rational arithmetic (sympy) for  $(N, D) = (1, 4), (2, 4), (2, 6)$ . (`scripts/qihara_general.py`, `outputs/qihara_general.txt`.)

| configuration                                      | parameters     | max. relative error  |
|----------------------------------------------------|----------------|----------------------|
| Gaussian Kraus operators, adjoint pairing          | $n = 3, D = 4$ | $2.1 \cdot 10^{-15}$ |
| canonical-form MPS tensors and adjoints            | $n = 3, D = 4$ | $2.0 \cdot 10^{-15}$ |
| generic real superoperators, involution (1 2)(3 4) | $n = 3, D = 4$ | $6.7 \cdot 10^{-16}$ |
| single Kraus pair, with the direct-sum check       | $n = 3, D = 2$ | $1.4 \cdot 10^{-15}$ |
| unitary Kraus, against Corollary 26.3              | $n = 3, D = 4$ | $1.3 \cdot 10^{-15}$ |

## 26.7 What this does and does not give

It gives the compression for *every* matrix product state transfer matrix  $\Sigma = \sum_i \text{Ad}(A_i)$ , with the physical index doubled by the adjoints. The price of dropping unitarity is that the degree term  $(D - 1)u^2$  becomes the operator  $\mathcal{D}(u)$  and the adjacency term  $u\Sigma$  becomes the deformed channel  $\mathcal{A}(u)$ , both depending on  $u$  through  $(1 - u^2 \text{Ad}(A_k^\dagger A_k))^{-1}$ .

It does not give a Ramanujan reading by itself. Without  $\mathcal{E}_i \mathcal{E}_i = 1$  there is no quadratic relation  $\mu\mu' = q$  pairing the eigenvalues of  $T$  with those of  $\Sigma$ . What replaces it, and whether canonical form (Definition 3.14) buys anything, is conjecture 35.1.

Side A is unconditionally positive:  $\text{Tr } T^\ell = \sum_P |\text{Tr } A_P|^2 \geq 0$  for any Kraus family, so the primes of this zeta are the primitive cyclically non-backtracking words in the  $A_k$  and  $A_k^\dagger$ , with real nonnegative multiplicities.

*Observation 26.7* (RH for the quantum Ihara zeta is Hastings' bound). [sketched] For unitary Kraus operators closed under adjoint, Corollary 26.3 attaches to each eigenvalue  $\lambda$  of  $\Sigma = D\Phi$  the quadratic  $1 - \lambda u + qu^2 = (1 - \mu u)(1 - \mu' u)$  with  $\mu\mu' = q = D - 1$  and  $\mu + \mu' = \lambda$ ; the eigenvalues  $\lambda$  are real because  $\Sigma$  is Hilbert–Schmidt self-adjoint for an adjoint-closed set. Both roots then have  $|\mu| = \sqrt{q}$  exactly when  $|\lambda| \leq 2\sqrt{q}$ . Hence every nontrivial pole of  $\zeta_\Phi$  lies on the circle  $|u| = q^{-1/2}$  if and only if  $D\lambda_2 \leq 2\sqrt{D - 1}$ , which is Hastings' bound  $\lambda_H$  [36] in the adjacency units of Section 2: the Riemann hypothesis for this zeta is exactly the statement that  $\Phi$  is a Ramanujan quantum expander (Definition 3.17). The note records this as an observation, not found in the literature in this form; it is tagged **sketched** because it has had no reviewer.

## 27 Weil Positivity for an Arbitrary Transfer Operator

This section answers the question TJO put on 2026-09-12: the reflection  $\rho \mapsto 1 - \bar{\rho}$  pairs the mode of the Riemann channel at a zero with the mode at its mirror image, RH says every mode is its own partner, and Weil's criterion is the positivity of that pairing written on the primes. What does this mean for a completely arbitrary Kraus family, finite-dimensional or continuous? The answer is in three layers. Without any duality, Weil positivity of the rescaled ring norms is exactly a one-sided bound, the spectral-gap half of RH. A duality of the spectrum, the functional equation, turns the bound into a circle and the Weil form into the pairing of each mode with its reflected partner; this identity is the finite form of the inflow identity. For Kraus families the two pairings of the Kraus index buy the two halves separately: inverse pairing gives the functional equation, adjoint pairing gives reality of the transfer spectrum, and only unitarity gives both.

*Status.* The theorems are the prover's, `codex:gpt-6-astra` (notes/weil-positivity/astra-proofs.md, Lamport structure, correction ledger in its T7), reviewed adversarially by `claude:opus` (notes/reviews/weil-positivity/astra-proofs.md), two model families; all verdicts **VALID**, two wording items fixed. The orchestrator's two observations in Section 28 were reviewed in the same file. Independent numerics by a third agent (numerical result 28.17) found the same three drafting errors the prover corrected.

### 27.1 Setting

Definitions 4.24 and 4.25 fix the objects: an operator  $X$  on a finite-dimensional space, a trivial sub-multiset  $S_{\text{triv}}$  of its spectrum, the retained multiset  $\mathcal{A}_{\text{ret}}$ , a critical radius  $r$ , the rescaled trace sequence  $\mathbf{n}_\ell = \sum_{\mu \in \mathcal{A}_{\text{ret}}} (\mu/r)^\ell$  with  $\mathbf{n}_{-\ell} = \bar{\mathbf{n}}_\ell$ , the Weil form  $W(c) = \sum c_\ell \bar{c}_k \mathbf{n}_{\ell-k}$ , the reflection  $J(\mu) = r^2/\bar{\mu}$  and the mode-pairing form  $M$ . In the Kraus case  $X = T$  and  $\text{Tr } T^\ell = \sum_w |\text{Tr } A_w|^2$  is the ring-norm sum of Corollary 26.2, so  $\mathbf{n}_\ell$  is side A rescaled and with the trivial modes subtracted, and  $W$  is a form on rings. Nothing below assumes diagonalisability until Proposition 28.4.

### 27.2 Positivity is the one-sided bound

**Theorem 27.1** (Finite Weil positivity). *[proved] For any  $X$ , any  $S_{\text{triv}}$  and any  $r > 0$ , the sequence  $\mathbf{n}_\ell$  is positive definite if and only if  $|\mu| \leq r$  for every  $\mu \in \mathcal{A}_{\text{ret}}$ , algebraic multiplicities included and zero eigenvalues allowed.*

The proof is the Poisson kernel and a growth lemma. A retained  $z = \mu/r$  with  $|z| < 1$  contributes to  $W(c)$  the integral of  $|f_c(e^{i\vartheta})|^2$  against the Poisson kernel  $(1 - |z|^2)/|1 - ze^{-i\vartheta}|^2 > 0$ , a point  $|z| = 1$  contributes  $|f_c(z)|^2$ , and  $z = 0$  contributes  $\sum |c_\ell|^2$ ; so the bound gives positivity. Conversely a positive definite sequence has  $|\mathbf{n}_\ell| \leq \mathbf{n}_0$  from its  $2 \times 2$  minors, while a retained mode outside the disc makes the sequence unbounded, because the modes of maximal modulus  $R > r$  contribute  $R^\ell$  times a trigonometric sum whose mean square over  $\ell < L$  tends to the sum of squared multiplicities. The point of the theorem is what it does not say: an eigenvalue inside the disc passes. Weil positivity alone is a spectral-gap statement, and the circle needs a second input.

### 27.3 A duality makes it two-sided: the inflow identity

**Theorem 27.2** (Reflection duality and the inflow identity). *[proved] If  $\mathcal{A}_{\text{ret}}$  is invariant under  $J$  (multiplicities preserved; in particular  $0 \notin \mathcal{A}_{\text{ret}}$ ), then  $W(c) = M(c)$  for every finitely supported  $c$ , and the following are equivalent:  $W \geq 0$ ;  $|\mu| = r$  for every  $\mu \in \mathcal{A}_{\text{ret}}$ ;  $J(\mu) = \mu$  for every  $\mu \in \mathcal{A}_{\text{ret}}$ .*

The identity  $W = M$  is the finite inflow identity: the left side is built from traces, i.e. from rings, the right side pairs each mode with its partner under the reflection. It holds whether or

not the modes are on the circle; what the circle adds is that every partner is the mode itself, so  $M(c) = \sum |f_c(\mu/r)|^2$ . The equivalence follows from Theorem 27.1:  $|\mu| \leq r$  and  $|J(\mu)| \leq r$  force  $|\mu| = r$ . This is exactly how RH follows from the one-sided bound  $\operatorname{Re} \rho \leq \frac{1}{2}$  together with the functional equation.

**Proposition 27.3** (Orbit contributions; an off-circle pair makes the form negative). [proved]  
*Under the hypotheses of Theorem 27.2, a J-fixed mode  $z$  of multiplicity  $m$  contributes  $m|f_c(z)|^2$  to  $W(c)$  and a two-element orbit  $\{z, Jz\}$  contributes  $2m \operatorname{Re}(f_c(z)\overline{f_c(Jz)})$ ; if any two-element orbit exists there is a  $c$  with  $W(c) < 0$ .*

This is Weil's off-line pair argument in finite form; the prover's proof interpolates  $f_c$  to vanish at every other retained value, which is what makes the single negative orbit dominate the whole form.

**Proposition 27.4** (Reciprocal symmetry without conjugation). [proved]  
*If  $\mathcal{A}_{\text{ret}}$  contains no zero and is invariant under the reciprocal map  $\mu \mapsto r^2/\mu$  alone, positivity is still equivalent to  $|\mu| = r$  throughout  $\mathcal{A}_{\text{ret}}$ . If  $\mathcal{A}_{\text{ret}}$  is also closed under complex conjugation it is J-invariant and  $W = M$ ; without conjugation symmetry  $W = M$  can fail. The reciprocal map is holomorphic on  $\mathbb{C}^\times$ , not linear.*

## 27.4 Kraus families: which pairing buys which half

**Proposition 27.5** (Conjugation symmetry needs no pairing). [proved] *For any family  $\mathcal{E}_i = \operatorname{Ad}(B_i)$  with arbitrary  $B_i \in M_n$  and any fixed-point-free reversal, the antiunitary involution  $x \mapsto x^\dagger$  on  $\mathcal{V} = M_n$ , and its componentwise version on  $\mathcal{W}$ , commute with  $\Sigma$  and with  $T$ . Hence  $\operatorname{spec}(\Sigma)$  and  $\operatorname{spec}(T)$  are closed under complex conjugation with multiplicities. The nonnegative ring trace  $\operatorname{Tr} T^\ell = \sum_w |\operatorname{Tr} B_w|^2$  of Corollary 26.2 likewise holds for every such family, with no pairing hypothesis: its proof uses only  $\operatorname{Ad}(A)\operatorname{Ad}(B) = \operatorname{Ad}(AB)$  and  $\operatorname{Tr} \operatorname{Ad}(A) = |\operatorname{Tr} A|^2$ .*

The draft sent to the prover asked for an inverse-paired counterexample to conjugation closure; none exists inside the Kraus class, and the independent numerics found the same. Likewise the trace formula  $\operatorname{Tr} T^\ell = \sum_w |\operatorname{Tr} B_w|^2$  of Corollary 26.2 never used the pairing: it is a property of the Ad form. So every Kraus-type family, however paired, has a positive side A and a real-structured side B.

**Theorem 27.6** (Inverse pairing gives the functional equation). [proved] *Let  $\mathcal{E}_i = \mathcal{E}_i^{-1}$  on an  $N$ -dimensional space,  $D = 2m \geq 4$ ,  $q = D - 1$ ,  $k = N(D - 2)/2$ . Then  $\operatorname{spec}(T)$  is the multiset  $\{+1^{[k]}, -1^{[k]}\}$  together with, for each eigenvalue  $\varkappa$  of  $\Sigma$  of multiplicity  $a_\varkappa$ , the two roots of  $\mu^2 - \varkappa\mu + q$  counted  $a_\varkappa$  times. The retained multiset after removing  $\{+1^{[k]}, -1^{[k]}\}$  has  $2N$  elements, no zero, and is invariant under  $\mu \mapsto q/\mu$ ; it is conjugation-invariant iff  $\operatorname{spec}(\Sigma)$  is. For inverse-paired Kraus matrices both hold, so the retained multiset is J-invariant for  $r = \sqrt{q}$ .*

The bookkeeping is where the draft was loose: a double root at  $\varkappa = \pm 2\sqrt{q}$  gives  $2a_\varkappa$  copies of  $\pm\sqrt{q}$ , and  $\varkappa = \pm D$  gives extra copies of  $\pm 1$  beyond the  $k$  from the  $(1 - u^2)$  factor.

**Proposition 27.7** (Both pairings at once is unitarity). [proved]  *$\operatorname{Ad}(B^\dagger) = \operatorname{Ad}(B)^{-1}$  iff  $B^\dagger B = 1$  iff  $B$  is unitary; a family is simultaneously adjoint-paired and inverse-paired iff every  $B_i$  is unitary. A scalar  $B^\dagger B = \kappa 1$  gives only  $\operatorname{Ad}(B^\dagger) = \kappa^2 \operatorname{Ad}(B)^{-1}$ ;  $B = 2$  disproves “unitary up to a scalar”.*

**Theorem 27.8** (The Weil criterion for Kraus families). [proved] *(i) For an adjoint-paired Kraus family, any  $S_{\text{triv}}$  and any  $r$ , Weil positivity of  $\mathfrak{n}_\ell(T; S_{\text{triv}}, r)$  is exactly the one-sided bound  $|\mu| \leq r$*

off  $S_{\text{triv}}$ ; nonnegativity of the ring traces alone does not give it. (ii) For unitary, adjoint-paired  $B_i$  with  $D \geq 4$ ,  $r = \sqrt{D-1}$ ,  $\Phi = \Sigma/D$ , and  $S_{\text{triv}} = \{+1^{[k]}, -1^{[k]}\}$  together with  $\{q, 1\}$  once for each eigenvalue  $+1$  of  $\Phi$  and  $\{-q, -1\}$  once for each eigenvalue  $-1$ : Weil positivity holds iff every other eigenvalue  $\lambda$  of  $\Phi$  satisfies  $|\lambda| \leq 2\sqrt{D-1}/D$  iff every retained mode is J-fixed. Here  $\lambda$  is real because  $\Phi$  is Hilbert–Schmidt self-adjoint.

Part (ii) is Hastings’ bound (cited fact 28.11) recovered as Weil positivity, and it is the quantum-channel case of observation 26.7: the Ramanujan property is the statement that the rescaled ring norms form a positive definite sequence.

**Proposition 27.9** (An adjoint-paired family with no duality). *[proved]* For  $n = 2$ ,  $D = 4$ ,  
 $B_1 = B_3 = \text{diag}(1, 2)$ ,  $B_2 = B_4 = 1$ , the retained spectrum of  $T$  is invariant under  $\mu \mapsto c/\mu$  for no nonzero  $c$ . The Weil form is nevertheless defined and its positivity is the one-sided bound.

So for a general channel “every mode is its own partner” has, literally, no partner to refer to: the reflection exists on the punctured plane but need not send a mode to a mode. What remains is Theorem 27.1: a spectral-gap statement, the analogue of  $\text{Re } \rho \leq \frac{1}{2}$  without the functional equation.

## 28 Weil Positivity, Continued: Continuous Time, Hilbert–Pólya, the Zeta Entry, Prior Art

This section continues Section 27 with the same objects (Definition 4.27); the status paragraph there applies.

### 28.1 Continuous time: cMPS and Lindblad generators

**Theorem 28.1** (Continuous Weil positivity is a half-plane bound). *[proved] For a linear operator  $\mathcal{L}$  on a finite-dimensional space, a trivial set  $S_{\text{triv}}$  and  $a \in \mathbb{R}$ , the rescaled continuous trace  $F$  of Definition 4.27 is continuous on  $\mathbb{R}$  and positive definite iff  $\text{Re } \varpi \leq a$  for every retained  $\varpi$ . A retained mode with  $\text{Re}(\varpi - a) = -b < 0$  and  $\text{Im } \varpi = \omega$  contributes the Lorentzian  $2b/(b^2 + (x - \omega)^2)$  to the spectral measure, a mode on the line a point mass, and positive definiteness forces  $|F| \leq F(0)$ , which a mode to the right of the line violates by continuous mean-square growth.*

**Theorem 28.2** (Line duality and the exact continuous pairing). *[proved] If the retained multiset is invariant under  $R_a(\varpi) = 2a - \bar{\varpi}$ , then for  $g = \sum_j c_j \delta_{t_j}$  and  $\hat{g}_{\mathcal{L}}(\varpi) = \sum_j c_j e^{t_j(\varpi - a)}$ ,*

$$\sum_{j,k} c_j \bar{c}_k F(t_j - t_k) = \sum_{\varpi \in \mathcal{A}_{\text{ret}}} \hat{g}_{\mathcal{L}}(\varpi) \overline{\hat{g}_{\mathcal{L}}(2a - \bar{\varpi})},$$

and  $F$  is positive definite iff  $\text{Re } \varpi = a$  for every retained  $\varpi$ .

This is the continuous inflow identity: the autocorrelation of the rescaled trace equals the pairing of each mode with its reflection in the line, Weil’s form in the form  $\sum_{\varpi} \hat{g}(\varpi) \overline{\hat{g}(1 - \bar{\varpi})}$ .

**Theorem 28.3** (Dyson expansion as continuous ring norms). *[proved] For  $\mathcal{L}(x) = Kx + xK^\dagger + \sum_j R_j x R_j^\dagger$  on  $M_n$ , with no trace preservation assumed,*

$$\text{Tr}_{M_n} e^{t\mathcal{L}} = \sum_{k \geq 0} \sum_{j_1, \dots, j_k} \int_{0 < t_1 < \dots < t_k < t} \left| \text{Tr} \left( e^{(t-t_k)K} R_{j_k} e^{(t_k-t_{k-1})K} \dots R_{j_1} e^{t_1 K} \right) \right|^2 dt_1 \dots dt_k \geq 0,$$

absolutely convergent, the  $k = 0$  term being  $|\text{Tr } e^{tK}|^2$ . Moreover  $\mathcal{L}$  commutes with  $x \mapsto x^\dagger$ , so  $\text{spec}(\mathcal{L})$  is conjugation-closed.

The free propagators between jumps were missing from the draft; without them the product is wrong for noncommuting  $K$  and  $R_j$ . This is the continuous side A: a cMPS with matrices  $K, R_j$  has ring norms given by this sum, and the transfer generator  $\mathcal{L}$  is its side B. For a Hamiltonian generator  $\mathcal{L} = -i[H, \cdot]$  with  $H = H^\dagger$  every eigenvalue is purely imaginary and the line statement with  $a = 0$  is trivial; that is the Hilbert–Pólya situation, and the content of RH in this language is the existence of a presentation in which the transfer generator is a scalar plus  $i$  times a Hermitian operator, which the next two propositions make precise.

### 28.2 The Hilbert–Pólya inner product, and what positivity does not see

**Proposition 28.4** (Unitarisability on the retained space). *[proved] Let  $S_{\text{triv}}$  be a union of full eigenvalue classes,  $V'$  the sum of the generalised eigenspaces of the retained classes,  $X' = X|_{V'}$ . Then  $X'$  is diagonalisable with all eigenvalues of modulus  $r$  iff there is an inner product on  $V'$  in which  $X'/r$  is unitary.*

**Proposition 28.5** (Skew-adjoint realisation). *[proved] With the same convention,  $\mathcal{L}' - a$  is skew-adjoint in some inner product on  $V'$  iff  $\mathcal{L}'$  is diagonalisable with all eigenvalues on  $\operatorname{Re} \varpi = a$ ; equivalently  $\mathcal{L}' = a + iH$  with  $H$  Hermitian in that inner product.*

**Proposition 28.6** (Weil forms do not see Jordan blocks). *[proved] For the Jordan block  $X = r \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$  with  $S_{\text{triv}} = \emptyset$ ,  $n_\ell = 2$  for all  $\ell$  and the sequence is positive definite, although  $X/r$  is not unitarisable; likewise  $\mathcal{L} = a + \begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix}$ . Weil positivity holds with multiplicities and is blind to Jordan structure; a Hilbert–Pólya inner product needs semisimplicity in addition.*

This separates two statements that are usually run together. “All modes on the critical circle” is a statement about the characteristic polynomial and is equivalent to Weil positivity. “There is a Hilbert space in which the transfer operator is  $r$  times a unitary” is stronger by exactly the absence of Jordan blocks. For the Riemann channel the distinction is the multiplicity of the zeros: a multiple zero of  $S$  gives a Jordan block of  $Z(t)$  on  $K_S$ .

### 28.3 The zeta instance and the prior art

*Observation 28.7* (The zeta dictionary entry). *[sketched] For the compressed semigroup  $Z(t)$  with modes  $\eta = -\bar{\rho}/2$ , the critical real part is  $a = -\frac{1}{4}$ , the reflection  $\rho \mapsto 1 - \bar{\rho}$  becomes  $\eta \mapsto -\frac{1}{2} - \bar{\eta}$ , the transform is  $\hat{g}_\zeta(s) = \int g(t) e^{(s-1/2)t/2} dt$ , and Weil’s form  $\sum_\rho \hat{g}(\rho) \overline{\hat{g}(1 - \bar{\rho})}$  is the autocorrelation of the centred trace distribution, whose full-line extension is  $T_Z(-u) = e^{u/2} \overline{T_Z(u)}$ . The trivial modes at 0 and  $-\frac{1}{2}$  and the Gamma-factor terms are not eigenmodes of  $Z(t)$ ; they stay on the geometric side of the explicit formula. The passage from finitely many modes to  $\zeta$  needs the completed-zeta data, a distributional trace realisation, the full explicit formula and the infinite Weil converse, with Bochner–Schwartz in place of Bochner.*

The prover’s T6 also derives that in the explicit-formula convention of Section 11 the prime part of the centred trace carries weight  $-2\Lambda(n)n^{-1/2}$  at  $\pm 2 \log n$ ; this is observation 11.6.

*Cited fact 28.8* (Weil’s criterion). *[cited] RH holds iff the Weil scalar product  $\sum_\rho F(\rho) \overline{G(1 - \bar{\rho})}$  is positive on the admissible test class [13, 55].*

Quoted (prov:cc20-weilcrit, prov:lagarias05-weil-scalar-product): In fact, following \cite{yoshida}, it is enough to prove the negativity of the right-hand side of \eqref{explicit form} for functions  $g$  with compact support in the (locally compact) multiplicative group  $\mathbb{R}_{>0}$ . Furthermore, given any finite set of complex numbers  $F \subset \{0, 1\}$ ,  $F \cap \mathbb{Z} = \emptyset$ , using the notation  $g^\sharp(x) := x^{-1} g(x^{-1})$ ,  $\overline{g(x)} = \overline{g(x)}$ , one has (Appendix \ref{apppositivity}, Proposition \ref{mainprop}) \begin{equation} \text{RH} \iff \sum\_v \sum\_{\mathcal{W}} (g \* \overline{g^\sharp}) \leq 0, \quad \text{for all } g \in C\_c(\mathbb{R}\_{>0}) \text{ with } \tilde{g}(z) = 0, \quad \text{for all } z \in F. \end{equation}

Given  $F(s), G(s) \in \mathcal{S}$  we define the Weil scalar product associated to the automorphic representation  $\pi$  by  $\langle F, G \rangle_{\pi} := \sum_{\rho \in Z(\pi)} F(\rho) \overline{G(1 - \bar{\rho})}$ . The sum on the right counts zeros with

*Cited fact 28.9* (Li’s criterion). *[cited] RH iff  $\lambda_n \geq 0$  for all  $n \geq 1$ ; each condition  $\lambda_n \geq 0$  encodes Weil positivity for one test function [84, 55].*

Quoted (prov:voros06-li-criterion, prov:lagarias05-li-is-weil): \emph{Li’s criterion} for the Riemann Hypothesis (RH) states that the latter is true if and only if a specific real sequence  $\{\lambda_n\}_{n=1,2,\dots}$  has \emph{all its terms positive} \cite{LI1,BL}.

\begin{theorem} \label{th31} Let  $\pi$  be an irreducible cuspidal unitary automorphic representation of  $GL(N)$ , with associated  $\xi$ -function  $\xi(s, \pi)$  and with Weil scalar

product  $\langle \cdot, \cdot \rangle_{\text{sw}(\pi)}$ . For the Li test functions  $G_n(s) = 1 - (1 - \frac{1}{s})^n$  there holds  $\langle G_n, G_m \rangle_{\text{sw}(\pi)} = \lambda_n(\pi) + \lambda_{-m}(\pi) - \lambda_{n-m}(\pi)$ .  $\square$

*Cited fact* 28.10 (Huang’s criterion for regular graphs). [cited]

A connected  $(q + 1)$ -regular graph on  $n$  vertices is Ramanujan iff  $h_k \geq 0$  for all  $k \geq 1$ , where for odd  $k$ ,  $h_k = 2(n - 1) + q^{k/2} + q^{-k/2} - q^{-k/2}N_k$  with  $N_k$  the number of cyclically non-backtracking closed walks of length  $k$  [39].

Quoted (prov:huang19-ramanujan-iff-positive, prov:huang19-hk-cycle-formula):  $\begin{thm}\label{thm:Li}$  The following are equivalent:  $\begin{enumerate}$   $\item X$  is Ramanujan.  $\item h_k \geq 0$  for all  $k \geq 1$ .  $\item h_k \geq 0$  for infinitely many even  $k \geq 2$ .  $\end{enumerate}$   $\end{thm}$

$2(n-1) + q^{\frac{k}{2}} + q^{-\frac{k}{2}} - q^{-\frac{k}{2}}N_k \geq 0$  for all odd  $k$ ,

*Cited fact* 28.11 (Hastings’ bound). [cited] For a quantum expander with  $D$  unitary Kraus operators the Ramanujan value of the second largest eigenvalue modulus is  $\lambda_H = 2\sqrt{D-1}/D$  [36].

Quoted (prov:hastings07-lambdaH): Let  $\lambda_2$  be the eigenvalue with the second largest absolute value of all eigenvalues other than  $\lambda_1$ . Let  $\lambda_H = \frac{2\sqrt{D-1}}{D}$ . The main result of this paper is that, in the Hermitian case, for any  $\epsilon > 0$  the probability that  $|\lambda_2|$  is within  $\epsilon$  of  $\lambda_H$  approaches unity as  $N \rightarrow \infty$ . Interestingly, this is the same as the recently proven tight bound [fried] in the classical case, but the proof in the quantum case is much simpler.

*Cited fact* 28.12 (Herglotz and Bochner). [cited] A sequence is positive definite iff its Toeplitz matrices are positive semidefinite iff it is the Fourier coefficient sequence of a positive measure on the circle [1]; a continuous positive definite function on  $\mathbb{R}$  is the Fourier transform of a finite positive measure [71].

Quoted (prov:acks14-herglotz, prov:norvidas20-bochner): A result of Herglotz, also known as Bochner’s theorem, asserts that:  $\begin{thm}\label{Tm:Oct27yt1}$  The function  $n \mapsto r(n)$  from  $\mathbb{Z}$  into  $\mathbb{C}^{\mathbb{N}}$  is positive definite if and only if there exists a unique positive  $\mathbb{C}^{\mathbb{N}}$ -valued measure  $\mu$  on  $[0, 2\pi]$  such that  $\begin{equation}\label{eq:Oct27nv1} r(n) = \int_0^{2\pi} e^{int} d\mu(t), \quad n \in \mathbb{Z}. \end{equation}$   $\end{thm}$

A function  $f: \mathbb{R} \rightarrow \mathbb{C}$  is said to be positive definite if  $\begin{equation}\label{1.1} \sum_{j,k=1}^n f(x_j - x_k) \overline{c_j} c_k \geq 0 \end{equation}$  holds for all finite sets of complex numbers  $c_1, \dots, c_n$  and points  $x_1, \dots, x_n \in \mathbb{R}$ . A survey about positive definite functions and its generalizations can be found in [9]. The Bochner theorem states that a continuous function  $f$  on  $\mathbb{R}$  is positive definite if and only if it is the Fourier transform of a finite nonnegative measure  $\mu$  on  $\mathbb{R}$ , i.e.  $f(x) = \hat{\mu}(x) = \int_{-\infty}^{\infty} e^{-ixt} d\mu(t)$ ,  $\square$

*Cited fact* 28.13 (Suzuki’s kernel form). [cited] RH is equivalent to nonnegative definiteness of the kernel  $G(t, u) = g(t - u) - g(t) - g(-u) + g(0)$  built from a screw function of  $\zeta$ , in the Herglotz–Nevanlinna framework, with Li’s criterion as a case of Weil positivity [80].

Quoted (prov:suzuki22-rh-iff-nnd, prov:suzuki22-weil-positivity):  $\begin{thm}\label{Thm_1_3}$  The RH is true if and only if the hermitian form  $\langle \cdot, \cdot \rangle_{G_g, a}$  is non-negative definite on  $\mathcal{C}_0(a)$ , that is,  $\langle \phi, \phi \rangle_{G_g, a} \geq 0$  for all  $\phi \in \mathcal{C}_0(a)$  for every  $0 < a < \infty$ . Moreover, assuming that the RH is true,  $\langle \cdot, \cdot \rangle_{G_g, a}$  is positive definite on  $L^2(-a, a)$ ,

that is,  $\langle \phi, \phi \rangle_{G,a} > 0$  for all non-zero  $\phi \in L^2(-a, a)$  for every  $0 < a < \infty$ . \end{theorem}

showed that the RH is true if and only if the distribution  $W$  is non-negative definite, that is,

*Cited fact* 28.14 (PT symmetry and real spectrum).

[cited] A PT-symmetric

non-Hermitian Hamiltonian has real spectrum when the symmetry is unbroken, i.e. its eigenstates are also PT eigenstates [6].

Quoted (prov:bbm17-pt-real): (iii) Although  $\hat{H}$  is not Hermitian,  $\hat{H}$  is  $\mathcal{CPT}$  symmetric; that is,  $\hat{H}$  is invariant under parity-time reflection (in the sense to be defined), which means that the eigenvalues of  $\hat{H}$  are either real or else occur in complex-conjugate pairs. If  $\hat{H}$  has *maximally broken*  $\mathcal{CPT}$  symmetry; that is, if all of its eigenvalues are *pure-imaginary complex-conjugate pairs*, then the eigenvalues of  $\hat{H}$  are real and the Riemann hypothesis follows. (iv) While  $\hat{H}$  is not Hermitian (symmetric)

Positivity criteria for RH are old (Weil; Li; the kernel form of Suzuki), and Huang’s criterion is the graph case. What is not in the literature found is the statement for an arbitrary transfer operator, the separation of the one-sided bound from the duality, and the Kraus dichotomy. Huang’s criterion is termwise rather than Toeplitz; the next observation reconciles the two.

## 28.4 Two observations by the orchestrator

*Observation* 28.15 (Huang’s criterion is the boundedness form). [proved] In finite dimensions, for a conjugation-closed retained multiset, the following are equivalent: positive definiteness of  $n_\ell$ ; boundedness;  $|n_\ell| \leq n_0$  for all  $\ell$ ; and the termwise criterion  $\operatorname{Re} n_\ell \leq n_0$  for all  $\ell \geq 1$ . For a connected non-bipartite  $(q+1)$ -regular graph with  $r = \sqrt{q}$  and  $S_{\text{triv}} = \{+1^{[m-n]}, -1^{[m-n]}, q, 1\}$  ( $m$  edges,  $n$  vertices), Huang’s sequence is exactly  $h_k = n_0 - n_k$  with  $n_0 = 2(n-1)$ , so Huang’s criterion is the boundedness form of Weil positivity. The Toeplitz form is what survives when the trace is a distribution and boundedness has no meaning.

*Observation* 28.16 (The Kraus dichotomy). [proved] For any Ad-type family the rings are positive and  $\operatorname{spec}(\Sigma)$  and  $\operatorname{spec}(T)$  are conjugation-closed. (b) Inverse pairing buys the functional equation: with  $S_{\text{triv}} = \{+1^{[k]}, -1^{[k]}\}$  and  $r = \sqrt{D-1}$ , “all retained modes on the circle” is equivalent to: every eigenvalue  $\varkappa$  of  $\Sigma$  is real with  $|\varkappa| \leq 2\sqrt{D-1}$ . (c) Adjoint pairing buys Hilbert–Schmidt self-adjointness of  $\Sigma$ , so  $\operatorname{spec}(\Sigma)$  is real for free;  $\operatorname{spec}(T)$  need not be real, and in general there is no functional equation (Proposition 27.9). (d) A family with both pairings is unitary; there the eigenvalue  $\varkappa = D$  of  $\Sigma$  (from  $\Phi(1) = 1$ ) always violates the band, so (b) is applied with the enlarged trivial set of Theorem 27.8(ii),  $S_{\text{triv}} = \{+1^{[k]}, -1^{[k]}\} \uplus \{q^{[d+]}, 1^{[d+]}\} \uplus \{(-q)^{[d-]}, (-1)^{[d-]}\}$ , and reduces to Hastings’ bound alone. Reality of  $\operatorname{spec}(\Sigma)$  is the Hilbert–Pólya half; the functional equation is the other half; a Kraus family gets the two from two different pairings, and gets both only by being unitary.

The dichotomy reopens conjecture 35.1 in a sharper form: inverse-paired non-unitary families (for instance  $B_i = GU_i G^{-1}$  with a common similarity  $G$ , which keeps  $\Sigma$  similar to a self-adjoint operator) have a genuine two-sided Ramanujan property with a positive side A, and their non-Hermitian  $\Sigma$  with real spectrum is the transfer-matrix version of an unbroken PT symmetry (cited fact 28.14).

*Numerical result* 28.17 (Numerical checks).

[numerical]

scripts/weil\_positivity.py (outputs/weil\_positivity.txt): random adjoint-paired families ( $n = 2, 3$ ;  $D = 4, 6$ ; 300 cases) satisfy “one-sided bound  $\Rightarrow$  Toeplitz positivity at  $L = 40$ ” without

exception, the threshold radius approaches  $\max |\mu|$  off  $S_{\text{triv}}$  from below like  $L^{-3/2}$  (relative gap  $3 \cdot 10^{-2}$ ,  $5 \cdot 10^{-3}$ ,  $4.7 \cdot 10^{-4}$  at  $L = 40, 160, 640$ ), the retained spectra are conjugation-closed to  $10^{-14}$  and not reciprocal-invariant; Haar-unitary families reproduce Corollary 26.3 to  $10^{-15}$ , the J-invariance, and “Weil positivity iff Hastings’ bound” on four passing and two violating families, with  $W = M$  to  $10^{-14}$ ; an inverse-paired non-unitary family has reciprocal-invariant, conjugation-closed spectrum and nonnegative ring traces to  $10^{-15}$ ; random Lindblad-type generators have  $\text{Tr } e^{t\mathcal{L}} \geq 0$  matching the Dyson expansion to  $10^{-7}$  and Gram positivity iff  $\max \text{Re } \varpi \leq a$ ; the Jordan block passes with  $n_\ell$  identical to its diagonal companion. Three of thirty checks fail: they are the three drafted statements the prover corrected.

Two remarks on the numerics. Positivity at a fixed finite  $L$  is weaker than the theorem: a mode slightly outside the circle is masked by the positive contribution of the interior modes until  $L$  is large. And the  $D = 4$  Pauli family  $\{X, Z, X^\dagger, Z^\dagger\}$  is an exact Ramanujan example with  $\Phi$ -spectrum  $\{1, 0, 0, -1\}$ , exercising the  $-1$  branch of the trivial set with retained modes  $\pm i\sqrt{3}$ .

## 29 An Ihara Zeta for Complexes I: the Literature and the Two Flows

TJO’s question of 2026-09-13 was whether the Hashimoto operator, the Bass identity and “RH  $\Leftrightarrow$  Ramanujan” survive when a graph becomes a finite simplicial complex, and whether the answer grades by cell dimension into an even and an odd sector. The literature answer is negative in that generality and positive on a thin class: finite quotients of affine buildings.

*Status.* The theorems here and in Sections 30 and 31 are the prover’s, `codex:gpt-6-astra` (notes/complex-zeta/astra-proofs.md, correction ledger T8.2, hypothesis register T8.3), carried as `sketched` pending the adversarial review; the numerics are an independent third agent’s. Each `asm`: row is an external theorem used but not reproved, byte-cited in `db/claims.tsv`.

### 29.1 The prototype: $\mathrm{PGL}_3$

*Cited fact* 29.1 (Kang–Li: the  $\mathrm{PGL}_3$  complex zeta and its Bass identity). [cited] For a torsion-free cocompact type-preserving  $\Gamma \subset \mathrm{PGL}_3(F)$ , the zeta of  $X_\Gamma$  counts tailless primitive closed geodesics made of same-type edges, by algebraic length (0804.2305:main.tex:215–220), and  $Z(X_\Gamma, u) = \frac{\{(1-u^3)^{-\chi(X_\Gamma)}\}}{\det(I - A_1u + qA_2u^2 - q^3u^3I)\det(I + L_Bu)}$ , (0804.2305:main.tex:236–239); equivalently  $1/D_E$  with  $D_E = \det(I - L_Eu)\det(I - L_E^t u^2)$  (:247–248). Kang–Li–Wang remove the regularity hypothesis and clear denominators (0809.1401v1:main.tex:1414–1416; operators at :342–368). Note the base  $(1 - u^3)$ , the *plus* sign in  $\det(I + L_Bu)$  and the  $u^2$  on the transposed edge factor [45, 46].

*Cited fact* 29.2 (Kang–Li–Wang: RH is four-way equivalent to Ramanujan, on three circles). [cited]  $X_\Gamma$  is a Ramanujan complex; \item [(2)] The nontrivial zeros of  $\det(I - A_1u + qA_2u^2 - q^3u^3I)$  have absolute value  $q^{-1}$ ; \item [(3)] The nontrivial zeros of  $\det(I + L_Bu)$  have absolute values  $1$ ,  $q^{-1/2}$  and  $q^{-1/4}$ ; and \item [(4)] The nontrivial zeros of  $\det(I - L_Eu)$  have absolute values  $q^{-1}$  and  $q^{-1/2}$ . (0804.2305:main.tex:285–295). Unlike the graph case RH is not a single circle: three for the chamber factor, two for the edge factor. In type  $\tilde{C}_2$  it degrades to bands,  $|u| = q^\alpha$  with  $-2 \leq \alpha \leq -1$  (1109.3854:zetagsp4-final.tex:145–157) [27].

### 29.2 All ranks: an alternating product over facet dimension

*Cited fact* 29.3 (Kang–Yu: the uniform  $\mathrm{PGL}_n$  identity is a torsion). [cited]  $(1-u^n)^{-\chi(X_\Gamma)}$ ,  $L(\Gamma, q^{(n-1)/2}u) = \prod_{k=1}^{n-1} Z_k^{-\epsilon_k(X, u)^{-1}}$ , (2607.21262:main.tex:2019–2023), each factor a reciprocal determinant  $\det(I - (-1)^{k+1}T_k)^{-1}$  of a successor operator (:2304–2312) with the parity sign  $\epsilon = (-1)^{(k+1)l_G}$  (:1971–1981). The Euler factor is literally a torsion: the alternating product of  $\det \Phi_i$  on cochains equals the same on cohomology, where  $\Phi_i$  acts by  $1 - u^n$  (:2463–2470), a strategy that It is inspired instead by Hoffman’s reformulation of Bass’s proof of the Ihara identity in terms of torsion of complexes~\cite{Hof}. (:354–356). The source states *no* RH [48].

*Cited fact* 29.4 (Lubetzky–Lubotzky–Parzanchevski: the branching geodesic flow). [cited] The higher-dimensional Hashimoto operator is multi-valued: The geodesic flow  $T$  on  $\mathrm{UT}^j(\mathcal{B})$  is defined as follows:  $T(\sigma)$  for  $\sigma = (v_0, (v_0, \dots, v_j))$  in  $\mathrm{UT}^j(\mathcal{B})$ , with  $(v_i) \equiv (v_0) + i$ , consists of all the cells  $\sigma' = (v_1, (v_1, \dots, v_j, w))$  in  $\mathrm{UT}^j(\mathcal{B})$  such that (1702.05452:rw\_ramanujan\_complex.tex:1009–1019), the two

conditions being consecutive colours and geodicty (the union must not be a cell); in dimension one this is non-backtracking. A  $k$ -out-regular digraph is Ramanujan when every eigenvalue has  $|\lambda| \leq \sqrt{k}$  or  $|\lambda| = k$  (:333-336) [60].

*Cited fact* 29.5 (The higher-rank RH is one-sided, and the interior really is occupied). [cited]Then  $Z_{\{T, fX\}}(u)$  satisfies the Riemann Hypothesis; that is, if  $Z_{\{T, fX\}}(u)$  has a pole at  $k^{-s}$  then  $|s|=1$  or  $\operatorname{Re} s \leq \frac{1}{2}$ . (1702.05452:rw\_ramanujan\_complex.tex:410-416), for every building. The inequality is not an equality: the source states that poles with  $|\operatorname{Re} s| < \frac{1}{2}$  occur, already with  $|\operatorname{Re} s| = 0$  in the graph case and with  $0 < |\operatorname{Re} s| < \frac{1}{2}$  in higher dimension (:1420-1424).

*Cited fact* 29.6 (Kamber: the converse, as an  $L_p$  criterion). [cited]With one Bernstein–Lusztig operator  $h_{\beta_i}$  per simple coweight and  $\zeta = \prod_i \det(1 - h_{\beta_i} u^{l(\beta_i)})^{-1}$  (1701.00154:L\_p\_Expander\_Complexes.tex:352-358) one gets an if-and-only-if: The complex  $XX$  is an  $L_p$ -expander if and only if every pole  $\lambda$  of  $\zeta_{\hat{B}}(\lambda)$  satisfies  $|\lambda| \leq q^{-(p-1)/p}$  or  $|\lambda| = q$ . (:359-363), Ramanujan meaning  $L_2$ -expander. “Any complex” there means any quotient of an affine building [44].

## 29.3 What happens off the buildings

*Cited fact* 29.7 (Storm: the hypergraph zeta is a bipartite graph zeta). [cited]Reading a complex as a hypergraph does give a zeta (`math/0608761:Ihara-SelbergofHypergraph.tex:173-174`), with a Bass identity (:463) and an if-and-only-if Ramanujan theorem (:627-630) — but all of it because  $[\zeta_{\text{hyperH}}(u) = Z_{\{B_{\text{hyperH}}\}}(\sqrt{u})]$  (:430): it is the Ihara zeta of the incidence bipartite graph in  $\sqrt{u}$ , blind to the simplicial structure beyond incidence [77].

*Cited fact* 29.8 (Benard–Chaubet–Dang–Schick: a general triangulation has a zeta, but no Ramanujan). [cited]On any triangulation of a compact oriented  $n$ -manifold the geodicty rule is exactly LLP’s — consecutive  $(n-1)$ -simplices must not bound a common  $n$ -simplex (2303.11226:zeta\_triangulations9.tex:165-172) — and The combinatorial zeta function  $\zeta_{\mathcal{T}}(z) = \prod_{\gamma \in \mathcal{P}} (1 - \varepsilon_{\gamma} z^{|\gamma|})$ ,  $\zeta_{\mathcal{T}}$  is a polynomial of degree  $|\mathcal{T}^{(n-1)}|$  vanishing to order  $b_1(M)$  at  $z = (n+2)^{-1}$  (:192-205). A Fried-type torsion theorem: that point is not  $q^{-1/2}$ , and there is no spectral-gap statement [5].

*Cited fact* 29.9 (No Ihara-type zeta of a general finite complex exists). [cited]Four sources say so in their own words: There is no Ihara-formula for higher rank up to date. (`math/0407509:main.tex:174`, 2004); Lubotzky’s 2018 ICM survey still calls attaching zetas to high-dimensional complexes a direction of research “with the hope that” Ramanujanness can be read off RH (1712.02526:main.tex:497); Hong–Kwon name “the lack of a unified zeta function” in higher rank (2411.15489:main.tex:354); and Kang–Yu claim only that theirs is To our knowledge, it is the first Ihara-type identity uniform in  $(n)$  for this family. (2607.21262:main.tex:249-250). This is a record of what was searched, not an impossibility theorem [16, 62, 37].

## 29.4 The graded reading, in the literature’s own words

*Cited fact* 29.10 (Deitmar: a Deligne-type alternating product, and a positive Euler exponent). [cited]For locally symmetric spaces the Ruelle zeta is an alternating product over exterior degree,  $Z_{\{P, \phi\}}^R(s) = \prod_{l=0}^{\dim N}$

$Z_{\{P, \sigma_1, \phi\}}(s+1|\alpha) \sim (-1)^1$ , (dg-ga/9511006:main.tex:1153-1168), and the divisor is a *degree-weighted* alternating sum  $-\sum_p p(-1)^p \dim H^p$  (:1194-1212), the plain Euler sum vanishing identically. Deitmar and Kang take the Euler factors with a positive exponent, because that exponent *is* an Euler number (1303.6848:main.tex:803) [15, 17].

*Cited fact* 29.11 (Dyatlov–Zworski: odd upstairs, even downstairs, order  $-\chi$ ). [cited] For a negatively curved oriented surface,  $\zeta_R(s) = \frac{\zeta_1(s)}{\zeta_0(s)\zeta_2(s)}$ ,  $\text{quad } s \in \mathbb{C}$ . so that the order at 0 is  $m_1(0) - m_0(0) - m_2(0)$  (1606.04560:zazi.tex:843-855), and  $s^{\chi(\Sigma)} \zeta_R(s)$  is holomorphic and nonzero at  $s = 0$  (:68-77). Even sectors contribute 1 each, the odd sector  $b_1$ : the continuous prototype of the graded reading [24].

*Cited fact* 29.12 (Hashimoto’s special value: the trees sit in the vertex determinant). [cited] With  $h_X(u) = \det(I - Au + (D - I)u^2)$  and  $\kappa(X)$  the tree number,  $h_X'(1) = -2 \chi(X) \kappa(X)$ , (2310.15619:main.tex:950-955). The complexity is carried by the Laplacian factor, not by the Euler factor  $(1 - u^2)^\chi$ , whose role is to cancel its zero at  $u = 1$  [49].

*Cited fact* 29.13 (Knill: graph analytic torsion is a super pseudodeterminant). [cited] Torsion  $\mathcal{A}(G)$  agrees with the super pseudo determinant  $\mathcal{S}(\text{rm SDet})(D) = \prod_k \text{rm Det}(D_k)^{(-1)^k}$  of the Dirac blocks  $D_k = d_k * d_k$  of the Dirac operator  $D = d + d^*$  and is the ratio of rooted spanning trees on even-dimensional simplices to those on odd simplices (2201.09412:reidemeister.tex:42-52). Knill never mentions the Ihara zeta, nor the Ihara literature  $\text{SDet}$  [51].

*Cited fact* 29.14 (Matsuura–Ohta: a fermionic proof of Bass’s identity on a graph). [cited] One Grassmann field on vertices and two per oriented edge, a Dirac operator off-diagonal between them, and two Schur evaluations of one determinant give  $(1 - t^2)^{n_E - n_V} \det \Delta$  (2501.08803:main.tex:750-755) and  $\zeta_{\Gamma}(q, u)^{-1} = (1 - t^2)^{n_E - n_V} \det \Delta_{\{q, u\}} = \det \left( (I_{2n_E} - qB_u) \right)$ , . (:811-826). The Euler exponent is the fermionic determinant of the edge-reversal involution,  $\det(I - tJ) = (1 - t^2)^{n_E}$  [66].

## 29.5 Hypothesis register

**Assumption 29.15** (H-POINT: building successor conventions). [assumed] Pointed facets, cyclically increasing ordered type, algebraic step length  $\lambda_0$ , link opposition, and quotient incidences, as in Definition 6.2.

**Assumption 29.16** (H-LLP: branching flows and the digraph bound). [assumed]  $UT^j$  with consecutive colours and the geodicty exclusion; and, for an equivariant collision-free  $b$ -regular branching operator on a Ramanujan complex,  $|\lambda| \leq \sqrt{b}$  off the peripheral  $|\lambda| = b$ , with interior radii genuinely occurring (cited facts 29.4 and 29.5).

**Assumption 29.17** (H-KL: the rank-three identity and its four-way RH). [assumed] With  $P_3(u) = I - A_1u + qA_2u^2 - q^3u^3I$ ,  $D_E = \det(I - uL_E) \det(I - u^2L_E^t)$ ,  $D_B = \det(I + uL_B)$ :  $D_B/D_E = (1 - u^3)^{\chi} / \det P_3$ , the four-way equivalence of cited facts 29.1 and 29.2, and the  $L$ -normalisation of Kang–Li–Wang [47].

**Assumption 29.18** (H-KY: the all-rank pointed cochain identity). [assumed] The deformed pointed cochains carry  $d(u)$ , a degree  $-1$  map  $\delta(u)$  and  $\Phi_i = (1 - u^n)I + d_{i-1}\delta_i + \delta_{i+1}d_i$ , with  $\det \Phi_0 = \det P_n = \det \sum_j (-1)^j q^{j(j-1)/2} u^j A_j$  and  $\det \Phi_i = (1 - u^n)^{if_i} \det(I - s_i T_i(u))$ ,  $i \geq 1$  (cited fact 29.3).

**Assumption 29.19** (H-KY-LOCAL: the local factorisation behind equivariant use). [assumed]  $\Phi_i = cW_i^{-1}J_i(I - s_i\Sigma_i)J_i^{-1}Q_i^{-1}$  with  $Q_i$  unipotent in the ordered-type filtration,  $J_i$  cyclic rotation,  $\Sigma_i$  carrying the same weighted closed paths as  $T_i$ , and  $\Phi_0 = P_n$  as an operator (2607.21262:main.tex:2516-2576, 2586-2640, 2930-2968, 2973-3091).

**Assumption 29.20** (H-KL-TABLE: the Iwahori-spherical constituents). [assumed] The unitary Iwahori-spherical types, their fixed-space dimensions and determinant roots, the Steinberg multiplicity  $3\chi-3$ , and the representation-wise conjugate-pair cancellations (0809.1401v1:main.tex:391-424, 1132-1154, 1171-1209, 1425-1461).

**Assumption 29.21** (H-LSV: the joint Hecke spectrum and the explicit complexes). [assumed] The colour-shift convention, commuting normality with  $A_k^* = A_{n-k}$ , the tempered joint spectrum  $\text{Aspec}_n$  with its type-trivial exceptions, and the explicit Ramanujan Cayley clique complexes on  $\text{PGL}_n(\mathbb{F}_{q^e})$  (Definitions 6.4 and 6.5).

**Assumption 29.22** (H-STRONG: Iwahori-spherical temperedness). [assumed] LLP's Ramanujan condition asks every infinite-dimensional *Iwahori-spherical* constituent of  $\ell^2(\Gamma \backslash G)$  to be tempered; this can be strictly stronger than the vertex condition, and the LSV examples satisfy it (1702.05452:rw\_ramanujan\_complex.tex:196-214, 674-701).

**Assumption 29.23** (H-KAMBER: the normalised one-sided criterion). [assumed]  $p$ -temperedness is equivalent to  $|\theta| \leq q^{l(\beta_i)(p-1)/p}$  for every eigenvalue  $\theta$  of  $h_{\beta_i}$  (cited fact 29.6); the bound is on eigenvalues, so it inverts to a *lower* bound  $|u| \geq q^{-(p-1)/p}$  on pole radii.

## 29.6 The two flows are not the same flow

**Theorem 29.24** (Ordered flow versus building flow, exactly, in rank three). [proved-conditional] *Let  $X$  be a simplicial type-preserving torsion-free finite quotient of the  $\text{PGL}_3$  building. Then (i) the colour-1 restriction of the edge successor relation is  $L_E$  and the colour-2 one is conjugate by edge reversal to  $L_E^t$ , so  $T_1(u) = uL_E \oplus u^2L_E^t$  and  $D_E(u) = \det(I - uL_E) \det(I - u^2L_E^t)$ ; (ii)  $T_2(u) = uL_B$  on the  $3f_2$  cyclically ordered pointed chambers; (iii) the unrestricted  $T_1$  of Definition 6.1 is not this block sum even after replacing  $u^2$  by  $u$  — its outdegree is  $2q^2 + q$  against  $q^2$  for  $L_E$ , and its directed-colour classes are not invariant; (iv)  $T_2$  has two invariant cyclic orientations of dimension  $3f_2$ , transposes of one another, so  $\det_{H_2}(I + uT_2) = D_B(u)^2$ . The  $u^2$  is an algebraic length, not a matrix square, and the  $3!$  orderings do not realise  $L_B$  once.*

**Theorem 29.25** (Which zeta reduces to which). [proved-conditional] *With the pointed data of Definition 6.2,  $Z_{\text{pt}} = D_B/D_E = (1 - u^3)^{\times L_{\text{vertex}}}$ , while  $Z_{\text{KL}} = 1/D_E$  and  $L_{\text{vertex}} = 1/\det P_3$ . The graded total is therefore the completed vertex  $L$ -function, not Kang–Li's edge-only zeta; with the unrestricted ordered operators it is neither. The orchestrator's draft asserted that the total graded zeta equals Kang–Li's  $Z(X_\Gamma, u)$ ; the prover refuted that, and the above is the correction.*

## 30 An Ihara Zeta for Complexes II: the Graded Geodesic Determinant

What survives on an arbitrary finite complex is a rational compression, not a polynomial Bass identity; what survives on a building is the full Hecke and torsion apparatus. Notation and the hypothesis rows **asm:h-\*** are those of Section 29; Section 31 carries the twisted and quantum layer, which sits on the building side, not the general one.

### 30.1 Torsion, compression, and the obstruction

**Theorem 30.1** (Finite cochain determinant cancellation). *[proved]* Let  $(C^\bullet, d)$  be a bounded finite-dimensional cochain complex over  $K = \mathbb{C}(u)$ ,  $h$  any degree  $-1$  map,  $c \in K^\times$ , and suppose each  $F_i = cI + d_{i-1}h_i + h_{i+1}d_i$  is invertible. Then  $\prod_i \det(F_i|C^i)^{(-1)^i} = \prod_i \det(F_i|H^i)^{(-1)^i} = c^{\sum_i (-1)^i \dim C^i}$ . Neither  $h^2 = 0$  nor  $h = d^*$  is required. The exponent is the dimension count of the complex used:  $\chi(X)$  for ordinary cochains,  $\sum_i (-1)^i (i+1)! f_i$  for all ordered cells,  $\sum_i (-1)^i (i+1) f_i$  for pointed cochains — already  $\chi = 1$  against 0 on a single edge.

**Theorem 30.2** (The ordinary Euler factor is a difference of two weighted sums). *[proved-conditional]* Under  $H$ -KY,  $\prod_i \det \Phi_i^{(-1)^i} = (1 - u^n)^{\chi_{\text{pt}}}$  with  $\chi_{\text{pt}} = \sum_i (-1)^i (i+1) f_i$ , and subtracting the local  $\sum_{i \geq 1} (-1)^i i f_i$  contributions leaves  $\mathcal{Z}_{\text{pt}}(X, u) = (1 - u^n)^{\chi(X)} / \det P_n(u)$ . The bare cochain superdeterminant carries the weighted exponent; only the final flow/vertex comparison carries the ordinary  $\chi$ . The draft's identification of the cochain product's exponent with  $\chi$  was wrong.

**Theorem 30.3** (Universal ordered-cell Bass–Schur identity). *[proved]* For every finite abstract simplicial complex and every  $1 \leq k \leq d$ , with  $R_k$  the first-vertex deletion,  $S_k$  the append map,  $C_k$  cyclic rotation,  $F_k = C_k + R_{k+1}S_{k+1}$  and  $K_k(z) = I + zF_k$ :  $T_k = S_k R_k - F_k$  and

$$\det_{H_k}(I - zT_k) = \det_{H_k} K_k(z) \det_{H_{k-1}}(I - zR_k K_k(z)^{-1} S_k).$$

This is an explicit rational compression by one dimension, retaining vertices, ordered edges and the upper-cell cyclic factor; it is not a degree- $(d+1)$  polynomial on vertices.

**Corollary 30.4** (Bass for every graph, with no regularity assumption). *[proved]* If  $d = 1$  then  $F_1 = J$  is edge reversal,  $\det(I + uJ) = (1 - u^2)^{f_1}$ ,  $R_1 S_1 = A$ ,  $R_1 J S_1 = \text{Deg}$ , and  $\det(I - uT_1) = (1 - u^2)^{f_1 - f_0} \det_{H_0}(I - uA + u^2(\text{Deg} - I))$ . Isolated vertices and trees are included by rational cancellation.

**Proposition 30.5** (The tetrahedron obstruction: no universal cubic collapse). *[proved]* On  $X = \partial\Delta^3$ ,  $(f_0, f_1, f_2) = (4, 6, 4)$ ,  $\chi = 2$ ,  $T_1 = 0$ ,  $\det(I + uT_2) = (1 - u^4)^6$  and  $\mathcal{Z}_{\text{ord}} = (1 - u^4)^6$ . There is no matrix polynomial  $P(u)$  of any size or degree with  $\mathcal{Z}_{\text{ord}} = (1 - u^3)^{\chi} / \det P(u)$ : that would force  $\det P = (1 - u^3)^2 / (1 - u^4)^6$ , which has poles at  $u = -1, \pm i$ . The orchestrator's draft asked for a universal cubic vertex Bass identity on every 2-complex with local coefficients  $Q_1, Q_2$ ; the prover refuted it here, and Theorem 30.3 is the surviving universal replacement.

### 30.2 The graded reading and what it does not say

**Theorem 30.6** (The exact graded identity). *[proved]* With odd part  $\bigoplus_{k \text{ odd}} H_k$ , even part  $\bigoplus_{k \text{ even}, k \geq 2} H_k$  and  $\Theta = \bigoplus_k s_k T_k$ ,  $\mathcal{Z}_{\text{ord}}(X, u) = \text{sdet}(I - u\Theta)^{-1}$  and  $\log \mathcal{Z}_{\text{ord}} = \sum_m \frac{u^m}{m} \text{str}(\Theta^m) = \sum_m \frac{u^m}{m} \sum_k s_k^{m+1} \text{Tr}(T_k^m)$ . The order at  $u_0 \neq 0$  is the odd minus the

even algebraic multiplicity of  $u_0^{-1}$ ; zero eigenvalues give no finite zero or pole, and common even/odd factors cancel. The draft's "the zeros are exactly the odd eigenvalues" is false without the reciprocal, sign, multiplicity and cancellation qualifications.

**Proposition 30.7** (Cancellation: the reduced vertex  $L$ -function has no finite zeros). *[proved-conditional]* Under  $H$ -KL,  $L_{\text{vertex}}(u) = D_B / ((1-u^3)^x D_E) = 1 / \det P_3(u)$ . In the factorised expression the chamber determinant is a numerator and the edge determinants denominators, but the reduced function is a reciprocal polynomial: every finite zero of  $D_B$  cancels, with multiplicity, against an edge or Euler factor. Calling the chamber zeros uncanceled zeros of  $L_{\text{vertex}}$ , as the draft did, is therefore wrong.

**Proposition 30.8** (The cohomology dictionary is parity only). *[proved]* The reindexing  $i = k-1$  matches the exponents of the Grothendieck–Lefschetz convention  $\prod_i \det(I - u \text{Fr} | H^i)^{(-1)^{i+1}}$ , but does not identify  $H_k$  or  $\mathbb{C}^{P_k}$  with cohomology of  $X$ , nor assign Weil weight  $k-1$ : a 3-cycle has six ordered-edge states and one-dimensional  $H^0$ , and the chamber radii are not those of a single pure weight.

**Proposition 30.9** (Which constituents supply the three chamber circles). *[proved-conditional]* Under  $H$ -KL and  $H$ -KL-TABLE, on a Ramanujan quotient: tempered principal series give chamber roots  $\pm q^{-1/2} \alpha_i^{-1/2}$  and edge roots  $q^{-1} \alpha_i^{-1}$ ; the tempered nonspherical type gives one chamber root at  $q^{-1/2}$  and two at  $q^{-1/4}$ ; Steinberg twists give the unit circle; one-dimensional constituents give trivial roots at  $q^{-1}$ . So the circle  $q^{-1/2}$  is not "the odd sector", and a single constituent can contribute to two circles at once.

### 30.3 An incidence matrix, and why the total is a Berezinian

**Theorem 30.10** (One explicit incidence matrix with two Schur evaluations). *[proved]* On  $H_{k-1} \oplus H_k \oplus H_{k+1}$  let  $\mathbb{D}_k(z)$  have diagonal  $(I, I + zC_k, I)$  and off-diagonal couplings  $-R_k, -zS_k, zR_{k+1}, -S_{k+1}$  between adjacent dimensions. Then  $\det \mathbb{D}_k(z) = \det_{H_k}(I - zT_k) = \det K_k(z) \det_{H_{k-1}} M_k(z)$ . Eliminating the edge mass first instead gives an honest coupled vertex/triangle block, which disappears when there are no triangles.

**Theorem 30.11** (The total object needs a ratio, not one fermion determinant). *[proved]* Giving the whole auxiliary space of  $\mathbb{D}_k(s_k u)$  parity  $k+1$  and setting  $\mathbb{D}_{\text{flow}} = \bigoplus_k \mathbb{D}_k(s_k u)$  gives  $\text{sdet } \mathbb{D}_{\text{flow}}(u) = Z_{\text{ord}}(X, u)^{-1}$ . An ordinary Berezin integral gives  $\det D$ , not a reciprocal or a superdeterminant merely because  $D$  has chiral blocks; the construction uses separate auxiliary copies of some cell spaces, so cell chirality and flow superparity are distinct gradings. The draft's universal ordinary fermion determinant with flow parity equal to cell chirality does not exist.

### 30.4 Arbitrary weights and Kraus rings

**Theorem 30.12** (Arbitrary weights preserve the compression). *[proved]* With the connection of Definition 6.6 and  $S_k, C_k$  replaced by their weighted versions,  $T_k^E = S_k^E R_k - F_k^E$  and the identity of Theorem 30.3 holds verbatim over  $V \otimes H_k$ , with no positivity, pairing, invertibility or flatness hypothesis and no inverse of any  $E_e$ . Any other successor rule with the same overlap admits the same construction, by letting  $F$  be the sum of its forbidden weighted transitions.

**Theorem 30.13** (Ring traces, primitive product, and positivity without pairing). *[proved]*  $\text{Tr}(T_k^E)^m = \sum \text{Tr}_V(E_m \cdots E_1)$  over based closed walks in the state digraph; for  $V = M_n$  and  $E_e = \text{Ad}(B_e)$  this is  $\sum |\text{Tr}(B_m \cdots B_1)|^2 \geq 0$ . Adjoint pairing is not needed: complete positivity of the step maps suffices, as in Corollary 26.2. The unsigned factor is  $\prod_{[\gamma]} \det_V(I - u^{|\gamma|} E_\gamma)^{-1}$  over primitive classes; for the signed factor replace  $E_\gamma$  by  $s_k^{|\gamma|} E_\gamma$ .

**Proposition 30.14** (The scope of the positive-ring no-go).

*[proved] A single finite transfer matrix contributes a reciprocal polynomial, so it has no nonconstant reduced numerator; in this finite model a numerator comes from net odd multiplicity between sectors. Positivity of a numerical ring sequence alone does not force this:  $(1-u)/(1-2u)$  has nonnegative coefficients and positive logarithmic coefficients  $(2^m - 1)/m$ , yet a genuine numerator.*

*Observation 30.15* (The natural object, and what “natural” means here). *[proved]* Definition 6.3 is functorial under isomorphisms preserving states, lengths, successor incidences and transports; it is not proved unique among possible choices of higher-dimensional geodesic. For a graph the default is Hashimoto’s operator and there is only an even flow sector; for a building quotient it is the completed vertex product of Theorem 30.2; the Kang–Li zeta is obtained by retaining only the edge factor.

### 30.5 The three smallest exact tests

*Numerical result 30.16* (Three smallest exact tests, on the ordered default). *[numerical]*  
 $C_3$  (unfilled):  $\det(I - uT_1) = (1 - u^3)^2$ , no chamber factor, total  $(1 - u^3)^{-2}$ . Filled  $\Delta^2$ : both flows vanish, total 1 — there is no forced  $(1 - u^3)$  factor.  $\partial\Delta^3$ :  $T_1 = 0$ ,  $\det(I + uT_2) = (1 - u^4)^6$ , total  $(1 - u^4)^6$ . The optional clique-only fourth test, the octahedron  $K_{2,2,2}$ , gives  $(1 - u^6)^8/(1 - u^4)^6$ , again with a pole at  $u = -1$  in the would-be vertex expression. Checked exactly with `sympy`.

## 31 An Ihara Zeta for Complexes III: Twists, Quantum Expanders, Numerics

The twisted layer of Section 30. Arbitrary weights already keep the compression (Theorem 30.12) and already give nonnegative ring traces (Theorem 30.13); what needs care is *which* twisted object is the Artin one, and what a Ramanujan complex buys for a channel.

### 31.1 Gauge triviality and the voltage quotient

**Proposition 31.1** (A representation connection on the full complex is pure gauge). *[proved]* On a Cayley complex the forward weights  $E_{(g,gs)} = \varrho(s)$  are generally not flat (flatness asks  $\varrho(t)\varrho(s) = \varrho(st)$ ); the covariant convention is  $\varrho(s^{-1})$ , and those weights are pure gauge,  $F_g = \varrho(g^{-1})$ , so  $\det(I - uT_k^E) = \det(I - uT_k)^{\dim V}$ : every closed straight walk of the full complex has identity holonomy. The four-cycle with  $G = \mathbb{Z}/4$  and one-dimensional  $\pi$  separates the full twist  $(1-u^4)^2$  from the trivial-block determinant  $(1-u)^2$ , refuting the draft's claim that the full twist is a representation block.

**Theorem 31.2** (The Artin decomposition of a voltage quotient). *[proved]* With the data of Definition 6.7 and a free action on states,  $\det(I - s_k T_k(u)) = \prod_{\varrho} \det(I - s_k T_{k,\varrho}(u))^{d_{\varrho}}$ . For  $R = \bigoplus_{\varrho} a_{\varrho} \varrho$  the Artin determinant has exponents  $a_{\varrho}$ , not  $d_{\varrho}$ :  $R = \pi \otimes \bar{\pi}$  need not embed in a single regular representation (for  $\pi = 1 \oplus 1$ ,  $R = 1^{\oplus 4}$ ). What always transfers is spectral support. An equivariant matrix identity restricts to every block; a scalar determinant equality need not.

**Theorem 31.3** (Exact Euler-factor bookkeeping on finite-group blocks). *[proved-conditional]* Under H-KY and H-KY-LOCAL, with  $w_{i,\varrho} = \det(W_i|C_{i,\varrho})$ ,  $\mathcal{Z}_{\varrho}(u) = F_{\varrho}(u)/\det(P_n|C_{0,\varrho})$  where  $F_{\varrho} = c^{m_{0,\varrho}} \prod_{i \geq 1} w_{i,\varrho}^{(-1)^i}$  and  $\prod_{\varrho} F_{\varrho}^{d_{\varrho}} = c^{\chi(X)}$ . Only if the action is free on unpointed simplices does this collapse to  $F_{\varrho} = c^{d_{\varrho}\chi/|G|}$ ; with simplex stabilisers the fractional power is not a rational determinant and must not be asserted.

### 31.2 Quantum expanders of type $\tilde{A}_{n-1}$

**Theorem 31.4** (Harrow-type transfer, with the correct exceptional space). *[proved-conditional]* Under H-LSV, for every finite-dimensional unitary  $\pi$ : (i) the  $\Phi_k$  are commuting normal unitary trace-preserving channels with  $\Phi_k^* = \Phi_{n-k}$ , whose common fixed space is the commutant  $\pi(G)'$ , of dimension one exactly when  $\pi$  is irreducible; (ii) they satisfy Definition 6.8 relative to  $\mathcal{E}_R$ ; (iii)  $P_{n,R}(u)$  restricted to  $\mathcal{E}_R^{\perp}$  has all zeros on  $|u| = q^{-(n-1)/2}$ ; (iv) under H-KL the rank-three quotient Artin flow factors inherit the radii of cited fact 29.2; (v) under H-KY, H-STRONG and H-LLP, general rank gives only the one-sided bound of Proposition 31.6. Removing only the channel fixed points, as the draft did, leaves type characters that violate the tempered bound.

**Proposition 31.5** (The exact converse criterion; one faithful  $\pi$  is not enough). *[proved]* Let  $\mathcal{B}_X$  be the irreducibles with a nonexceptional joint eigenvalue outside  $\text{Aspec}_n$ . Then  $R$  is Ramanujan iff  $\mathcal{B}_X \cap \text{supp}(R) = \emptyset$ , and  $X$  is vertex-Ramanujan iff  $\mathcal{B}_X = \emptyset$ ; coverage of every nonexceptional irreducible by  $\pi \otimes \bar{\pi}$  is sufficient for the converse. Faithfulness is not: for  $G = \mathbb{Z}/28$ ,  $S = \{\pm 1, \pm 3\}$ , the Cayley graph is not Ramanujan yet a faithful character has  $\pi \otimes \bar{\pi} = 1$  and a vacuous quantum condition.

**Proposition 31.6** (One-sided RH, in the correct variable). *[proved-conditional]* Under H-LLP, for a  $b$ -regular flow with zeta  $\det(I - uT)^{-1}$ , a pole at  $u = b^{-s}$  has  $\text{Re } s = 1$  when  $|\lambda| = b$

and  $\operatorname{Re} s \leq \frac{1}{2}$  when  $|\lambda| \leq \sqrt{b}$ , the imaginary part being defined only modulo  $2\pi/\log b$ ; the source's printed  $|s| = 1$  cannot literally replace  $\operatorname{Re} s = 1$ . Under *H-KAMBER* the actual nontrivial pole radii satisfy  $|u| \geq q^{-(p-1)/p}$ : the quoted eigenvalue bound inverts.

**Proposition 31.7** (Which rank-three factors have a duality). [proved-conditional] With  $A_2 = A_1^*$  and  $p_a(u) = 1 - au + q\bar{a}u^2 - q^3u^3$ , the zero multiset is invariant under  $u \mapsto q^{-2}/\bar{u}$ , before imposing Ramanujan; the principal-series parts of the edge and chamber factors inherit dualities at  $q^{-2}$  and  $q^{-1}$ . The full edge and chamber factors do not: a tempered nonspherical block has one root at radius  $q^{-1/2}$  and two at  $q^{-1/4}$ , so no  $u \mapsto c/\bar{u}$  or  $u \mapsto c/u$  can fix or exchange them. Higher-rank *RH* has no universal functional equation.

**Proposition 31.8** (Weil positivity on a retained block is a bound until a duality is supplied). [proved] For a retained finite block  $M$  with algebraic eigenvalue multiset  $\{\mu_j\}$  and  $r > 0$ , the sequence  $\nu_m = \sum_j (\mu_j/r)^m$  is positive definite exactly when  $|\mu_j| \leq r$  throughout; invariance of the multiset under  $\mu \mapsto r^2/\bar{\mu}$  then forces  $|\mu| = r$ . This is the finite statement of Theorem 27.1, applied to each retained ordinary block, and it does not transfer automatically to an alternating product with negative spectral multiplicities.

### 31.3 Numerics on the smallest $\tilde{A}_2$ Ramanujan complex

*Numerical result 31.9* (The smallest  $\tilde{A}_2$  Ramanujan complex). [numerical] `scripts/a2_complex_zeta.py` (`outputs/a2_complex_zeta.txt`), LSV algorithm at  $q = 3$ ,  $d = 3$ ,  $e = 1$ :  $|S_1| = 13$ ,  $\langle S_1 \rangle = \operatorname{PGL}_3(\mathbb{F}_q[3])$  of order 5616,  $S_2 = S_1^{-1}$ ,  $f = (5616, 73008, 97344)$ ,  $\chi = 29952 = |G| \cdot 16/3$ , every vertex link the incidence graph of  $\operatorname{PG}(2, 3)$  (26 vertices, 4-regular, bipartite, girth 6). Two facts not anticipated in the brief: the complex is *not* 3-colourable ( $\operatorname{PGL}_3(\mathbb{F}_q[3])$  is simple), so Kang–Li's type-preservation hypothesis (I) fails on it; and all 13 generators have order 3, so the 24336 triangles carry  $\mathbb{Z}/3$  stabilisers and  $G$  is not free on unpointed 2-cells. A type-preserving 3-fold cover with  $f_0 = 16848$ ,  $\chi = 89856$  is built alongside.

*Numerical result 31.10* (The vertex spectrum is tempered). [numerical]  $A_2 = A_1^t$  exactly,  $A_1$  normal,  $[A_1, A_2] = 0$  (exact sparse check). Of the 5616 eigenvalues of  $A_1$  exactly one is trivial ( $\lambda = 13$ ); all 5615 others are tempered by Cohn's self-inversive criterion, maximum defect  $2.0 \cdot 10^{-14}$ , and  $\max |\lambda| = 7.126860684 < 3q = 9$ . So the complex is Ramanujan and cited fact 29.2(2) holds outright.

*Numerical result 31.11* (The corrected identity, and the drafted one failing). [numerical] Traces computed exactly as integers by the free  $G$ -action:  $\operatorname{Tr} L_E^m = 0$  for  $m \leq 5$ , then 657072 at  $m = 6$  up to  $1.5 \cdot 10^{17}$  at  $m = 18$ ;  $\operatorname{Tr} L_B^m = 0$  for  $m \leq 14$ , then 16426800 at  $m = 15$ . Verified exactly to  $u^{18}$ , on the 5616-vertex base *and* on the 16848-vertex type-preserving cover:  $\det P_3(u) \det(I + uL_B) = (1 - u^3)^\chi \det(I - uL_E) \det(I - u^2L_E^t)$ , with degree 308880 on both sides and the residual fitting  $\chi$  exactly (29952, resp. 89856) at every order. The placement written in the orchestrator's brief fails at  $u^3$  (residual  $-59904$ ) and fails the degree count (398736 against 219024);  $\det(I - uL_B)$  in place of  $\det(I + uL_B)$  first fails at  $u^{15}$ . The identity holds on the base even though type preservation fails there.

*Numerical result 31.12* (Flow operators: the prover's cross-checks). [numerical]  $L_E$  has outdegree exactly  $q^2 = 9$  and  $L_B$  outdegree  $q = 3$ , whereas the unrestricted ordered edge rule has outdegree  $2q^2 + q = 21$ : it is not Kang–Li's operator. On all six orderings of a chamber,  $\operatorname{Tr} T_2^m = 2 \operatorname{Tr} L_B^m$  for  $m = 1, \dots, 18$ , i.e.  $\det(I + uT_2) = \det(I + uL_B)^2$ . On  $\partial\Delta^3$ ,  $T_1 = 0$  and  $T_2$  is a permutation of cycle type  $4^6$ . All three confirm Theorem 29.24 and Proposition 30.5.

*Numerical result 31.13* (The three circles, exactly, on two blocks). [numerical] Trivial block:  $|\lambda(L_E)| \in \{9 (\times 1), \sqrt{3} (\times 12)\}$  and  $|\lambda(L_B)| \in \{3, \sqrt{3} (\times 12), 3^{1/4} (\times 24), 1 (\times 15)\}$ . Twelve-dimensional block:  $|\lambda(L_E)| \in \{3 (\times 36), \sqrt{3} (\times 120)\}$  and  $|\lambda(L_B)| \in \{\sqrt{3} (\times 192), 3^{1/4} (\times 240), 1 (\times 192)\}$ . Exactly the radii  $\{1, 3^{1/4}, 3^{1/2}\}$  for  $L_B$  and  $\{3, \sqrt{3}\}$  for  $L_E$  predicted by cited fact 29.2, and nothing else; on the full space ARPACK’s forty largest moduli for  $L_E$  are 9 once and 3 forty times.

*Numerical result 31.14* (The twelve-dimensional quantum twist). [numerical]  $\pi$  the 12-dimensional irrep (the permutation action on the 13 points of  $\mathbb{P}^2(\mathbb{F}_q[3])$  minus the constant). The channels  $\Phi_k(X) = \frac{1}{13} \sum_{s \in S_k} \pi(s) X \pi(s)^\dagger$  on  $M_n[12]$  are unital, trace-preserving and commuting, with exactly one eigenvalue 1; the largest nontrivial  $|13\mu| = 6.481723$ , below the classical deltoid radius  $3q = 9$  and below Hastings’ bound  $2\sqrt{D-1} = 6.928203$  at  $D = 13$ . All 143 nontrivial joint eigenvalues lie inside the tempered deltoid (Cohn defect  $2.2 \cdot 10^{-15}$ ). This is a Ramanujan quantum expander of type  $\tilde{A}_2$ , and it cannot fail:  $\Phi_1 = A_1^{(\pi \otimes \bar{\pi})}/13$ , so its spectrum is a sub-multiset of  $\text{spec}(A_1)$ .

*Numerical result 31.15* (Gauge triviality, voltage blocks, and the Euler bookkeeping). [numerical] The forward weights  $\pi(s)$  are not flat (39 of the 52 chambers fail  $\pi(t)\pi(s) = \pi(st)$ ); the covariant weights  $\pi(s^{-1})$  on the full complex are pure gauge,  $\det(I - uT_E^\pi) = \det(I - uL_E)^{12}$ . The genuine Artin object is the voltage quotient — 13 states, 9 voltaged successors each — whose block traces match the character sums and differ from the gauge-trivial twist, confirming Proposition 31.1 and Theorem 31.2. Writing the residual as  $(1-u)^a(1-u^3)^c$ : trivial block (13, 1),  $\nu = 1 \oplus \pi$  block (13, 65),  $\pi$  block (0, 64) — so for  $\pi$  the naive  $c = \chi \dim \pi / |G| = 64$  is exactly right, while the whole anomaly sits in the trivial block, where  $\chi/|G| = 16/3$  is replaced by  $(1-u)^{13}(1-u^3)^1$ . For the trivial block this is an exact polynomial identity of degree 55 on both sides. This is Theorem 31.3’s stabiliser caveat, seen.

### 31.4 What this does and does not give

The natural object is Definition 6.3: a graded geodesic determinant of a complex *with specified geodesic data* — states, successor incidences, algebraic lengths, transports — and not an unqualified “Thara zeta of every complex”. On an arbitrary complex what is universal is Theorem 30.3, a rational compression that still mentions the forbidden transitions; the polynomial vertex collapse is a *building* phenomenon, and Proposition 30.5 shows it fails on a four-vertex sphere. Opposition, not non-incidence, is the building’s successor rule (Theorem 29.24), and the  $u^2$  is a length.

The odd sector is real as a graded *presentation* and unreal as a divisor: Theorem 30.6 is exact, but on the building vertex side the chamber factor cancels completely (Proposition 30.7), so “fermionic zeros” names a net odd multiplicity that must be checked after cancellation. The parallel with the continuous Ruelle picture (cited fact 29.11) and with Knill’s supertrace (cited fact 29.13) is parity and alternating structure, not an identification of spaces or weights (Propositions 30.8 and 30.9).

The quantum version is the voltage-quotient Artin block, not a connection on the full cover, which is pure gauge (Proposition 31.1). Harrow transfer then gives Ramanujan quantum expanders of type  $\tilde{A}_2$  for free from a Ramanujan complex (Theorem 31.4, numerical result 31.14) — the channel adds no spectral values. But higher-rank RH is one-sided (Proposition 31.6), the interior of the band is genuinely occupied (cited fact 29.5), and there is no universal duality to close it (Proposition 31.7): what one has is Weil-positivity-as-bound, Theorem 27.1 applied block by block (Proposition 31.8). None of this touches the Riemann side directly. It is a supply of exactly-solvable graded transfer operators with a Ramanujan property, not a channel whose zeta is  $\zeta(s)$ .

**Conjecture 31.16** (Characterising vertex collapse for a general complex). [conjectured]

*There is a combinatorial characterisation of the finite complexes  $X$  for which  $B_X(u) = (1 - u^{d+1})^{\chi(X)} / \mathcal{Z}_{\text{ord}}(X, u)$  is the determinant of a matrix polynomial  $I - uA + \sum_{j \geq 2} u^j Q_{j-1}$  with  $Q_{j-1}$*

*built from local link data. Building quotients are sufficient (H-KY); Proposition 30.5 gives a necessary polynomiality condition; matching generalised-polygon link parameters is not known to suffice, and no recognition theorem for Bruhat–Tits universal covers is assumed here.*

## 32 Prior Art for the Quantum Ihara Zeta

The search of 2026-09-12 used three literature agents (graph zetas and  $L$ -functions; quantum expanders and free probability; direct phrase search), after which every load-bearing sentence was rechecked against the arXiv T<sub>E</sub>X source under `refs/src/` (fetched by `refs/fetch_sources.sh`, hashes in `refs/manifest.sha256`). Each quote below is reproduced verbatim from `db/provenance.tsv` and carries its locus; sources with no local T<sub>E</sub>X are unverified and are used for attribution only.

### 32.1 The object under examination

The object is the one fixed in Definition 3.18: for a unital channel  $\Phi(X) = \frac{1}{D} \sum_i U_i X U_i^\dagger$  with unitary Kraus operators closed under adjoint, the non-backtracking edge superoperator  $T$  on  $M_n \otimes \mathbb{C}^D$  and the zeta  $\zeta_\Phi(u) = \det(1 - uT)^{-1}$ . The founding session made four claims about it. First, an Ihara–Bass identity,

$$\det(1 - uT) = (1 - u^2)^{n^2(D-2)/2} \det(1 - u D\Phi + (D-1)u^2). \quad (4)$$

Second, that all nontrivial poles of  $\zeta_\Phi$  lie on  $|u| = (D-1)^{-1/2}$  if and only if  $\Phi$  is a Ramanujan quantum expander (Definition 3.17). Third, that  $\zeta_\Phi$  is the Artin–Ihara  $L$ -function (Definition 3.10) of the bouquet with  $D/2$  loops twisted by  $\varrho: g_i \mapsto \text{Ad}(U_i)$ , with  $\text{Tr } T^\ell = \sum_w |\text{Tr } U_w|^\ell$  over cyclically reduced words  $w$  of length  $\ell$ . Fourth, that the Weil representation of  $\text{SL}_2(\mathbb{F}_p)$  with LPS generators gives an instance where the analogue of the Riemann hypothesis holds exactly. Alongside these sits the matrix-product-state reading (channel as transfer matrix,  $\zeta_\Phi$  as a generating function of ring norms, one correlation length as the Riemann hypothesis, AKLT giving  $K_4$ ).

### 32.2 Verdict

| claim                                                  | status                        | where                                                                                                                                                                                                                                                   |
|--------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1, Ihara–Bass with $\text{Ad}(U_i)$ weights            | known                         | Matsuura–Ohta 2022, eqs. (2.8) to (2.11) with $X_e = U_e \otimes U_e^\dagger$ , loops allowed; the arbitrary-weight formula of Watanabe–Fukumizu 2011 for loopless graphs; twisted zetas back to Sunada 1986 and Hashimoto 1989 (unverified, secondary) |
| 3, trace formula, free-group $L$ -function             | known                         | Matsuura–Ohta 2022 eq. (2.10); Matsuura–Ohta 2026, where the object is named an Artin–Ihara $L$ -function for a representation extended to $U(N_c)$                                                                                                     |
| 2, poles on the critical circle iff Hastings–Ramanujan | not found                     | zero hits for any zeta or Ihara combined with channel, Kraus, CP map or expander; Matsuura–Ohta contain none of the words Ramanujan, expander, Riemann hypothesis; Bordenave–Collins have the operator but no determinant and no zeta                   |
| 4, exactly Ramanujan channels from LPS                 | mechanism known, instance not | Harrow 2008 plus LPS; Iyer, Jain, Jordan and Somma (Feb 2026) state it with $SU(2)$ irreps and call explicit Ramanujan quantum expanders a longstanding open problem; no Weil-representation instance and no zeta                                       |
| MPS and ring-norm reading, AKLT = $K_4$                | not found                     | no hits for Ihara or zeta combined with matrix product state, transfer matrix or AKLT                                                                                                                                                                   |

### 32.3 Matsuura and Ohta

*Cited fact* 32.1 (Extended Ihara zeta with invertible matrix weights). [cited] Suppose  $\{X_e\}_{e \in E}$  are invertible matrices of size  $K$  living on each edge. (2204.06424:main.tex:476), with  $\zeta_G(q; X) \equiv \det \Bigl( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr) - q W_X \Bigl( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr) - q A_X + q^2 (D - \sum_{V} \sum_{V \in C} \frac{1}{|C|} \sum_{C \in R_k} \text{Tr } P_C(X)) \Bigl( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr)^{-1}$  (2204.06424:main.tex:479-482) for the edge-matrix determinant,  $\zeta_G(q; X) = \exp \left( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \right) - q W_X \left( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \right) - q A_X + q^2 (D - \sum_{V} \sum_{V \in C} \frac{1}{|C|} \sum_{C \in R_k} \text{Tr } P_C(X)) \left( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \right)^{-1}$  (2204.06424:main.tex:503-505) for the trace formula over reduced cycles, and the three-term identity  $\zeta_G(q; X) = (1 - q^2)^{-(n_V - n_E)K} \det \Bigl( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr) - q A_X + q^2 (D - \sum_{V} \sum_{V \in C} \frac{1}{|C|} \sum_{C \in R_k} \text{Tr } P_C(X)) \Bigl( \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr)^{-1}$  (2204.06424:main.tex:516-519), proved by the classical route: The following proof is parallel to the one given in [bass1992ihara] for the original Ihara zeta function (2204.06424:main.tex:533-534). This is claim 1 with arbitrary invertible matrix weights in place of  $\text{Ad}(U_i)$ , and claim 3 in the shape of a sum over cyclically reduced classes.

*Cited fact* 32.2 (Adjoint weights and the name Artin–Ihara  $L$ -function). [cited] The weights they apply the formula to are the adjoint ones, with  $X_e = U_e \otimes U_e^\dagger$  (2204.06424:main.tex:696); the 2026 paper names the resulting object an Artin–Ihara  $L$ -function for a representation  $R$  of a finite group extended to  $U(N_c)$ , taking  $R$  to be the adjoint representation with  $\varrho_{\text{adj}}(U_e) = U_e \otimes U_e^\dagger$  (2607.27935:main.tex:508-510, with the  $L$ -function at 2607.27935:main.tex:381 and its Bartholdi-deformed three-term formula at 2607.27935:main.tex:401-404). So the twisting representation of claim 3 is theirs, up to the column-stacking convention of Section 2.

Their construction admits loops and multiple edges: a loop contributes  $X_e + X_e^{-1}$  to the weighted adjacency matrix and 2 to the degree. Specialising to the bouquet with  $n_V = 1$ ,  $n_E = D/2$ ,  $K = n^2$  and graph degree  $D$ , the exponent  $(n_V - n_E)K$  of their eq. (2.11) becomes  $-n^2(D - 2)/2$ , the bracket becomes  $1 - uA_X + (D - 1)u^2$  with  $A_X = \sum_i \bar{U}_i \otimes U_i = D\Phi = \Sigma$ , and inverting eq. (2.8) turns  $\zeta_X^{-1}$  into  $\det(1 - uT)$ . That is Equation (4), so the note records the verdict SAME for claims 1 and 3. The companion paper [64] carries the same adjoint weights into a matrix-weighted Bartholdi zeta, of which the Ihara case is one value of the Bartholdi parameter, and is SAME by the same route; [65] and [67] are the loci above.

### 32.4 Watanabe and Fukumizu

*Cited fact* 32.3 (Ihara–Bass for arbitrary matrix weights, loopless). [cited] The edge zeta is built from matrix weights  $\{u_e\}_{e \in E}$  with  $u_e \in \text{Mat}_{r_t(e) \times r_o(e)}$ , with the cycle weight  $\pi(\frac{p}{q}) := u_{e_1} \cdots u_{e_k}$  (1103.0605:section3.tex:307-310 and 1103.0605:section3.tex:315), and the corollary `cor:IBfornonhyper` reads  $Z_G(\{u\})^{-1} = \det(I + \hat{\mathcal{M}}_{\{D\}}(\{u\}) - \hat{\mathcal{M}}_{\{A\}}(\{u\})) \prod_{e \in E} \det(I - u_e u_{\bar{e}})$  (1103.0605:section3.tex:333-335), of which the authors say Corollary `cor:IBfornonhyper` gives the extension of the result to graphs with arbitrary weights. (1103.0605:section3.tex:373). No inverse relation is imposed between  $u_e$  and  $u_{\bar{e}}$ , which is the feature Theorem 26.1 needs.

Their graphs are hypergraphs, A *hypergraph*  $\mathcal{H} = (V, F)$  consists of a set of *vertices*  $V$  and a set of *hyperedges*  $F$ . (1103.0605:section2.tex:17-18), whose hyperedges are subsets of  $V$  of size two: no loops and no multiple edges, so a bouquet is not one of their graphs. Theorem 26.1 is the loop-admitting analogue of the corollary, of the same shape after  $u_e = u\mathcal{E}_e$ , but the corollary does not cover the bouquet and its scalar reduction does

not evaluate correctly there; the note also records that the hypergraph-free proof it points to (the NeurIPS 2009 supplement) was not fetched. Verdict RELATED: the closest prior art for arbitrary weights, not a special case, and with no representation-theoretic or quantum reading [85].

### 32.5 Bordenave and Collins

*Cited fact* 32.4 (Matrix-valued non-backtracking operator, no zeta). [cited]The operator is  $B = \sum_{j \neq i} a_j \otimes S_i \otimes E_{\{ij\}}$ , (1801.00876:tenseur9.tex:784), introduced because Our proof relies on the development of a matrix version of the non-backtracking operator theory and a refined trace method. (1801.00876:tenseur9.tex:158), and aimed at this application: This extension is especially relevant in the context of quantum expanders. (1801.00876:tenseur9.tex:165). The only place the tradition names the identity is in the later work, This formula is closely related to various spectral identities known as Ihara-Bass identities (2304.05714:PT-RC\_reloaded.tex:2568). The same operator as  $T$ , then, and the spectral correspondence, but as a resolvent identity rather than a determinant: no zeta and no Riemann hypothesis.

The verdict is RELATED. The follow-ups are [9] for general unitaries and [8] for the operator-valued identity; [7] also records, at 1801.00876:tenseur9.tex:608–610, that a family of tensor-squared permutations is a nearly optimal quantum expander in Hastings’ sense.

### 32.6 Hastings and Harrow

*Cited fact* 32.5 (Transfer of the spectral gap from a Cayley graph). [cited] $\lambda_2(\mathcal{E}) \leq \lambda_2(W_\Gamma)$ . (0709.1142:main.tex:213), under the dictionary degree becomes the number of Kraus operators, the spectral gap becomes the gap of the quantum operation (0709.1142:main.tex:72–73). Here  $\lambda_2$  of the channel is its second largest singular value as a map on density matrices and  $\lambda_2$  of  $W_\Gamma$  that of the Cayley graph transition matrix, so taking  $\Gamma$  to be LPS generators in  $\text{PSL}_2(\mathbb{F}_p)$  gives exactly Ramanujan channels from any irreducible representation.

The bound itself,  $\lambda_H = 2\sqrt{D-1}/D$ , is Hastings’ [36], quoted in Definition 3.17; that paper has no zeta and no non-backtracking operator, and the word Ramanujan does not occur in [34]. Verdict SAME for the mechanism behind claim 4, NOT for anything zeta-shaped.

### 32.7 Iyer, Jain, Jordan and Somma

*Cited fact* 32.6 (Exact Ramanujan quantum expanders from LPS). [cited]it is possible to construct *exact* Ramanujan quantum expanders from  $SU(2)$  unitaries derived from the LPS construction~\cite{lubotzky1988ramanujan,harrow2007quantum}. (2602.15180:main.tex:1508), and, of the explicit construction, Our quantum circuits can be used to construct explicit Ramanujan quantum expanders, a longstanding open problem. (2602.15180:main.tex:241). The mechanism is the one of cited fact 32.5, carried by  $SU(2)$  irreducible representations rather than by the Weil representation, and no zeta is attached.

Verdict RELATED for claim 4 [41]. The note records that their Appendix C example,  $p = 5$  and  $D = 6$ , uses the same norm-5 quaternions as the script behind Section 17. Separately, [42] amplifies arbitrary expanders to almost Ramanujan ones, short of the bound and with no zeta: verdict NOT.

## 32.8 Quantum-walk zetas and the secondary sources

*Cited fact* 32.7 (Attribution of the twisted zeta and of the three-term formula). [cited] A zeta function of a regular graph  $G$  associated with a unitary representation of the fundamental group of  $G$  was developed by Sunada [Sunada1, Sunada2]. (2105.02677:main.tex:87–89); for graph coverings, Stark and Terras further introduced notions of *Galois coverings* and *Artin-Ihara  $L$ -functions* for graphs (2309.15873:AIM\_Final.tex:131); and for the determinant, the three-term determinant formula from [Theorem 18.15]{Terras:2011} applied to the Artin-Ihara  $L$ -function (2405.04361:v3.tex:427). The twisted zeta is therefore Sunada’s [78], sharpened by Hashimoto [35] and Bass [3]; the  $L$ -function of a covering is Stark and Terras’ [76], with the three-term formula at Theorem 18.15 of [81]. None of these primary sources is local.

In Stark and Terras the representation is one of the Galois group of a covering of finite graphs, hence of a finite group, whereas  $\text{Ad}(U_i)$  for generic unitaries generates an infinite group: the object of Definition 3.18 is not literally a Stark–Terras  $L$ -function, although the determinant formula is the same. The secondary sources [25] and [30] serve this attribution only; verdict RELATED.

The quantum-walk zeta line [54], [53], [52] builds zetas from the unitary evolution matrix of a quantum walk rather than from Kraus operators, [53] extending to open quantum random walks on the torus by Fourier analysis; verdict NOT for claims 1 to 4, the value here being the Sunada pointer. Ben-Aroya, Schwartz and Ta-Shma [4] is the one unverified source near claim 2: an agent read from the publisher PDF that they believe their first construction has the potential of being improved to a quantum Ramanujan expander.

## 32.9 What was not found, and the limits of the search

*Observation* 32.8 (Verdict of the prior-art search). [sketched]

No source found attaches a zeta function, or an analogue of the Riemann hypothesis, to a quantum channel; none states that the poles of  $\zeta_\Phi$  lie on the critical circle if and only if  $\Phi$  meets the Hastings bound  $\lambda_H$ ; none gives the matrix-product-state ring-norm reading or the identification of AKLT with  $K_4$ . The Ad-weighted Ihara–Bass formula and the free-group  $L$ -function reading are known (cited facts 32.1 and 32.2), arbitrary matrix weights are known for loopless graphs (cited fact 32.3), and the matrix-valued non-backtracking operator is known (cited fact 32.4).

The consequence for this book: [65], [85] and the Sunada–Hashimoto line are cited for the object and the determinant, while the Ramanujan reading of Section 26, the dictionary of Section 18 and the Weil–LPS instance of Section 17 are this notebook’s own. Theorem 26.1 is the bouquet analogue of cited fact 32.3, proved there from scratch; no local source covers arbitrary non-invertible weights on a bouquet.

These arXiv metadata queries returned zero hits: "Ihara zeta" AND "quantum channel"; "Ihara zeta" AND "completely positive"; "zeta" AND "quantum expander"; "zeta" AND "Kraus operators"; abs:"zeta function" AND abs:"superoperator"; "Ruelle zeta" AND "quantum channel"; "Ihara" AND "matrix product state"; "AKLT" AND "zeta"; "non-backtracking" AND "superoperator"; "Ihara" AND ("quantum Ramanujan" OR "Ramanujan quantum"). Web and Scholar searches for the phrase quantum Ihara zeta, for Ihara zeta with noncommutative or quantum graphs (Weaver, Duan–Severini–Winter), for Ihara zeta with matrix product states or tensor networks, and a MathOverflow search for Ihara zeta and quantum expanders, returned nothing on point. The standing limit: arXiv search is metadata-only, so a formula inside a paper body would not surface, and the absence claims of observation 32.8 are weaker than its presence claims.

### 32.10 Corrections

One agent attributed arXiv:2309.15873 to Kudo and Li. The  $\text{\TeX}$  header of the local source, at 2309.15873:AiM\_Final.tex:86,90, gives the authors as Mason Eyler and Jaiung Jun; [25] and every citation of it were corrected on the evening of 2026-09-12.

## 33 The Continuous Dictionary

TJO's question: replace channels by Lindblad generators. The simplest continuous instance is the generator of rotations on a circle of circumference  $\log p$ , whose trace is a Poisson comb at multiples of  $\log p$ ; the Selberg analogue should be a Lindbladian built from the vector fields on  $\mathrm{PSL}_2(\mathbb{R})$ , with the Laplacian as the small operator, and a formal analogue of the comb. This section records what survives when the statements are written down and proved. The proofs are in `notes/selberg/astra-proofs.md`, authored by the second model family (codex, `gpt-6-astra`, reasoning effort xhigh) as a constructive prover from the brief `notes/selberg/astra-brief.md`, and reviewed adversarially by a Claude Opus refuter; the symbolic and numerical checks are in `scripts/selberg_lindblad.py`. Conventions: Definitions 4.18 to 4.23;  $\Delta = -y^2(\partial_x^2 + \partial_y^2)$  is positive;  $\Gamma$  is torsion-free and cocompact except in the modular computation.

### 33.1 Standard inputs

Six facts are used as assumptions, printed as such because no local source quotes them in full; every claim below that depends on one prints `-conditional`.

**Assumption 33.1** (Fuchsian geometry).

[assumed]

A torsion-free cocompact Fuchsian group acts freely and properly discontinuously on  $\mathbb{H}$ ; primitive hyperbolic conjugacy classes correspond to primitive oriented closed geodesics with period equal to translation length; hyperbolic balls have area  $2\pi(\cosh R - 1)$ .

**Assumption 33.2** (Laplace spectrum of a compact surface). [assumed]  $\Delta$  on a connected compact hyperbolic surface has discrete spectrum  $0 = \lambda_0 < \lambda_1 \leq \dots \rightarrow \infty$ , written  $\lambda_j = \frac{1}{4} + r_j^2$  with  $r_j \geq 0$  real for  $\lambda_j \geq \frac{1}{4}$ ,  $r_j = i\nu_j$  with  $0 < \nu_j < \frac{1}{2}$  for  $0 < \lambda_j < \frac{1}{4}$ , and  $r_0 = i/2$ .

**Assumption 33.3** (Selberg zeta is entire with known divisor).

[assumed]  $Z_{\mathrm{Sel}}$  extends

to an entire function with zero divisor  $\sum_{j \geq 0} ([\frac{1}{2} + ir_j] + [\frac{1}{2} - ir_j]) + (2g - 2) \sum_{n \geq 0} (2n + 1)[-n]$  and no other zeros or poles.

**Assumption 33.4** (Closed realisations).

[assumed] Complete volume-preserving flows

have skew-adjoint unitary generators, and  $\Delta$  on a compact manifold has its self-adjoint realisation and heat semigroup.

**Assumption 33.5** (Standard facts about  $\zeta$ ).

[assumed]  $\zeta$  continues meromorphically

with  $\zeta(0) = -\frac{1}{2}$ ,  $\zeta(1 + w) = 1/w + \gamma_E + O(w)$ , the functional equation  $\zeta(z) = \chi(z)\zeta(1 - z)$  with  $\chi(z) = \pi^{z-1/2}\Gamma((1 - z)/2)/\Gamma(z/2)$ , no zeros on  $\mathrm{Re} z = 1$ , and  $|\zeta'/\zeta(1 + \epsilon + iv)| \leq C \log^2(3 + |v|)$  for  $|v| \geq 1$ ,  $0 \leq \epsilon \leq \epsilon_0$ .

**Assumption 33.6** (Standard facts about  $\Gamma$ ).

[assumed]  $\Gamma$  is meromorphic

and zero-free with simple poles at the nonpositive integers,  $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ , and  $(\Gamma'/\Gamma)$  has the Gauss integral representation and  $O(\log)$  vertical growth in right half-planes.

### 33.2 Cited facts

*Cited fact 33.7* (Guillemin flat trace formula).

[cited] Dyatlov and Zworski

[23] state, for an Anosov flow,  $\mathrm{tr} \, e^{-itP} = \sum_{\gamma} \frac{T_{\gamma} \, \delta(t - T_{\gamma})}{|\det(I - \mathcal{P}_{\gamma})|}$ ,  $\forall t > 0$  (1306.4203:zeta.tex:260-264, sum over closed orbits including repetitions,  $T_{\gamma}^{\#}$  the primitive period). In our notation this is Definition 4.20.

*Cited fact 33.8* (Band structure of the resonances). [cited] Dyatlov, Faure and Guillarmou [22], Theorem 1: Assume that  $M$  is a compact hyperbolic surface ( $n=1$ ) and the spectrum of the positive Laplacian on  $M$  is (see Figure~\ref{f:dim2})  $\text{\Spec}(\Delta) = \{s_j(1-s_j)\}, \{q_j(1-q_j)\}, s_j \in [0,1] \cup \text{\Big}(\frac{1}{2} + i\mathbb{R} \text{\Big})$ . Then Pollicott-Ruelle resonances for the geodesic flow on  $SM$  in  $\mathbb{C} \setminus (-1 - \frac{1}{2} \mathbb{N}_0)$  are

$$\lambda_{j,m} = -m - 1 + s_j, \quad m \in \mathbb{N}_0.$$

lie on the lines  $\text{Re} = -\frac{1}{2} - k$  and the first band is in bijection with the Laplace spectrum (1403.0256:RuelleResonForHn).

*Cited fact 33.9* (Ruelle zeta is a Selberg quotient). [cited] Dyatlov and Zworski [24]:  $\zeta_S(s) := \prod_{\gamma \in \mathcal{G}} \prod_{m=0}^{\infty} (1 - e^{-(m+s)\ell_\gamma})$ ,  $\zeta_R(s) = \frac{\zeta_S(s)}{\zeta_S(s+1)}$ , (1606.04560:zazi.tex:117-119).

*Cited fact 33.10* (Selberg trace formula, hyperbolic term). [cited] Marklof [63]:  $\sum_{j=0}^{\infty} h(\rho_j) = \frac{\text{Area}(M)}{4\pi} \int_{-\infty}^{\infty} h(\rho) \tanh(\pi \rho) \, d\rho + \sum_{\gamma \in H_*} \sum_{n=1}^{\infty} \frac{\ell_\gamma}{\sinh(n\ell_\gamma/2)}$ , (math/0407288:selberg07.tex) the hyperbolic term over primitive classes with repetition index  $n$ .

*Cited fact 33.11* (Casimir and Laplacian). [cited] Fleig, Gustafsson, Kleinschmidt and Persson [28]: The Casimir operator~\eqref{eq:SL2cas} then becomes a second order differential operator, namely the Laplacian 
$$\Delta_{\text{SL}(2, \mathbb{R})} = y^2 \left( \partial_x^2 + \partial_y^2 \right) - y \partial_x \partial_\theta.$$
 (1511.04265:AutomorphicBook.tex:3107-3110), with the Casimir  $\frac{1}{4}h^2 + \frac{1}{2}(ef + fe)$  in the Chevalley basis (line 2851) reducing to  $y^2(\partial_x^2 + \partial_y^2)$  on  $\mathbb{H}$  (lines 409-411).

*Cited fact 33.12* (Scattering determinant of  $\text{PSL}_2(\mathbb{Z})$ ). [cited] Friedman, Jorgenson and Smajlović [29]: In the case when  $\Gamma$  is the modular group  $\text{PSL}(2, \mathbb{Z})$ , the scattering determinant is given by  $\phi(s) = \sqrt{\pi} \frac{\Gamma(s-1/2)}{\Gamma(s)} \frac{\zeta(2s-1)}{\zeta(2s)}$ , (1607.08053:main.tex:592-593).

*Cited fact 33.13* (Continuous-spectrum term of the trace formula). [cited] Momeni and Venkov [70]: 
$$C = C(\overset{\sim}{h}(r); \Gamma; \chi) = -\frac{1}{4\pi} \int_{-\infty}^{\infty} \frac{1}{\varphi'} \frac{1}{\varphi} \left( \frac{1}{2} + ir; \Gamma; \chi \right) h(r^2 + \frac{1}{4}) dr + \frac{1}{4} h(\frac{1}{4})$$
 (1108.5659:main.tex:2128-2129).

### 33.3 The circle comb and why it is blind to zeros

**Proposition 33.14** (One circle). [proved] For  $L > 0$  and  $U_t$  translation on  $L^2(\mathbb{R}/L\mathbb{Z})$ , the smeared operator  $\int h(t) U_t dt$  is trace class for Schwartz  $h$  and  $\text{Tr} U(h) = \sum_n \hat{h}(-2\pi n/L) = L \sum_k h(kL)$ ; hence  $\text{Tr} U_t = L \sum_{k \in \mathbb{Z}} \delta(t - kL)$  as a tempered distribution.

**Proposition 33.15** (All primes: an orbital trace, not a Hilbert-space trace). [proved] For the direct sum over primes of the circles of circumference  $\log p$ , the locally summed component traces give, in  $\mathcal{D}'((0, \infty))$ ,  $\sum_p \sum_{k \geq 1} \log p \delta(t - k \log p) = \sum_{n \geq 2} \Lambda(n) \delta(t - \log n)$ . The smeared direct-sum operator is never trace class (the constant modes give a nonzero eigenvalue of infinite multiplicity), and the positive prime comb is not tempered (its mass  $\sum_p \log p / (1 + \log p)^N$  diverges for every  $N$ ).

**Proposition 33.16** (Finite sets of primes see only Euler poles). [proved-conditional] For a finite set  $S$  of primes the generator of the direct-sum flow has spectrum  $\bigcup_{p \in S} (2\pi i / \log p) \mathbb{Z}$ , exactly the pole set of  $\prod_{p \in S} (1 - p^{-s})^{-1}$  in the spectral variable  $s$ ; the finite product has no zeros, all nonzero poles are simple, and no zero of the continued  $\zeta$  lies in this spectrum.

The correction to the worklog’s wording: the prime comb is an orbital (locally finite) trace, and the phrase “Poisson trace of the direct sum” does not name a trace-class operator. The missing ingredient is transverse expansion: a circle has no Poincaré map, so no Jacobian, so no resonances.

### 33.4 The Lindbladian of the vector fields is the Casimir, and the Laplacian on the $K$ -invariant sector

**Proposition 33.17** (Multiplication observables see a sum of squares). *[proved]* For real vector fields  $X_j$  acting as derivations of a smooth function algebra and jump operators  $L_j = -iX_j$ , on a common invariant domain,  $\sum_j (L_j M_f L_j - \frac{1}{2}\{L_j^2, M_f\}) = -\frac{1}{2} \sum_j [L_j, [L_j, M_f]] = M_{\frac{1}{2} \sum_j X_j^2} f$ ; with the plus sign, since the  $X_j$  are formally anti-self-adjoint.

**Proposition 33.18** (Casimir on the  $K$ -invariant sector). *[proved]* For right- $K$ -invariant  $f(g) = F(g \cdot i)$ :  $\mathbf{W}f = 0$ ,  $\Omega f = y^2(F_{xx} + F_{yy})$  (the constant is  $c = 1$ ), and  $\frac{1}{2}(\mathbf{H}^2 + \mathbf{E}^2)f = 2\Omega f = -2\Delta F$ . On all functions  $\frac{1}{2}(\mathbf{H}^2 + \mathbf{E}^2) = 2\Omega + \frac{1}{2}\mathbf{W}^2$ : the Casimir carries the compact direction with a minus sign and is not a diffusion Lindbladian on all of  $L^2(\Gamma \backslash G)$ ; the two-jump Lindbladian with  $L_1 = -i\mathbf{H}$ ,  $L_2 = -i\mathbf{E}$  preserves the  $K$ -invariant sector and equals  $-2\Delta$  there.

The second-order computation is the whole content: along  $g_{x,y} \exp(u\mathbf{H}) \cdot i = x + iye^{2u}$  and  $g_{x,y} \exp(u\mathbf{E}) \cdot i = x + y \tanh 2u + iy \operatorname{sech} 2u$ , one finds  $\mathbf{H}^2 f = 4y^2 F_{yy} + 4y F_y$  and  $\mathbf{E}^2 f = 4y^2 F_{xx} - 4y F_y$ , and the first-order drifts cancel. Squaring the projected tangent vectors alone would miss the drift and give the wrong operator.

**Proposition 33.19** (The quantum Lindbladian of a representation). *[proved]* For a unitary representation  $\pi$  with anti-self-adjoint generators  $A_j = d\pi(X_j)$  and jumps  $L_j = -iA_j$ , on finite-rank smooth operators,  $\sum_j (L_j X L_j - \frac{1}{2}\{L_j^2, X\}) = \frac{1}{2} \sum_j B_j^2 X$  with  $B_j X = [A_j, X]$  the generators of  $\pi \otimes \bar{\pi}$  on Hilbert–Schmidt operators; for  $\mathfrak{sl}_2$  and the two non-compact jumps it is  $2\Omega_{\pi \otimes \bar{\pi}} + \frac{1}{2}B_W^2$ , equal to  $2\Omega_{\pi \otimes \bar{\pi}}$  on diagonal- $K$ -invariant operators.

This is the continuous Harrow construction: a channel from a representation becomes a Lindbladian from a representation, and its gap is a question about  $\pi \otimes \bar{\pi}$ .

**Conjecture 33.20** (Gap of the representation Lindbladian). *[open]* For an irreducible unitary  $\pi$  of  $\mathrm{SL}_2(\mathbb{R})$ , or of  $\mathrm{SL}_2(\mathbb{F}_p)$ , the two-jump Lindbladian  $2\Omega_{\pi \otimes \bar{\pi}} + \frac{1}{2}B_W^2$  has a spectral gap governed by the decomposition of  $\pi \otimes \bar{\pi}$ ; for the Weil representation this is the continuous shadow of the Weil–LPS channels of Section 17.

## 34 The Continuous Dictionary II: Selberg Tower and the Cusp Comb

Continuation of Section 33, same conventions and the same proof file `notes/selberg/astra-proofs.md`.

### 34.1 The geodesic flow: continuous Ihara–Bass

**Proposition 34.1** (Poincaré Jacobian).

[proved]

For a closed geodesic of length  $\ell$  the linearised Poincaré map on  $\mathfrak{sl}_2/\mathbb{R}H$  is  $\text{Ad}(a_\ell^{-1})$  with multipliers  $e^\ell, e^{-\ell}$ , so  $|\det(1 - P_\gamma^k)| = (e^{k\ell} - 1)(1 - e^{-k\ell}) = 4 \sinh^2(k\ell/2) = e^{k\ell}(1 - e^{-k\ell})^2$ ; the circle has an empty transverse space and factor 1.

**Proposition 34.2** (Flat trace, Selberg tower, sign).

[proved-conditional] For

$\text{Re } \varsigma > 0$ ,  $\Lambda_{\text{fl}}(\varsigma) = \sum_\gamma \sum_{k \geq 1} \ell_\gamma e^{-\varsigma k \ell_\gamma} / (4 \sinh^2(k \ell_\gamma / 2))$  converges absolutely, the tower  $D_{\text{tow}}(\varsigma) = \prod_{j \geq 1} Z_{\text{Sel}}(\varsigma + j) = \prod_\gamma \prod_{m \geq 0} (1 - e^{-(\varsigma+m+1)\ell_\gamma})^{m+1}$  converges absolutely without zeros, and  $\Lambda_{\text{fl}} = +D'_{\text{tow}}/D_{\text{tow}}$ , equivalently  $D_{\text{tow}}(\varsigma) = \exp \left( - \int_0^\infty t^{-1} e^{-\varsigma t} \text{Tr}^b e^{-t\mathcal{X}} dt \right)$ .

The tower comes from  $1/(4 \sinh^2(x/2)) = \sum_{m \geq 0} (m+1) e^{-(m+1)x}$ : the Jacobian of the two transverse directions is what a graph does not have. The sign was drafted as a minus and is a plus; the graph counterpart of  $D_{\text{tow}}$  is  $\det(1 - uT)$ , not the zeta.

**Proposition 34.3** (Divisor of the tower and the first band). [proved-conditional]  $D_{\text{tow}}$  continues to an entire function whose zeros are  $\varsigma = -\frac{1}{2} - k \pm ir_j$  ( $k, j \geq 0$ ) and  $\varsigma = -N$  ( $N \geq 1$ ) with order  $(2g-2)N^2 + 2$ ;  $\Lambda_{\text{fl}}$  has simple poles there with residue the order. The nonconstant first band  $\mathcal{B}_0$  is exactly the pole multiset in  $-1 < \text{Re } \varsigma < 0$ , and  $\mathcal{B}_0 \subset \{\text{Re } \varsigma = -\frac{1}{2}\}$  iff every  $r_j$ ,  $j \geq 1$ , is real iff  $\Delta$  has no eigenvalue in  $(0, \frac{1}{4})$ .

The constant mode has  $r_0 = i/2$  and contributes  $\varsigma = 0$  and  $-1$ ; it must be removed, exactly as the constant adjacency mode is removed in the Ramanujan condition of a graph. With it removed, the continuous Ramanujan property is Selberg's  $\frac{1}{4}$  property, and the band structure is the one Dyatlov, Faure and Guillarmou prove for the operator-theoretic resonances (cited fact 33.8); Proposition 34.3 makes no claim about operator realisations, only about the poles of the continued  $\Lambda_{\text{fl}}$ .

**Proposition 34.4** (Graph comparison and the Ruelle quotient).

[proved-conditional] For a

finite graph with Hashimoto matrix  $T$ ,  $\text{Tr } T^m = \sum_\gamma \sum_{k \ell_\gamma = m} \ell_\gamma$  and  $\det(1 - uT) = \prod_\gamma (1 - u^{\ell_\gamma})$ , with no Jacobian and no tower, and  $\partial_\varsigma \log \det(1 - e^{-\varsigma} T) = \sum_m \text{Tr } T^m e^{-\varsigma m}$  has the sign of  $\partial_\varsigma \log D_{\text{tow}}$ . For the surface,  $\zeta_R(s) = Z_{\text{Sel}}(s)/Z_{\text{Sel}}(s+1)$  (telescoping the product) and  $D_{\text{tow}}(\varsigma)/D_{\text{tow}}(\varsigma+1) = Z_{\text{Sel}}(\varsigma+1)$ .

### 34.2 The cusp of the modular surface is the prime comb

**Proposition 34.5** (Exact Fourier identity for the scattering multiplier). [proved-conditional]  $m_{\text{sc}}$  is real, even, smooth and of polynomial growth, and

$$C_{\text{cusp}} = -P + A_{\text{arch}}, \quad P(t) = \sum_{n \geq 2} \frac{\Lambda(n)}{n} [\delta(t - 2 \log n) + \delta(t + 2 \log n)],$$

where  $\langle A_{\text{arch}}, h \rangle = \frac{1}{2} \int h - (\gamma_E + \log \pi) h(0) + \int_0^\infty \frac{e^{-u} h(0) - \frac{1}{2} e^{-u/2} (h(u) + h(-u))}{1 - e^{-u}} du$ , so  $A_{\text{arch}}(t) = \frac{1}{2} - 1/(4 \sinh(|t|/2))$  away from 0. The prime atoms are dips; the constant  $\frac{1}{2}$  is the pole of  $\zeta$  at 1 (the ordinary boundary value of  $\zeta'/\zeta(1+2ir) + \zeta'/\zeta(1-2ir)$  differs from its Abel boundary distribution by  $\pi \delta_0$ ), not a digamma term.

The route:  $\varphi'/\varphi(s) = (\Gamma'/\Gamma)(s - \frac{1}{2}) - (\Gamma'/\Gamma)(s) + 2\zeta'/\zeta(2s - 1) - 2\zeta'/\zeta(2s)$ ; the functional equation moves  $\zeta'/\zeta(2ir)$  to  $-\zeta'/\zeta(1 - 2ir)$  plus Gamma terms, and the  $(\Gamma'/\Gamma)(ir)$  terms cancel exactly, leaving  $m_{\text{sc}} = a + b$  with  $b(r) = \zeta'/\zeta(1 + 2ir) + \zeta'/\zeta(1 - 2ir)$ ; the Dirichlet series of  $b$  at  $1 + \epsilon$  transforms termwise to the comb, and the  $\epsilon \downarrow 0$  limit produces  $-\pi\delta_0$  on the frequency side.

*Numerical result* 34.6 (Symbolic and numerical checks).

[numerical]

`scripts/selberg_lindblad.py` (output `outputs/selberg_lindblad.txt`): exact sympy, Lindbladian  $= \frac{1}{2}X^2$ ; the left-invariant fields derived from the Iwasawa coordinates satisfy the  $\mathfrak{sl}_2$  brackets, and on  $K$ -invariant  $F$ :  $Hf = 2yF_y$ ,  $Ef = 2yF_x$ ,  $Wf = 0$ ,  $(H^2 + E^2)f = 4y^2(F_{xx} + F_{yy})$ , constant  $c = +1$ , Lindblad constant 2; quantum tensor identity to  $7 \cdot 10^{-15}$ , and for the three-dimensional  $\mathfrak{sl}_2$  irrep the Lindbladian spectrum is  $\{0, -4^{\times 3}, -12^{\times 5}\} = -2j(j + 1)$ ; Poincaré determinant  $2 - 2\cosh(k\ell)$ , absolute value  $4\sinh^2(k\ell/2)$ ; cusp transform with a Gaussian window ( $G_w = 40$ ,  $R = 6G_w$ ): dips at  $2\log n$  with amplitude ratio to  $\Lambda(n)/n$  of mean  $-31.91545$ , spread  $10^{-3}$ , against the predicted  $-2G_w/\sqrt{2\pi} = -31.91538$ ; residual after the exact archimedean background and the additive constant is  $3 \cdot 10^{-8}$  over  $t \in [0.5, 7]$ . The constant came out as 1 because the script's multiplier is  $-\text{Re}(\varphi'/\varphi)$ , twice  $m_{\text{sc}}$ .

**Proposition 34.7** (What the rate- $\frac{1}{4}$  damping identifies).

[proved-conditional] On

$(0, \infty)$ ,  $e^{-t/4}P_{\text{ch}} = P_+ = -(C_{\text{cusp}} - A_{\text{arch}})|_{(0, \infty)}$ . This is an equality of prime atomic measures after subtracting  $A_{\text{arch}}$  and choosing positive weights; it is not an equality of the full cusp distribution with a semigroup trace, nor of operators or spectra, and it implies nothing about the Riemann hypothesis.

The sentence in the previous session, “the cusp term of the modular trace formula is the trace of the Riemann channel damped at rate  $\frac{1}{4}$ ”, is therefore corrected: what is proved is that the prime measure written in the Riemann-channel note, damped at rate  $\frac{1}{4}$ , is the prime part of the cusp distribution. The full trace of  $Z(t)$  is not established as a pure prime comb (its own explicit formula has pole, archimedean and sign bookkeeping), and the rate  $\frac{1}{4}$  is the uniform rate only under the Riemann hypothesis.

### 34.3 The dictionary

*Observation* 34.8 (Continuous dictionary).

[sketched-conditional] Row by row: edge

space  $\leftrightarrow \Gamma \backslash G$ ; Hashimoto operator  $T \leftrightarrow$  geodesic generator  $\mathcal{X}$ ; adjacency  $A \leftrightarrow$  Casimir  $\Omega$ , equal to  $-\Delta$  on the  $K$ -invariant sector; Ihara–Bass  $\det(1 - uT) = (1 - u^2)^{|E| - |V|} \det(1 - uA + qu^2) \leftrightarrow$  the tower  $D_{\text{tow}} = \prod_{j \geq 1} Z_{\text{Sel}}(\zeta + j)$  with zeros in the bands  $-\frac{1}{2} - k \pm ir_j$ ; Ramanujan  $\leftrightarrow$  no eigenvalue in  $(0, \frac{1}{4})$ ; the primes of  $\zeta$  enter only through the cusp of the modular surface, as the scattering term, not as closed geodesics.

| discrete                       | continuous                                                       | where                      |
|--------------------------------|------------------------------------------------------------------|----------------------------|
| edge space $\mathcal{W}$       | $\Gamma \backslash \text{PSL}_2(\mathbb{R})$                     | Definition 4.18            |
| $T$                            | $\mathcal{X} = H/2$                                              | Definition 4.18            |
| $\text{Tr } T^m$ (no Jacobian) | $\text{Tr}^b e^{-t\mathcal{X}}$ , weights $1/4 \sinh^2$          | Propositions 34.1 and 34.4 |
| $A$ , $\Sigma = D\Phi$         | $\Omega$ , $-2\Delta$ on $K$ -invariants                         | Proposition 33.18          |
| $\det(1 - uT)$                 | $D_{\text{tow}}(\zeta)$ , sign $+D'_{\text{tow}}/D_{\text{tow}}$ | Proposition 34.2           |
| $\mu\mu' = q$                  | bands $-\frac{1}{2} - k \pm ir_j$                                | Proposition 34.3           |
| Ramanujan                      | $\text{spec } \Delta \cap (0, \frac{1}{4}) = \emptyset$          | Proposition 34.3           |
| mixed-unitary channel          | two-jump Lindbladian of $\pi \otimes \bar{\pi}$                  | Proposition 33.19          |
| primes as rings                | cusp term, dips $\Lambda(n)/n$ at $2\log n$                      | Proposition 34.5           |

### 34.4 Not established

A literal operator-theoretic compression from  $\Gamma \backslash G$  to the  $K$ -invariant sector that turns  $D_{\text{tow}}$  into a Laplacian determinant (the continuous Ihara–Bass as an identity of operators rather than of divisors); an anisotropic-space realisation of the trace resonances; a trace formula for the full compressed semigroup  $Z(t)$ ; any gap for conjecture [33.20](#); anything about the Riemann hypothesis or the  $\frac{1}{4}$  property of a given surface.

## 35 Open Problems and Dead Routes

Dead routes are results. Both lists mirror `HANDOFF.md`, which is the live version.

### 35.1 Open problems, in order

**Conjecture 35.1** (Ramanujan without unitarity). *[open]* *There is a Ramanujan-type reading of the poles of  $\zeta_T$  for non-unitary Kraus families, replacing the relation  $\mu\mu' = q$  of the unitary case, and canonical form  $\sum_k A_k^\dagger A_k = 1$  constrains the poles.*

What is known: Theorem 26.1 replaces  $qu^2$  by the operator  $\mathcal{D}(u)$  and  $u\Sigma$  by the deformed channel  $\mathcal{A}(u)$ , both built from  $(1 - u^2 \text{Ad}(A_k^\dagger A_k))^{-1}$ , so no quadratic relation pairs the eigenvalues of  $T$  with those of  $\Sigma$ . Cheap next step: a random-MPS pole-radius hunt (bond dimension 2 to 4, physical dimension 2) to see whether the poles cluster on one circle at all.

**Stinespring form of the compressed semigroup.** Conjecture 11.10 (stated in Section 11): a Stinespring form of  $Z(t)$  on  $K_S$  whose Kraus (jump) operators are the prime dilations.

The single-Kraus lift exists trivially. A Holevo-form covariant Lindbladian with the von Mangoldt jump measure lives on the full regular representation, has symbol  $\zeta^t$ , and is not the compressed object. A negative answer with a reason is a result. Suggested route (HANDOFF step 2): write  $Z(t)$  explicitly through Cauchy kernels and test a generator of Holevo jump form with jumps at  $k \log p$  on  $K_S$ , blind, in a second model family.

**Hecke identification of the Weil–LPS spectra.** Conjecture 17.4 (stated in Section 17): the joint spectra of the commuting channels  $\Phi_q^\pm$  are Hecke eigenvalues of weight-2 forms for the quaternion algebra ramified at  $\{2, \infty\}$ , via Jacquet–Langlands.

Needs LMFDB and the Jacquet–Langlands bookkeeping; the visible structure (joint eigenvalue pairs, small-degree algebraic integers, multiplicities equal to those of irreducibles of  $\text{PSL}_2(\mathbb{F}_p)$  in  $W_\pm \otimes \overline{W}_\pm$ ) is consistent with it.

**Conjecture 35.2** (Assembling over  $p$ ). *[open]* *The Weil–LPS channels for different  $p$  assemble into one object whose rings are the primes and whose transfer spectrum is the zeros of  $\zeta$ .*

This is the parent campaign’s DG-GLOBAL question in this guise. Nothing is known beyond the obstruction recorded below.

**Conjecture 35.3** (The Phantasm has both halves). *[open]* *There is a  $\mathbb{Z}_2$ -graded Lindbladian (or a type III generalisation at  $\beta = 1$ ) whose stationary state is the Bost–Connes KMS state, with entanglement Hamiltonian  $\log N$ , and whose odd relaxation modes are the nontrivial zeros: the object having both halves of observation 13.17. The rebound-state equation of the SHW renewal picture with a mixed rebound state is its finite-dimensional shadow; the vacuum rebound state is excluded by Proposition 13.8.*

This replaces item 0b of the reframe programme. The fixed point constrains the Lindbladian only partially, and at  $\beta = 1$  the fixed point is not normal (Proposition 13.10), so  $K_S$  with its pure vacuum cannot be the whole bond; the identification of  $K_S$  with the Bost–Connes bond that Section 11 records as missing is the first obstacle. A second open item from the same campaign is the unrestricted infinite parity question, observation 12.13.

**Conjecture 35.4** (Galois-graded bond at finite level).

[open] The model space of a congruence subgroup  $\Gamma_0(N)$ , whose scattering determinant involves the Dirichlet  $L$ -functions mod  $N$ , carries the zeros of  $L(s, \chi)$  organised by the characters  $\chi$  of  $(\mathbb{Z}/N)^\times$ , with the Shor map  $U_a$  as the level- $N$  Galois symmetry acting on the bond; the graded bond of observation 13.18 is realised at finite level.

This is a computation, not yet done; it is the Riemann-side test of the trivial-versus-nontrivial-character reading of the grading that the Artin–Schreier super-transfer matrix exhibits.

## 35.2 Dead routes

*Observation 35.5* (Prime-by-prime ansätze are blind to zeros).

[sketched] The direct sum over primes of rotations on circles of circumference  $\log p$  has trace  $\sum_n \Lambda(n) \delta(s - \log n)$  by Poisson summation and dynamical zeta equal to the Euler product; for any finite set  $S$  of primes its spectrum is the set of poles of the partial Euler product. The same holds for product states and commuting dilations. Zeros exist only in the infinite product, so no prime-by-prime ansatz can see them.

Recorded 2026-09-12 from the circle-flow discussion. Two remarks kept: in the Witt convention  $L_1 - L_{-1}$  is the dilation flow (source and sink) and  $L_0$  the rotation; the only relation between primes is additive.

*Observation 35.6* (Symmetry protection fixes signs, not moduli).

[sketched] The content of an  $L$ -function that has the three properties of a symmetry-protected topological invariant (a phase, quantised, robust under symmetric perturbation) is its sign data: the root number as a product of local  $\epsilon$ -factors with the product formula as anomaly cancellation, the sign of a quadratic Gauss sum (the Arf invariant over  $\mathbb{F}_2$ , the Gauss–Milgram central-charge phase over  $\mathbb{Z}/p$ ), and Hilbert–symbol reciprocity as a local-to-global cocycle condition. The Riemann hypothesis is modulus data. Moduli are not protected: a Ramanujan graph ceases to be Ramanujan under a small symmetric perturbation, and the cubic Cayley graphs of  $\mathrm{PSL}_2(\mathbb{F}_p)$  on the generators of  $\mathrm{PSL}_2(2, \mathbb{Z})$  fail the Ramanujan bound for  $p \geq 11$  while keeping every symmetry. Cohomological protection therefore cannot deliver RH.

Recorded 2026-09-12 from the SPT side quest. What survives of the bulk–boundary lesson is the positivity form of Weil’s criterion: the reflection  $\rho \mapsto 1 - \bar{\rho}$  pairs the mode at a zero with the mode at its mirror image, RH says every mode is its own partner, and Weil’s form is the pairing of each mode with its partner, written through the explicit formula as a form on the primes. Its finite-dimensional, arbitrary-Kraus version is the subject of Section 27. Two relabellings, not new content: the Weil–LPS channels live on the edge Hilbert space of the  $\mathbb{Z}/p \times \mathbb{Z}/p$  cluster-state phase (the Heisenberg group is that edge’s projective representation and  $\mathrm{SL}_2(\mathbb{F}_p)$  its automorphism group, acting by the Weil representation); and Bost–Connes at  $\beta = 1$  is a critical point of a spontaneous breaking of the Galois symmetry, not a symmetry-protected phase.

Further routes closed or flagged in the founding session: the functional integral framing of Selberg as a continuum limit (TJO: too far); calling  $-\log E$  the entanglement Hamiltonian (an error, corrected in observation 10.7); an algebraic-integer recognition of the Weil–LPS joint eigenvalues from five-digit output (tautological, since the adjacency matrix is integral); the Dwork transfer operator as a route to the Riemann hypothesis for curves (observation 18.4).

## 36 Reproducibility Map

Every numerical claim in this book points to a script under `scripts/` whose standard output is captured under `outputs/`. All scripts run from the repository root with `python3` and need only `numpy`, `scipy`, `mpmath` and `sympy`. Random data are seeded. The local CI re-runs the fast scripts and diffs their output against the captured files.

| script                    | what it establishes                                                                                                                                                                                           | output, claims                                                                                              |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| qihara.py                 | Quantum Ihara–Bass identity for Haar-random unitary Kraus sets, $(n, D) = (3, 4), (4, 6)$ , at complex $u$ ; Ramanujan check of $\Sigma$                                                                      | outputs/qihara.txt;<br>num:qihara-unitary                                                                   |
| bcmpo.py                  | Bost–Connes phase operators $e(a/b)$ as MPOs over the prime chain against Hurwitz-zeta sums; character-sector transfer eigenvalues against $L(\beta, \bar{\chi})/\zeta(\beta)$                                | outputs/bcmpo.txt;<br>num:bc-mpo                                                                            |
| scat.py                   | Unimodularity of $S(\tau)$ , its zeros at $\gamma_n/2 - i/4$ , contractivity in the lower half plane, prime-phase identity                                                                                    | outputs/scat.txt;<br>num:scattering-symbol                                                                  |
| ringnorm.py               | Windowed sum over the first 3000 zeros against the full explicit formula; dips at $\log n$ of depth $\Lambda(n)/\sqrt{n}$                                                                                     | outputs/ringnorm.txt;<br>num:ringnorm-3000                                                                  |
| weil_lps.py               | Weil–LPS channels for $(p; q)$ in seven cases: intertwining check, Ramanujan bound, Harrow containment, Hecke relations, commutation, joint spectra                                                           | outputs/weil_lps_p*.txt;<br>num:weil-lps-ramanujan                                                          |
| weil_lps_hashimoto.py     | Direct edge superoperator for $(13, 17)$ , odd block: Ihara–Bass to $10^{-14}$ , moduli $\{17, \sqrt{17}, 1\}$                                                                                                | outputs/weil_lps_hashimoto_p13_q17_odd.t<br>num:weil-lps-rh                                                 |
| artin_schreier_mps.py     | Brute-force exponential sums against $\text{Tr } E^n$ for quadratic Artin–Schreier data; $EE^\dagger = qI$                                                                                                    | outputs/artin_schreier_mps.txt;<br>num:artin-schreier-mps                                                   |
| qihara_general.py         | Theorem 26.1 in both forms, five configurations, float to $10^{-15}$ and exact rational arithmetic                                                                                                            | outputs/qihara_general.txt;<br>num:qihara-general                                                           |
| artin_schreier_super.py   | Sign law, endpoint criteria, corrected Frobenius blocks, super-transfer supertraces against point counts, symplectic matrices and the Weil character formula, Prony on the counts; 1969 checks                | outputs/artin_schreier_super.txt;<br>num:artin-schreier-super                                               |
| riemann_graded_trace.py   | Windowed supertrace of the Riemann graded generator against $2P_+$ with 3000 zeros; ladder alternatives; flipped-pair negativity; finite vacuum-decay spectra; prime-chain M/M/1 bands                        | outputs/riemann_graded_trace.txt;<br>num:riemann-graded-trace                                               |
| bc_entropy.py             | Entropy of the Bost–Connes Gibbs density and its prime and bond cutoffs; Hagedorn level count                                                                                                                 | outputs/bc_entropy.txt;<br>num:bc-entropy                                                                   |
| cmps_parity_supertrace.py | Explicit Jordan–Wigner Fock vectors of lattice boson–fermion MPS rings against the overlap, parity and twisted-supertrace formulas; translations; continuum Dyson, Laplace and Frullani identities; 89 checks | outputs/cmps_parity_supertrace.txt;<br>num:cmps-parity-supertrace                                           |
| weil_ring_norm_check.py   | Elliptic-curve counts as twisted ring norms of the $\mathbb{C}^{1 1}$ tensor, explicit vectors, Hilbert–Schmidt chain, genus-two searches, the $y^3 - y = x^2$ indicator state                                | outputs/weil_ring_norm_check.txt;<br>num:weil-ring-norm-check                                               |
| ring_norm_tensor.py       | Toral Frobenius counts, Lefschetz, Fourier orbits, Lenstra structure, refutation of the drafted trace bound, structured genus-two solutions, supersingularity of the quadratic family, cut ranks (562 s)      | outputs/ring_norm_tensor.txt;<br>num:ring-norm-tensor,<br>num:cut-rank-binary,<br>num:genus-two-cm-diagonal |
| ring_norm_certificate.py  | Exact rational contradiction certificate of the $\mathbb{F}_5$ three-letter genus-two tensor; nine-letter tensor at $q = 25, 17, 16$ ; Jordan–Wigner norms                                                    | outputs/ring_norm_certificate.txt;<br>num:ring-norm-certificate                                             |
| ring_norm_examples.py     | One explicit MPS per genus 0, 3; spec                                                                                                                                                                         | outputs/ring_norm_examples.txt;                                                                             |

## Data

`data/zeros3000.npy`: the first 3000 imaginary parts of the zeta zeros, from `mpmath.zetazero`.  
`data/ringnorm.npz`: the arrays of the ring-norm test.

## Reference sources

`refs/fetch_sources.sh` downloads the arXiv TeX source of every cited paper into `refs/src/<id>/` (not redistributed) and writes `refs/manifest.sha256`. Every quote in `db/provenance.tsv` is byte-checked by the gate at its stated `<id>:<file>:<line>`.

## Reviews

`notes/reviews/` holds the adversarial review reports; a `proved` claim names its review file in `db/claims.tsv` and the gate checks that the file contains `VERDICT <id>: VALID`.

## 37 Lab Log

The dated worklogs under `docs/worklog/` are the primary chronology; `HANDOFF.md` is the live state. This section is their mirror at the level of decisions.

### 2026-09-10/11: founding session

One conversation in the parent repository `arithmetic-quantum-mechanics`. Trajectory: Bost–Connes basics; why not Wick-rotate to Hilbert–Pólya; an elliptic curve over  $\mathbb{F}_2$  as a finite instance; permutations and the one determinant identity; automata and rationality; the actual Riemann hypothesis as a flow with orbit periods  $\log p$ ; the Ihara zeta of regular graphs as the vestigial Selberg zeta; Shor’s unitary as the level- $N$  Galois symmetry of Bost–Connes; quantum expanders and the quantum Ihara zeta; matrix-product states and the ring-norm reading; the Bost–Connes system as a matrix-product operator over the prime chain; the Holevo–Werner lift; and the computation of the scattering symbol, its zeros, and the 3000-zero ring-norm test.

Corrections TJO made along the way, kept as working rules: stay elementary, no jargon dumps, one step at a time; no curves or field extensions when a permutation will do; “QFT” did not mean functional integrals; length quantisation should come from a flatness-type condition.

Afternoon: the Weil–LPS channels computed for seven primes. Bugs found and fixed: quaternion enumeration starting at an odd coordinate (caught by the divisor-sum count); parity as a second fixed point on the full space (resolved by splitting into the two Weil blocks); the Hecke relation failing on  $SL_2$  by signs (resolved on  $PSL_2$ ).

Evening: the matrix-product formulation of the Weil conjectures for quadratic Artin–Schreier curves. Bugs: a wrong irreducibility test for composite  $n$ ; the even- $n$  sign, which was the  $L$ -function normalisation, not a bug.

### 2026-09-12

Morning: circle flows discussed (no code); the expository note on Deligne’s proof written; repository made public under AGPL-3.0.

Afternoon: prior-art search on the quantum Ihara zeta, three Opus literature agents, then byte-verification against arXiv TeX sources (`refs/`). Finding: the object and its determinant formula are known (Matsuura–Ohta; Watanabe–Fukumizu for loopless graphs; Sunada); the RH-for-channels reading, the MPS dictionary and the Weil-representation instance were not found. Then the Ihara–Bass formula for arbitrary Kraus families proved in Lamport style and taken through two rounds of adversarial review by Opus reviewers; promoted to **proved**. Standing preference recorded: TeX source, not PDF, for every quotation.

Evening: this lab book set up following the parent repository’s sharded pattern, with the notation, definitions, claims and provenance databases under `db/`, the generated macro and status files, the gate `scripts/labbook_check.py`, the local CI `scripts/ci_local.sh` and its pre-commit hook. Notation and definitions across the seven notes de-duplicated into the databases; the clashes and their resolutions are listed in Section 2.

Night: the Selberg dictionary campaign (codex prover, Opus refute review, shards 09b/09c), and the Weil-positivity campaign of the second afternoon session (shards 08b/08c). Evening: TJO set the Bost–Connes reframe as the central priority and named the chased object the Phantasm.

## 2026-09-12, night: the Phantasm campaign

Orchestration: one codex prover session (gpt-6-astra, xhigh; brief `notes/riemann-cmps/astra-brief.md`, proofs `notes/riemann-cmps/astra-proofs.md`, 1356 lines, labels T1–T8), three Opus lanes in parallel (Artin–Schreier numerics, Riemann-side numerics, TeX sources with 57 byte-verified quotes and 17 new arXiv sources), then an Opus REFUTE review. Results: Section 12, Section 13, Section 19, the definitions shard Section 5, and the restatements of Proposition 11.5 (HANDOFF item 1c) and Proposition 18.1. Corrections along the way: the founding session’s  $\alpha_i = -\lambda_i(E)$  was found false by the numerics lane; the orchestrator’s replacement (a conjugated law) was found false by the same lane in two further cases; the prover produced the exact law (radical sign, Stickelberger on the complement) and the numerics lane confirmed it in 624 further checks. TJO’s steering during the campaign: the fixed point of the bond Lindbladian is literally the entanglement spectrum, so the Gibbs entropy and its cutoffs are entanglement data (recorded; the Hagedorn level density rules out a CFT reading); and the Phantasm must carry the Bost–Connes symmetry  $\hat{\mathbb{Z}}^\times$ , which led to the trivial-versus-nontrivial-character reading of the grading and to conjecture 35.4.

## 2026-09-13: finite BC symmetry constraints

TJO clarified that the BC state is the prescribed stationary state of an unknown Lindbladian, constrained by Galois symmetry and adelic symplectic structure. The finite inverse problem was formulated as a linear section of the conditional Choi cone in Section 14. At criticality its dimensions were computed for Galois, linear Weil, and additional Weyl covariance; the last must be distinguished from merely carrying the symplectic representation. The finite BC phase marginal does not specify a matrix-state extension, and an exact finite representation of the full BC algebra cannot retain nontrivial phase observables.

An explicit  $p = 7$  covariant GKLS generator has unequal odd decay rates. Compatible two-prime models retain a free coupling even with fixed local semigroups. The numerical script passes 427 checks and is included in local CI. Full written derivations were supplied by Codex but no independent review was performed, so the eight new analytic claim rows are **sketched**; the ninth is **numerical**. The arithmetic map to the scattering bond and the adelic limit remain open.

## 2026-09-13: the fermion parity of a cMPS ring

TJO was cautious about reading the ring norm of a boson–fermion cMPS as a supertrace, asked whether it had been proved (it had not: the book only cited the lattice fermionic-MPS contraction), and asked for a direct, rigorous calculation of the expectation value of  $(-1)^F$  ending in a supertrace and a superdeterminant. Done alone by Opus at TJO’s instruction, without subagents, in Sections 15 and 16. Outcome: for arbitrary data the parity expectation is a trace of the sign-twisted generator; with a bond grading it is identically  $\pm 1$ ; the supertrace, graded on the doubled bond, is the norm of the ring closed with the bond parity (the periodic fermion ring), and its ring-length zeta is  $1/\text{sdet}$ . Unreviewed, so the rows are **sketched**; the Jordan–Wigner script passes 89 checks.

## 2026-09-14: two Lindblad papers, mixing times

At TJO’s request two Opus readers read arXiv:2609.13121 (Becker–Zworski, optimal relaxation of Witten Lindbladians) and arXiv:2609.12284 (Shang, a dissipative quantum linear-system solver) from TeX; reading notes are in `notes/extract/`. Neither bears directly on Ramanujan properties, cMPS or KMS states. Both readers independently argue that the compressed Riemann semigroup

has operator norm one for all times, independently of RH, so a mixing-time form of the Ramanujan statement must restrict to regular data. This is unreviewed and not registered as a claim.

## 2026-09-14: the ring-norm-tensor campaign

TJO asked for an educated guess for the MPS tensor of the absolute simplest variety without an obvious Polyá–Hilbert Hamiltonian, with the wish list (1) natural tensor, (2) graded decomposition and why, (3) ring norms equal counts, (4) the PH operator, (5) Ramanujan, all modulo gauge; a codex `gpt-6-astra` prover, an Opus numerics lane, an Opus sources lane and an Opus REFUTE reviewer. Morning: the previous session’s genus-one ring-norm claim was double-checked on real curves and extended (genus two needs fermionic letters; found numerically). The campaign read “obvious” as the quadratic Artin–Schreier unitary, proved that family supersingular, and took the ordinary elliptic curve as Tier A (lifted Frobenius on the flat torus: count = Lefschetz number, grading = form degree, ket/bra = Hodge decomposition, PH unitary = pullback on harmonic forms, Ramanujan = degree of the lift, gauge = lattice basis / ideal class) and genus two as Tier B (block equations, the vacuum ansatz impossible, an explicit nine-letter tensor for  $q \geq 16$ , a rationally certified three-letter tensor over  $\mathbb{F}_5$ , Rosati as the PH metric). Correction recorded: the orchestrator’s drafted Hilbert–Schmidt/trace bound was false; the genus bound was re-proved by the all-power word-trace inequality. Shards 02g, 06c–06f; `docs/worklog/2026-09-14.md`.

Evening: TJO asked how hard it is to write down the actual matrices for one curve of each genus, with entanglement properties and parent Hamiltonians. Shard 06g: genus 0 to 3 explicit (the genus-three example over  $\mathbb{F}_{37}$  needs only three brute-force counts), correlation lengths ( $2/\log q$  for every odd mode is RH), block Schmidt spectra from the doubled bond, and parent Hamiltonians whose periodic-fermion ground state is selected by a parity-twisted boundary term.

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